

GAUGE AND GRAVITATIONAL RESPONSES OF A FIELD

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ABSTRACT. A field configuration is ordinarily specified as a section or connection only after its base and bundle have been fixed. A field source $\mathfrak{R}$ combines a local algebra, its restrictions, derivations, and Dirichlet datum with a finite-projective metric internal module, its declared gauge group, and finite assignments of tensor types and roles. In a Riemannian bundle realization, the Dirichlet datum determines the base metric and its torsion-free connection, while the internal module becomes a finite-rank bundle N with its fiber metric. The realized configuration space consists of pairs

$$\Psi = (\nabla, K), \quad K \in \Gamma^s(\text{Sym}^2 T^*M \otimes N),$$

where ∇ is a metric connection on N . Together with the base connection, each pair forms a single metric connection on $TM \oplus N$, whose off-diagonal blocks K and $-K^\dagger$ are forced by metricity. Its squared curvature defines the positive energy Φ of the resulting field $\mathfrak{F}(\mathfrak{R})$ at configuration Ψ , and the differential of that energy has the unique metric-dual response

$$D\Phi_\Psi(v) = G_\Psi(\mathcal{Y}_\Psi, v), \quad \mathcal{Y}_\Psi = (\mathcal{Y}_{\nabla, \Psi}, \mathcal{Y}_{K, \Psi}).$$

The internal rank fixes which response types the field possesses, and the configuration fixes their values. The connection component is the gauge response and exists from rank two onward; the K -variation response exists at every rank, and on the oriented rank-one class with positive Riemannian base metric it is the gravitational response, with the scalar Gauss and one-form Codazzi contractions of the assembled curvature as its readouts. Because both responses are variations of one energy, their mixed second variations agree as continuous weak bilinear forms on the stated Sobolev branch.

The same response is exact on phase space, as the Hamiltonian derivation of the signed Legendre function $H_R = T_{\text{kin}} - \Phi$, and at each finite label, as the common-core commutator with its symmetrically ordered constraint in a gauge-compatible representation. Where the curvature-energy motion form is positive definite it fixes the Cameron–Martin norm of a nonempty quasifree class, and the adjoint of the core constraint yields the maximal weak constraint H_* and the positive Master operator $H_*^* H_*$ with no chosen self-adjoint extension. In a flat-torus model, locally gauge-equivalent flat configurations are distinguished by a weighted heat trace of a translation-invariant restriction of the curvature Hessian; its holonomy dependence has nonzero derivative along the preserved rank-one gravitational summand. The bounded images of the source operations generate the smallest unital C^* -algebras containing bounded functions of the Master operator, rank-one curvature contractions, bounded cylinder functions, and positive products of charge-sector transitions.

Independently specified indexed operational data recover $\mathfrak{R}$ up to structural equivalence, and the recovered source determines the field and its law. States and representations, clocks and temporal developments, measurement models, and empirical outcomes remain additional data.

1. INTRODUCTION

Gauge theory and gravitational initial-data geometry differ in what they take to be prior to variation. Gauge theory begins with an internal connection on a chosen base, while gravitational initial data begins with a metric and a second fundamental form, as in the ADM formulation [1]. Yet locality, internal directions, admissible configurations, and the governing law remain distinguishable structures in both descriptions. The problem considered here is to determine which of these structures must be supplied independently and which follow from data recording local variation, internal organization, and their structural roles.

The relation between geometry and internal connection has taken several forms. Weyl attached electromagnetism to a scale connection [50]; Kaluza–Klein theory obtained an internal connection from a higher-dimensional metric [27, 29]; gauge formulations of gravity represented gravitational variables by connections on a given manifold [49, 28, 21, 2]; and larger-connection constructions placed gravitational and internal variables in different blocks of one connection [35, 37, 34]. Spectral geometry instead recovers a Riemannian manifold from algebraic and spectral data and derives gravitational and internal terms from the spectral action [12, 10]. Each program establishes a different dependence among metric, connection, internal structure, and field content. The result developed here begins from local, variational, internal, and finite assignment data from which the geometric realization, admissible field configurations, and response law are determined.

Several established results recover geometric or internal structure from data specified before a field configuration. A commutative algebra represents functions on its spectrum [18]. A regular strongly local Dirichlet form defines an intrinsic metric and, under the corresponding compatibility and completeness hypotheses, a length and geodesic structure [45, 44], while its first-order module records directions of variation [11]. Finite projective modules realize vector bundles through the Serre–Swan correspondence [42, 46], and categorical sector data recover compact gauge structure [15]. The present work places these results under common localization, restriction, regularity, and compatibility data, determines the field and its response from their combined output, and proves recovery of the primitive data from independently specified finite and local data.

The primitive object is a source presentation

$$(1) \quad \mathfrak{R} = (\mathcal{A}, \Lambda, \text{Der}(\mathcal{A}), \mathcal{E}, \mathcal{N}_g, \mathcal{G}_{\mathcal{N}_g}, \vartheta).$$

The commutative algebra $\mathcal{A}$ represents local quantities, and Λ specifies their localization and restriction. The module $\text{Der}(\mathcal{A})$ gives their directions of variation, while the Dirichlet datum $\mathcal{E}$ supplies the corresponding energy, measure, positive local integration functionals, and regularity scale. The finite-projective metric module $\mathcal{N}_g = (\mathcal{N}, g_{\mathcal{N}})$ supplies the internal directions and their metric, and $\mathcal{G}_{\mathcal{N}_g}$ is the declared subgroup of metric-module automorphisms. The finite datum ϑ labels the tangential and internal factors, specifies the permitted tensor expressions, and records the paired assignments used by the subsequent constructions. Together these seven components constitute the source presentation $\mathfrak{R}$. When they satisfy the regularity and compatibility conditions of Section 3, $\mathfrak{R}$ is a field source and determines the full system $\mathfrak{F}(\mathfrak{R})$, while a field configuration is a point of $\mathcal{C}^s(\mathfrak{R})$.

Suppose that the localization data are spatial, the algebra is smoothable, the carré-du-champ pairing is nondegenerate, and the metric internal module admits

its Serre–Swan realization. The Dirichlet datum then determines the base metric h , and the Koszul formula determines its torsion-free metric connection. The module $\mathcal{N}_g$ becomes a finite-rank vector bundle $N \rightarrow M$ with fiber metric g_N , while the declared gauge group acts by metric bundle automorphisms. Thus the Riemannian base and the internal bundle realize structures already determined at the algebraic and module levels of the source.

On this realization, $\mathfrak{R}$ determines the affine configuration space

$$\mathcal{C}^s(\mathfrak{R}) = \{\Psi = (\nabla^{\mathcal{N}}, K)\}, \quad K \in \Gamma^s(\text{Sym}^2 T^*M \otimes N),$$

where $\nabla^{\mathcal{N}}$ is a metric connection on N . Together with the source-determined torsion-free connection on TM , each pair forms one metric connection on $TM \oplus N$. Metricity fixes its off-diagonal blocks as K and $-K^\dagger$. The resulting curvature has tangential, mixed, and internal components, and its squared norm defines a positive curvature-energy functional Φ .

The source also determines a configuration-space metric G on the connection and K -variation spaces. For every relatively compact smooth domain U and integer $s > \dim(U)/2 + 2$, the differential of Φ has a unique L^2 -Riesz representative

$$(2) \quad D\Phi_\Psi(v) = G_\Psi(\mathcal{Y}_\Psi, v), \quad \mathcal{Y}_\Psi = G_\Psi^{-1} D\Phi_\Psi.$$

Here v is an admissible variation, $D\Phi_\Psi$ is the resulting variational covector, and $\mathcal{Y}_\Psi$ is its metric-dual response. The representatives are compatible under relatively compact inclusions, so the assignment $\Psi \mapsto \mathcal{Y}_\Psi$ is the response law of the field $\mathfrak{F}(\mathfrak{R})$. Its zeros are the stationary configurations of the curvature energy.

The orthogonal decomposition of the variation space gives

$$\mathcal{Y}_\Psi = (\mathcal{Y}_{\nabla, \Psi}, \mathcal{Y}_{K, \Psi}).$$

The connection component is the gauge response, namely the metric dual of the curvature-energy variation in the internal-connection directions, while the second component is the corresponding response in the K -variation directions. Internal rank determines the dimensions of these variation spaces. On a connected domain of constant positive internal rank, $\mathfrak{so}(1) = 0$ eliminates the connection-variation summand at rank one and leaves the K -variation summand. On the oriented rank-one branch with positive Riemannian base metric, the surviving component is the gravitational response, whose associated geometric quantities are the scalar Gauss and one-form Codazzi contractions of the assembled curvature. From rank two onward both variation summands have positive local rank, although either response value may vanish at a particular configuration. Under the Sobolev hypotheses of Theorem 5.3, the two mixed second variations agree as continuous weak bilinear forms. Rank therefore determines which response components exist, while the field configuration determines their values.

The source determines this response independently of its phase-space and quantum expressions. On each local or cylinder label, the kinetic pairing and curvature energy determine the signed Legendre function

$$H_R = T_{\text{kin}} - \Phi$$

and its Hamiltonian derivation. Where the curvature-energy motion form is positive definite, it supplies the Cameron–Martin norm for a nonempty class of quasifree representations. Within a gauge-compatible representation, total symmetrization is the unique finite-label quantization of real polynomials of degree at most four

satisfying the leading-symbol and linear symplectic-equivariance conditions. The resulting symmetric constraint implements the quantized phase-space derivation by commutators on its common core. The adjoint of that core constraint gives the maximal weak constraint, and its positive product gives the Master operator. Each phase-space, commutator, and operator-theoretic expression retains its specified domain [8, 38, 22, 48].

At every stable critical configuration, the curvature Hessian determines a nonnegative self-adjoint operator on the Hilbert-space quotient by the closed infinitesimal gauge directions. In the flat rank-three torus model of Section 9.5, this quotient has an infinite-dimensional kernel remaining after gauge reduction, while the rank-one tangential factor of the nonzero shape tensor determines a translation-invariant reducing subspace. On that subspace, $P_\eta L_{\theta,k} e^{-tL_{\theta,k}}$ is trace class, and its trace distinguishes flat configurations that are gauge equivalent on every contractible region but globally holonomy distinct. Its holonomy dependence has a nonzero derivative under variation within the preserved rank-one gravitational summand. The Hessian therefore retains global information that is absent from the local curvature of the background configuration.

The bounded observable content is determined componentwise by declared operations and norm closure. On represented and local-operation components, their bounded images generate the unique smallest unital C^* -subalgebras of the specified operator algebras. Transition objects carry the corresponding generated subalgebras of their endomorphism algebras, and source isomorphisms transport the completed category of charge sectors and transitions. Graph-norm commutator ideals and finite operator families depend on separately specified represented operators and their common domains. These distinctions identify the bounded observable algebras determined by the field source and the represented operator data required for the additional constructions.

A system of indexed operational data consists of local operation algebras and their restrictions, positive quadratic forms and integration functionals with compatible carré-du-champ data, rigid metric tensor categories with finite-projective probe functors and coherent restrictions, and finite structural assignments distinguishing laws from variable inputs. Under conditions (R1)–(R9), the forward and recovery functors give an equivalence between the groupoid of regular source presentations and the groupoid of regular indexed data,

$$\mathbf{S} \simeq \mathbf{B}.$$

For the field-source subgroupoid, operational determination therefore has the factorization

$$\mathcal{B} \xrightarrow{R} \mathfrak{R} \xrightarrow{\mathfrak{F}} \mathfrak{F}(\mathfrak{R}).$$

The first arrow recovers $\mathfrak{R}$ up to structural equivalence, and the second determines $\mathfrak{F}(\mathfrak{R})$, whose field law contains the response and source-generated bounded observable algebras.

Within the fixed seven-datum source class, the source-incidence matrix has full column rank, and deleting any one component removes an incidence required by the field law. Each source component therefore has a determined downstream role. A field configuration selects a point of the source-determined configuration space, a representation realizes an observable algebra on a Hilbert space, a state assigns

expectation values, and a clock together with a temporal development orders configurations. Measurement models and empirical maps supply the further data required to connect the intrinsic law with empirical outcomes. Source determination reaches $\mathfrak{F}(\mathfrak{A})$ and its field law, while these assignments retain their distinct mathematical roles.

The paper follows this order of determination. Section 2 compares its primitive data and logical direction with existing programs. Section 3 defines the source class and states the main theorem, and Section 4 derives the configuration space, assembled connection, curvature, and positive energy. Section 5 proves the response theorem, followed by the rank-one gravitational response in Section 6. Section 7 establishes the phase-space and quantum expressions, and Section 8 determines the source-generated bounded observable algebras and the combined field law. Section 9 then realizes the response and operator-theoretic conclusions in explicit compact models, including the global distinction of locally gauge-equivalent flat configurations by their response spectrum. Section 10 proves the fixed-class irredundancy, automorphism, sharpness, and recovery-limit results. Section 11 establishes operational determination. The appendices contain the analytic, algebraic, categorical, and model arguments in proof-dependency order.

2. RELATION TO EXISTING THEORIES AND PROGRAMS

The programs considered below differ principally in where they place the boundary between primitive and determined structure. Some begin with a manifold, metric, connection, phase space, or algebra of observables already specified. Others recover geometry or gauge structure from spectral, categorical, discrete, or operator-theoretic data. The present construction begins with localized quantities, their directions and energy of variation, finite-rank internal data, and structural typing. Its Riemannian realization, configuration space, curvature response, constraint expressions, and source-generated bounded observable algebras arise in that order, while a state and temporal development require further data. The comparisons are organized by this direction of determination.

2.1. Geometry and internal structure. Several established mathematical results supply individual steps of the construction. A commutative algebra represents functions on its spectrum [18]. A regular strongly local Dirichlet form defines an intrinsic metric and, under the corresponding compatibility and completeness hypotheses, a length and geodesic structure [45, 44], and its first-order module records directions of variation [11]. Finite projective modules realize vector bundles through the Serre–Swan correspondence [42, 46]. These results recover separate geometric ingredients. The source presentation places them under common localization, restriction, regularity, and compatibility data, after which the assembled connection, curvature energy, and response are determined from their combined output.

Geometric theories relating gravity and internal fields assign these ingredients different prior roles. Weyl incorporated a scale connection into spacetime geometry [50], while Kaluza–Klein theory obtained an internal connection as a component of a higher-dimensional metric [27, 29]. Gauge formulations of gravity express gravitational variables through connections on a given differentiable manifold [49, 28, 21, 2]. Larger-connection theories place gravitational and internal variables in different blocks of a connection for a larger group, with the splitting controlled by a group embedding and symmetry breaking [35, 37, 34]; connection formulations

can also recover a metric from the curvature of an internal connection [30]. In the construction here, the tangential and internal modules are distinguished by the source typing, the Dirichlet datum determines the realized base metric, and metricity fixes the off-diagonal blocks of the connection on $TM \oplus N$ as K and $-K^\dagger$. The similar block form therefore arises from a different order of determination.

Spectral geometry begins from algebraic and spectral data that encode the Riemannian metric through the Dirac operator, while the spectral action produces gravitational and internal terms [12, 10]. The field source used here begins from a Dirichlet form recording localized variation, together with separate internal metric, gauge, and structural typing data. Under the smooth nondegenerate realization hypotheses, these data determine the Riemannian base and internal bundle; the response and its later expressions then follow from the curvature energy. The two constructions consequently differ at the level of their primitive geometric data as well as in the structures subsequently determined from them.

Matrix models, causal sets, and causal fermion systems place the primitive boundary before a smooth manifold. Matrix models obtain metric and gauge fluctuations from a matrix substrate in a large- N regime [25, 43], causal sets recover continuum geometry statistically from a discrete causal order [5], and causal fermion systems derive spacetime, causal structure, and Dirac-type dynamics from a variational principle for measures on operators [16]. Their relation to continuum geometry is formulated through a large- N regime, statistical approximation, or minimizing measure and continuum limit. The realization theorem used here gives conditional exact determination within the smooth Riemannian branch. It identifies the geometry determined by a source that satisfies the realization hypotheses, while the operational theorem later recovers that source from independently specified indexed data. Once the smooth realization has supplied the base and configuration space, the next comparison concerns the law carried by that configuration space.

2.2. Response and gravitational initial data. Variational field theory ordinarily begins with a configuration space and an action or energy functional, whose first variation gives the Euler–Lagrange covector and whose vanishing selects stationary fields [22]. Because the present source determines both the admissible configuration space and the metric G on its variations, the curvature-energy differential has the metric-dual response

$$D\Phi_\Psi(v) = G_\Psi(\mathcal{Y}_\Psi, v), \quad \mathcal{Y}_\Psi = G_\Psi^{-1} D\Phi_\Psi.$$

Its zero set consists of the stationary configurations, while away from that set $\mathcal{Y}_\Psi$ retains the direction and magnitude of the energy variation. The orthogonal connection and K -variation summands give the gauge and shape components of one response law on the configuration space, whose value is determined by the selected configuration.

The positive-Riemannian oriented rank-one realization is the point at which this response meets gravitational initial-data geometry. In the ADM formulation, a spatial metric and second fundamental form constitute canonical initial data whose scalar and vector constraints are the Gauss–Codazzi relations [1]. Here the base metric is obtained from the realized Dirichlet datum, and the rank-one tensor K belongs to the source-determined configuration space. The scalar Gauss and one-form Codazzi contractions of the assembled curvature are then the geometric quantities

accompanying the shape response, which on this branch is the gravitational response. The curvature-energy estimate for the Einstein initial-data defect makes this comparison quantitative. Lapse, shift, Lorentzian embedding, and hypersurface development enter with temporal development data specified after the response law.

Hypersurface-deformation kinematics begins at that later stage. The Hojman–Kuchař–Teitelboim characterization determines the gravitational constraint algebra from a geometrodynamical phase space and a representation of hypersurface deformations [23]. Its hypotheses already include the canonical geometric data on which the deformation algebra acts. The source considered here determines the curvature variation and its metric dual before a hypersurface-deformation representation or temporal development has been specified.

2.3. Constraint and quantum expressions. The Wheeler–DeWitt equation imposes the Hamiltonian constraint on quantum states of gravitational geometry [13, 24, 31]. In the present construction, the signed Legendre function $H_R = T_{\text{kin}} - \Phi$ represents the already determined response on its response-detecting cotangent phase components. The positive curvature-energy motion form supplies the Cameron–Martin norm on the branch where that form is positive definite, while the signed function retains the constraint role. This separates the positive metric used for the response and quantum carrier from the indefinite DeWitt supermetric associated with Lorentzian geometrodynamics.

The Master-constraint programme replaces a family of constraints by a nonnegative operator [48]. The same order is realized here by starting from the signed constraint on its common core, taking its adjoint as the maximal weak constraint, and forming the positive Master operator $H_*^* H_*$, which completes a constraint whose phase-space derivation already represents the response. Loop quantization takes connection and holonomy variables as primary quantum data [2, 40]. Here the internal connection is one component of the source-determined configuration space, and bounded holonomy operations enter the generated observable algebras where the source declares them. The finite-label commutator and maximal-constraint results express the response on their specified domains after the response law has been fixed.

2.4. Bounded observables and operational recovery. Algebraic quantum field theory organizes observables into local algebras on a specified spacetime [20, 8], and locally covariant quantum field theory assigns such algebras functorially to spacetimes and their embeddings [9]. Doplicher–Roberts reconstruction recovers a compact gauge group from a category of localized transportable sectors over a given net [15]. In the present source, localization and restriction are primitive data, while declared bounded operations generate the smallest unital C^* -algebras containing their images. The category of charge sectors and original transitions is completed separately and transported by source isomorphisms. Representations and states then realize and evaluate these algebras as additional data.

The operational converse completes the direction of determination. Within the regularity class defined by conditions (R1)–(R9) in Section 11, independently specified indexed data recover the field source $\mathfrak{R}$ up to structural equivalence, after which the source determines the field $\mathfrak{F}(\mathfrak{R})$ and its law. The theorem therefore

factors indexed data through the recovered source to the field, with the common regularity class as its domain and structural equivalence as its recovery relation.

These relations locate the present theory among the existing programs. Dirichlet-form, module, and categorical results supply its mathematical recovery steps; geometric and spectral theories provide neighboring dependencies among metric, connection, and internal structure; canonical gravity supplies the precise rank-one initial-data comparison; and quantum constraint and algebraic field theory supply the later representational and observable settings. The resulting determination order begins with the field source, passes through the field and its response, and reaches its constraint, quantum, and bounded-observable expressions before the separate choices of state and temporal development.

3. THE FIELD SOURCE AND THE MAIN THEOREM

A source presentation contains local quantities, their directions of variation, the energy that measures that variation, and finite-rank internal data. These data determine a configuration space, curvature energy, response, constraints, and bounded observable algebras. The source, the field it determines, and a configuration of that field are distinct objects.

3.1. Source Presentations.

Definition 3.1 (Source presentation). *A source presentation is a tuple*

$$(3) \quad \mathfrak{R} = (\mathcal{A}, \Lambda, \text{Der}(\mathcal{A}), \mathcal{E}, \mathcal{N}_g, \mathcal{G}_{\mathcal{N}_g}, \vartheta).$$

Here $\mathcal{A}$ is a commutative algebra and Λ gives its localization and restriction maps. The $\mathcal{A}$ -module $\text{Der}(\mathcal{A})$ consists of derivations of $\mathcal{A}$, and $\mathcal{E}$ is the Dirichlet datum described below. The metric module $\mathcal{N}_g = (\mathcal{N}, g_{\mathcal{N}})$ is finite projective, $\mathcal{G}_{\mathcal{N}_g}$ is the declared subgroup of its metric automorphisms, and ϑ records the tensor types and role assignments used by the subsequent constructions.

Elements of $\mathcal{A}$ are the local quantities of the source, while Λ specifies the regions to which they restrict. The derivation module gives the directions in which they vary, and $\mathcal{E}$ measures that variation. The module $\mathcal{N}_g$ supplies finite-rank internal directions and their metric, with $\mathcal{G}_{\mathcal{N}_g}$ acting as the declared internal gauge group.

Write $\Omega_{\mathfrak{R}}^1$ for the one-form module dual to $\text{Der}(\mathcal{A})$. The last component of the source fixes the tensor assignment for the second configuration variable and the formal role assignments used later.

Definition 3.2 (Tensor and role assignments). *The datum ϑ is a pair*

$$\vartheta = (\mu, \rho_{\Sigma}).$$

The first component consists of a bijection

$$\mu : \{\text{Der}(\mathcal{A}), \mathcal{N}_g\} \longrightarrow \{\text{T}, \text{N}\}$$

together with a nonempty finite set $\mathcal{K}_{\mu}$ of admissible tensor expressions built over $\mathcal{A}$ from the two labeled modules and their duals. The source class considered here uses

$$\mu(\text{Der}(\mathcal{A})) = \text{T}, \quad \mathcal{K}_{\mu} = \{\text{Sym}_{\mathcal{A}}^2(\Omega_{\mathfrak{R}}^1) \otimes_{\mathcal{A}} \mathcal{N}\}.$$

For the four two-element sets

$$\begin{aligned} L_{\kappa} &= \{\text{C}, \text{Q}\}, & L_{\rho} &= \{\text{R}, \text{O}\}, \\ L_{\varsigma} &= \{\text{V}, \text{H}\}, & L_{\iota} &= \{\text{L}, \text{I}\}, \end{aligned}$$

The codes T, N mean tangential and internal. In the four separately typed two-point sets, C, Q mean carrier and constraint, R, O mean rank and readout, V, H mean visible and hidden, and L, I mean law and input. Let $\Sigma = \{\kappa, \rho, \varsigma, \iota\}$. The second component is a family of bijections

$$\rho_\Sigma = (\rho_\sigma)_{\sigma \in \Sigma}, \quad \rho_\sigma : L_\sigma \longrightarrow L_\sigma.$$

The elements of the sets L_σ are formal names. Thus ρ_Σ records only assignments fixed before the energy, constraint, readout, and bounded law are formed.

The declared tensor expression in $\mathcal{K}_\mu$ is the module of the second configuration variable K . The remaining assignments distinguish the positive curvature energy from its signed constraint, the internal rank from the curvature contractions read from it, the projective label classes entering the bounded law from their complementary classes, and the determined law from separately supplied inputs. Sections 4–8 construct the objects to which these names apply. Because ϑ is declared as part of the source, two source presentations may agree in their first six components and differ in ϑ .

The Dirichlet datum $\mathcal{E}$ consists of a regular strongly local Dirichlet form on $\mathcal{A}$, its reference measure, the positive local integration functionals $\langle \cdot \rangle_\lambda$ on the compact and cylinder regions indexed by λ , and the stated regularity scale. Its carré du champ determines the metric differential structure used throughout the paper.

Lemma 3.3 (Metric and connection determined by the Dirichlet form). *The carré-du-champ structure of a regular, strongly local Dirichlet form determines its canonical first-order bimodule and metric pairing, up to unitary bimodule isomorphism [11, Theorems 8.1–8.3]. In the source class, this module is identified with $\Omega_{\mathfrak{R}}^1$, dual to $\text{Der}(\mathcal{A})$; on the nondegenerate branch its metric induces the derivation pairing g_D . The derivation module, with its commutator bracket and tautological anchor, is a Lie–Rinehart algebra in the sense of [39, Section 2], and*

$$\begin{aligned} 2g_D(\nabla_X^D Y, Z) &= Xg_D(Y, Z) + Yg_D(X, Z) - Zg_D(X, Y) \\ &\quad + g_D([X, Y], Z) - g_D([X, Z], Y) - g_D([Y, Z], X) \end{aligned}$$

determines the unique torsion-free metric connection ∇^D .

Proof. By the cited representation theorem, the canonical derivation takes values in a Hilbert bimodule whose pairing is the polarized carré-du-champ form in the commutative strongly local case. Identification with $\Omega_{\mathfrak{R}}^1$ and nondegenerate duality induce g_D on $\text{Der}(\mathcal{A})$. The displayed formula is the algebraic Levi–Civita construction, analogous to the Riemannian Lie–algebroid formula of [6, Section 3.1]. Appendix A.1 proves existence and uniqueness in the present Lie–Rinehart setting and shows why nondegeneracy is required. $\square$

The form, measure, local integration functionals, and regularity scale remain components of $\mathcal{E}$, whereas g_D and ∇^D are determined from them.

3.2. Field Sources and Their Equivalence. The source class is fixed by conditions on the seven primitive data. The structures obtained from them enter only after this class has been defined.

Definition 3.4 (Field source). *A source presentation is a field source when the following conditions hold.*

- (1) *The localization datum Λ has compatible restriction maps and separates the stated local and cylinder regions.*
- (2) *Each local algebra has a compatible smooth regular core, dense in the Dirichlet graph norm and invariant under $\text{Der}(\mathcal{A})$, and the derivations commute with restriction and are closed under their bracket.*
- (3) *The Dirichlet form is closed and strongly local, and its symbol is elliptic, with nondegenerate induced pairing g_D .*
- (4) *The reference measure and local integration functionals in $\mathcal{E}$ are positive and compatible with restriction.*
- (5) *The metric internal module $\mathcal{N}_g$ is finite projective of locally constant positive rank.*
- (6) *The action of $\mathcal{G}_{\mathcal{N}_g}$ is metric and compatible with the source restrictions.*
- (7) *The assignments in ϑ are well typed and are preserved by restriction.*
- (8) *The local and cylinder systems, including their admitted transition maps, are continuous and projectively separated on the stated cofinal families.*

On each relatively compact smooth domain U , the response theorem uses an integer $s > \dim U/2 + 2$, support-compatible coefficients, and H_0^s variations whose trace jets of order below $s - 1/2$ vanish. The finite-dimensional phase spaces determined by the source have Weyl algebras whose restriction maps admit a nonempty compatible family of regular states. Every member of this family is admitted. Selecting a particular state and its GNS representation supplies additional data.

The finite-projective metric categories used for transitions are obtained from $\mathcal{N}_g$, its declared gauge action, and the restrictions in Λ . They are therefore determined by the source tuple, so their determining data already occur in the seven components above.

These hypotheses are checkable properties of the primitive data and of the restriction systems they generate. Theorem 3.8 is consequently a determination theorem for a previously defined class, and Section 10 tests the necessity of the eight componentwise conditions within that class.

Definition 3.5 (Structure-preserving source isomorphism). *A structure-preserving isomorphism of source presentations consists of isomorphisms of the algebra and its localization data, the derivation module, the Dirichlet datum, and the metric internal module. It conjugates the declared metric-gauge subgroup to its counterpart, preserves ϑ , commutes with all local and cylinder restriction maps, and is continuous for their projective topologies. The cofinal separatedness condition makes such an isomorphism determined by its local and finite restrictions. Two source presentations are structurally equivalent when they are related by such an isomorphism.*

Let $\mathbf{S}^F$ be the groupoid whose objects are the field sources just defined and whose arrows are their structure-preserving isomorphisms. The compact models of Section 9 show that this groupoid is nonempty, and Section 11 identifies it as a replete full subgroupoid of the regular source groupoid reconstructed there. Every source-level uniqueness statement in the paper is taken up to this structural equivalence.

The sources with an oriented positive-Riemannian realization and an internal bundle of rank one form the full replete subgroupoid $\mathbf{S}^{\text{GC}} \subset \mathbf{S}^F$. Section 6 establishes the gravitational comparison on this subgroupoid, while sources of higher internal rank remain objects of $\mathbf{S}^F$.

The internal automorphisms left after the declared gauge group are recorded by the normalizer quotient

$$D_G = N_{\text{Aut}(\mathcal{N}_g)}(\mathcal{G}_{\mathcal{N}_g})/\mathcal{G}_{\mathcal{N}_g}.$$

Every construction obtained functorially from the algebra, Dirichlet form, metric module, curvature, functional calculus, and projective restrictions is equivariant under source isomorphisms, so its action descends to D_G after passing to gauge-equivalence classes. The full source-automorphism classification is given in Section 10.

3.3. The Field Determined by a Source.

Definition 3.6 (Field determined by a source). *For $\mathfrak{R} \in \mathbf{S}^F$, the field $\mathfrak{F}(\mathfrak{R})$ is the system consisting of the configuration and variation spaces, the connection and curvature assembled from each configuration, the curvature energy and its metric-dual response, the signed constraint, and the source-generated bounded operations. It also contains, as a separate component, the completed real metric C^* -category of sectors and bounded covariant transitions. On their stated domains it assigns the phase, finite-label, and represented quantum expressions of the response. Its field law is $\mathcal{L}_{\mathfrak{R}}$, defined in Section 8.3.*

Thus $\mathfrak{F}(\mathfrak{R})$ denotes the complete law-bearing field, while

$$\Psi \in C^s(\mathfrak{R})$$

selects one of its configurations. A Riemannian realization represents the source modules by geometric bundles, and a compatible Hilbert-space representation represents one member of the source-determined quantum family. Both choices preserve the distinction between the field and its configuration.

3.4. Riemannian Bundle Realization. A Riemannian bundle realization of $\mathfrak{R}$ consists of a Riemannian manifold (M, h) and a finite-rank vector bundle $N \rightarrow M$ with fiber metric g_N . It also supplies compatible identifications

$$\mathcal{A} \cong C^\infty(M), \quad \text{Der}(\mathcal{A}) \cong \Gamma(TM), \quad \Omega_{\mathfrak{R}}^1 \cong \Gamma(T^*M), \quad \mathcal{N} \cong \Gamma(N).$$

Under these identifications, g_D becomes h , $g_{\mathcal{N}}$ becomes g_N , Λ becomes restriction to the declared regions of M , and $\mathcal{G}_{\mathcal{N}_g}$ becomes the declared group of metric bundle automorphisms. The Dirichlet form, measure, local integration functionals, and assignments in $\mathcal{V}$ are carried to their corresponding geometric data. The localization is spatial when its regions identify with a basis of open subsets of M and its maps become the ordinary restrictions of smooth functions.

Theorem 3.7 (Riemannian realization of a field source). *Suppose the localization of a field source is spatial in this sense, its algebra and derivation module have the displayed smooth realization, and its Dirichlet/carré-du-champ data give a regular nondegenerate metric-measure structure compatible with the local integration functionals. Suppose also that $\mathcal{N}_g$ has the displayed Serre–Swan realization and that the restriction maps, gauge action, and assignments in $\mathcal{V}$ are preserved. Then these data form a Riemannian bundle realization of $\mathfrak{R}$, unique up to a diffeomorphism of M and a metric bundle isomorphism of N carrying the declared gauge and assignment data.*

Proof. The intrinsic-metric theorem of [45, Assumption (A) and Theorem 1] and the length-space result of [44, Theorem 4.1] supply the metric and length stages of the Dirichlet realization; the full smooth Riemannian identification is among the hypotheses above. Isomorphisms of the resulting smooth-function algebras arise from diffeomorphisms [36, Theorem 1]. The metric and measure are fixed by the Dirichlet and integration data, while the finite-projective metric module is represented by the Serre–Swan correspondence [42, Section 50][46, Section 3, Theorem 2]. Their compatibility with Λ , $\mathcal{G}_{\mathcal{N}_g}$, and ϑ gives the stated structural equivalence. $\square$

The realization supplies a geometric representation of structures already defined from $\mathfrak{R}$. In particular, the base connection is the realization of ∇^D , and the internal bundle N becomes an oriented metric line on the rank-one Gauss–Codazzi branch. The compact T^3 model of Section 9 satisfies the realization hypotheses.

3.5. Determination of the Field and Recovery of Its Source. The remaining sections construct the field response, its phase and quantum expressions, and the bounded law generated by the source. The following theorem states their common conclusion. Its proof is completed in Section 11, after the independently specified indexed local and finite data and the source-recovery functor have been defined.

Theorem 3.8 (Field response and source recovery). *For every $\mathfrak{R} \in \mathbf{S}^F$, the field $\mathfrak{F}(\mathfrak{R})$ has the following properties.*

- (1) *Each configuration $\Psi = (\nabla^{\mathcal{N}}, K)$ determines a metric connection on the direct sum of $\text{Der}(\mathcal{A})$ and $\mathcal{N}$, its curvature energy Φ , and the unique metric-dual response*

$$D\Phi_{\Psi}(v) = G_{\Psi}(\mathcal{Y}_{\Psi}, v), \quad \mathcal{Y}_{\Psi} = (\mathcal{Y}_{\nabla, \Psi}, \mathcal{Y}_{K, \Psi}).$$

The two components are the connection and shape responses, and the first is the gauge response.

- (2) *On connected domains of constant positive internal rank, rank one has no connection-response type and retains the shape-response type, while both types are present from rank two onward. Either available response may vanish at a particular configuration. On $\mathbf{S}^{\text{GC}}$, the rank-one shape response has the Gauss–Codazzi gravitational interpretation of Section 6. The mixed second variations agree as continuous weak bilinear forms on the stated Sobolev branch.*
- (3) *The Hamiltonian vector field of the signed constraint, the local and finite-label response derivations, their finite-label commutator representations, and the bounded common-core restriction represent the same response on their respective domains. The bounded common-core domain may be empty.*
- (4) *On each represented or local-operation component and at each transition object, the declared bounded operations generate the unique smallest unital C^* -algebra in the specified ambient algebra. Together with the response and the completed real metric C^* -category of sectors and bounded covariant transitions, these algebras define the source-determined field law of Section 8.*
- (5) *The indexed local and finite data satisfying the conditions of Section 11 recover $\mathfrak{R}$ up to structural equivalence, after which $\mathfrak{R}$ determines the response and field law. Within the fixed seven-component source class, every source*

component enters at least one required incidence of $\mathcal{L}_{\mathfrak{R}}$, and deleting that component removes the incidence.

The source determines the compatible represented laws indexed by all admitted regular states and the maps relating their local and finite components. Choosing a representation, state, self-adjoint extension, graph-domain family, clock, temporal development, history, measurement model, apparatus, or empirical map adds data to this field law.

4. INTRINSIC GEOMETRY AND CURVATURE ENERGY

4.1. Configuration Space and Riemannian Realization. Fix a field source $\mathfrak{R} \in \mathbf{S}^F$ and a regularity index s in its declared regularity scale. The source and the field $\mathfrak{F}(\mathfrak{R})$ remain fixed throughout this section, while $\Psi = (\nabla^{\mathcal{N}}, K)$ denotes a variable field configuration.

The differences of two regularity- s metric connections form the module

$$\mathcal{V}_{\nabla}^s(\mathfrak{R}) = \Omega_{\mathfrak{R}}^1 \otimes_{\mathcal{A}} \mathfrak{so}(\mathcal{N}_g),$$

and the second configuration variable belongs to

$$\mathcal{V}_K^s(\mathfrak{R}) \subseteq \text{Sym}_{\mathcal{A}}^2(\Omega_{\mathfrak{R}}^1) \otimes_{\mathcal{A}} \mathcal{N}.$$

Thus K is an $\mathcal{N}$ -valued symmetric bilinear form on $\text{Der}(\mathcal{A})$. Its symmetry is a condition on the configuration space and will remain independent of the metricity of the block connection below.

Let $\text{Conn}^s(\mathcal{N}_g)$ denote the regularity- s metric connections on $\mathcal{N}_g$.

Definition 4.1 (Configuration space). *The regularity- s configuration space determined by $\mathfrak{R}$ is the affine space*

$$\mathcal{C}^s(\mathfrak{R}) = \text{Conn}^s(\mathcal{N}_g) \times \mathcal{V}_K^s(\mathfrak{R}),$$

with translation module

$$T\mathcal{C}^s(\mathfrak{R}) = \mathcal{V}_{\nabla}^s(\mathfrak{R}) \oplus \mathcal{V}_K^s(\mathfrak{R}).$$

Lemma 4.2 (Affine structure of the metric connections). *Under the field-source regularity conditions, $\text{Conn}^s(\mathcal{N}_g)$ is nonempty and is an affine torsor for $\mathcal{V}_{\nabla}^s(\mathfrak{R})$. A reference metric connection $\nabla_{0,s}^{\mathcal{N}}$ therefore gives the coordinate expression*

$$\mathcal{C}^s(\mathfrak{R}) \cong (\nabla_{0,s}^{\mathcal{N}} + \mathcal{V}_{\nabla}^s(\mathfrak{R})) \times \mathcal{V}_K^s(\mathfrak{R}),$$

and changing the reference connection translates the same affine space.

Proof. Write the finite-projective module as the range of an idempotent in a finite free module. Projecting the componentwise derivation gives a connection ∇^0 . Its metric defect is the symmetric bilinear form

$$C_X(\nu, \eta) = Xg_{\mathcal{N}}(\nu, \eta) - g_{\mathcal{N}}(\nabla_X^0 \nu, \eta) - g_{\mathcal{N}}(\nu, \nabla_X^0 \eta).$$

Nondegeneracy of $g_{\mathcal{N}}$ determines a self-adjoint endomorphism S_X by $2g_{\mathcal{N}}(S_X \nu, \eta) = C_X(\nu, \eta)$. Then $\nabla^0 + S$ is metric. The regularity of the smooth core, its derivations, and the metric makes it a regularity- s connection. The difference A of two metric connections satisfies

$$g_{\mathcal{N}}(A_X \nu, \eta) + g_{\mathcal{N}}(\nu, A_X \eta) = 0,$$

so $A \in \Omega_{\mathfrak{R}}^1 \otimes \mathfrak{so}(\mathcal{N}_g)$. Conversely every such translation preserves metricity. This proves the torsor statement, and translation by the difference of two reference connections proves independence of the coordinate origin. $\square$

4.1.1. *Riemannian realization and gauge action.* Under a Riemannian bundle realization from Theorem 3.7, the source modules become

$$\text{Der}(\mathcal{A}) \cong \Gamma(TM), \quad \Omega_{\mathfrak{R}}^1 \cong \Gamma(T^*M), \quad \mathcal{N} \cong \Gamma(N),$$

and the two configuration variables become a metric connection on the internal bundle N and a symmetric N -valued two-tensor. Hence

$$\mathcal{C}^s(\mathfrak{R})|_{(M,h,N,g_N)} \cong \text{Conn}^s(N, g_N) \times \Gamma^s(\text{Sym}^2 T^*M \otimes N).$$

This is an identification of affine spaces, with translation space

$$\mathcal{V}_{\nabla^N}^s(M, N) \oplus \Gamma^s(\text{Sym}^2 T^*M \otimes N).$$

Here N is the internal bundle. On the rank-one Gauss–Codazzi class of Section 6, it is an oriented metric line.

Let $\mathcal{G}_{\mathcal{N}_g}^s$ be the subgroup of the declared metric automorphisms preserving these regularity spaces. Its action is

$$u \cdot (\nabla^{\mathcal{N}}, K) = (u \nabla^{\mathcal{N}} u^{-1}, uK).$$

Conjugation acts on $\nabla^{\mathcal{N}}$ as a differential operator, so a nonconstant u produces the usual inhomogeneous term in the connection transformation.

Theorem 4.3 (Configuration-space realization and gauge covariance). *The source determines $\mathcal{C}^s(\mathfrak{R})$, every Riemannian bundle realization gives the affine identification above, and the action of $\mathcal{G}_{\mathcal{N}_g}^s$ preserves the configuration space. Under realization this action is the standard action of regularity-preserving metric bundle automorphisms of N on (∇^N, K) .*

Proof. The realization carries metric connections and their skew-adjoint differences to the corresponding bundle connections and one-forms with values in $\mathfrak{so}(N, g_N)$. It carries the covariant symmetric tensor module to $\Gamma^s(\text{Sym}^2 T^*M \otimes N)$. Metric automorphisms preserve both factors, and the displayed action transfers to the bundle action. $\square$

4.2. Block Metric Connection and Curvature. The source fixes the torsion-free g_D -metric connection ∇^D on $\text{Der}(\mathcal{A})$. A configuration supplies the internal connection and the remaining mixed term. Together these data form one connection on $\text{Der}(\mathcal{A}) \oplus \mathcal{N}$.

For $X \in \text{Der}(\mathcal{A})$, contraction of K gives

$$K_X : \text{Der}(\mathcal{A}) \longrightarrow \mathcal{N}, \quad K_X Y = K(X, Y).$$

The two metric pairings determine its adjoint $K_X^\dagger : \mathcal{N} \rightarrow \text{Der}(\mathcal{A})$ through

$$g_{\mathcal{N}}(K_X Y, \nu) = g_D(Y, K_X^\dagger \nu).$$

Theorem 4.4 (Block metric connection and determination of its mixed blocks). *For every $\Psi = (\nabla^{\mathcal{N}}, K) \in \mathcal{C}^s(\mathfrak{R})$,*

$$(4) \quad \nabla_{\Psi, X}^\oplus = \begin{pmatrix} \nabla_X^D & -K_X^\dagger \\ K_X & \nabla_X^{\mathcal{N}} \end{pmatrix}$$

defines a connection on $\text{Der}(\mathcal{A}) \oplus \mathcal{N}$ that preserves $g_D \oplus g_{\mathcal{N}}$. Conversely, if a connection preserving this split metric has lower-left mixed block K_X , its upper-right mixed block is uniquely $-K_X^\dagger$.

Proof. The diagonal blocks are metric. The two mixed contributions cancel because

$$g_D(-K_X^\dagger \nu, Y) + g_N(\nu, K_X Y) = 0.$$

The Leibniz rule follows from the two connection Leibniz rules and the $\mathcal{A}$ -linearity of K_X . Conversely, if the upper-right block is A_X , metricity on $Y \oplus 0$ and $0 \oplus \nu$ gives

$$g_D((A_X + K_X^\dagger)\nu, Y) = 0$$

for every Y and ν . Nondegeneracy of g_D yields $A_X = -K_X^\dagger$. $\square$

Metricity determines the relation between the two mixed blocks. It permits both the symmetric and skew parts of a general bilinear form, so the symmetry of K remains the independent configuration-space condition imposed above.

4.2.1. *Curvature decomposition.* Write

$$\mathcal{F}_\Psi^\oplus(X, Y) = [\nabla_{\Psi, X}^\oplus, \nabla_{\Psi, Y}^\oplus] - \nabla_{\Psi, [X, Y]}^\oplus.$$

The connections ∇^D and ∇^N induce a connection on K , given by

$$(\nabla_X K)(Y, Z) = \nabla_X^N(K(Y, Z)) - K(\nabla_X^D Y, Z) - K(Y, \nabla_X^D Z).$$

Theorem 4.5 (Curvature of the block metric connection). *The curvature of (4) has the block form*

$$(5) \quad \mathcal{F}_\Psi^\oplus(X, Y) = \begin{pmatrix} B_T(\Psi)(X, Y) & -B_K(\Psi)(X, Y)^\dagger \\ B_K(\Psi)(X, Y) & B_\perp(\Psi)(X, Y) \end{pmatrix},$$

where

$$(6) \quad B_T(\Psi)(X, Y) = R^D(X, Y) - K_X^\dagger K_Y + K_Y^\dagger K_X,$$

$$(7) \quad B_K(\Psi)(X, Y) = (\nabla_X K)_Y - (\nabla_Y K)_X,$$

$$(8) \quad B_\perp(\Psi)(X, Y) = R^N(X, Y) - K_X K_Y^\dagger + K_Y K_X^\dagger.$$

These are respectively the tangential, mixed, and internal curvature blocks of the single connection $\nabla_\Psi^\oplus$.

Proof. Insert the block matrix (4) into the curvature commutator. The upper-left product gives R^D and the two displayed quadratic K -terms. The lower-left block is the covariant exterior derivative of K , and the lower-right block gives R^N and its quadratic K -terms. Curvature of a metric connection is skew-adjoint for the split metric, which fixes the remaining mixed block as $-B_K^\dagger$. Appendix A.4 gives the full block multiplication. $\square$

The tangential block therefore combines the curvature determined by the Dirichlet datum with a quadratic expression in K . The mixed block measures the covariant variation of K , while the internal block combines the curvature of ∇^N with a second quadratic expression in K . All three are block projections of the curvature of the single connection $\nabla_\Psi^\oplus$.

4.2.2. *Realization and gauge covariance.* Under a Riemannian bundle realization, ∇^D becomes the Levi-Civita connection of h , $\nabla^{\mathcal{N}}$ becomes ∇^N , and (4) becomes a metric connection on $TM \oplus N$. Equations (6)–(8) become the corresponding bundle curvature formulas. In particular, $B_{\perp}$ contains the curvature R^N of the internal connection together with its quadratic K -term.

For $u \in \mathcal{G}_{\mathcal{N}_g}^s$, put $\tilde{u} = \text{id}_{\text{Der}(\mathcal{A})} \oplus u$. Then

$$\nabla_{u \cdot \Psi}^{\oplus} = \tilde{u} \nabla_{\Psi}^{\oplus} \tilde{u}^{-1}, \quad \mathcal{F}_{u \cdot \Psi}^{\oplus} = \tilde{u} \mathcal{F}_{\Psi}^{\oplus} \tilde{u}^{-1},$$

and consequently

$$B_T \longmapsto B_T, \quad B_K \longmapsto u B_K, \quad B_{\perp} \longmapsto u B_{\perp} u^{-1}.$$

Theorem 4.6 (Realization and gauge covariance of the curvature). *Riemannian bundle realizations carry the block metric connection and all three curvature formulas to their counterparts on $TM \oplus N$. The internal gauge action intertwines the connection and curvature and preserves every metric contraction of the three curvature blocks.*

Proof. The realization maps preserve direct sums, pairings, adjoints, covariant derivatives, composition, and commutators. The gauge formulas follow by conjugation with $\tilde{u}$, and metricity of u preserves the three component norms and their contractions. $\square$

On the oriented positive-Riemannian internal-rank-one class, Section 6 applies scalar and one-form contractions to B_T and B_K and supplies their Gauss–Codazzi interpretation.

4.3. **Curvature Energy and the Configuration-Space Metric.** The metric pairings on the two summands induce the Hilbert–Schmidt norm of the full curvature endomorphism. At a configuration Ψ , define

$$(9) \quad \phi_{\mathfrak{R}}(\Psi) = \frac{1}{2} (\|B_T(\Psi)\|^2 + 2\|B_K(\Psi)\|^2 + \|B_{\perp}(\Psi)\|^2).$$

The mixed block and its adjoint have equal Hilbert–Schmidt norms, which accounts for the coefficient 2.

Lemma 4.7 (Positivity of the curvature-energy density). *For every $\Psi \in \mathcal{C}^s(\mathfrak{R})$, $\phi_{\mathfrak{R}}(\Psi) \geq 0$. It vanishes exactly when $B_T(\Psi) = B_K(\Psi) = B_{\perp}(\Psi) = 0$.*

Proof. Equation (9) is one half of the squared Hilbert–Schmidt norm of $\mathcal{F}_{\Psi}^{\oplus}$. The source metrics are positive and nondegenerate, so the sum vanishes exactly when all three blocks vanish. $\square$

4.3.1. *Local and finite-label curvature energies.* For a label $\lambda \in \Lambda$, write τ_{λ} for the positive integration functional carried by the Dirichlet datum on that label, and let $\phi_{\mathfrak{R},\lambda}$ be the restriction of (9) to its configuration and coefficient data. The curvature energy is the function

$$(10) \quad \Phi_{\mathfrak{R},\lambda}(\Psi_{\lambda}) = \tau_{\lambda}(\phi_{\mathfrak{R},\lambda}(\Psi_{\lambda})).$$

The argument Ψ may be suppressed after the label λ and its configuration domain have been fixed.

For $\lambda \subseteq \mu$, let $r_{\lambda\mu}$ denote the source restriction of coefficients, density support, configuration and variation slots. Its joint action with the integration functional is

$$r_{\lambda\mu}^{\mathfrak{R}}[\tau_{\mu}(f_{\mu})] = \tau_{\lambda}(r_{\lambda\mu} f_{\mu}).$$

Theorem 4.8 (Restriction-compatible curvature energies). *The field source determines a nonnegative family $\lambda \mapsto \Phi_{\mathfrak{R},\lambda}$ that is invariant under label-preserving internal gauge transformations and satisfies*

$$(11) \quad r_{\lambda\mu}^{\mathfrak{R}} \Phi_{\mathfrak{R},\mu} = \Phi_{\mathfrak{R},\lambda} \quad \lambda \subseteq \mu.$$

If λ is realized by a relatively compact Riemannian region $U \subset M$, then

$$(12) \quad \Phi_U(\Psi) = \frac{1}{2} \int_U (\|B_T(\Psi)\|^2 + 2\|B_K(\Psi)\|^2 + \|B_{\perp}(\Psi)\|^2) dV_h.$$

For a finite cylinder label, $\Phi_{\mathfrak{R},\lambda}$ is obtained by applying the label's positive integration functional to its restricted curvature density and satisfies the same joint restriction law.

Proof. Assembly of the block connection, curvature, adjoints, and metric contractions consists of finite-order operations that commute with the source restriction maps. Therefore $r_{\lambda\mu} \phi_{\mathfrak{R},\mu}(\Psi_{\mu}) = \phi_{\mathfrak{R},\lambda}(r_{\lambda\mu} \Psi_{\mu})$. Applying the restricted integration functional proves (11), including background and degree-zero density terms. Composition follows from composition of the source restrictions. Gauge invariance follows from conjugation of the curvature and preservation of the metrics and label functional. On a Riemannian region, τ_U becomes integration against dV_h , which gives (12). $\square$

The compatible energy family is continued through its weak Hessian in Appendix B.2–B.5.

The refinement operation in (11) restricts the local expression and its integration datum together. Section 7 distinguishes this joint source restriction from the metric-dual phase inclusion and proves their compatibility with the response derivations.

4.3.2. Metric on configuration variations. Restrict the two variation modules to a fixed label λ . Their fixed pairings are

$$G_{\nabla,\lambda}(\alpha, \beta) = \tau_{\lambda} \left(g_D^{ij} \operatorname{tr}_{\mathcal{N}}(\alpha_i^{\dagger} \beta_j) \right),$$

$$G_{K,\lambda}(K, L) = \tau_{\lambda} \left(g_D^{ia} g_D^{jb} g_{\mathcal{N}}(K_{ij}, L_{ab}) \right).$$

The first uses the finite-projective trace on $\operatorname{End}_{\mathcal{A}}(\mathcal{N})$, and the second is the tensor-product metric restricted to symmetric covariant two-tensors.

Theorem 4.9 (Orthogonal direct-sum metric on the configuration space). *The two pairings determine*

$$G_{\mathfrak{R},\lambda} = G_{\nabla,\lambda} \oplus G_{K,\lambda}$$

on the labelwise configuration tangent module. The two canonical inclusions are isometries with orthogonal images. Among metrics natural under source automorphisms and restricting to the displayed pairings on the two summands, $G_{\mathfrak{R},\lambda}$ is unique.

Proof. The direct-sum pairing has the two stated restrictions and a zero mixed term. The source automorphism induced by $-\operatorname{id}_{\mathcal{N}}$ fixes a connection variation by conjugation and reverses a K -variation. Naturality therefore makes every competing mixed pairing B satisfy

$$B(\alpha, K) = B(\alpha, -K) = -B(\alpha, K),$$

so $B = 0$. The fixed diagonal restrictions then determine the direct-sum metric. Appendix A.7 gives the metric-module construction and its inverse. $\square$

When the label is understood, write $G_{\mathfrak{R}} = G_{\mathfrak{R},\lambda}$. The metric is translation invariant on the affine configuration space, so the notation G_{Ψ} used in the response equation denotes this same metric at the point Ψ .

Embedding a K -variation into the block connection produces both K and $-K^{\dagger}$. Consequently the pullback of the endomorphism Hilbert–Schmidt metric is

$$G^{\oplus} = G_{\nabla} \oplus 2G_K,$$

whereas the configuration-space metric counts the single independent K -variation once.

Let $G_{\mathfrak{R},\lambda}^b$ be the Riesz morphism of the configuration metric and define its metric cotangent module by

$$T_{G,\lambda}^* \mathcal{C}^s(\mathfrak{R}) = \text{ran } G_{\mathfrak{R},\lambda}^b.$$

The inverse $G_{\mathfrak{R},\lambda}^{\sharp}$ is defined on this metric dual. Under the Sobolev hypotheses of Section 5, the pairing extends between the stated response and variation spaces, and the first-variation theorem proves that $D\Phi_U(\Psi)$ has a unique L^2 -representative. Thus the local curvature energy and the orthogonal configuration metric supply exactly the two objects used to define the response in the next section.

5. THE RESPONSE OF A FIELD

For every relatively compact Riemannian domain, the curvature energy of a field configuration has a differential on the connection and K -variation spaces defined in Section 4.3. Their orthogonal metric identifies this covector with a unique vector, and the resulting assignment from configurations to metric-dual vectors is the response law. Its two components are the gauge response in the internal-connection directions and the shape response in the K -variation directions.

5.1. Metric-Dual Response of the Curvature Energy. Let U be a relatively compact smooth Riemannian domain, with smooth boundary when present, and fix an integer $s > \dim(U)/2 + 2$. Write $\mathcal{C}^s(U; \mathfrak{R})$ for the affine H^s configuration chart obtained by restricting $\mathcal{C}^s(\mathfrak{R})$ to U , and let

$$\mathcal{V}_0^s(U) = H_0^s(T^*U \otimes \mathfrak{so}(N, g_N)) \oplus H_0^s(\text{Sym}^2 T^*U \otimes N)$$

be its zero-boundary Sobolev variation space. A variation will be written as $v = (\alpha, \dot{K})$. Here H_0^s is the H^s -closure of compactly supported smooth sections, and its vanishing trace jets eliminate the boundary term in Green’s identity after zero extension [33, Chapter 1, Theorems 11.4–11.5].

The configuration metric of Theorem 4.9 restricts to

$$G_U = G_{\nabla,U} \oplus G_{K,U}$$

on $\mathcal{V}_0^s(U)$. Denote its Hilbert completion by

$$\mathcal{V}_G(U) = \overline{\mathcal{V}_0^s(U)}^{\|\cdot\|_{G_U}}.$$

On the maximal Green-adjoint domain established in Appendix C.1, put

$$Z_{\Psi} = D_{\nabla_{\Psi}^{\oplus}, \max}^{\dagger} \mathcal{F}_{\Psi}^{\oplus}.$$

Let $E_{\nabla}(\Psi)$ be its internal-connection block and define the unprojected lower-left tensor by

$$E_K^0(\Psi)(X, Y) = \text{pr}_N(Z_{\Psi}(X)(Y, 0)).$$

This tensor need not be symmetric. Its component in the symmetric K -variation space is the orthogonal projection

$$E_K^+(\Psi) = P_+ E_K^0(\Psi), \quad P_+ = \frac{1}{2}(1 + \varsigma),$$

where ς exchanges the two tangential covector arguments.

Theorem 5.1 (Metric-dual field response). *For U , s , and $\mathcal{C}^s(U; \mathfrak{A})$ as above, the local curvature energy Φ_U is continuously Fréchet differentiable in the H^s configuration chart. For every Ψ , its bulk differential extends to a continuous covector on $\mathcal{V}_G(U)$ and has the unique Riesz representative*

$$\mathcal{Y}_U(\Psi) = (\mathcal{Y}_{\nabla, U}(\Psi), \mathcal{Y}_{K, U}(\Psi)) \in H^{s-2} \subset L^2$$

satisfying

$$D\Phi_U(\Psi)[\alpha, \dot{K}] = G_{\nabla, U}(\mathcal{Y}_{\nabla, U}(\Psi), \alpha) + G_{K, U}(\mathcal{Y}_{K, U}(\Psi), \dot{K}).$$

Its components are

$$(13) \quad \mathcal{Y}_{\nabla, U}(\Psi) = E_{\nabla}(\Psi), \quad \mathcal{Y}_{K, U}(\Psi) = 2E_K^+(\Psi) = 2P_+ E_K^0(\Psi).$$

The first is the gauge response, the metric dual of the curvature-energy variation in the internal-connection directions, and the second is the shape response, the corresponding metric dual in the K -variation directions.

For every H^{s+1} metric gauge transformation u preserving these Sobolev spaces,

$$\mathcal{Y}_{\nabla, U}(u\Psi) = u\mathcal{Y}_{\nabla, U}(\Psi)u^{-1}, \quad \mathcal{Y}_{K, U}(u\Psi) = u\mathcal{Y}_{K, U}(\Psi).$$

If $U \subseteq V$ carries compatible source backgrounds, configurations, coefficients, and bundle data, zero extension $E_{VU} : \mathcal{V}_0^s(U) \rightarrow \mathcal{V}_0^s(V)$ and its L^2 -adjoint satisfy

$$(14) \quad D\Phi_U(\Psi_U) = E_{VU}^* D\Phi_V(\Psi_V), \quad \mathcal{Y}_U(\Psi_U) = E_{VU}^{*, L^2} \mathcal{Y}_V(\Psi_V),$$

where $\Psi_U = \Psi_V|_U$.

Proof. The curvature map is a continuous polynomial from the affine H^s chart to $H^{s-1} \subset L^2$, which gives the stated Fréchet differentiability. For the block-connection increment

$$\mathcal{A}(\alpha, \dot{K})_X = \begin{pmatrix} 0 & -\dot{K}_X^\dagger \\ \dot{K}_X & \alpha_X \end{pmatrix},$$

the affine curvature identity and Green's identity give

$$D\Phi_U(\Psi)[\alpha, \dot{K}] = \langle \mathcal{A}(\alpha, \dot{K}), Z_{\Psi} \rangle_{L^2},$$

because the full vanishing trace jet of $\mathcal{V}_0^s(U)$ removes the boundary term. The internal block contributes $G_{\nabla, U}(E_{\nabla}, \alpha)$. The lower-left mixed block and its adjoint upper-right block make equal Hilbert–Schmidt contributions, while the configuration metric counts the independent symmetric K -variation once. Since $\dot{K}$ is symmetric and P_+ is the orthogonal projection onto its tensor space,

$$D\Phi_U(\Psi)[\alpha, \dot{K}] = G_{\nabla, U}(E_{\nabla}, \alpha) + 2G_{K, U}(P_+ E_K^0, \dot{K}).$$

Both representatives lie in $H^{s-2} \subset L^2$, so this formula extends the differential continuously to $\mathcal{V}_G(U)$. Density and Riesz uniqueness give (13). Gauge transformations intertwine the block connection, its Green adjoint, and the metric pairings;

P_+ acts only on the tangential covector arguments and hence commutes with their internal action. For compatible $U \in V$, the full trace-jet condition makes zero extension commute with the local differential expression, and Riesz uniqueness gives (14). Appendix C.1–C.2 contains the graph-domain, trace, and extension arguments. $\square$

The theorem defines the local response law

$$\mathcal{Y}_U : \mathcal{C}^s(U; \mathfrak{R}) \longrightarrow \mathcal{V}_G(U), \quad \Psi \longmapsto \mathcal{Y}_U(\Psi).$$

A variation is an argument of $D\Phi_U(\Psi)$, this differential is the variational covector, and $\mathcal{Y}_U(\Psi) = G_U^\# D\Phi_U(\Psi)$ is its response value. Nondegeneracy of G_U gives

$$\mathcal{Y}_U(\Psi) = 0 \iff D\Phi_U(\Psi)[v] = 0 \quad (\forall v \in \mathcal{V}_0^s(U)).$$

Thus the zero set of the response law is the stationary locus of the local curvature energy relative to the declared variations.

5.2. Rank Dependence of the Gauge and Shape Responses.

Theorem 5.2 (Rank dependence of the response components).

Let U be a nonempty connected domain in a positive-dimensional Riemannian base, and suppose that the internal bundle has constant positive rank r . At $r = 1$, the connection-variation summand is zero and the K -variation summand has positive local fiber rank. For $r \geq 2$, both summands have positive local fiber rank. A Riesz component may vanish at a configuration even when its summand has positive rank.

Source isomorphisms preserve the internal rank, the two variation summands, and their response values. Restriction to a nonempty constant-rank subdomain preserves the zero or positive local rank of each summand. For a compatible supported inclusion $U \in V$, the response restriction law implies

$$\mathcal{Y}_V(\Psi_V) = 0 \implies \mathcal{Y}_U(\Psi_V|_U) = 0.$$

Proof. The connection-variation fiber contains the factor $\mathfrak{so}(r)$, whose dimension is $r(r-1)/2$. It is therefore zero at $r = 1$ and has positive rank for $r \geq 2$. If $d = \dim U > 0$, the K -variation fiber has rank

$$\frac{d(d+1)}{2}r > 0.$$

On a trivializing open set, multiplying a nonzero tensor by a compactly supported bump function produces a nonzero smooth variation, so positive local fiber rank gives a nonzero variation summand. The value of its Riesz component is determined separately by the chosen configuration and may be zero. Source isomorphisms preserve the ranks, summands, and response vectors, while constant-rank restriction preserves the fiber calculation. The final implication follows directly from (14). $\square$

Internal rank therefore determines which response directions are available, while the field configuration determines their values. On the oriented positive-Riemannian rank-one Gauss–Codazzi class, Section 6 identifies the surviving shape response as the gravitational response.

5.3. Reciprocity of the Mixed Variations. For α in the connection-variation summand and $\dot{K}$ in the K -variation summand, define the mixed susceptibility forms by

$$\begin{aligned}\chi_{\nabla K, \Psi}^U(\alpha, \dot{K}) &= D^2\Phi_U(\Psi)[(\alpha, 0), (0, \dot{K})], \\ \chi_{K\nabla, \Psi}^U(\dot{K}, \alpha) &= D^2\Phi_U(\Psi)[(0, \dot{K}), (\alpha, 0)].\end{aligned}$$

Theorem 5.3 (Symmetry of the mixed second differential). *Under the affine H^s configuration and H_0^s variation hypotheses above, Φ_U is twice Fréchet differentiable and*

$$(15) \quad \chi_{\nabla K, \Psi}^U(\alpha, \dot{K}) = \chi_{K\nabla, \Psi}^U(\dot{K}, \alpha).$$

This equality holds between continuous bilinear forms in the H_0^s topology and is preserved by source isomorphisms and H^{s+1} metric gauge transformations. If $U \subseteq V$ has compatible backgrounds, configurations, coefficients, and bundle data, the full-trace-jet zero extension satisfies

$$\chi_{\nabla K, \Psi_U}^U(\alpha, \dot{K}) = \chi_{\nabla K, \Psi_V}^V(E_{VU}\alpha, E_{VU}\dot{K}).$$

The conclusion is equality of continuous bilinear forms in the stated Sobolev topology. A bounded L^2 -adjoint identity for the mixed blocks of the linearized response requires additional domain and boundedness results.

Proof. In the affine H^s chart, put $L_{\Psi, v} = D_{\nabla \oplus \Psi} \mathcal{A}(v)$. The curvature satisfies

$$\mathcal{F}(\Psi + v) = \mathcal{F}(\Psi) + L_{\Psi, v} + \frac{1}{2}[\mathcal{A}(v) \wedge \mathcal{A}(v)].$$

Sobolev multiplication makes this a continuous polynomial into $H^{s-1} \subset L^2$, so $\Phi_U = \frac{1}{2}\|\mathcal{F}\|_{L^2}^2$ is smooth. Its second differential is

$$D^2\Phi_U(\Psi)[v, w] = \langle L_{\Psi, v}, L_{\Psi, w} \rangle_{L^2} + \langle \mathcal{F}(\Psi), [\mathcal{A}(v) \wedge \mathcal{A}(w)] \rangle_{L^2}.$$

The first term is symmetric. Exchanging the two Lie-algebra-valued one-forms in the second changes both the exterior sign and the bracket sign, so that term is symmetric as well. Taking $v = (\alpha, 0)$ and $w = (0, \dot{K})$ proves (15). Source and gauge maps act affinely with constant tangent maps, so the second-order chain rule gives covariance. For compatible $U \subseteq V$, support-compatible zero extension and the full vanishing trace jet identify the two mixed derivatives without boundary distributions. $\square$

5.4. The Response Law Determined by the Source. Let $\mathbf{R}^V$ be the groupoid whose objects are compatible indexed families of configuration spaces, connection- and K -variation Hilbert spaces, their orthogonal metrics, the curvature energies Φ , the covectors $D\Phi$, their responses $\mathcal{Y}$, and the zero-extension and adjoint maps for the supported inclusions above. Its arrows consist of invertible configuration maps and summand-preserving isometries that preserve Φ , $D\Phi$, and $\mathcal{Y}$, and commute with the zero-extension maps and the adjoint maps in (14). Its arrow class contains every isomorphism satisfying these conditions, including those not induced by source isomorphisms.

Theorem 5.4 (Source determination of the response law). *The assignment of the preceding data to a field source and their transport by source isomorphisms define a functor*

$$(16) \quad \mathbb{R}^V : \mathbf{S}^F \longrightarrow \mathbf{R}^V.$$

For every $\mathfrak{R}$, the object $\mathbb{R}^V(\mathfrak{R})$ contains the compatible local maps $\Psi \mapsto \mathcal{Y}_U(\Psi)$, and hence the response law of $\mathfrak{F}(\mathfrak{R})$.

Proof. Sections 4.1 and 4.3 determine the configuration spaces, their orthogonal metrics, and the compatible curvature energies. Theorem 5.1 supplies the covectors, Riesz representatives, component decomposition, and supported-inclusion squares. A source isomorphism transports configurations affinely, acts by an isometry on each variation summand, preserves the curvature energy, and intertwines its differential and Riesz representative. These are precisely the defining equations for arrows of $\mathbf{R}^V$. Identities and compositions are preserved componentwise, while the local squares compose because zero extensions compose. $\square$

Theorem 5.2 gives the internal-rank dependence of this law, and Theorem 5.3 gives its weak mixed reciprocity. Temporal developments require clocks, causal cones, foliations, lapse and shift data, and temporal comparison maps in addition to the source-determined response law.

Section 6 now specializes the surviving rank-one shape response to the gravitational Gauss–Codazzi class. Section 7 then derives the phase-space and finite-label quantum expressions of the response law on their respective domains.

6. THE GRAVITATIONAL RESPONSE AT INTERNAL RANK ONE

Let $\mathbf{S}^{\text{GC}}$ be the nonempty replete full subcategory of field sources admitting an oriented positive-Riemannian internal-rank-one Gauss–Codazzi realization. For $\mathfrak{R} \in \mathbf{S}^{\text{GC}}$, write the realized split bundle as

$$TM \oplus N$$

with base metric h and oriented metric line N . The metric h is determined by the source and remains fixed on the declared configuration space. A local oriented unit section n of N identifies the symmetric internal-valued tensor with

$$K = k \otimes n, \quad k \in \Gamma(\text{Sym}^2 T^*M),$$

so k represents the K -component at rank one.

6.1. Rank-One Split Geometry.

The connection-variation module has fiber $T^*M \otimes \mathfrak{so}(1) = 0$, whereas the symmetric k -variation module retains positive rank when $\dim M > 0$. At internal rank one, the response law therefore has no connection-variation summand and hence no gauge-response type, while its shape-variation summand remains. The assembled metric connection on $TM \oplus N$ remains defined and continues to depend on k .

Lemma 6.1 (Rank-one curvature reduction). *On the oriented metric-line realization, the internal curvature component of the assembled connection vanishes:*

$$B_{\perp} = 0.$$

The tangential and mixed components B_T and B_K retain respectively the quadratic and covariant-derivative dependence on k .

Proof. The curvature formula of Section 4.2 gives

$$B_{\perp}(X, Y) = R^N(X, Y) - K_X K_Y^{\dagger} + K_Y K_X^{\dagger}.$$

The curvature R^N of a metric connection on a real line takes values in $\mathfrak{so}(1)$ and hence vanishes. For a local unit section n , put $k_X = k(X, \cdot)$. Then

$$(K_X K_Y^\dagger)n = h(k_X^\sharp, k_Y^\sharp)n = (K_Y K_X^\dagger)n,$$

so the two quadratic line endomorphisms cancel. The formulas for B_T and B_K in Section 4.2 then give the remaining assertion. $\square$

This is an intrinsic statement about the split bundle and its metric connection. A hypersurface embedding or a Lorentzian development is a further geometric assignment.

6.2. Scalar Gauss and One-Form Codazzi Contractions. Let

$$\mathcal{B}_{\mathfrak{R}} = \mathcal{B}_T^{\mathfrak{R}} \oplus \mathcal{B}_K^{\mathfrak{R}} \oplus \mathcal{B}_\perp^{\mathfrak{R}}$$

be the direct sum containing the three curvature components. In internal rank one, complete contraction with the base and internal metrics gives

$$\mathfrak{s} : \mathcal{B}_T^{\mathfrak{R}} \longrightarrow \mathcal{A}, \quad \mathfrak{v} : \mathcal{B}_K^{\mathfrak{R}} \longrightarrow \Omega_{\mathfrak{R}}^1.$$

These maps define the rank-one curvature-contraction map

$$\mathcal{R}_1 : \mathcal{B}_{\mathfrak{R}} \longrightarrow \mathcal{A} \oplus \Omega_{\mathfrak{R}}^1, \quad \mathcal{R}_1(B_T, B_K, B_\perp) = (\mathfrak{s}(B_T), \mathfrak{v}(B_K)).$$

Theorem 6.2 (Rank-one Gauss–Codazzi contractions). *With $K = k \otimes n$ and the Riemannian split sign convention of the block connection,*

$$\mathfrak{s}(B_T) = \text{Scal}_h - (\text{tr}_h k)^2 + \|k\|_h^2,$$

and

$$\mathfrak{v}(B_K) = \text{div}_h k - \text{d}(\text{tr}_h k).$$

Here the second expression is the one-form whose local components are $\nabla^i k_{ij} - \nabla_j(\text{tr}_h k)$. Consequently

$$\mathcal{R}_1 = 0$$

is the contracted Gauss–Codazzi system of the positive-Riemannian split realization.

Proof. The orthonormal-frame contraction is given in Lemma F.1. It traces the tangential curvature component to the displayed scalar and contracts the normal coefficient of the mixed component to the displayed one-form. $\square$

6.3. The Rank-One Gravitational Response. At each configuration, assemble the shape response and the two curvature contractions as

$$\mathfrak{G}_{\mathfrak{R}} = (\mathcal{Y}_K, \mathfrak{s}(B_T), \mathfrak{v}(B_K)).$$

The first entry is the K -component of the metric-dual response law and takes values in the symmetric K -variation space. The other entries are configuration-valued scalar and one-form contractions of the assembled curvature. Their different codomains preserve their distinct mathematical roles.

Let $\mathbf{G}^C$ be the groupoid whose objects carry

$$(G_K, \mathcal{Y}_K, \mathfrak{s}(B_T), \mathfrak{v}(B_K)).$$

Its arrows are all invertible isomorphisms whose shape component is an isometry, which preserve the four entries, and which commute with the declared restriction and zero-extension maps. Let U_K forget the two curvature contractions, and let pr_K project a variational response object to $(G_K, \mathcal{Y}_K)$.

Theorem 6.3 (Gauss–Codazzi response comparison). *The rank-one assignment defines a functor*

$$\mathbb{G}_{\text{GC}} : \mathbf{S}^{\text{GC}} \longrightarrow \mathbf{G}^C,$$

and the comparison with the variational response is strict on objects and arrows:

$$(17) \quad \text{pr}_K \mathbb{R}^V|_{\mathbf{S}_{\text{GC}}} = U_K \mathbb{G}_{\text{GC}}.$$

Thus the Gauss–Codazzi data carry the same shape metric and shape response as the rank-one restriction of the variational response law. Consequently $\mathcal{Y}_K$ is the gravitational response on this branch, while $\mathfrak{s}(B_T)$ and $\mathfrak{v}(B_K)$ remain its accompanying curvature contractions. The Gauss–Codazzi zero condition, the stationarity equation, and the signed phase-space constraint retain their respective domains.

Proof. The curvature-response theorem supplies $(G_K, \mathcal{Y}_K)$, and Theorem 6.2 supplies the scalar and one-form contractions. Source isomorphisms transport the Riesz response and both complete metric contractions tensorially, so the assignment lands in $\mathbf{G}^C$ and preserves identities and composition. On every source object, both sides of (17) have the same shape metric and Riesz response. On every source arrow, both sides assign the same source-induced shape isometry and response transport. The equality is therefore strict on objects and arrows. The field source on the flat three-torus with its standard locality and Dirichlet data, trivial oriented metric line, reducing Fourier labels, and compatible Gaussian Weyl states provides an object of $\mathbf{S}^{\text{GC}}$. $\square$

6.4. Comparison with Lorentzian Initial Data.

For comparison with the choice of normal signature, define

$$\mathfrak{s}_\varepsilon(B_T) = \text{Scal}_h - \varepsilon((\text{tr}_h k)^2 - \|k\|_h^2), \quad \varepsilon \in \{1, -1\}.$$

Lemma 6.4 (Normal-sign comparison). *The positive-Riemannian source realization gives $\varepsilon = 1$. The Lorentzian ADM normal-sign convention gives $\varepsilon = -1$ [1] and hence*

$$\text{Scal}_h + (\text{tr}_h k)^2 - \|k\|_h^2.$$

The two scalar expressions differ by

$$2((\text{tr}_h k)^2 - \|k\|_h^2),$$

which is not identically zero.

Proof. Substitution of $\varepsilon = 1$ gives the scalar expression in Theorem 6.2, while $\varepsilon = -1$ reverses the sign of its quadratic extrinsic term. Subtracting the two expressions gives the displayed difference. $\square$

For the corresponding comparison of symmetric-tensor pairings, let $m = \dim M$ and define

$$G_t(k, k') = \langle k, k' \rangle_h - t(\text{tr}_h k)(\text{tr}_h k').$$

Lemma 6.5 (Trace-coupling and signature comparison). *The pairing G_t is positive definite exactly for $t < 1/m$ and positive semidefinite at $t = 1/m$, where its kernel is the pure-trace direction. The source-induced K -variation metric is the positive member G_0 . The ADM/DeWitt trace coupling is G_1 , which is indefinite for every $m \geq 2$ [1, 13].*

Proof. Write

$$k = k^\circ + \frac{\text{tr}_h k}{m} h$$

for the orthogonal trace-free and pure-trace decomposition. Then

$$G_t(k, k) = \|k^\circ\|_h^2 + (\text{tr}_h k)^2 \left(\frac{1}{m} - t \right).$$

This proves the positive-definite and semidefinite classifications. For the pure-trace tensor $k = h$,

$$G_1(h, h) = m - m^2 < 0$$

when $m \geq 2$, whereas any nonzero trace-free tensor has positive square. Thus G_1 is indefinite in those dimensions. $\square$

Passing from the source-induced values $(\varepsilon, t) = (1, 0)$ to the Lorentzian comparison values $(\varepsilon, t) = (-1, 1)$ requires a normal-sign choice, a trace-coupling choice, and temporal-development data in addition to the positive-Riemannian source structures.

The oriented metric-line branch therefore has no gauge-response variation, retains the shape response, and supplies scalar and one-form Gauss–Codazzi contractions of the same assembled curvature. Section 7 now gives the phase-space and quantum expressions of the response and compares the zero sets of H_R and $\mathcal{R}_1$.

7. PHASE-SPACE AND QUANTUM EXPRESSIONS OF THE RESPONSE

Section 5 determines the response on configuration space, and Section 6 identifies its gravitational branch. This section expresses that same response first as a vector field on cotangent phase space, then as a derivation of a local CCR algebra, and finally as a finite-label operator commutator. The phase function, operator, and bounded commutator domain introduced along the way retain their distinct domains.

7.1. The Response on Cotangent Phase Space. For each local or cylinder label λ , the classical phase space is the affine cotangent bundle

$$T_G^* \mathcal{C}_\lambda^s(\mathfrak{R}) = \mathcal{C}_\lambda^s(\mathfrak{R}) \times T_{G,\lambda}^* \mathcal{C}^s(\mathfrak{R}),$$

whose points are written $z = (\Psi, \pi)$. Its translation space is

$$\mathcal{P}_\lambda^s(\mathfrak{R}) = T\mathcal{C}_\lambda^s(\mathfrak{R}) \oplus T_{G,\lambda}^* \mathcal{C}^s(\mathfrak{R}),$$

with vectors $x = (q, \eta)$. The cotangent pairing gives the nondegenerate symplectic form

$$\sigma_{\mathfrak{R},\lambda}((q, \eta), (Q, \Pi)) = \langle \Pi, q \rangle - \langle \eta, Q \rangle.$$

In a positive-Riemannian split realization this is the usual cotangent symplectic form. When configuration polynomials are required, a source background Ψ_λ^0 gives the affine coordinate $q = \Psi - \Psi_\lambda^0$. A change of background acts by a canonical translated $*$ -isomorphism on the configuration and response algebras and does not add an origin to the field source. Appendix D.1 proves this chart independence.

The inverse Riesz map $G_{\mathfrak{R},\lambda}^\sharp$ identifies the metric cotangent module with the configuration translation module. The fiberwise Legendre dual of $\frac{1}{2}G_{\mathfrak{R},\lambda}(v, v)$ is

$$T_{\text{kin},\lambda}^{\mathfrak{R}}(\Psi, \pi) = \frac{1}{2} \langle \pi, G_{\mathfrak{R},\lambda}^\sharp \pi \rangle,$$

because $\pi = G_{\mathfrak{R},\lambda}^b v$. The factor $1/2$ is therefore fixed by the Legendre pairing.

For the affine-linear configuration observable $Q_\lambda(\ell)$ and the momentum observable $P_\lambda(v) = \langle \pi, v \rangle$, define

$$\delta_{\mathfrak{R},\lambda} Q_\lambda(\ell) = \ell(G_{\mathfrak{R},\lambda}^\sharp \pi), \quad \delta_{\mathfrak{R},\lambda} P_\lambda(v) = D\Phi_{\mathfrak{R},\lambda}(\Psi)[v] = G_{\mathfrak{R},\lambda}(\mathcal{Y}_\Psi, v).$$

The first entry is the cotangent duality between momentum and configuration velocity. The second is the curvature-energy response covector already determined in Section 4.3. The Leibniz and Poisson rules extend the assignment to the polynomial phase algebra and define a vector field on $T_G^* \mathcal{C}_\lambda^s(\mathfrak{R})$ without a choice of clock.

Here and below, a component of the label category is a connected component of its underlying undirected graph.

Theorem 7.1 (Hamiltonian representation of the response). *Within the declared two-dimensional kinetic-curvature family*

$$H_{a,b} = aT_{\text{kin}} + b\Phi,$$

there is exactly one member whose Hamiltonian derivation equals $\delta_{\mathfrak{R}}$ on every component containing a label λ with $D\Phi_\lambda \neq 0$. It is

$$H_R = T_{\text{kin}} - \Phi, \quad X_{H_R} = (G_{\mathfrak{R}}^\sharp \pi, G_{\mathfrak{R}}^b \mathcal{Y}).$$

On a component where $D\Phi_\lambda \equiv 0$ at every label, equality of the derivations fixes $a = 1$ and gives no equation for b . The signed representative declared by the source has $b = -1$ on that component as well. The conclusion determines a phase-space vector field and no temporal development.

Proof. The cotangent symplectic form gives

$$X_{H_{a,b}} = (aG_{\mathfrak{R}}^\sharp \pi, -bG_{\mathfrak{R}}^b \mathcal{Y}).$$

Equality on configuration generators forces $a = 1$, and equality on any label with nonzero curvature response forces $b = -1$. Naturality carries these coefficients across its label component. When every response covector on a component vanishes, the momentum-generator equation contains no term in b , so only the kinetic coefficient is determined by the derivation. The source declaration supplies the signed representative there. Appendix C.3 proves the classification in the stated competitor family and the corresponding augmentation statement. $\square$

The theorem's competitor family is essential. If a configuration-independent source volume $\mathcal{V}$ is admitted as a third generator, then

$$H_{R,c} = T_{\text{kin}} - \Phi + c\mathcal{V}$$

has the same response derivation whenever $D\mathcal{V} = 0$, although its coefficient can alter the constraint shell, the Master operator, and the associated spectral data.

7.1.1. Positive energy, signed constraint, and signature. The curvature energy Φ is nonnegative, as is T_{kin} on the positive metric branch. Their difference

$$H_{R,\lambda}^{\mathfrak{R}} = T_{\text{kin},\lambda}^{\mathfrak{R}} - \Phi_{\mathfrak{R},\lambda}$$

is a restriction-compatible and gauge-covariant signed constraint. The metric-dual phase inclusions preserve its kinetic term, and joint source restriction preserves its curvature term. A concrete Riemannian split realization carries both terms to their realized counterparts.

Theorem 7.2 (Signature restrictions of the positive realization data). *Every realization preserving the positive Dirichlet form, internal metric, curvature pairing, and Hilbert-space inner product has a Riemannian base, a positive internal metric, and nonnegative curvature energy. A Lorentzian base, an indefinite internal or curvature pairing, or a Krein-space representation changes one of these data and requires the corresponding signature structure. Temporal interpretation requires temporal data in addition to the phase vector field.*

Proof. Positivity of the Dirichlet form gives the Riemannian base metric, positivity of the internal datum gives its metric and compact metric-gauge type, and the positive curvature pairing gives $\Phi \geq 0$. The representation pairing is positive by the Hilbert-space hypothesis. Appendix D.2.1 gives finite-dimensional witnesses showing that each conclusion fails when its corresponding positivity datum is replaced by an indefinite one. $\square$

Lemma 7.3 (Unboundedness of the signed constraint). *If a label contains phase points with zero kinetic term and unbounded curvature energy, then $H_{R,\lambda}^{\mathfrak{R}}$ is unbounded below.*

Proof. At every such point, $H_{R,\lambda}^{\mathfrak{R}} = -\Phi_{\mathfrak{R},\lambda}$. Values of $\Phi_{\mathfrak{R},\lambda}$ above any prescribed constant therefore give values of $H_{R,\lambda}^{\mathfrak{R}}$ below its negative. $\square$

Thus H_R represents the response as a signed phase constraint. It is not asserted to be a lower-bounded generator of real-time evolution.

7.2. Gauss–Codazzi Contractions and the Zero Set of H_R . On the oriented positive-Riemannian internal-rank-one branch, three zero conditions occur on two different spaces. The condition $\mathcal{Y}_K = 0$ is the symmetric-tensor-valued stationarity equation for the curvature energy on configuration space. The condition $\mathcal{R}_1 = 0$ sets the scalar and one-form Gauss–Codazzi contractions to zero, also on configuration space. The scalar equation $H_R = 0$ defines the signed kinetic–curvature shell on cotangent phase space. Their common origin in the assembled curvature does not identify their zero sets.

Write

$$e_1 = |\mathfrak{s}(B_T)|^2 + \|\mathfrak{v}(B_K)\|^2$$

for the squared pointwise norm of the rank-one contraction pair $\mathcal{R}_1$.

Theorem 7.4 (Curvature-energy bound for the rank-one contractions). *On every internal-rank-one realized label of base dimension d ,*

$$e_1 \leq (2d^2 + d) \phi_{\mathfrak{R}}$$

pointwise. Consequently $\Phi_{\mathfrak{R},\lambda} = 0$ implies $\mathcal{R}_1 = 0$ on λ . On the shell $H_{R,\lambda}^{\mathfrak{R}} = 0$,

$$\langle e_1 \rangle_{\lambda} \leq (2d^2 + d) T_{\text{kin},\lambda}^{\mathfrak{R}}.$$

The contraction pair therefore vanishes on the intersection of this shell with $T_{\text{kin},\lambda}^{\mathfrak{R}} = 0$.

Proof. In a realized orthonormal frame, the scalar contraction is a sum of at most d^2 components of B_T , and each component of the one-form contraction is a sum of d components of B_K . Cauchy–Schwarz gives $|\mathfrak{s}(B_T)|^2 \leq d^2 \|B_T\|^2$ and $\|\mathfrak{v}(B_K)\|^2 \leq d \|B_K\|^2$. The curvature-energy density gives $\|B_T\|^2 \leq 2\phi_{\mathfrak{R}}$ and $\|B_K\|^2 \leq \phi_{\mathfrak{R}}$,

proving the pointwise bound. Positive integration gives the labelwise consequences. On $H_R = 0$, the equality $\Phi_{\mathfrak{A},\lambda} = T_{\text{kin},\lambda}^{\mathfrak{A}}$ gives the displayed estimate. $\square$

Lemma 7.5 (Mutual noncontainment of the zero sets). *Neither $\{H_R = 0\}$ nor $\{\mathcal{R}_1 = 0\}$ contains the other. A configuration with a nonzero curvature contraction and momentum scaled so that $T_{\text{kin}} = \Phi$ lies in $\{H_R = 0\}$ with $\mathcal{R}_1 \neq 0$. A flat configuration with nonzero momentum has $\mathcal{R}_1 = 0$ and $H_R = T_{\text{kin}} > 0$.*

Proof. A symmetric shape field with sufficiently large traceless part makes

$$\text{Scal}_h - (\text{tr}_h k)^2 + \|k\|_h^2$$

positive on a realized label. Positivity of the kinetic form then allows a momentum whose kinetic term equals the positive curvature energy. For the second witness, zero curvature annihilates $\mathcal{R}_1$ and Φ , while nonzero momentum retains a positive kinetic term. $\square$

Lemma 7.6 (Euler–Lagrange equation and moment-map comparison). *Let $\mathcal{F}(\Psi) = (B_T, B_K, B_\perp)$ and $W = \text{diag}(1, 2, 1)$. The interior Euler–Lagrange equation of Φ_λ on compactly supported variations is*

$$D\mathcal{F}_\Psi^* W\mathcal{F}(\Psi) = 0.$$

It is a variational stationarity equation, whereas $\mathcal{R}_1 = 0$ is an algebraic curvature contraction. The internal Gauss moment map is momentum-linear and therefore has a third mathematical type.

Proof. The first variation of $\Phi_\lambda = \frac{1}{2}\langle \mathcal{F}, W\mathcal{F} \rangle_\lambda$ in the direction $\dot{\Psi}$ is

$$\langle D\mathcal{F}_\Psi[\dot{\Psi}], W\mathcal{F}(\Psi) \rangle_\lambda.$$

Green’s identity moves the linearized operator onto $W\mathcal{F}(\Psi)$, with the boundary current vanishing under the response-domain hypotheses of Section 5. This gives the displayed equation. It differentiates the curvature, while $\mathcal{R}_1$ contracts it algebraically. A cotangent-lift moment map is momentum-linear, completing the type separation. $\square$

Section 8.1 retains this separation when it defines the corresponding bounded joint ideal and computes its vector annihilator.

7.3. The Response Derivation on the CCR Algebra and Its Representations. The quantum expression begins before a Hilbert-space representation is chosen. Fix a relatively compact region U , and let $\mathcal{A}_{\text{CCR}}(U)$ be the polynomial CCR $*$ -algebra generated by supported configuration observables $Q_U(\ell)$ and momenta $P_U(v)$. In an affine chart, let $\mathcal{C}_\Phi(U)$ be the smallest unital commutative $*$ -algebra of functions on the smooth configuration core that contains the cylindrical polynomials and the functions

$$\Psi \longmapsto D^{k+1}\Phi_U(\Psi)[v_0, \dots, v_k],$$

and is closed under pointwise products and supported directional derivatives. Equal functions on the smooth core are identified. The canonical translated isomorphisms described above make this definition independent of the chart origin.

The compactly supported variation space acts on $\mathcal{C}_\Phi(U)$ by directional differentiation. Define

$$\mathcal{B}_\Phi(U) = \mathcal{C}_\Phi(U) \rtimes U(\mathcal{V}_c(U)_\mathbb{C})$$

with cross relation

$$P_U(v)F - FP_U(v) = -i\partial_v F.$$

The polynomial CCR algebra embeds in this crossed product.

Theorem 7.7 (Response derivation on a local CCR algebra). *The assignment*

$$\Delta_U Q_U(\ell) = P_U(G_U^\sharp \ell), \quad \Delta_U P_U(v) = D\Phi_U(v)$$

*descends to a *-derivation*

$$\Delta_U : \mathcal{A}_{\text{CCR}}(U) \longrightarrow \mathcal{B}_\Phi(U).$$

For every buffered inclusion $U \subseteq V$, the induced maps satisfy

$$K_{VU}\Delta_U = \Delta_V \iota_{VU}.$$

Metric gauge transformations intertwine the derivations, and the compatible local maps induce

$$\Delta_{\mathfrak{R}} : \mathcal{A}_c(\mathfrak{R}) \longrightarrow \mathcal{B}_c(\mathfrak{R})$$

between the algebraic colimits.

Proof. Symmetry of $G_U^\sharp$ annihilates the image of the configuration commutator relation, while symmetry of $D^2\Phi_U$ annihilates the image of the momentum commutator relation. Configuration functions commute, and the cross relation gives the canonical CCR pairing. These identities establish descent through the CCR ideal. Finite-order locality, zero extension of smearings, and restriction-compatible source coefficients give the buffered-inclusion identity. The crossed-product universal property gives composition and gauge covariance. Appendix C.4–C.5 proves equality on overlaps, injectivity, and naturality under inclusion of regions. $\square$

The codomain $\mathcal{B}_\Phi(U)$ contains the curvature-energy derivative functions needed by Δ_U . The theorem does not select a Hilbert-space representation or assert an endoderivation of the whole crossed product.

7.3.1. Local Weyl algebras. For $x \in \mathcal{P}_\lambda^s(\mathfrak{R})$, let $W(x)$ be the Weyl generator. The relations

$$W(x)^* = W(-x), \quad W(x)W(y) = e^{-\frac{i}{2}\sigma_{\mathfrak{R},\lambda}(x,y)}W(x+y)$$

define the algebraic Weyl algebra $\mathcal{W}_\lambda^\circ(\mathfrak{R})$; its C^* -completion is $\mathcal{W}_\lambda(\mathfrak{R})$. Associativity follows from the cocycle identity for $\sigma_{\mathfrak{R},\lambda}$, and label restrictions induce the corresponding homomorphisms. Concrete Riemannian split realizations preserve the phase space, symplectic form, and Weyl relations. The standard Weyl, quasifree, and GNS results used here are recalled from [8, Section 5.2.2.2, Theorem 5.2.8, and Section 5.2.4].

7.3.2. Quasifree representations and the internal gauge action. At every configuration, the second differential of the curvature energy is a continuous symmetric form on the declared Sobolev variation domain and is the weak linearization of the response. At a critical configuration it descends through the admitted infinitesimal hard gauge directions. When this quotient is a form core for a closed nonnegative hard realization, it has the unique configuration-indexed self-adjoint representative $L_{\mathfrak{R},\Psi}$. Appendix B.5 states the exact quotient, closure, label, and common-domain hypotheses; an individual source may have no configuration satisfying them.

On the positive hard quotient, write $L = L_{\mathfrak{R},\Psi}^+$. An operator-regular full-phase quasifree covariance is admissible when the configuration pullback of its inverse full

covariance form has the same domain and values as the closed curvature-stiffness form $Q_{\mathfrak{R},\Psi}$ on every stiffness-reducing envelope. This class allows configuration–momentum cross terms and is nonempty. For every $c \geq \frac{1}{4}$, one compatible family is represented by

$$C_{\mathfrak{R},\Psi}^{(c)} = \begin{pmatrix} L^{-1} & 0 \\ 0 & cL \end{pmatrix},$$

whose regular-core covariance satisfies $\Gamma_{\mathfrak{R},\Psi}^{(c)} + \frac{i}{2}\sigma \geq 0$. The compatible state sets form a nonempty fibration over the stable critical source-configuration pairs. The construction determines the fibration and selects no critical configuration, covariance, state, GNS representation, purity condition, or vacuum. Appendix B.6 proves these statements; Appendix D.5 supplies the finite-label and gauge-implementation details.

The internal gauge group acts symplectically on the local Weyl algebras. After an admitted covariance family has been chosen, the same appendix identifies the subgroup preserving that representation package, its unitary implementation and Gauss generators, the invariant subspace, and the static Ward identity. At internal rank one, $\mathfrak{so}(1) = 0$, so the connection-response type disappears while the shape response may remain.

7.4. Finite-Label Constraints and Completion. For the finite labels considered below, π_λ represents the response derivation Δ_λ by commutation with a symmetric constraint C_λ on an invariant polynomial or Schwartz core. These operators are defined label by label. A single operator on an infinite-label Hilbert space is a further analytic structure and is introduced only when its common core and action have been specified. The operator-theoretic background is the adjoint and spectral calculus of unbounded operators, within the usual constraint-system setting [38, 22, 48].

7.4.1. Symmetric finite-label constraints. Fix a finite label λ with nondegenerate phase form, a regular representation, and a dense invariant polynomial or Schwartz core $\mathcal{D}_{0,\lambda}$. In an affine chart the curvature term is a configuration polynomial of degree at most four, while the kinetic term is quadratic in momentum. The separated form of H_R depends on this polarization, whereas a general linear symplectic change of phase coordinates mixes configuration and momentum. The ordering prescription must therefore be equivariant under the full linear symplectic group rather than only under transformations preserving the chosen polarization.

Theorem 7.8 (Totally symmetrized finite-label quantization). *On each finite label, the totally symmetrized product of the field and momentum operators is the unique linear quantization map on real polynomials of degree at most four that has the classical polynomial as its leading symbol and is equivariant under every linear symplectic transformation of the label’s phase module. This map sends $H_{R,\lambda}^{\mathfrak{R}}$ to a symmetric operator C_λ on $\mathcal{D}_{0,\lambda}$, obeying a number-operator bound, and the implemented internal gauge unitaries fix it on every label they fix. For a label inclusion with isometry $\rho_{\mu\lambda}$, it satisfies the compression identity*

$$\rho_{\mu\lambda}^* C_\mu \rho_{\mu\lambda} = C_\lambda.$$

If $P_{\mu\lambda} = \rho_{\mu\lambda} \rho_{\mu\lambda}^*$ commutes with C_μ on the fine core and

$$P_{\mu\lambda} \mathcal{D}_{0,\mu} = \rho_{\mu\lambda} \mathcal{D}_{0,\lambda},$$

then the core operator splits exactly along the range of $\rho_{\mu\lambda}$. The maximal adjoint constraints, Master operators, and bounded functions of the Master operators are compatible under the same restriction.

Proof. In the degree-at-most-four universe of the theorem, complexify the real representation and the candidate equivariant map. The real symplectic Lie algebra complexifies to the complex symplectic Lie algebra, so invariance extends to the connected complex symplectic group [19, Proposition 1.7.7]. Every equivariant map between tensor powers is then formed from pairwise symplectic contractions [19, Theorem 5.3.5 and Lemma 5.5.4]. On symmetric powers, such a contraction vanishes when applied to two indices of an already-symmetric tensor. No degree-lowering correction therefore survives symmetrization, and the totally symmetrized product is the only assignment reproducing the classical polynomial at leading order under this constraint. $H_{R,\lambda}^{\mathfrak{A}}$ is real and, on each finite label, a quadratic form in the momentum variables minus an at-most-quartic polynomial in the configuration variables, with source-fixed coefficients. The symmetrized product of finitely many field operators built from these variables carries every vector of the smooth core to a vector obeying the required number-operator bound, since finitely many creation and annihilation factors raise the number operator by a bounded finite power. The curvature-energy form is invariant, and the kinetic pairing is built from metrics the internal gauge action preserves, so the classical family is itself invariant on every label the gauge action fixes; equivariance of total symmetrization therefore transfers this classical invariance to the represented operators. Applying the same equivariance to the label isometry gives the displayed compression identity. Under the exact reducing-core hypothesis, the core operator is an orthogonal direct sum. Its adjoint splits along the same range projection, and the same is true of the Master operators and their bounded functional calculi. $\square$

Remark 7.9. *The Born–Jordan ordering has the same leading classical symbol and fails full symplectic equivariance; a generic rotation of one mode leaves an exact nonzero residual already on quartic monomials. The symplectic-equivariance condition therefore fixes the symmetrized product.*

Lemma 7.10 (Ordering dependence). *Any other assignment of a symmetric operator to the same classical quartic, obtained by resolving a literal operator ordering through the canonical commutation relation, differs from the symmetrized assignment by a quadratic form plus a constant, both fixed by the symplectic pairing alone; for the kinetic quadratic form the same resolution differs from the symmetrized assignment by a constant alone. Neither difference reopens a free parameter, since each coefficient is fixed, once the ordering convention is fixed, by the source-determined symplectic pairing.*

Proof. Exchanging two adjacent factors in a literal operator product replaces the pair by its reversed order plus the symplectic pairing of the two arguments, a source-fixed number, and removes the pair. Iterating this exchange on any starting order reduces it to the symmetrized product plus a sum of such removals, each of even degree drop and fixed coefficient. $\square$

7.4.2. Finite-label commutator representation. For each finite source label λ , let $\mathcal{E}_{\Phi}(\lambda)$ be the unital $\ast$ -algebra generated by cylindrical configuration expressions and iterated directional derivatives of the curvature energy, closed under products

and further directional derivatives. For $\nu \subseteq \lambda$, the source restriction $R_{\nu\lambda}^0$ restricts coefficients, density support, configuration and variation slots, and integration data together. Define

$$\mathcal{N}_\Phi(\lambda) = \bigcap_{\nu \subseteq \lambda} \ker(\text{ev}_\nu R_{\nu\lambda}^0)$$

and set

$$\mathcal{C}_\Phi^\sharp(\lambda) = \mathcal{E}_\Phi(\lambda) / \mathcal{N}_\Phi(\lambda).$$

Evaluation at λ gives a possibly nonfaithful homomorphism from this quotient to functions on the finite configuration space; the intersection of all restricted-evaluation kernels defines the quotient. Set

$$\mathcal{B}_\Phi(\lambda) = \mathcal{C}_\Phi^\sharp(\lambda) \rtimes U(V_{\lambda, \mathbb{C}}).$$

Define the labelwise derivation by

$$\Delta_\lambda Q_\lambda(\ell) = P_\lambda(G_\lambda^\sharp \ell), \quad \Delta_\lambda P_\lambda(v) = D\Phi_\lambda(v).$$

Metric symmetry and Hessian symmetry make this assignment descend through the polynomial CCR ideal.

Let λ now be a finite label with nondegenerate phase form for which configuration expressions act by multiplication and momenta act by directional derivatives on a common polynomial/Schwartz core $\mathcal{D}_{0,\lambda}$. Denote the resulting, possibly nonfaithful representation

$$\pi_\lambda : \mathcal{B}_\Phi(\lambda) \longrightarrow \text{End}(\mathcal{D}_{0,\lambda}),$$

and let C_λ be the totally symmetrized operator representing $H_{R,\lambda}^{\mathfrak{R}}$.

Theorem 7.11 (Finite-label commutator representation of the derivation). *For every $A \in \mathcal{A}_{\text{CCR}}(\lambda)$,*

$$\pi_\lambda(\Delta_\lambda A) = i[C_\lambda, \pi_\lambda(A)]$$

on the common core. In particular,

$$\begin{aligned} i[C_\lambda, Q_\nabla(\ell_\nabla)] &= \ell_\nabla(\widehat{G_\nabla^{-1} \pi_\nabla}), & i[C_\lambda, P_\nabla(\alpha)] &= G_\nabla(\widehat{\mathcal{Y}_\nabla}, \alpha), \\ i[C_\lambda, Q_K(\ell_K)] &= \ell_K(\widehat{G_K^{-1} \pi_K}), & i[C_\lambda, P_K(\dot{K})] &= G_K(\widehat{\mathcal{Y}_K}, \dot{K}). \end{aligned}$$

These identities hold exactly for the totally symmetrized quantization on the common core.

Proof. The kinetic term commutes with every momentum generator and the curvature potential commutes with every configuration generator. The canonical commutation relations give the four displayed identities. In the Moyal commutator, every correction beyond the Poisson term differentiates the linear operand at least twice and therefore vanishes. Both sides are derivations, so equality on the phase generators gives equality on the polynomial CCR algebra. $\square$

The exact statement concerns the response derivation generated from the linear configuration and momentum observables on the fixed common core. It does not identify the commutator of two arbitrary independently quantized polynomials with the quantization of their Poisson bracket.

Refinement is covariant on phase generators and contravariant on coefficient functions and iterated derivatives of the curvature energy. For $\lambda \subset \mu$, let $j_{\mu\lambda} : V_\lambda \rightarrow V_\mu$ be the tangent inclusion and define the metric-dual covector lift

$$L_{\mu\lambda} = G_\mu^\flat j_{\mu\lambda} G_\lambda^\sharp.$$

It determines the symplectic phase inclusion

$$I_{\mu\lambda}(v, \ell) = (j_{\mu\lambda}v, L_{\mu\lambda}\ell).$$

Joint restriction of source coefficients, integration data, and curvature derivatives sends $\mathcal{N}_\Phi(\mu)$ into $\mathcal{N}_\Phi(\lambda)$ and therefore descends to the quotient algebras. Together with $P_\mu(j_{\mu\lambda}v) \mapsto P_\lambda(v)$, it determines $K_{\lambda\mu}$ on the crossed-product subalgebra generated by $\mathcal{C}_\Phi^\sharp(\mu)$ and the included momenta. It makes no assignment to momenta outside $j_{\mu\lambda}V_\lambda$.

Theorem 7.12 (Compatibility of phase inclusions, restrictions, and derivations). *The metric-dual lifts and joint source restrictions compose along label chains and satisfy*

$$K_{\lambda\mu}\Delta_\mu I_{\mu\lambda} = \Delta_\lambda.$$

This identity requires neither reducing operator transport nor a common unbounded constraint.

Proof. The identity $G_\mu^\sharp L_{\mu\lambda} = j_{\mu\lambda}G_\lambda^\sharp$ proves the assertion on configuration generators. Joint restriction on the quotient gives $K_{\lambda\mu}D\Phi_\mu(j_{\mu\lambda}v) = D\Phi_\lambda(v)$, which proves it on momenta. Derivation and composition laws give the full claim. Appendix D.1 proves that the metric-dual lift is required for nonorthogonal inclusions and that source coefficients, densities, variation slots, and integration functionals must be restricted together. $\square$

7.4.3. *Common core, adjoint constraint, and Master operator.* Suppose, in addition to the finite-label construction, that a Hilbert space $\mathcal{H}_C$, a dense invariant core $\mathcal{D}_0$, and a symmetric operator

$$C_0 : \mathcal{D}_0 \subset \mathcal{H}_C \longrightarrow \mathcal{H}_C$$

have been specified so that C_0 represents the compatible labelwise constraints on cylindrical vectors. This common operator is a separate analytic hypothesis. Compression compatibility of the finite-label operators alone does not produce it.

Theorem 7.13 (Adjoint constraint and Master operator). *Let*

$$H_* = C_0^*, \quad M_* = H_*^* H_*.$$

Then H_ is closed and densely defined, M_* is positive and self-adjoint, and its closed quadratic form has domain $\mathcal{D}(H_*)$ and value*

$$\mathfrak{m}_*[\psi] = \|H_*\psi\|^2.$$

Consequently

$$\ker M_* = \ker H_*.$$

Moreover,

$$H_*\psi = 0 \iff \langle C_0\phi, \psi \rangle = 0 \quad (\phi \in \mathcal{D}_0),$$

so the common-core Master operator detects the weak constraint kernel. The construction uses the Hilbert adjoint of the specified symmetric core operator and selects no self-adjoint extension of C_0 .

Proof. The symmetric densely defined operator C_0 is closable, so its adjoint H_* is closed and densely defined. The standard theorem for A^*A , applied to $A = H_*$, gives the positivity and self-adjointness of M_* [41, Proposition 3.18(ii)]. Its associated closed form is $\|H_*\psi\|^2$, which vanishes exactly on $\ker H_*$. Appendix E.1–E.3

gives the full domain statement and the reducing restriction theorem. Starting from the separately supplied common operator, it also constructs the unique minimal reducing envelope generated by each label seed; raw label subspaces need not themselves reduce the constraint. $\square$

Bounded functions of M_* , including

$$\mathfrak{b}_* = \frac{M_*}{1 + M_*}, \quad E_\epsilon = \frac{\epsilon}{M_* + \epsilon},$$

retain constraint information without adding an operator domain. Appendix E.4–E.6 treats normal functionals supported on $\ker M_*$ and the character at zero when zero belongs only to the continuous spectrum. It also proves that the compact-torus label family of Section 9 has no single vacuum-containing Fock-space constraint with the required label restrictions. In that model the finite-label operators and their bounded functional calculi remain the valid objects.

7.4.4. The bounded commutator domain. Fix a finite label λ with nondegenerate phase coordinates, represented Hilbert space $\mathcal{H}_\lambda$, invariant polynomial/Schwartz core $\mathcal{D}_\lambda$, and symmetric constraint C_λ . The directions on which the curvature-energy motion form vanishes remain label data and are absent from these represented coordinates. Independently fix a specified represented unital C^* -algebra $\mathfrak{B}_\lambda \subset \mathcal{B}(\mathcal{H}_\lambda)$. This algebra is chosen independently of the source-generated bounded algebra $\mathfrak{B}_x^0$; equality between them requires a separate identification. Write

$$\mathfrak{P}_\lambda = \pi_\lambda(\mathcal{A}_{\text{CCR}}(\lambda))$$

for the represented scalar polynomial CCR algebra. Define the bounded first-commutator domain relative to $\mathfrak{B}_\lambda$ by

$$\mathfrak{A}_\lambda^1 = \left\{ R \in \mathfrak{B}_\lambda : R\mathcal{D}_\lambda, R^*\mathcal{D}_\lambda \subset \mathcal{D}_\lambda, \quad \exists B_R \in \mathcal{B}(\mathcal{H}_\lambda), \quad B_R|_{\mathcal{D}_\lambda} = i[C_\lambda, R]|_{\mathcal{D}_\lambda} \right\}.$$

For $R \in \mathfrak{A}_\lambda^1$, denote the unique bounded extension by

$$\delta_\lambda^v(R) = \overline{i[C_\lambda, R]|_{\mathcal{D}_\lambda}}.$$

The core $*$ -algebra generated by the polynomial and bounded domains is

$$\mathfrak{J}_\lambda = \text{Alg}_{\mathcal{D}_\lambda}^*(\mathfrak{P}_\lambda \cup \mathfrak{A}_\lambda^1).$$

Theorem 7.14 (Common-core derivation and its polynomial and bounded restrictions). *The common-core commutator*

$$\partial_\lambda(A) = i[C_\lambda, A], \quad A \in \mathfrak{J}_\lambda,$$

defines an algebraic $*$ -derivation into the adjointable endomorphisms of $\mathcal{D}_\lambda$. Its polynomial and bounded restrictions are

$$\partial_\lambda(\pi_\lambda A) = \pi_\lambda(\Delta_\lambda A), \quad \overline{\partial_\lambda(R)} = \delta_\lambda^v(R).$$

The algebra $\mathfrak{A}_\lambda^1$ is the maximal unital $*$ -subalgebra of $\mathfrak{B}_\lambda$ whose elements and adjoints preserve the fixed core and whose constraint commutators extend boundedly. Moreover,

$$\mathfrak{P}_\lambda \cap \mathfrak{A}_\lambda^1 = \mathbb{C}1.$$

Let $\lambda \subset \mu$ be a compatible reducing refinement. Assume the Hilbert isometry $J_{\mu\lambda}$ maps $\mathcal{D}_\lambda$ into $\mathcal{D}_\mu$ and satisfies

$$C_\mu J_{\mu\lambda} = J_{\mu\lambda} C_\lambda, \quad R_\mu J_{\mu\lambda} = J_{\mu\lambda} R_\lambda, \quad R_\mu^* J_{\mu\lambda} = J_{\mu\lambda} R_\lambda^*,$$

for $R_\lambda \in \mathfrak{A}_\lambda^1$ and $R_\mu \in \mathfrak{A}_\mu^1$. Then

$$K_{\lambda\mu}\Delta_\mu I_{\mu\lambda} = \Delta_\lambda, \quad \delta_\mu^v(R_\mu)J_{\mu\lambda} = J_{\mu\lambda}\delta_\lambda^v(R_\lambda).$$

Compatible reducing refinements compose and include strict nonidentity refinements.

Proof. Core and adjoint-core invariance make sums, products, and involutions well-defined on $\mathcal{D}_\lambda$. The commutator product rule and symmetry of C_λ give the Leibniz and involution identities, while density of the core makes every bounded extension unique. The finite-label commutator theorem supplies the polynomial restriction and its descent through the represented polynomial quotient. For coherent vectors $W(tz)\Omega$, the expectation of a nonconstant represented scalar CCR polynomial grows as a nonconstant polynomial in t , so such an operator cannot extend boundedly. This leaves only $\mathbb{C}1$ in the intersection. The three displayed intertwining identities give the second refinement identity on the core, and uniqueness carries it to the Hilbert spaces. Appendix D.4 proves preservation of the domains, compatibility on overlaps, existence of strict refinements, and the counterexamples showing why each domain hypothesis is required. $\square$

The bounded restriction is noncentral in the untruncated three-mode torus Fock representation. If $\mathfrak{B}_\lambda$ contains the vacuum projection $P_\Omega = |\Omega\rangle\langle\Omega|$, then

$$\|[C_F, P_\Omega]\| = \frac{|a|}{2\sqrt{2}} > 0 \quad (a \neq 0).$$

Thus $\mathfrak{A}_\lambda^1$ can contain noncentral bounded response content. The example uses a separately chosen Fock representation and vacuum. The theorem determines the derivation on $\mathfrak{I}_\lambda$ and its bounded restriction on $\mathfrak{A}_\lambda^1$, relative to the specified represented algebra and core. Extension to a larger generated algebra or an infinite-label operator domain requires corresponding domain hypotheses.

Remark 7.15 (Role of spatial diffeomorphisms). *Spatial diffeomorphisms enter through equivalence of Riemannian realizations and through source automorphisms. For a fixed presentation, Section 3 realizes the base automorphisms as isometries, and Section 11 carries the same equivalence on the operational side. Internal gauge transformations have the Gauss generators of Section 7.3.2, while the signed response is represented by C_λ and, when a common core operator is specified, by H_* and M_* . Spatial diffeomorphisms remain kinematic equivalences in this paper. A represented spatial-diffeomorphism constraint and a hypersurface-deformation algebra would require further structure.*

Section 8.1 now takes the bounded Master functions, the finite-label commutator domains, and the rank-one curvature contractions as typed inputs to the source-generated bounded observable algebras.

8. BOUNDED OBSERVABLES, CHARGE SECTORS, AND THE FIELD LAW

Sections 5–7 establish the response structures that determine three kinds of bounded data. Source-supplied bounded operations generate a smallest unital C^* -algebra on every represented or local-operation component. Charge sectors and source-supplied transitions form a separately completed metric C^* -category, whose endomorphism algebras contain the positive products of those transitions. States and positive functionals evaluate the resulting operators, while graph domains,

finite operator families, relational clocks, and temporal developments require additional data of their respective types.

8.1. Bounded Observable Algebras Generated by the Source. The represented components, local-operation components, and transition-sector objects form the disjoint index

$$\mathcal{I}^0 = \{0\} \times \mathcal{I}^{(0)} \sqcup \{1\} \times \mathcal{I}^{(1)} \sqcup \{2\} \times \mathcal{I}^{(2)}.$$

The first two kinds of component carry specified bounded operator algebras. The transition-sector objects carry endomorphism algebras in the completed category introduced below.

For a represented or local-operation component x , let B_x denote its specified bounded operator algebra. Five source-supplied families may occur there. They consist of represented units, declared bounded functions of the Master operator, declared bounded functions of the rank-one curvature contractions, declared bounded cylinder functions, and the positive source and target products of source-supplied transitions. We index these families by $V0$, VM , VR , VH , and VT , respectively. For $r \in \{V0, VM, VR, VH\}$ present at x , the source supplies a parameter set and a map

$$\tilde{\sigma}_x^r : P_x^r \longrightarrow B_x.$$

The source-typed presence predicate records whether that parameter set and map occur at x . In particular, VR occurs only on labels carrying the rank-one curvature contractions, while VH occurs only where a bounded cylinder function has been declared as a local operation. Define

$$\mathfrak{V}_x^0 = C_{B_x}^* \left(1, \bigcup_{r \in \{V0, VM, VR, VH\}} \tilde{\sigma}_x^r(P_x^r) \right),$$

and write σ_x^r for the same generating map with codomain restricted to $\mathfrak{V}_x^0$.

If $T : i \rightarrow j$ is a source-supplied transition, its two positive products have different endomorphism types. Their VT parameters are elements

$$v_T^- \in \mathfrak{V}_{U,i}^0, \quad v_T^+ \in \mathfrak{V}_{U,j}^0$$

whose faithful representations satisfy

$$\iota_{U,i}(v_T^-) = T^*T, \quad \iota_{U,j}(v_T^+) = TT^*.$$

Thus a transition supplies one positive generator at its source object and one at its target object. Categorical completion adds morphisms while these source-supplied parameter sets remain fixed.

Theorem 8.1 (Generated bounded C^* -algebras). *For every represented or local-operation component x , $\mathfrak{V}_x^0$ is the unique smallest unital C^* -subalgebra of B_x containing the source-supplied generators at x . Source isomorphisms transport these algebras isomorphically. A noninvertible map defined for one generator family transports that family, while transport of the full algebra requires compatible target images for every generator present at the source. At every transition-sector object i , the VT generators define a faithful unital inclusion*

$$\iota_{U,i} : \mathfrak{V}_{U,i}^0 \hookrightarrow \text{End}_{\mathbf{C}_U}(i),$$

which may be proper.

Proof. The intersection of the unital C^* -subalgebras of B_x containing the displayed generators is again a unital C^* -subalgebra and is their unique minimum. Naturality of the generating maps gives the assertion for source isomorphisms. A map specified on one family extends to its generated algebra, and the universal property applies to $\mathfrak{V}_x^0$ once the map is specified compatibly on the complete generating set. At a transition-sector object, the positive endomorphisms T^*T and TT^* generate the stated subalgebras, and faithful-probe completion supplies the inclusions $\iota_{U,i}$. $\square$

Where the Master and rank-one functional calculi coexist, their bounded generators also define a joint ideal. This ideal records simultaneous vector support at the zero of the Master constraint and the zero of the rank-one curvature contractions, while the constraint and curvature readout retain their different types. On such a represented rank-one component x , put

$$b_x = M_{*,x}(1 + M_{*,x})^{-1}, \quad G_{EH,x} = \{b_x\} \cup \{f(\mathcal{R}_{1,x}) : f \in C_b, f(0) = 0\},$$

so that

$$\mathfrak{V}_x^0 = C^*(1, b_x, \{\epsilon(M_{*,x} + \epsilon)^{-1} : \epsilon > 0\}, \{h(\mathcal{R}_{1,x}) : h \in C_b\}),$$

$$\mathcal{I}_{EH,x}^0 = \overline{\text{span}(\mathfrak{V}_x^0 G_{EH,x} \mathfrak{V}_x^0)}^{\|\cdot\|}.$$

Theorem 8.2 (Annihilator of the generated Master–Gauss–Codazzi ideal). *The common vector annihilator of $\mathcal{I}_{EH,x}^0$ is*

$$\text{Ann}(\mathcal{I}_{EH,x}^0) = \ker H_{*,x} \cap \bigcap_{f(0)=0} \ker f(\mathcal{R}_{1,x}).$$

Source isomorphisms preserve this ideal. A noninvertible map carries it into the corresponding target ideal whenever a unital $$ -homomorphism on $\mathfrak{V}_x^0$ transports both the VM and VR generating families.*

Proof. Let K_x denote the displayed intersection. The unit preserves K_x , the operator b_x vanishes on it, each $\epsilon(M_{*,x} + \epsilon)^{-1}$ acts as the identity, and every $h(\mathcal{R}_{1,x})$ acts by the scalar $h(0)$. Hence K_x is invariant under $\mathfrak{V}_x^0$, and every member of $G_{EH,x}$ annihilates it. Conversely, every member of $G_{EH,x}$ lies in the ideal, so the ideal's annihilator lies in K_x . Finally, $\ker b_x = \ker M_{*,x} = \ker H_{*,x}$. Unitary source transport preserves the two functional calculi, and a $*$ -homomorphism carrying both generating families sends every algebraic ideal generator into the target ideal. $\square$

8.2. Charge Sectors and Source-Supplied Transitions. The charge sectors and source-supplied covariant transitions form the projective diagram

$$\widehat{\mathbf{C}}_\bullet = \{\widehat{\mathbf{C}}_U, \widehat{s}_{VU} : \widehat{\mathbf{C}}_V \longrightarrow \widehat{\mathbf{C}}_U\}_{U \leq V}$$

of completed real metric C^* -categories. The norm used in the completion is determined by the specified faithful finite-projective probes. This categorical completion and the componentwise generation of $\mathfrak{V}_{U,i}^0$ are separate operations. The residual source-symmetry group D_G acts on objects and morphisms and hence on T^*T and TT^* . On an object fixed by a subgroup, the invariant endomorphisms form the corresponding fixed-point C^* -algebra; the remaining objects and morphisms are carried to their symmetry-related images.

Theorem 8.3 (Transition categories and their endomorphism observables). *The source-supplied transitions, their adjoints and composites, and the restriction functors generate $\widehat{\mathbf{C}}_\bullet$ after faithful-probe completion. At every object i , their positive products generate a unital algebra $\mathfrak{V}_{U,i}^0$ and a natural faithful inclusion*

$$\iota_{U,i} : \mathfrak{V}_{U,i}^0 \hookrightarrow \text{End}_{\widehat{\mathbf{C}}_U}(i).$$

*The inclusion is proper on the weight-one real $SO(2)$ torus object. A source-supplied transition remains a morphism between its source and target sectors, while T^*T and TT^* are endomorphism observables in the two corresponding object algebras.*

Proof. Adjoint, composition, and restriction extend continuously in the norm defined by the faithful probes. The two positive products associated with each source-supplied transition have the required source and target types, and their unital C^* -algebras give $\mathfrak{V}_{U,i}^0$. The transition and probe comparison maps prove naturality. On the weight-one real $SO(2)$ torus object, the completed endomorphism algebra contains a nonscalar endomorphism outside the algebra generated by these positive products, which proves that the inclusion is strict. $\square$

8.3. The Bounded Field Law. Fix $\mathfrak{R} \in \mathbf{S}^F$. The common field-law data $L_{\mathfrak{R}}$ contain the configuration and variation spaces, the assembled metric connection and its curvature, the curvature energy and response, the signed phase-space constraint, the local quantum derivations and finite-label expressions, and the curvature readout on the oriented positive-Riemannian internal-rank-one components. Its represented coordinates range over the entire admitted indexed family and do not select a single Hilbert-space representation or state.

Adjoining the component index, the source-supplied generating maps, their generated algebras, the completed transition category and its faithful inclusions, and the defined component maps gives the bounded field-law package

$$\mathcal{L}_{\mathfrak{R}} = \left(L_{\mathfrak{R}}, \mathcal{I}^0, \{P^r, \sigma^r\}_{r=V0,VM,VR,VH,VT}, \mathfrak{V}^0, \widehat{\mathbf{C}}_\bullet, \iota, \mathcal{M} \right).$$

Here $\mathcal{M}$ records the component maps and restriction comparisons that are defined between the displayed families.

Appendix G constructs the probe-norm transition completion and its generated endomorphism algebras, proves the strict torus witness, and defines independently the groupoid $\mathbf{L}^B$ of packages of this type and all isomorphisms preserving their generator origins, component types, transition diagrams, inclusions, and restriction maps. If $\mathbf{L}$ denotes the groupoid of the common field-law data, the projection

$$U : \mathbf{L}^B \longrightarrow \mathbf{L}$$

forgets the added bounded coordinates. The source assignments are the functors

$$L : \mathbf{S}^F \longrightarrow \mathbf{L}, \quad \mathbb{L} : \mathbf{S}^F \longrightarrow \mathbf{L}^B.$$

Theorem 8.4 (Source determination of the bounded field law). *Within $\mathbf{S}^F$, the source presentation determines $\mathcal{L}_{\mathfrak{R}}$ functorially, and*

$$U\mathbb{L} = L$$

on objects and arrows. The component algebras, transition category, and their maps are fixed before a state evaluation, a represented graph-domain family, a finite operator family, a selected self-adjoint extension, or a temporal development is supplied.

Proof. Sections 4–7 determine $L_{\mathfrak{A}}$. The preceding subsections determine the component index, generating maps, componentwise C^* -algebras, completed transition category, faithful inclusions, and defined component maps. Source isomorphisms preserve the component types and generator domains, carry every generator to its corresponding target generator, and intertwine the transition restrictions and positive-product identities. These maps define $\mathbb{L}$, while projection to the first coordinate gives the displayed equality with L . $\square$

The dependency order within this package is fixed by the domains of its entries. The geometric, variational, phase-space, and algebraic quantum data are determined before a Hilbert-space representation is chosen. The indexed represented part records every admitted constraint realization, its maximal adjoint and Master operator when defined, and the associated bounded functional calculi. The local-operation algebras and completed transition category retain their own specified bounded ambient algebras. Section 11 combines this independently defined bounded package with the full response package over their common field-law data.

8.4. States, Filtered Evaluations, and Additional Structures. Let

$$U_S : \Omega \longrightarrow \mathbf{S}^F$$

be the compatible-state fibration. Its fiber over $\mathfrak{A}$ consists of the normalized compatible regular states admitted by the represented branch. A state or positive functional restricts to every generated component algebra in its domain and may evaluate invariant transition endomorphisms in the corresponding endomorphism algebras.

Two such functionals are equivalent on the declared bounded observables when their restrictions agree on every common component algebra $\mathfrak{V}_x^0$ and on every invariant positive transition product in their common domains. The residual group D_G acts on these evaluations by pullback along its action on $\mathcal{L}_{\mathfrak{A}}$, so one D_G -orbit consists of symmetry-related evaluations of the same law.

Theorem 8.5 (Operators and their evaluations). *For $R \in \mathfrak{V}_x^0$, the source determines R as an element of the fixed component algebra, while a compatible functional ω supplies the scalar $\omega(R)$. The charge sectors, energy, response, constraint, rank-one curvature operators, transition morphisms, and invariant transition products retain their corresponding object, morphism, or endomorphism types before those scalar values are assigned.*

If an additional tensor factor acts trivially or by scalars on every declared operator, functionals that differ only on that factor have identical restrictions to the declared bounded algebras. Declaring a nonscalar operator on the additional factor enlarges the bounded observable data.

Proof. Membership of R follows from the source-supplied generating maps and C^* -closure. A functional evaluates that already determined operator and does not change its algebraic relations. The same distinction holds between a transition endomorphism and its functional value. Scalar or trivial action on an additional tensor factor leaves every stated restriction unchanged, while a declared nonscalar action contributes a new bounded generator. $\square$

8.4.1. *Master-filtered rank-one curvature evaluations.* On a finite, local, or cylindrical rank-one label λ , let

$$A_\lambda = F_\lambda(\mathcal{R}_{1,\lambda}), \quad F_\lambda \in C_b(\text{Spec } \mathcal{R}_{1,\lambda}).$$

This joint continuous functional calculus is the source-generated rank-one readout family. Bounded Borel functions belong only to a separately declared represented attachment. The following estimates apply to either family once the observable and the Master calculus lie in a common represented von Neumann algebra.

Given a normal state ω_λ and $\epsilon > 0$, define

$$E_{\epsilon,\lambda} = \epsilon(M_{*,\lambda} + \epsilon)^{-1}.$$

Whenever $\omega_\lambda(E_{\epsilon,\lambda}^2) > 0$, its Master-filtered value on A_λ is

$$\omega_{\lambda,\epsilon}(A_\lambda) = \frac{\omega_\lambda(E_{\epsilon,\lambda} A_\lambda E_{\epsilon,\lambda})}{\omega_\lambda(E_{\epsilon,\lambda}^2)}.$$

If $P^{M_{*,\lambda}}$ is the spectral measure of the Master operator, set

$$\nu_\lambda(B) = \omega_\lambda(P^{M_{*,\lambda}}(B)), \quad \omega_\lambda[M_{*,\lambda}] = \int_{[0,\infty)} \mu \, d\nu_\lambda(\mu).$$

The filtered Master evaluation is the bounded-calculus expression

$$\omega_{\lambda,\epsilon}[M_{*,\lambda}] = \frac{\omega_\lambda(M_{*,\lambda} E_{\epsilon,\lambda}^2)}{\omega_\lambda(E_{\epsilon,\lambda}^2)}.$$

Theorem 8.6 (Compatible Master-filtered rank-one evaluations). *Suppose $\|A_\lambda\| \leq R$ and*

$$\omega_\lambda[M_{*,\lambda}] \leq \delta\epsilon, \quad 0 < \delta < \frac{1}{2}.$$

Then

$$\omega_\lambda(E_{\epsilon,\lambda}^2) \geq 1 - 2\delta, \quad \omega_{\lambda,\epsilon}[M_{*,\lambda}] \leq \frac{\epsilon}{4\omega_\lambda(E_{\epsilon,\lambda}^2)},$$

and

$$|\omega_{\lambda,\epsilon}(A_\lambda) - \omega_\lambda(A_\lambda)| \leq 2R \left(\sqrt{\delta} + \frac{2\delta}{1 + \sqrt{1 - 2\delta}} \right).$$

For $\lambda \subseteq \lambda'$, suppose a normal unital $$ -homomorphism*

$$r_{\lambda'\lambda} : \mathcal{A}_{\lambda'}^{\text{vN}} \longrightarrow \mathcal{A}_\lambda^{\text{vN}}$$

preserves the bounded Borel Master calculus and the rank-one observable,

$$r_{\lambda'\lambda}(f(M_{*,\lambda'})) = f(M_{*,\lambda}), \quad r_{\lambda'\lambda}(A_{\lambda'}) = A_\lambda,$$

and suppose $\omega_{\lambda'} = \omega_\lambda \circ r_{\lambda'\lambda}$. Whenever the filter denominators are positive,

$$\omega_{\lambda',\epsilon}(A_{\lambda'}) = \omega_{\lambda,\epsilon}(A_\lambda).$$

If both reference states have positive zero-window mass, the corresponding normal kernel-supported limits satisfy the same identity at $\epsilon = 0$.

Proof. The scalar functional-calculus inequalities

$$1 - E_{\epsilon,\lambda}^2 \leq \frac{2M_{*,\lambda}}{\epsilon}, \quad M_{*,\lambda} E_{\epsilon,\lambda}^2 \leq \frac{\epsilon}{4}$$

give the filtered-mass and constraint-suppression bounds. Normalizing $E_{\epsilon,\lambda}\Omega_\lambda$ and applying the vector-state estimate gives the displayed expectation bound. Preservation of the two bounded calculi carries the filter, the observable, and both terms

of the normalized evaluation through $r_{\lambda'\lambda}$. The normal zero-width identity follows by taking the kernel-supported limits. Appendix F.2 gives the spectral-moment formulation and the full estimates. $\square$

An exact tensor-product cancellation is also available when

$$A_{\lambda'} = A_\lambda \otimes 1, \quad \omega_{\lambda'} = \omega_\lambda \otimes \omega_z, \quad E_{\epsilon, \lambda'} = E_{\epsilon, \lambda} \otimes E_{\epsilon, z}.$$

The factorization of the Master filter is an explicit hypothesis. When $\omega_z(E_{\epsilon, z}^2) > 0$, its contribution cancels from the numerator and denominator. This special identity does not replace the joint Master-and-curvature comparison map required for a general label extension.

Theorem 8.7 (Finite-label approximation of filtered rank-one expectations). *Let A be a bounded rank-one curvature observable and A_F a finite-label observable with limiting image $\iota_F(A_F)$. Their evaluations obey*

$$\begin{aligned} |\omega(A) - \omega_{F, \epsilon}(A_F)| &\leq \|A - \iota_F(A_F)\| \\ &\quad + |\omega(\iota_F(A_F)) - \omega_F(A_F)| \\ &\quad + |\omega_F(A_F) - \omega_{F, \epsilon}(A_F)|. \end{aligned}$$

Consequently the filtered values converge to $\omega(A)$ when these three errors tend to zero. If $\|A_F\| \leq R$ and $\omega_F[M_{*, F}] \leq \delta_F \epsilon$, the last term is bounded by the stability estimate in Theorem 8.6; it therefore suffices that the first two errors and δ_F tend to zero.

Proof. Insert $\omega(\iota_F(A_F))$ and $\omega_F(A_F)$ between the endpoint evaluations and apply the triangle inequality. Appendix F.4 records the resulting convergence criterion. $\square$

8.4.2. Natural state selections.

Theorem 8.8 (The state fibration and natural state selections). *The bounded field-law functor determines the compatible-state fibration and its transport. The groupoid of natural sections over the full source groupoid is empty if some connected source component has no automorphism-invariant admitted state and is noncontractible otherwise. Hence the law supplies no distinguished state, vacuum, pure GNS representation, irreducible representation, or D_G -orbit.*

Over the stable critical source-configuration pairs for which $L_{\mathfrak{R}, \Psi}$ is strictly positive on $\mathcal{H}_{\mathfrak{R}, \Psi}$, the covariances

$$C_{\mathfrak{R}, \Psi}^{(c)} = \text{diag}(L_{\mathfrak{R}, \Psi}^{-1}, cL_{\mathfrak{R}, \Psi}), \quad c \geq \frac{1}{4},$$

give distinct natural quasifree sections. Uncertainty saturation selects $c = \frac{1}{4}$ within this family only after the saturation condition is added.

Proof. Natural sections are classified by automorphism-fixed states on the connected components of the source groupoid. The absence of such a state on one component makes the section groupoid empty. Otherwise the inhabited strictly positive stiffness component contains at least the distinct sections $c = \frac{1}{4}$ and $c = \frac{1}{2}$, so the section groupoid is noncontractible. On a spectral value $\ell > 0$, quasifree positivity reduces to positivity of

$$\begin{pmatrix} \ell^{-1} & i/2 \\ -i/2 & c\ell \end{pmatrix},$$

whose determinant is $c - \frac{1}{4}$. Saturation therefore gives $c = \frac{1}{4}$, but the saturation predicate is additional to the bounded field law and does not supply a state on every source component. $\square$

8.4.3. Relational clocks and temporal developments. On a finite-dimensional cylindrical phase space, or on a smooth local phase chart carrying the required Hamiltonian flow, let φ_s be the characteristic flow of X_{H_R} . A chosen local function T satisfying $X_{H_R}(T) \neq 0$ and a nearby value τ determine the local solution of

$$T(\varphi_{s(z,\tau)}z) = \tau.$$

Every local phase function f then gives the relational observable

$$O_{f,T}(\tau)(z) = f(\varphi_{s(z,\tau)}z).$$

Appendix F.5 proves the local flow invariance and slice recovery. The clock, its selected reading, and a quantum representation of the resulting classical observables are additional data over the fixed law.

Let

$$U_D : \mathbf{D} \longrightarrow \mathbf{S}^F$$

denote the declared oriented development fibration.

Theorem 8.9 (No natural oriented development). *The fibration U_D has no natural section. At an oriented internal-rank-one configuration with $K = 0$, central normal inversion fixes the source and the field configuration while exchanging the two normal orientations. Naturality would require a selected orientation to equal its opposite.*

Proof. A natural section evaluated at the fixed source configuration would have to be invariant under every source automorphism fixing that configuration. Central normal inversion belongs to that stabilizer and acts freely on the two-point orientation fiber. The fiber has no fixed point, so no natural section exists. $\square$

A temporal development begins after an orientation, clock, causal cone, lapse, shift, and the remaining development data have been supplied. Similarly, a graph-norm algebra begins with a separately specified represented constraint, invariant domain, and generating family, while a finite operator family carries its own declared membership. The following models make such represented choices and compute their spectra, commutator kernels, and graph ideals while the source-generated bounded law remains fixed.

9. MODELS AND EXPLICIT CALCULATIONS

The models below exhibit the response components and their domain-specific expressions before turning to the represented structures that require additional choices. A rank-two configuration has simultaneously nonzero connection and shape responses, while two rank-one configurations exhibit the gravitational response, its finite-label commutator expression, and a calculation extending from curvature energy to a bounded commutator and a local relational observable. Two further torus calculations distinguish a fixed holonomy class carried by transition data from the global holonomy dependence of a configuration-indexed response spectrum. The final subsection fixes additional graph-domain and finite-family data and computes the resulting ideals and commutator kernels. Each calculation uses only the source, configuration, representation, operator family, state, or clock stated for that model.

9.1. Simultaneous Gauge and Shape Responses at Rank Two. Let $T^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ have its flat metric and standard volume, and write $V = (2\pi)^3$. On the trivial oriented Euclidean plane bundle, choose an orthonormal frame (e_1, e_2) , let J generate the standard $SO(2)$ -action, and write the metric connection as $\nabla^{\mathcal{N}} = d + A$. With the harmonic coefficients α_i fixed, consider the two-parameter representative slice

$$A = (\alpha_1 dx^1 + \alpha_2 dx^2 + \alpha_3 dx^3 + a \sin x^1 dx^2)J, \quad K = k dx^1 \otimes dx^1 \otimes e_1.$$

The configuration-space metric restricts to $V da^2 + V dk^2$. Appendix J.1 computes

$$\Phi(a, k; \alpha) = V k^2 \left(\alpha_2^2 + \alpha_3^2 + \frac{a^2}{2} \right) + \frac{V}{2} a^2$$

and gives the Riesz coordinates of the response restricted to this slice as

$$\mathcal{Y}_a = a(1 + k^2), \quad \mathcal{Y}_k = 2k \left(\alpha_2^2 + \alpha_3^2 + \frac{a^2}{2} \right), \quad \frac{\partial^2 \Phi}{\partial a \partial k} = 2Vak.$$

At

$$\alpha_2 = \frac{1}{4}, \quad \alpha_3 = 0, \quad a = k = 1,$$

these coordinates are $\mathcal{Y}_a = 2$ and $\mathcal{Y}_k = 9/8$, and the mixed derivative is $2V$. Their nonvanishing proves that the full connection and shape responses are both nonzero at this configuration, as stated in Theorem J.1. The second component is the shape response at internal rank two; the gravitational interpretation belongs to the oriented positive-Riemannian rank-one branch.

9.2. A One-Mode Rank-One Gravitational Response. On the same flat base, replace the normal plane by an oriented metric line with unit section n and trivial normal connection. Define

$$\kappa = \sin(x^2) dx^1 \otimes dx^1 \otimes n, \quad e = \frac{\kappa}{\|\kappa\|_G}, \quad K(q) = qe,$$

and set

$$\lambda = 2\Phi(e) > 0.$$

The mode e has unit configuration norm. Appendix J.2 computes the restricted curvature energy and its metric-dual shape response as

$$\Phi(q) = \frac{\lambda}{2} q^2, \quad \mathcal{Y}_K(q) = \lambda q.$$

The internal gauge algebra is $\mathfrak{so}(1) = 0$, so the connection-variation summand is absent. On this oriented positive-Riemannian rank-one realization, the surviving shape response is the gravitational response and is nonzero for $q \neq 0$.

Let $\mathcal{H} = L^2(\mathbb{R}, dq)$, and on the Schwartz core $\mathcal{S}(\mathbb{R})$ set

$$(Q\psi)(q) = q\psi(q), \quad (P\psi)(q) = -i\psi'(q).$$

The symmetrically represented constraint on this core and its response commutator satisfy

$$C_0 = \frac{1}{2}(P^2 - \lambda Q^2), \quad i[C_0, P] = \lambda Q.$$

The operator C_0 is essentially self-adjoint; write $C = \overline{C_0}$. The common-core commutator is the finite-label quantum expression of the gravitational response, while spectral statements refer to the closure C and its Master operator C^2 .

9.3. Curvature, Constraint, and a Bounded Commutator in a Rank-One Field. For a second rank-one model, let

$$S = d\theta^2 \otimes d\theta^2 - d\theta^3 \otimes d\theta^3, \quad K_{q,x} = (qh + x \sin \theta^1 S) \otimes n$$

on the flat three-torus with its oriented metric normal line and trivial normal connection. Denote the resulting two-parameter family of field configurations by

$$\iota(q, x) = (\nabla^{\mathcal{N}} = d, K_{q,x}).$$

Direct contraction of the assembled curvature gives

$$\Phi(q, x) = 3q^4 + x^2 + \frac{3}{8}x^4.$$

Since the shape metric restricts to $3dq^2 + dx^2$, the pullback of the response is

$$\iota^* \mathcal{Y}_K = 4q^3 \partial_q + \left(2x + \frac{3}{2}x^3\right) \partial_x.$$

The corresponding signed Legendre function is

$$H_R = \frac{p_q^2}{6} + \frac{p_x^2}{2} - 3q^4 - x^2 - \frac{3}{8}x^4.$$

In the normalized coordinates

$$Q_q = \sqrt{3}q, \quad Q_x = x, \quad P_q = \frac{p_q}{\sqrt{3}}, \quad P_x = p_x,$$

the symmetric two-mode constraint on the Schwartz core of $L^2(\mathbb{R}^2, dQ_q dQ_x)$ is

$$\widehat{C}_{\lambda_2} = \frac{1}{2}(\widehat{P}_q^2 + \widehat{P}_x^2) - \frac{1}{3}\widehat{Q}_q^4 - \widehat{Q}_x^2 - \frac{3}{8}\widehat{Q}_x^4.$$

Define

$$\gamma(Q_q, Q_x) = \pi^{-1/2} e^{-(Q_q^2 + Q_x^2)/2}, \quad P_\gamma = |\gamma\rangle\langle\gamma|.$$

The projection P_γ is separately specified in the represented algebra $\mathfrak{B}_{\lambda_2}$. Its core-invariance properties and bounded commutator place it in $\mathfrak{A}_{\lambda_2}^1$, and direct calculation gives

$$\|[\widehat{C}_{\lambda_2}, P_\gamma]\| = \sqrt{\frac{475}{96}}.$$

The declared generator list for this source contains no operation with output P_γ ; the conclusion is membership in the represented first-commutator domain and does not place the projection in $\mathfrak{V}_x^0$.

The positive-Riemannian rank-one contractions are

$$(\mathfrak{s}(B_T), \mathfrak{v}(B_K)) = (-6q^2 + 2x^2 \sin^2 \theta^1, 0).$$

On the region $p_q \neq 0$, choose the relational clock $T = q$. If φ_s is the characteristic flow of H_R , the equation

$$q(\varphi_s z) = \tau$$

has a unique local solution $s = s(z, \tau)$, and

$$O_x(\tau)(z) = x(\varphi_{s(z, \tau)} z)$$

is the corresponding local relational observable. Its leading expansion is

$$O_x(\tau) = x + \frac{3p_x}{p_q}(\tau - q) + O((\tau - q)^2).$$

Here τ is the selected reading of the chosen clock. Appendix J.3 gives the complete curvature formulas, operator-domain verification, commutator calculation, and local characteristic expansion.

9.4. Constraint Spectrum and Holonomy on the Three-Torus. Return to the rank-two flat-torus source and set $K = 0$. Fix a harmonic holonomy class $[\alpha] \in (\mathbb{R}/\mathbb{Z})^3$ and take

$$A = \alpha_{\text{harm}} + a \sin x^1 dx^2 J, \quad \alpha_{\text{harm}} = \sum_{i=1}^3 \alpha_i dx^i J, \quad a \neq 0.$$

For $n \geq 1$, let E_n be the real coexact Fourier span of the nonzero cubic modes with $|k_i| \leq n$. The stiffness $L_n = d^*d$ is positive on E_n , while the harmonic coefficients remain outside these represented coordinates. On the finite-dimensional Schrödinger label, the signed function

$$H_n(q, p) = \frac{1}{2} \|p\|^2 - \frac{1}{2} \langle q + A_b, L_n(q + A_b) \rangle, \quad A_b = a \sin x^1 dx^2 J,$$

has symmetric/Weyl quantization $C_{0,n}$ on $\mathcal{S}(E_n) \subset L^2(E_n)$. Write

$$C_n = \overline{C_{0,n}}, \quad M_n = C_n^2.$$

Theorem J.2 gives

$$\sigma(C_n) = \sigma_{\text{ess}}(C_n) = \mathbb{R}$$

with purely absolutely continuous spectral type and

$$\sigma(M_n) = \sigma_{\text{ess}}(M_n) = [0, \infty), \quad \ker M_n = \{0\}.$$

Thus zero belongs to the continuous spectral support of the Master operator and is not a normalizable zero mode.

The harmonic form satisfies

$$d\alpha_{\text{harm}} = 0, \quad d^*\alpha_{\text{harm}} = 0, \quad d^*d\alpha_{\text{harm}} = 0.$$

Its holonomy class remains fixed in the transition data while the coexact configuration modes are represented on E_n . The declared local operations contain no bounded holonomy cylinder function, so the source-generated bounded observable algebra acquires no holonomy generator.

Although the declared bounded generators contain no holonomy cylinder function, holonomy can enter the spectrum of the self-adjoint curvature Hessian at a configuration with nonzero K . The next model computes this dependence.

9.5. Holonomy Dependence of the Curvature Hessian on the Flat Three-Torus. Let the internal bundle over the flat three-torus be the trivial oriented rank-three metric bundle. In an oriented orthonormal frame, let J_{23} generate rotations of $\text{span}\{e_2, e_3\}$, and consider

$$A_\theta = \theta dx^1 J_{23}, \quad K_k = k dx^1 \otimes dx^1 \otimes e_1, \quad \theta \in \mathbb{R}/\mathbb{Z}, \quad k \neq 0.$$

These configurations have zero assembled curvature and are critical. For fixed k , different values of θ are gauge equivalent on every contractible region, while their holonomies around the x^1 -circle may be globally distinct. The image line of K_k is preserved by the connection and defines an oriented rank-one subsystem, so variation in the k -direction is gravitational in the sense of Section 6.

Let $L_{\theta,k}$ be the nonnegative self-adjoint representative of the curvature Hessian on the Hilbert-space quotient by the closed infinitesimal internal-gauge directions.

The quotient has an infinite-dimensional kernel remaining after gauge reduction. Consequently its resolvent is not compact and $e^{-tL_{\theta,k}}$ is not trace class. The covector dx^1 , the rank-one tangential factor of K_k , determines the Haar projector P_η onto fields invariant under translations in the x^2 - and x^3 -directions. This projector reduces $L_{\theta,k}$, and

$$\Xi_{\theta,k}(t) = \text{Tr}(P_\eta L_{\theta,k} e^{-tL_{\theta,k}})$$

is finite for every $t > 0$.

Theorem 9.1 (Holonomy-dependent weighted Hessian trace on the flat torus). *For the family $A_\theta = \theta dx^1 J_{23}$ above,*

$$\Xi_{\theta,k}(t) = \sum_{m \in \mathbb{Z}} [6\mathcal{A}_2(m, t) \cos(2\pi m\theta) + 8\mathcal{A}_1(m, t) \cos(2\pi m\theta) \cos(2\pi mk)],$$

where

$$\mathcal{A}_c(m, t) = \sqrt{\frac{\pi}{c}} t^{-3/2} \left(\frac{1}{2} - \frac{\pi^2 m^2}{ct} \right) e^{-\pi^2 m^2 / (ct)}.$$

At $k = 1/6$ and $t = 1$, the locally gauge-equivalent configurations $\theta = 0$ and $\theta = 1/2$ have different values of Ξ . The derivative $\partial_k \Xi_{\theta, 1/6}(1)$ is nonzero on a nonempty open subset of the holonomy circle. Let $W_m^{\theta k} = 8\mathcal{A}_1(m, t) \cos(2\pi m\theta) \cos(2\pi mk)$ denote the mixed holonomy contribution to the m -th Fourier coefficient. At $m = 1$ and $\theta = k = 1/6$, its normalization by $8\mathcal{A}_1(1, t)$ is $1/4$, while the normalization of its k -derivative by $-16\pi\mathcal{A}_1(1, t)$ is $\sqrt{3}/4$.

Proof. Appendix J.5 derives the Hessian on the combined connection–shape gauge quotient, identifies the kernel remaining after gauge reduction, diagonalizes the translation-invariant subspace determined by the rank-one tangential factor of K_k , and applies Poisson summation to the weighted heat trace. It also proves that the complete Fourier coefficient distinguishes the stated pair. $\square$

The two character factors in the mixed holonomy contribution are the transverse internal holonomy and the holonomy of the preserved rank-one assembled connection. Local curvature therefore does not determine the curvature Hessian in this model: configurations with identical curvature on every contractible region have different weighted heat traces. This trace belongs to the spectral calculus of the represented Hessian, while the source-generated observable algebra retains its declared generators. The result supplies neither an ordinary heat trace nor a mass gap, selected state, or empirical realization.

9.6. Graph-Domain Ideals and Finite Operator Families. At a fixed untruncated finite-mode torus representation, the response squares of the declared graph generators generate precisely the compact occupation ideal. Fix the Fock carrier $\mathcal{H}_x$ at such a label and its self-adjoint represented constraint A_x , and let

$$\delta_x(T) = i[A_x, T]$$

when the commutator extends boundedly. The graph norm is

$$\|T\|_{1,x} = \|T\| + \|\delta_x(T)\|.$$

The adjoint-closed family Σ_x^1 consists of all rank-one occupation projections, configuration multipliers with bounded commutator, bounded functions of A_x^2 , represented complete observables commuting with A_x , and compact-gauge averages with

graph-norm-continuous orbits. Define

$$\begin{aligned}\mathcal{A}_{1,x} &= \overline{\text{Alg}^*(\Sigma_x^1)}^{\|\cdot\|_{1,x}}, & \mathfrak{V}_x &= C^*(\mathcal{A}_{1,x}), \\ D_x &= C^*(\mathfrak{V}_x, \delta_x(\mathcal{A}_{1,x})), & I_x &= \text{Ideal}_{D_x}\{P_\nu : \nu \in \mathbb{N}_0^{d_x}\}, \\ J_x &= \text{Ideal}_{D_x}\{\delta_x(T)^*\delta_x(T) : T \in \mathcal{A}_{1,x}\}.\end{aligned}$$

For a selected subfamily $S_x \subseteq \Sigma_x^1$, put

$$\begin{aligned}\mathcal{A}_{1,x,S_x} &= \overline{\text{Alg}^*(S_x)}^{\|\cdot\|_{1,x}}, \\ \mathfrak{V}_{x,S_x} &= C^*(\mathcal{A}_{1,x,S_x}), \\ D_{x,S_x} &= C^*(\mathfrak{V}_{x,S_x}, \delta_x(\mathcal{A}_{1,x,S_x})), \\ J_{x,S_x} &= \text{Ideal}_{D_{x,S_x}}\{\delta_x(T)^*\delta_x(T) : T \in \mathcal{A}_{1,x,S_x}\}.\end{aligned}$$

Let $W : \mathcal{H}_y \rightarrow \mathcal{H}_x$ be an isometry satisfying

$$W \text{Dom } A_y = \text{Dom } A_x \cap W\mathcal{H}_y, \quad A_x W = W A_y.$$

Suppose that every $s \in S_x$ has a partner $r_W(s) \in S_y \cup \{0\}$ satisfying

$$sW = W r_W(s), \quad s^*W = W r_W(s)^*.$$

Compression then defines

$$\rho_W^D : D_{x,S_x} \longrightarrow D_{y,S_y}, \quad \rho_W^D(X) = W^* X W.$$

Theorem 9.2 (The graph-domain ideal on the untruncated torus label). *On every fixed full untruncated torus label covered by these data,*

$$J_x = I_x = D_x \cap \mathcal{K}(\mathcal{H}_x).$$

The common ideal is proper, essential, and not centrally complemented in D_x . On a selected subfamily satisfying the displayed reduction and generator-and-adjoint intertwining conditions,

$$\rho_W^D(J_{x,S_x}) \subseteq J_{y,S_y}.$$

The natural Fourier-mode inclusion across an interaction edge fails the required constraint intertwining relation.

Proof. Appendix J.6 proves the full fixed-label ideal equality, properness, essentiality, noncentral complementation, the selected-subfamily containment, and the interaction-crossing failure. $\square$

The finite-family calculation uses a separately declared list

$$\mathcal{F} = \{p_a : a \in A\} \dot{\cup} \{z_b : b \in Z\},$$

where the p_a are mutually orthogonal occupation projections and the nonprojection operators satisfy $[C, z_b] = 0$ for the fixed represented constraint C . Define

$$\begin{aligned}G &= \sum_{h \in \mathcal{F}} [C, h]^* [C, h], \\ K^{(1)} &= \bigcap_{h \in \mathcal{F}} \ker[C, h], \\ K^{(\infty)} &= \bigcap_{w \in \text{Alg}^*(\mathcal{F})} \ker[C, w].\end{aligned}$$

For the connection with one nonzero harmonic, the corresponding family has a uniform positive lower bound on the nonzero spectrum of G . For the real-analytic profile

$$f(x^1) = \sum_{m \geq 1} \frac{a_m}{m} \sin(mx^1), \quad a_m = a 2^{-(m-1)},$$

the spectrum of G on the sparse two-mode labels $\lambda_M = \{0, Me_1\}$ is $a_M^2/2$, which tends to zero at the rate $4^{-(M-1)}$. These are response-detection bounds for the declared finite families; the constraint and Master spectra remain those of the preceding subsection.

Theorem 9.3 (Commutator kernels of the specified finite families). *For the single-harmonic and decaying-harmonic families,*

$$K^{(1)} = K^{(\infty)}.$$

The single-harmonic family has a uniform spectral gap above zero for G , whereas the decaying-harmonic family has no uniform gap. As a declared list, $\mathcal{F}$ is strictly smaller than every algebraic, graph, norm-linear, operator-system, or C^ -generated family formed from it.*

Proof. Appendix J.6 proves the kernel equality, the two lower-bound conclusions, and the strict membership distinctions. $\square$

These finite-family data determine the displayed commutator kernels and Gram bounds. A central decomposition requires separately specified central data and is outside this finite-family theorem.

Theorem 9.4 (Incompatibility of the four specified operator-family roles). *Consider assignments in which one membership family is required simultaneously to be the norm-generated C^* -algebra, the graph-domain family whose response-square ideal equals the occupation ideal, the specified finite operator family above, and a projective graph family transported by the natural Fourier-mode inclusions. No assignment in this class satisfies all four requirements on the fixed torus representations.*

Proof. Appendix J.6 gives the infinite diagonal separating norm and graph closure, the finite diagonal separating generated and declared finite-family membership, and the interaction-crossing failure of graph transport. The three calculations exclude every assignment in the stated four-role class. $\square$

The rank-two and rank-one configurations realize the response-channel conclusions, while the spectral and commutator calculations realize their domain-specific expressions on fixed representations. The flat coupled model further shows that locally indistinguishable curvature data can give different weighted heat traces of the curvature Hessian. The graph algebras and finite operator families require the additional represented data stated above. The next section returns to the seven source components and determines their individual roles and the limits of recovery from weaker finite data.

10. IRREDUNDANCY AND RECOVERY LIMITS OF THE SOURCE DATA

Section 3 fixes the seven components of a field source, and Sections 4–9 determine the structures that follow from them. The present section identifies a representative role for each component and proves single-deletion irredundancy within this

fixed source architecture. It then classifies source automorphisms, tests the eight conditions defining the field-source class, and determines the reach of a specified finite transport model. The final comparison describes how six additional time, signature, coupling, or lift data affect the fixed field law. The field-level conclusions are taken within $\mathbf{S}^F$ and up to the structural equivalence of Section 3; the one-condition tests use the explicitly stated pre-source comparison classes.

10.1. Roles and Single-Deletion Irredundancy. Order the components of the source presentation as

$$P = (\mathcal{A}, \Lambda, \text{Der}(\mathcal{A}), \mathcal{E}, \mathcal{N}_g, \mathcal{G}_{\mathcal{N}_g}, \vartheta).$$

Each component has a distinct representative role in the field law. The algebra $\mathcal{A}$ supplies the local quantities, coefficients, and module operations, while Λ supplies their localization, restriction, and projective labels. The module $\text{Der}(\mathcal{A})$ gives the directions of variation and the first-order calculus, and the Dirichlet datum $\mathcal{E}$ determines the base-direction metric and torsion-free connection together with the integration, regularity, and motion data.

The metric module $\mathcal{N}_g$ supplies the internal values and metric contractions, including the target of the shape field and the adjoints used in the assembled connection. The subgroup $\mathcal{G}_{\mathcal{N}_g}$ gives the declared internal gauge action, and ϑ fixes the tensor assignment together with the carrier/constraint, rank/readout, visible/hidden, and law/input roles. These are representative downstream roles; the same source component may enter several later constructions.

Appendix H.1 fixes eight formal dependence profiles $Q_1, \dots, Q_8$ for the finite single-deletion test. Their incidence matrix $J_g = (j_{\alpha i}) \in M_{8 \times 7}(\mathbb{Z})$ has columns indexed by P , with $j_{\alpha i} = 1$ exactly when Q_α includes the i -th source component. The representative roles above supply the downstream interpretation, while the appendix records the complete matrix and its row order.

Theorem 10.1 (Single-deletion irredundancy of the source data). *Within the fixed seven-component source class and this eight-row test, every source component has a determined downstream role, and deleting any one component removes at least one incidence required by $\mathcal{L}_{\mathfrak{R}}$. The matrix J_g has full column rank.*

Proof. Before the internal module and its metric are combined, the corresponding matrix J_8 has rank seven and

$$\ker J_8 = \text{span}\{(0, 0, 0, 0, -1, 1, 0, 0)^T\}.$$

The kernel vector is the formal difference between the internal-module and internal-metric columns. The source signature carries the pair $\mathcal{N}_g = (\mathcal{N}, g_{\mathcal{N}})$ as one typed component, and a source isomorphism preserves both entries of that pair. Replacing the two artificial bookkeeping columns by the one $\mathcal{N}_g$ column gives J_g . Appendix H.1 computes a determinant-one minor from its first seven rows and verifies that every column is nonzero. The minor proves full column rank, and each nonzero column records an incidence removed by the corresponding single deletion. The source presentation is defined before $\mathcal{L}_{\mathfrak{R}}$, so these incidences record downstream uses of the primitive data. $\square$

10.2. Source Automorphisms and Residual Internal Symmetry. Structural equivalence identifies source presentations through isomorphisms preserving their local, derivation, Dirichlet, metric-module, gauge, and assignment data. For a fixed

source, the internal part of such an automorphism must carry the declared gauge subgroup into itself and therefore belongs to its metric-module normalizer. Quotienting by transformations already contained in the declared gauge group leaves the residual internal symmetry D_G .

Theorem 10.2 (Source automorphisms). *Let $\text{Aut}_\vartheta(\mathfrak{R})$ be the group of structure-preserving automorphisms of the source presentation. Then*

$$\text{Aut}_\vartheta(\mathfrak{R}) \cong \text{Aut}(\mathcal{A}, \Lambda, \text{Der}(\mathcal{A}), \mathcal{E}, \vartheta) \ltimes \text{N}_{\text{Aut}(\mathcal{N}_g)}(\mathcal{G}_{\mathcal{N}_g}),$$

where the first factor preserves the algebra, localization, derivation, Dirichlet, and assignment data. Its semidirect action is the transport of the metric-module and gauge data. The residual internal symmetry is

$$D_G = \text{N}_{\text{Aut}(\mathcal{N}_g)}(\mathcal{G}_{\mathcal{N}_g})/\mathcal{G}_{\mathcal{N}_g}.$$

Under the Riemannian realization hypotheses, the realized automorphism group is

$$\text{Isom}_\vartheta(M, h, dV_h) \ltimes \text{N}_{\text{O}(N, g_N)}(\mathcal{G}_N),$$

with residual quotient $\text{N}_{\text{O}(N, g_N)}(\mathcal{G}_N)/\mathcal{G}_N$.

Proof. A structure-preserving automorphism preserves all seven components of the source. On a smooth realization, an automorphism of $C^\infty(M)$ is composition with a unique diffeomorphism [36, Theorem 1]. Preservation of the Dirichlet form, carré du champ, and positive integration data preserves the realized metric and measure, and the torsion-free metric derivation connection is then determined functorially.

The internal component preserves the source exactly when it normalizes $\mathcal{G}_{\mathcal{N}_g}$, which gives the second factor and the quotient D_G . Projective separatedness makes a source automorphism determined by its restrictions to the stated cofinal family, while preservation of ϑ fixes the module labeling and each declared role assignment. Together these actions give the displayed semidirect product. The realized statement follows from the Riemannian realization and the Serre–Swan correspondence [42, Section 50][46, Section 3, Theorem 2]. Appendix H.2 gives representative normalizer calculations for D_G . $\square$

Every algebraic, Dirichlet, metric-module, curvature, functional-calculus, and projective construction in the field law is equivariant under source automorphisms. On gauge-equivalence classes, the declared gauge subgroup acts trivially and the induced internal action factors through D_G .

Appendix H.5 also gives a finite preservation model whose complete 25-variable linearized system has rank 24 and kernel equal to its embedded $\mathfrak{so}(2)$ gauge line. This determines the infinitesimal stabilizer of that finite model. A disconnected reflection remains in its exact stabilizer modulo connected $SO(2)$, and the appendix’s nonlinear calculation treats only its four declared competitor families. The field-level automorphism theorem above has its independent structure-preserving proof.

10.3. Independent Tests of the Field-Source Conditions. Number the eight componentwise conditions in the definition of a field source by P1–P8. They concern localization and restriction, smooth cores and derivation naturality, ellipticity and nondegeneracy of the Dirichlet symbol, positive integration, finite projectivity of the metric module, metric-gauge compatibility, the structural assignments, and continuity and separatedness of the generated projective systems.

Theorem 10.3 (Admitted-source sufficiency and comparison-relative sharpness). *For every $\mathfrak{R} \in \mathbf{S}^F$, P1–P8 and the remaining regularity and compatibility hypotheses in the field-source definition determine the field and law of Theorem 3.8.*

Separately, each P1–P8 condition has a one-condition witness in the corresponding pre-source comparison class of Appendix H.4. The witness satisfies the other seven predicates and loses the named structural guarantee or acquires the named ambiguity, up to structural equivalence.

Proof. The first assertion applies Theorem 3.8 to an already admitted field source. Membership in $\mathbf{S}^F$ includes P1–P8 together with the Sobolev and support conditions, the Weyl-algebra conditions, and the existence of compatible regular states stated in Section 3.

For the second assertion, the appendix defines the pre-source class, structural equivalence, and all eight comparison classes before evaluating them. Duplicating a localization-index object produces an ambiguity in that index description while preserving joint faithfulness of the original fiber restrictions. Scaling the derivation restriction maps on a two-level star breaks their naturality with the unchanged algebra restrictions. The zero regular strongly local Dirichlet form retains the standard smooth core but has degenerate symbol, and sign-reversed local integrals retain their linear compatibilities while losing positivity.

A countable direct-sum metric module, together with its induced local, cylinder, and transition systems, remains projectively separated but loses finite-projective duality. A restriction-compatible dilation action is nonmetric, a nonbijective assignment supplies no internal target for the shape field, and a two-component pre-source with visible-only restrictions has a nonidentity hidden-sign arrow whose stated cofinal restrictions are all the identity. In each row the other seven predicates and the invariant distinguishing the witness from its control are verified in the appendix. $\square$

10.4. What Finite Transport Data Determine. A marked finite family is defined in Appendix H.3 by the coordinates

$$(a, \lambda, d, e, n, g, \theta).$$

They record a semisimple-factor count, a chart count, the dimension of an actual derivation subspace, a quadratic-energy coefficient, the rank of a supported metric module, the order of a cyclic orthogonal gauge subgroup, and one of two structural assignments. The six finite transport evaluations are

$$\mathfrak{T}(a, \lambda, d, e, n, g, \theta) = (a\lambda, a, d, de, n, ng).$$

Theorem 10.4 (Finite transport obstruction to the assignment data). *In the specified marked family, $\mathfrak{T}$ determines the six scalar incidence coordinates (a, λ, d, e, n, g) . Its multiplicative incidence matrix J_T has rank six and its assignment column vanishes. Two structurally inequivalent assignments have the same value of $\mathfrak{T}$, so these finite transport data leave the assignment coordinate undetermined.*

Proof. The six evaluations give the inverse formulas

$$a = T_2, \quad \lambda = T_1/T_2, \quad d = T_3, \quad e = T_4/T_3, \quad n = T_5, \quad g = T_6/T_5.$$

Their exponent vectors form the matrix

$$J_T = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix},$$

which has rank six and a one-dimensional kernel in the assignment coordinate. The two assignments in the marked family differ by exchanging the named carrier and constraint roles. Structural equivalence preserves that assignment, while none of the six evaluations contains it. $\square$

The six coordinates are scalar counts, dimensions, a trace, a module rank, and a group order in this marked family. The result does not reconstruct six complete source components or place the finite proxy in $\mathbf{S}^F$. The independently specified data of Section 11 contain indexed local algebras, closed quadratic forms, transition categories, and compatible role and law/input assignments; those data recover ϑ together with the other source components.

10.5. Additional Time, Signature, Coupling, and Lift Data. Consider six additions to the fixed source. They are a time covector used in a Lorentzian time-direction construction, an ADM lapse and shift, a Lorentzian causal cone, the replacement of the positive-Riemannian normal sign $\varepsilon = 1$ by $\varepsilon = -1$, a trace-coupling parameter t , and an Eisenhart–Brinkmann null-lift coordinate. The comparison uses each datum only through the standard construction just named.

Theorem 10.5 (Effect of the six additional data on the fixed law). *In the stated constructions, a time covector, lapse and shift, causal cone, or null-lift coordinate entering the field law replaces a positive base or extended pairing by an indefinite one. Changing ε changes the rank-one scalar contraction, and every $t \neq 0$ changes the kinetic pairing determined by the source. Each such use therefore defines an enlarged source and changes the corresponding component of $\mathcal{L}_{\mathfrak{R}}$.*

If an added datum occurs in no component of the fixed law, its column in the extended incidence matrix is zero. The operational data generated in the fixed signature of Section 11 then remain unchanged, and their recovery functor returns $\mathfrak{R}$, up to structural equivalence, while leaving the added datum undetermined. This conclusion concerns the six listed data, the field law of this paper, and its operational recovery target. An added datum that defines a further output enlarges that target.

Proof. A time covector used to define a Lorentzian time direction selects a negative direction. An ADM lapse N and shift N^i give the metric

$$-N^2 dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt),$$

which is congruent to a form with one negative direction for every shift [1]. A Lorentzian causal cone with timelike interior likewise requires an indefinite quadratic form. On the two added directions, an Eisenhart–Brinkmann null lift contains the block

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

whose eigenvalues are 1 and -1 . These four constructions therefore change the positive base or extended pairing to an indefinite one.

The normal-sign comparison of Section 6 shows that passing from $\varepsilon = 1$ to $\varepsilon = -1$ changes the rank-one scalar contraction by

$$2((\mathrm{tr}_h K)^2 - \|K\|_h^2),$$

which is not identically zero. For $m = \dim M$, the trace-coupled kinetic form

$$G_t(K, K') = \langle K, K' \rangle_h - t(\mathrm{tr}_h K)(\mathrm{tr}_h K')$$

is indefinite for $t > 1/m$, positive semidefinite at $t = 1/m$, and positive definite for $t < 1/m$. The source-induced pairing is G_0 , so every $t \neq 0$ changes that pairing, including the positive-definite range $0 < t < 1/m$.

Suppose instead that an added datum X occurs in no component of $\mathcal{L}_{\mathfrak{R}}$. Every entry in its extended incidence column then vanishes, and deleting X changes none of the eight tested incidences. The fixed operational signature contains no X -coordinate, so the operational data generated by the unchanged law recover the original seven-component presentation and contain no value from which to determine X . $\square$

Remark 10.6 (State-generated automorphism groups). *A chosen state or positive functional may determine a modular or thermal-time automorphism group on its algebra. Such a group depends on the chosen functional and is additional state data of a different type from the six geometric comparisons above.*

10.6. Consequences Within the Fixed Source Class. The seven source components are irredundant under the declared single-deletion test, and their structure-preserving automorphisms have the normalizer classification above. For sources already admitted to $\mathbf{S}^F$, the complete field-source hypotheses determine the field and its law. Separately, the eight componentwise conditions have comparison-relative sharpness in the stated pre-source classes. The marked finite transport family recovers six scalar incidence coordinates while leaving its assignment coordinate unresolved. The six additional data change the source and field law when they enter through their stated constructions; the fixed operational signature leaves an unused datum undetermined.

These results delimit the fixed source class and the reach of finite transport. Section 11 begins from independently specified indexed local, variational, transition, and assignment data and proves that, under its stated regularity and compatibility hypotheses, those data recover the complete source up to structural equivalence. The recovered source then determines the field and its law.

11. RECOVERY OF THE FIELD SOURCE FROM OPERATIONAL DATA

At each finite or local index one may record the available operations and their restrictions, the quadratic form and positive functional that measure their variation, and the internal transition objects detected by finite-projective metric modules. A compatible directed system of these records is the input of this section. Its algebraic, variational, and categorical data are specified before a source presentation, and the theorem below recovers the source from them. The recovered source then determines the field $\mathfrak{F}(\mathfrak{R})$ and its law.

11.1. Indexed Local Data and Their Isomorphisms.

Definition 11.1 (Indexed operational data). *A system of indexed operational data B begins with a directed category $\mathbf{I}_B$ whose objects are finite or local indices and whose arrows are restrictions. At each index W , it contains a real response domain D_W , a distinguished regular operation core $A_W^0 \subseteq D_W$ with declared normed algebra completion A_W , a positive quadratic form Q_W on D_W , a positive linear functional τ_W on A_W , and a symmetric local square field*

$$\Gamma_W : A_W^0 \times A_W^0 \longrightarrow A_W.$$

The restrictions are linear on the response domains, preserve the regular cores, and carry the operation, quadratic, integration, and square-field data according to the compatibility conditions stated below.

At each index U , the system also contains an additive, idempotent-complete, rigid symmetric metric tensor category $\mathbf{C}_U$ of internal transitions and a finite-projective metric probe functor

$$\omega_U : \mathbf{C}_U \longrightarrow \mathbf{Proj}_g(A_U),$$

where $\mathbf{Proj}_g(A_U)$ is the category of finite-projective metric modules over A_U . For $U \leq V$, transition restriction $s_{VU} : \mathbf{C}_V \rightarrow \mathbf{C}_U$ and coefficient base change $b_{VU} : \mathbf{Proj}_g(A_V) \rightarrow \mathbf{Proj}_g(A_U)$ are related by a strong symmetric monoidal natural isomorphism

$$\chi_{VU} : b_{VU}\omega_V \Longrightarrow \omega_U s_{VU},$$

and these comparisons satisfy the cocycle under successive restrictions. Finally, B contains finite structural labels for the module types and paired role assignments of Section 3, with the assignments fixed by the law distinguished from those supplied as variable input.

The local algebras A_W , forms Q_W , functionals τ_W , square fields Γ_W , transition categories, and probe diagrams are indexed objects rather than components of a global source presentation. Reconstruction assembles from them the source algebra and localization system, Dirichlet and derivation data, internal metric module and gauge group, and structural datum. The functionals τ_W belong to the fixed local integration data and are distinct from states subsequently chosen on bounded observable algebras.

Definition 11.2 (Indexed-data isomorphism). *A structure-preserving morphism $B \rightarrow B'$ assigns the objects and restriction arrows of $\mathbf{I}_B$ bijectively to those of $\mathbf{I}_{B'}$. It intertwines the response domains and operation algebras by compatible linear and algebra isomorphisms preserving Q_W , τ_W , and Γ_W ; intertwines the transition categories by metric tensor equivalences preserving their finite-projective probe functors and the comparison maps χ_{VU} ; and preserves the structural labels and the distinction between law assignments and variable inputs. The morphism is continuous for the projective topology, and equality is detected on every cofinal family of finite or local restrictions. An isomorphism is such a morphism with a structure-preserving inverse, and we write $B \cong B'$ when one exists.*

Lemma 11.3 (Indexed-data isomorphism). *The relation defined by $B \cong B'$ is reflexive, symmetric, and transitive.*

Proof. Identity, inverse, and componentwise composition preserve every listed datum and coherence diagram. $\square$

11.2. Hypotheses for Source Recovery. Let $\mathcal{B}$ denote the class of indexed operational data satisfying the following regularity conditions, in the precise forms recorded in Appendix I:

$$\mathcal{B} = \left\{ B : B \models \bigwedge_{j=1}^9 (R_j) \right\}$$

Here the numbered conditions are:

- (R1) projective separatedness and compatible restriction maps that are unital homomorphisms of algebras;
- (R2) Cauchy/tail control for the seminorm induced by Q_W , together with the Markov contraction property;
- (R3) regular-core density in the declared response-domain completion;
- (R4) positive integration, compatible local carré-du-champ identities, and polarization;
- (R5) first-order symbol pairing and completion;
- (R6) finite-projective probe functors and coherent tensor restriction;
- (R7) positive Hom-space pairing and a faithful tensor-generating family;
- (R8) compatibility of structural labels and separation of fixed laws from variable inputs;
- (R9) real function-algebra structure on the completed operation cores A_W .

These are hypotheses on the indexed data under which the forward construction and source reconstruction have matching domains. Condition (R9) states that each A_W is a commutative unital real Banach algebra in which $\|f^2\| = \|f\|^2$ and $1 + f^2$ is invertible for every element; such an algebra is isometrically an algebra of real-valued continuous functions on its compact Hausdorff spectrum [32, Theorem 1.2.9, Corollary 1.2.12, and the strict-reality argument on pp. 26–27]. The algebra clause of (R1) states that restriction maps are unital algebra homomorphisms for this structure. The contraction clause of (R2) states that Q_W does not increase under the unit contraction $f \mapsto f \wedge 1$. Condition (R4) requires the functionals τ_W and square fields Γ_W to restrict compatibly and to satisfy

$$Q_W(f) = \tau_W(\Gamma_W(f, f))$$

on the regular core. Conditions (R6)–(R7) require the probe functors and comparison maps above to preserve metrics, tensor products, duals, evaluations, and coevaluations, with positive-definite Hom-space pairings. The specified transition objects generate the tensor categories faithfully, so a tensor-natural metric automorphism of the indexed probe functor that acts trivially on every generator is the identity.

These hypotheses separate the reconstruction into four steps. Conditions (R9) and (R1) recover the local spectra, source algebra, and localization system. Conditions (R2)–(R5), together with the positive functionals, recover the Dirichlet datum and generated first-order calculus, whose nondegenerate dual is the derivation module declared in Section 3. Conditions (R6)–(R7) recover the finite-projective metric internal module and its restriction-compatible gauge group, while (R8) recovers ϑ . This base reconstruction uses the complete indexed probe datum. A particular carrier-role point belongs to the graph-and-point refinement stated below and is not selected by the unpointed operational datum.

Let $\mathbf{S}$ be the groupoid of source presentations satisfying the source regularity conditions, with invertible structure-preserving morphisms. Let $\mathbf{B}$ be the groupoid whose objects are the indexed data in $\mathcal{B}$ and whose arrows are the isomorphisms above. The componentwise forward assignment defines

$$F : \mathbf{S} \longrightarrow \mathbf{B}.$$

Theorem 11.4 (Forward soundness). *If $\mathfrak{R}$ is a source presentation satisfying the field-source conditions of Section 3, then the indexed data $F(\mathfrak{R})$ belong to $\mathcal{B}$.*

Proof. The source algebra and its localization data give the indexed operation algebras, regular cores, and separated unital restrictions required by (R1) and (R9). The Dirichlet datum gives Q_W , τ_W , Γ_W , the Markov contraction, the closed response completion, and the first-order symbol pairing required by (R2)–(R5). The finite-projective metric internal module, its declared gauge action, and their restrictions give the transition tensor categories, probe functors, coherent comparison maps, positive Hom-space pairings, and faithful generators required by (R6)–(R7). The structural datum ϑ gives the finite labels and the separation of law assignments from variable inputs required by (R8). Appendix I matches these hypotheses to the analytic and categorical reconstruction results used below. $\square$

11.3. Equivalence with the Field Source.

Theorem 11.5 (Reconstruction of the source data). *For every $B \in \mathcal{B}$, the reconstruction gives a source presentation $R(B)$. Isomorphisms of indexed operational data induce invertible structure-preserving source morphisms and define a functor*

$$R : \mathbf{B} \longrightarrow \mathbf{S}.$$

Proof. The completed commutative algebras and their restriction homomorphisms recover the source algebra $\mathcal{A}$ and its localization datum Λ by Gelfand reconstruction. The Riesz–Markov representation of each τ_W , together with Q_W and Γ_W , recovers the reference measure, positive local integration, and closed Dirichlet form in $\mathcal{E}$. The square-field and polarization identities reconstruct the generated first-order module by the canonical derivation representation in its commutative, strongly local specialization [11, Theorems 8.1–8.3]. Its nondegenerate dual recovers $\text{Der}(\mathcal{A})$ in the source class of Section 3.

The tensor categories, finite-projective probes, coherent comparison maps, positive Hom-space pairings, and faithful generators recover $\mathcal{N}_g$ and $\mathcal{G}_{\mathcal{N}_g}$ by the indexed metric-gauge reconstruction of Appendix I.4. The finite-projective realization uses the Serre–Swan correspondence [42, Section 50][46, Section 3, Theorem 2]. This is the primitive internal-source component of the base reconstruction; the supplied carrier point introduced below is transported separately. Compatibility of the finite structural labels with the law and input assignments recovers ϑ . These seven components assemble into $R(B)$.

An isomorphism of indexed data preserves the local algebra products, functionals, square fields, first-order pairings, transition tensor functors, probe comparisons, and structural labels. Functoriality of Gelfand and Serre–Swan reconstruction, together with the restriction-natural first-order and indexed metric-gauge reconstruction of the appendix, therefore produces an invertible structure-preserving source morphism. Identity and composition follow componentwise, so R is a functor. $\square$

Lemma 11.6 (Preservation of source isomorphisms). *If $\varphi : \mathfrak{R} \rightarrow \mathfrak{R}'$ is an invertible structure-preserving morphism of source presentations satisfying the source regularity conditions, then $F(\varphi)$ is an isomorphism in $\mathbf{B}$ from $F(\mathfrak{R})$ to $F(\mathfrak{R}')$. Structurally equivalent presentations in this class therefore determine isomorphic indexed operational data.*

Proof. The morphism φ commutes with the local and cylinder restrictions, so it carries the index category of $F(\mathfrak{R})$ bijectively to that of $F(\mathfrak{R}')$. Preservation of the source algebra and Dirichlet datum intertwines A_W , D_W , Q_W , τ_W , and Γ_W at every index. Preservation of the metric internal module and declared gauge action intertwines the transition tensor categories, finite-projective probe functors, comparison maps, Hom-space pairings, and faithful generators. Preservation of ϑ carries the finite structural labels and the law/input distinction to their counterparts. The inverse source morphism supplies the inverse indexed morphism, so $F(\varphi)$ is an isomorphism in $\mathbf{B}$. The source-automorphism theorem of Section 10 identifies the base, gauge, and residual internal components of these structure-preserving isomorphisms. $\square$

Theorem 11.7 (Equivalence of the source and operational-data groupoids). *The functors $F : \mathbf{S} \rightarrow \mathbf{B}$ and $R : \mathbf{B} \rightarrow \mathbf{S}$ are quasi-inverse. There are natural isomorphisms*

$$\epsilon : RF \Longrightarrow \text{Id}_{\mathbf{S}}, \quad \eta : FR \Longrightarrow \text{Id}_{\mathbf{B}},$$

and hence an equivalence of groupoids

$$\boxed{\mathbf{S} \simeq \mathbf{B}}.$$

In particular,

$$R(F(\mathfrak{R})) \cong \mathfrak{R}, \quad F(R(B)) \cong B,$$

naturally in $\mathfrak{R}$ and B .

Proof. For a source $\mathfrak{R}$, Gelfand and Riesz–Markov evaluation, the canonical comparison map for the first-order module and its dual, Serre–Swan evaluation, indexed metric-gauge reconstruction, and the identity on the recovered structural labels define $\epsilon_{\mathfrak{R}} : R(F(\mathfrak{R})) \rightarrow \mathfrak{R}$. Every component commutes with restriction, so compatibility with the restriction maps and projective separatedness determine a unique source isomorphism. For indexed data B , the corresponding comparison maps return the operation algebras, response domains, quadratic forms, positive functionals, square fields, regular cores, generated first-order modules, transition categories, probe functors, comparison maps, Hom pairings, faithful generators, structural labels, assignments, and restriction maps. They define $\eta_B : F(R(B)) \rightarrow B$. The restriction and probe-coherence identities in Appendix I.5 make both families natural. $\square$

Corollary 11.8 (Correspondence of isomorphism classes). *The equivalence induces a bijection*

$$\pi_0(\mathbf{B}) \cong \pi_0(\mathbf{S})$$

between isomorphism classes of indexed operational data and structural isomorphism classes of regular source presentations.

Proof. The natural isomorphisms identify the composites with the two identity functors, so passing to isomorphism classes gives mutually inverse maps. The source

arrows are the structure-preserving isomorphisms of Section 3; their base, declared-gauge, and residual internal components are classified in Section 10. $\square$

11.4. Closed-Form and Carrier-Point Refinement. For the positive functional τ_W , let $H_{\tau,W}$ be the Hilbert completion of

$$A_W / \{a : \tau_W(a^2) = 0\}$$

with inner product $\tau_W(ab)$. An indexed datum is graph admitting when its response domain has an injective map

$$j_W : D_W \hookrightarrow H_{\tau,W}, \quad j_W(a) = [a] \quad (a \in A_W^0),$$

is complete for

$$\|u\|_{D,W}^2 = \|j_W u\|_{H_{\tau,W}}^2 + Q_W(u),$$

has A_W^0 as a graph-dense core, and carries bounded restriction maps in the Hilbert and graph norms. Write $\mathbf{B}^D$ and $\mathbf{S}^D$ for the corresponding replete behavior and source subgroupoids.

A compatible carrier-role point is a family

$$x_{g,W} \in \mathbf{C}_W, \quad \xi_{VU} : s_{VU} x_{g,V} \xrightarrow{\cong} x_{g,U},$$

whose probes are positive finite-projective metric modules of positive locally constant rank. The comparisons satisfy the restriction cocycle, and exactly one isomorphism class carries the type $t_g = (N, C)$. Fiberwise, the action of $\text{Aut}^{\otimes,*}(\omega_W)$ on the point has trivial kernel. Let $\widehat{\mathbf{B}}^{D,g}$ be the graph-admitting behavior groupoid with this supplied point, and let $\widehat{\mathbf{S}}^{D,g}$ be the corresponding source groupoid whose point lies in the transition category of its forward behavior.

Theorem 11.9 (Graph-and-point refinement of source recovery). *The source-operational-data equivalence restricts and lifts to*

$$\widehat{\mathbf{S}}^{D,g} \simeq \widehat{\mathbf{B}}^{D,g}.$$

The response-domain comparison is the unique surjective graph isometry extending the regular-core comparison, and the carrier point is transported through the transition-category comparison. On the canonical-carrier forward subgroupoid the probed point is isomorphic to the primitive $\mathcal{N}_g$. An arbitrary supplied point need not have this identification.

Proof. Appendix I.5 proves that the actual source and behavior comparison maps preserve τ_W, Q_W, j_W , extend uniquely from the graph-dense cores, and are natural under restriction. It then normalizes the base equivalence to an adjoint equivalence and transports a supplied point through the transition-category components of its unit and counit. Theorem I.10 gives a one-window behavior with primitive $\mathcal{N}_g = \mathbb{R}$ and supplied point $\mathbb{R}^2$, which proves the final assertion by rank. $\square$

Corollary 11.10 (Recovery of the realized geometric data). *Let $B \in \mathcal{B}$. Suppose the reconstructed source presentation $R(B)$ satisfies the realization hypotheses of Section 3, with spatial locality, smoothability, Dirichlet/carré-du-champ metric-measure reconstruction, positive integration, Serre–Swan realization of the metric module as an internal bundle, and reconstruction of the declared internal gauge group. Then B determines the Riemannian and internal-bundle data*

$$(M, h, N, g_N)$$

up to diffeomorphism of M and bundle automorphism of N preserving the fiber metric, with the residual source symmetry recorded by D_G .

Proof. The reconstruction theorem gives $F(R(B)) \cong B$. Theorem 3.7 assigns (M, h, N, g_N) to $R(B)$, uniquely up to diffeomorphism and bundle automorphism preserving the fiber metric. Composing the two results gives the claim. $\square$

11.5. The Field Law Determined after Source Recovery. The equivalence above completes the converse recovery of the source. The field law is then obtained by applying the source-to-law assignments already proved in Sections 4–8. Operational determination therefore has the order

$$\mathcal{B} \xrightarrow{R} \mathfrak{R} \xrightarrow{L} L_{\mathfrak{R}}.$$

The source groupoid $\mathbf{S}^F$ fixed in Section 3 is a nonempty full replete subgroupoid of $\mathbf{S}$. Let $\mathbf{B}^F$ be the full subgroupoid of $\mathbf{B}$ on the objects

$$\mathbf{B}^F = \{B \in \mathbf{B} : R(B) \in \mathbf{S}^F\}.$$

This subgroupoid is replete because R carries isomorphisms of indexed data to source isomorphisms and $\mathbf{S}^F$ is replete. It is nonempty because the natural isomorphism $R(F(\mathfrak{R})) \cong \mathfrak{R}$, for $\mathfrak{R} \in \mathbf{S}^F$, places $F(\mathfrak{R})$ in $\mathbf{B}^F$. Write $F|_{\mathbf{S}^F}$ and $R|_{\mathbf{B}^F}$ for the corresponding restrictions of F and R , and write

$$\mathbb{L} : \mathbf{S}^F \longrightarrow \mathbf{L}^B$$

for the source-to-bounded-law functor of Sections 8.1 and 8.3.

Corollary 11.11 (Law after source recovery). *Define*

$$\mathbb{L}^B := \mathbb{L} R|_{\mathbf{B}^F}.$$

Then there is a natural isomorphism

$$\mathbb{L}^B F|_{\mathbf{S}^F} \xrightarrow{\cong} \mathbb{L}.$$

Thus the indexed data in $\mathbf{B}^F$ reconstruct a source, after which the source determines its bounded law. The displayed composite and natural isomorphism identify that law on the forward essential image of $\mathbb{L}$.

Proof. The subgroupoids are full and replete, so the reconstruction and forward functors restrict to $R|_{\mathbf{B}^F}$ and $F|_{\mathbf{S}^F}$. Apply $\mathbb{L}$ to the natural isomorphism $R|_{\mathbf{B}^F} F|_{\mathbf{S}^F} \cong \text{Id}_{\mathbf{S}^F}$. Naturality is inherited by functoriality of $\mathbb{L}$. $\square$

The field law has two independently defined packages. The groupoid $\mathbf{R}^V$ records the variational response, which the functor

$$\mathbb{R}^V : \mathbf{S}^F \longrightarrow \mathbf{R}^V$$

assigns to each source. The groupoid $\mathbf{R}$ adds its separately defined phase-space, local quantum, bivariate finite-label, finite-commutator, and bounded common-core expressions. The corresponding theorems define a functor

$$(18) \quad \mathbb{R} : \mathbf{S}^F \longrightarrow \mathbf{R}, \quad \pi_V \mathbb{R} = \mathbb{R}^V.$$

On a phase component where the variational response vanishes identically, the phase theorem of Section 7 retains $(\vartheta, [1 : -1])$ as specified phase data, and the common operator domain may be empty.

The groupoid $\mathbf{L}^B$ records the source-generated bounded observable algebras and completed transition data. The functors $U_R : \mathbf{R} \rightarrow \mathbf{L}$ and $U_L : \mathbf{L}^B \rightarrow \mathbf{L}$ forget the

additional response and bounded-law entries, respectively. The following pullback records compatible response and bounded law packages with the same underlying field law.

Definition 11.12 (Combined field-law target). *An object of $\mathbf{L}^F$ is a triple*

$$(\mathcal{P}, V, \theta), \quad \theta : U_R(\mathcal{P}) \xrightarrow{\cong} U_L(V),$$

where $\mathcal{P}$ is an object of $\mathbf{R}$, V is an object of $\mathbf{L}^B$, and θ is an isomorphism in $\mathbf{L}$. An arrow (p, v) is any pair of isomorphisms in the two target groupoids satisfying

$$(19) \quad \theta' U_R(p) = U_L(v) \theta.$$

Pairs satisfying (19) are closed under identity, componentwise composition, and componentwise inverse. Hence $\mathbf{L}^F$ is a groupoid with two projection functors.

Theorem 11.13 (Field-law functor after source recovery). *The response functor and the source-generated bounded-law functor define a strict source functor*

$$(20) \quad \mathbb{F} : \mathbf{S}^F \longrightarrow \mathbf{L}^F, \quad \pi_R \mathbb{F} = \mathbb{R}, \quad \pi_L \mathbb{F} = \mathbb{L}.$$

The equivalence of the source and operational-data groupoids then defines

$$(21) \quad \mathbb{F}^B := \mathbb{F} R \downarrow_{\mathbf{B}^F} : \mathbf{B}^F \longrightarrow \mathbf{L}^F$$

and a natural isomorphism

$$(22) \quad \mathbb{F}^B F \downarrow_{\mathbf{S}^F} \xrightarrow{\cong} \mathbb{F}.$$

Thus the indexed operational data recover a source and that source determines the response and source-generated bounded law of $\mathfrak{F}(\mathfrak{A})$. The displayed factorization through $\mathbf{S}^F$ and its natural comparison on the forward essential image constitute the operational determination proved here.

Proof. The two functors have the same underlying law on each source, $U_R \mathbb{R} = U_L \mathbb{L} = L$. Assign

$$\mathbb{F}(\mathfrak{A}) = (\mathbb{R}(\mathfrak{A}), \mathbb{L}(\mathfrak{A}), 1_{L(\mathfrak{A})}).$$

On a source isomorphism, the two forgotten arrows are the same $L(f)$, so (19) is the identity square. This proves (20). Equation (21) is composition with the independently defined reconstruction functor. Applying $\mathbb{F}$ to the natural isomorphism $R \downarrow_{\mathbf{B}^F} F \downarrow_{\mathbf{S}^F} \cong \text{Id}_{\mathbf{S}^F}$ gives (22). $\square$

Corollary 11.14 (Operational recovery and source irredundancy). *Within the fixed seven-datum source class, the composite*

$$(23) \quad \mathbf{B}^F \xrightarrow{R \downarrow_{\mathbf{B}^F}} \mathbf{S}^F \xrightarrow{\mathbb{F}} \mathbf{L}^F$$

recovers the source from its indexed operational data and then determines the field-law object associated with $\mathfrak{F}(\mathfrak{A})$. Every source component in this class has a determined downstream role, and deleting any one component removes at least one incidence required by that law.

Proof. The equivalence of groupoids gives the first arrow and Theorem 11.13 gives the second. The full-rank and single-deletion incidence theorem of Section 10 gives the stated role of every source component within the fixed source class. $\square$

Functoriality also shows that isomorphic indexed operational data reconstruct isomorphic sources, whose source isomorphisms transport the bounded-law object, its generated observable algebras, the completed transition category, and the inclusion maps of Section 8.1.

11.6. Domain, Nonvacuity, and Comparison-Relative Failure Tests. The domain $\mathbf{B}$ consists of the indexed systems whose operation algebras and restrictions satisfy (R1) and (R9), whose quadratic, integration, square-field, and first-order data satisfy (R2)–(R5), whose finite-projective tensor probes satisfy (R6)–(R7), and whose structural labels satisfy (R8). These hypotheses give the exact domain on which the reconstruction functor assigns a source object.

The compact model of Section 9 shows that this domain is nonempty. Its Fourier restrictions and flat Dirichlet data satisfy (R1)–(R5), its rank-two $SO(2)$ tensor probes satisfy (R6)–(R7), and its structural labels satisfy (R8). The local operation algebras satisfy (R9), while the weight-one $SO(2)$ object is a faithful generator. The resulting indexed data B therefore belong to $\mathcal{B}$, and reconstruction gives $F(R(B)) \cong B$.

For the graph-and-point refinement, Appendix I.6 fixes a pre-regular universe containing every primitive indexed datum together with the response-domain injection and the supplied carrier point. Before constructing any witness, it defines

$$\mathbf{C}_i = \{B : B \models (R_9), \bigwedge_{j \neq i} (B \models (R_j))\}, \quad i = 1, \dots, 8.$$

Theorem 11.15 (Comparison-relative necessity of (R1)–(R8)). *For every $i = 1, \dots, 8$, there is a pre-regular indexed system $B_i \in \mathbf{C}_i$ that fails (Ri) and lies outside the essential image of the enhanced forward functor*

$$\widehat{F} : \widehat{\mathbf{S}}^{D,g} \longrightarrow \widehat{\mathbf{B}}^{D,g}.$$

Thus each of (R1)–(R8) is individually necessary for membership in that image relative to the fixed comparison classes.

Proof. The eight constructions in Theorem I.11 retain (R9) and the other seven conditions. Their respective invariant defects are a noninjective declared restriction family, an incomplete graph domain, a core whose graph closure is proper, failure of the identity $Q = \tau\Gamma$, an ungenerated first-order summand, a nonunitary and nonmonoidal probe comparison, a nonzero probe-invisible morphism, and a collapsed law/input assignment. Each defect is preserved by enhanced operational isomorphism and absent from every forward object, which proves exclusion from the essential image. $\square$

This theorem concerns the fixed classes $\mathbf{C}_i$ and the enhanced forward essential image. Condition (R9) is retained in every witness. No necessity result for (R9), universal axiom minimality, or necessity for every possible recovery architecture follows.

The groupoid equivalence determines the source presentation from the indexed operational data, and the composite functor determines the law of the corresponding field $\mathfrak{F}(\mathfrak{R})$. A state, representation, clock, temporal development, and selected self-adjoint extension of H_R supply additional data.

11.7. Assembly of the Main Theorem.

Proof of Theorem 3.8. Theorem 5.1 gives the unique metric-dual response, Theorem 5.2 gives its rank-dependent connection and shape components, and Theorem 5.3 gives the mixed bilinear identity on its stated Sobolev branch. The Hamiltonian phase-space, local CCR, finite-label commutator, and common-core theorems represent this response on their respective domains. On the positive-Riemannian oriented rank-one branch, Theorem 6.3 supplies the gravitational comparison.

The theorems on generated bounded C^* -algebras, transition-endomorphism subalgebras, and functorial source determination of the bounded-law data define $\mathbb{L}$, while Theorem 11.13 combines it with the response functor. The equivalence of the source and indexed-data groupoids, together with Corollary 11.14, gives recovery from the indexed data and the fixed-class incidence conclusion.

The graph-norm commutator ideal and finite operator families retain the additional represented data stated in Section 9.6, while the state-fibration and operator-domain theorems establish the remaining additional-data statements. Applying each result on its stated source groupoid, branch, and domain gives the properties of $\mathfrak{F}(\mathfrak{R})$ asserted in the theorem. $\square$

12. CONCLUSION

Within the class considered here, a commutative algebra with its localization and restriction maps, its derivation module and Dirichlet datum, a finite-projective metric internal module with its declared gauge group, and the tensor and role assignments in ϑ form a field source $\mathfrak{R}$. The complete law-bearing system determined by that source is the field $\mathfrak{F}(\mathfrak{R})$, while a configuration Ψ is one point of its configuration space. When the Riemannian realization hypotheses hold, $\mathfrak{R}$ determines (M, h, N, g_N) up to the stated geometric equivalence. At $\Psi = (\nabla^{\mathcal{N}}, K)$, the two configuration variables combine with the torsion-free metric connection on TM to form a single metric connection on $TM \oplus N$. Its squared curvature defines the positive energy Φ , and the configuration-space metric identifies the differential of that energy with the unique metric-dual response

$$D\Phi_{\Psi}(v) = G_{\Psi}(\mathcal{Y}_{\Psi}, v).$$

The response splits orthogonally into the components obtained by varying $\nabla^{\mathcal{N}}$ and K . The first is the gauge response and the second is the shape response. On a connected region of constant internal rank one, no connection-response type exists, while the shape-response type remains. On the oriented positive-Riemannian rank-one branch, the shape response is the gravitational response, accompanied by the scalar Gauss and one-form Codazzi contractions of the assembled curvature. From internal rank two onward both response types exist, although either value may vanish at a particular configuration. Under the stated Sobolev hypotheses, their mixed second variations agree as continuous weak bilinear forms. The internal rank therefore fixes which response components are available, the configuration fixes their values, and both arise from the differential of the same curvature energy.

The same response has phase-space and quantum expressions on their specified domains. The Hamiltonian vector field of its signed Legendre function, the local and finite-label response derivations, and the bounded common-core restriction represent it in their respective settings. At each finite label, symmetric ordering

represents the response derivation by the constraint commutator without a remainder term. A separately specified common symmetric core determines the maximal weak constraint and positive Master operator through its Hilbert adjoint, with no choice of self-adjoint extension; its zero eigenspace may be trivial even when zero belongs to the continuous spectrum. Source-supplied bounded operations generate the smallest unital C^* algebras containing their images, and the source-supplied transitions generate the completed metric C^* categories. Graph-domain ideals and finite operator families depend on separately specified representations and remain represented attachments.

The flat coupled torus model shows that this hierarchy carries information beyond the local curvature of a configuration. Its curvature Hessian on the gauge quotient has an infinite-dimensional kernel, so neither a compact resolvent nor an ordinary heat trace is available. The nonzero shape tensor nevertheless determines a translation-invariant reducing subspace on which $P_\eta L_{\theta,k} e^{-tL_{\theta,k}}$ is trace class. Its exact winding formula distinguishes locally gauge-equivalent flat configurations with different global holonomy and varies nontrivially within the preserved rank-one gravitational summand. The field therefore contains its configuration space and local curvature together with the law governing admissible variations, whose Hessian can retain global holonomy information. This trace belongs to the spectral calculus of the represented Hessian, while the source-generated observable algebra retains its declared generators.

Conversely, on the regular classes the forward and recovery functors establish the equivalence

$$\mathbf{S} \simeq \mathbf{B}.$$

Indexed local and finite data B therefore determine $R(B)$ up to structural equivalence. Whenever $R(B)$ lies in the field-source class, it determines $\mathfrak{F}(R(B))$ and its field law. Within the fixed seven-datum source class, every source component has a determined downstream role, and deleting any one component removes at least one incidence required by that law. The source is thus both the datum from which the field law follows and a structure recoverable from indexed data defined without first choosing a source presentation.

States, Hilbert-space representations, clocks, temporal developments, measurement models, apparatus, and empirical maps enter as additional data with distinct roles. A configuration selects a point of the configuration space, and the stationarity condition identifies the zero set of the response. A representation realizes a bounded observable algebra, a state assigns expectation values, and a clock together with a temporal development supplies an ordering and a history. Measurement models and empirical maps relate these mathematical structures to possible outcomes. Accordingly, a field in this class is the complete law-bearing structure $\mathfrak{F}(\mathfrak{A})$ determined by a recoverable source. Its configuration space, realized geometry under the stated hypotheses, response law, constraint and quantum expressions, and generated bounded observable algebras are determined independently of any selected configuration, state, or temporal development.

APPENDIX A. CONNECTION, CURVATURE, AND THE CONFIGURATION-SPACE METRIC

Fix a field source $\mathfrak{R} \in \mathbf{S}^F$ as defined in Section 3. On its nondegenerate branch, the metric on $\text{Der}(\mathcal{A})$ determines the torsion-free metric connection on the derivation module. A configuration $\Psi = (\nabla^\mathcal{N}, K)$ supplies the internal connection and the mixed tensor from which a single metric connection and its curvature are formed. The calculations below establish the connection and curvature formulas, their Riemannian realization and internal gauge covariance, and the orthogonal metric on the two independent configuration variations used in Section 4.

A.1. The torsion-free metric connection on the derivation module. A Lie–Rinehart algebra over a real commutative algebra A is an A -module L with a real Lie bracket and an A -linear anchor

$$\rho : L \longrightarrow \text{Der}_\mathbb{R}(A)$$

satisfying

$$[X, fY] = f[X, Y] + (\rho(X)f)Y$$

for $f \in A$. The module $\text{Der}(\mathcal{A})$, with its commutator bracket and the tautological anchor into the derivations of $\mathcal{A}$, has this structure. Its bracket is the bracket that enters the curvature term $\nabla_{\Psi, [X, Y]}^\oplus$.

For a symmetric A -bilinear pairing g on L , define the Koszul covector $\mathcal{K}_g(X, Y) : L \rightarrow A$ by

$$\begin{aligned} \mathcal{K}_g(X, Y)(Z) &= \rho(X)g(Y, Z) + \rho(Y)g(X, Z) - \rho(Z)g(X, Y) \\ &\quad + g([X, Y], Z) - g([X, Z], Y) - g([Y, Z], X). \end{aligned}$$

Lemma A.1 (Koszul covector). *For every $X, Y \in L$, the map $\mathcal{K}_g(X, Y)$ is A -linear in Z .*

Proof. Expanding $\mathcal{K}_g(X, Y)(fZ)$ by the Leibniz identities and collecting the derivatives of f leaves

$$(\rho(X)f)g(Y, Z) - (\rho(X)f)g(Z, Y) + (\rho(Y)f)g(X, Z) - (\rho(Y)f)g(Z, X),$$

which vanishes by symmetry of g . $\square$

Assume that g is nondegenerate in the module sense that its metric map

$$g^\flat : L \longrightarrow L^\vee = \text{Hom}_A(L, A), \quad g^\flat(X) = g(X, -),$$

is an isomorphism. The formula

$$\nabla_X Y = \frac{1}{2}(g^\flat)^{-1}(\mathcal{K}_g(X, Y))$$

then defines an element of L satisfying

$$2g(\nabla_X Y, Z) = \mathcal{K}_g(X, Y)(Z)$$

for every $Z \in L$. Expanding the Koszul covector in its first two entries gives

$$\mathcal{K}_g(fX, Y) = f\mathcal{K}_g(X, Y), \quad \mathcal{K}_g(X, fY) = f\mathcal{K}_g(X, Y) + 2(\rho(X)f)g(Y, -).$$

Inversion by $g^\flat$ therefore yields

$$\nabla_{fX} Y = f\nabla_X Y, \quad \nabla_X(fY) = f\nabla_X Y + (\rho(X)f)Y,$$

so ∇ is a connection on L .

Theorem A.2 (Koszul determination). *Let L be a Lie–Rinehart algebra over A , and let g be a symmetric A -bilinear pairing for which $g^\flat : L \rightarrow L^\vee$ is an isomorphism. The connection defined above is the unique torsion-free, g -metric connection on L .*

Proof. Subtracting $\mathcal{K}_g(Y, X)$ from $\mathcal{K}_g(X, Y)$ leaves $2g([X, Y], Z)$ for every Z , and hence $\nabla_X Y - \nabla_Y X = [X, Y]$. Adding the Koszul expressions with second entries Y and Z gives

$$g(\nabla_X Y, Z) + g(Y, \nabla_X Z) = \rho(X)g(Y, Z),$$

so the connection is metric. Conversely, torsion-freeness and metricity for any connection ∇' reproduce

$$2g(\nabla'_X Y, Z) = \mathcal{K}_g(X, Y)(Z).$$

The isomorphism $g^\flat$ then forces $\nabla' = \nabla$. $\square$

Applying the theorem to $L = \text{Der}(\mathcal{A})$ and $g = g_D$ determines the unique torsion-free g_D -metric connection ∇^D on the nondegenerate branch.

Two coordinate examples isolate the role of nondegeneracy. First consider, on a two-dimensional coordinate patch, the metric

$$g_{11} = 1 + (x^1)^2, \quad g_{12} = x^1 x^2, \quad g_{22} = 1 + (x^2)^2.$$

Its determinant is $1 + (x^1)^2 + (x^2)^2$, and the Koszul formula gives the torsion-free metric Christoffel connection everywhere. For

$$g = \text{diag}(1, x^1),$$

the equations for $\Gamma_{12}^\bullet$ are

$$\begin{pmatrix} 1 & 0 \\ 0 & x^1 \end{pmatrix} \begin{pmatrix} \Gamma_{12}^1 \\ \Gamma_{12}^2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1/2 \end{pmatrix}.$$

Away from $x^1 = 0$, they give

$$\Gamma_{12}^1 = 0, \quad \Gamma_{12}^2 = \frac{1}{2x^1}.$$

At $x^1 = 0$, the coefficient matrix has rank one while the augmented matrix has rank two, so the Koszul system is inconsistent. The divergent component therefore reaches its pole at the same locus where the metric map ceases to be invertible.

A.2. Metric compatibility of the block connection. Fix a configuration $\Psi = (\nabla^\mathcal{N}, K)$ and $X \in \text{Der}(\mathcal{A})$. Consider a connection on $\text{Der}(\mathcal{A}) \oplus \mathcal{N}$ whose diagonal blocks are the two metric connections and whose lower-left block is K_X ,

$$\Gamma_X = \begin{pmatrix} \nabla_X^D & C_X \\ K_X & \nabla_X^\mathcal{N} \end{pmatrix}.$$

Lemma A.3 (Adjoint relation between the mixed connection blocks). *The connection Γ_X is metric for $g_D \oplus g_\mathcal{N}$ if and only if*

$$C_X = -K_X^\dagger.$$

Proof. Metricity on $Y \oplus 0$ and $0 \oplus \nu$ gives

$$g_D(C_X \nu, Y) + g_\mathcal{N}(\nu, K_X Y) = 0$$

for every $Y \in \text{Der}(\mathcal{A})$ and $\nu \in \mathcal{N}$. The definition of the adjoint turns this equation into

$$g_D((C_X + K_X^\dagger)\nu, Y) = 0.$$

Nondegeneracy of g_D gives $C_X = -K_X^\dagger$. Conversely, this identity cancels the mixed terms, while metricity of the two diagonal connections supplies the remaining terms. $\square$

Metric compatibility therefore determines the connection

$$\nabla_{\Psi, X}^\oplus = \begin{pmatrix} \nabla_X^D & -K_X^\dagger \\ K_X & \nabla_X^{\mathcal{N}} \end{pmatrix}.$$

A.3. The symmetry condition on K . Metric compatibility determines the adjoint relation between the two mixed blocks and permits both the symmetric and skew parts of an internal-valued bilinear form. Write

$$K = K^+ + K^-.$$

The connection

$$\begin{pmatrix} \nabla_X^D & -(K_X^+ + K_X^-)^\dagger \\ K_X^+ + K_X^- & \nabla_X^{\mathcal{N}} \end{pmatrix}$$

remains metric for the split pairing.

Lemma A.4 (Symmetry of K as a configuration-space condition). *The configuration space imposes*

$$K \in \text{Sym}_{\mathcal{A}}^2(\Omega_{\mathfrak{R}}^1) \otimes_{\mathcal{A}} \mathcal{N}.$$

Metric compatibility of the block connection imposes only the adjoint relation between its off-diagonal blocks.

Proof. The skew component enters the lower-left block and its negative adjoint enters the upper-right block, so the same mixed cancellation proves metricity. The symmetry of K is consequently the independent restriction declared in $\mathcal{C}^s(\mathfrak{R})$. $\square$

A.4. Curvature of the assembled connection. For $X, Y \in \text{Der}(\mathcal{A})$, define

$$\mathcal{F}_{\Psi}^\oplus(X, Y) = [\nabla_{\Psi, X}^\oplus, \nabla_{\Psi, Y}^\oplus] - \nabla_{\Psi, [X, Y]}^\oplus.$$

The two diagonal connections induce the covariant derivative

$$(\nabla_X K)(Y, Z) = \nabla_X^{\mathcal{N}}(K(Y, Z)) - K(\nabla_X^D Y, Z) - K(Y, \nabla_X^D Z).$$

With respect to $\text{Der}(\mathcal{A}) \oplus \mathcal{N}$, the curvature has the block form

$$\mathcal{F}_{\Psi}^\oplus(X, Y) = \begin{pmatrix} B_T(\Psi)(X, Y) & -B_K(\Psi)(X, Y)^\dagger \\ B_K(\Psi)(X, Y) & B_\perp(\Psi)(X, Y) \end{pmatrix},$$

where

$$\begin{aligned} B_T(\Psi)(X, Y) &= R^D(X, Y) - K_X^\dagger K_Y + K_Y^\dagger K_X, \\ B_K(\Psi)(X, Y) &= (\nabla_X K)_Y - (\nabla_Y K)_X, \\ B_\perp(\Psi)(X, Y) &= R^{\mathcal{N}}(X, Y) - K_X K_Y^\dagger + K_Y K_X^\dagger. \end{aligned}$$

Lemma A.5 (Curvature blocks of the metric connection). *The tangential, mixed, and internal blocks displayed above are the canonical block projections of the curvature endomorphism $\mathcal{F}_{\Psi}^\oplus$.*

Proof. Insert the block forms of $\nabla_{\Psi,X}^{\oplus}$ and $\nabla_{\Psi,Y}^{\oplus}$ into the curvature commutator. The upper-left block is the curvature of ∇^D plus the two products of mixed blocks, which gives

$$R^D - K_X^{\dagger} K_Y + K_Y^{\dagger} K_X.$$

The lower-left block is the covariant exterior derivative of K , the lower-right block is

$$R^{\mathcal{N}} - K_X K_Y^{\dagger} + K_Y K_X^{\dagger},$$

and metricity of the curvature fixes the upper-right block as the negative adjoint of the lower-left block. $\square$

Thus the tangential block combines the source-determined base curvature with quadratic terms in K , the mixed block records the covariant exterior derivative of K , and the internal block combines the curvature of the internal connection with its quadratic K -terms. They are three projections of one curvature at the selected configuration.

A.5. Riemannian bundle realization. Under the Riemannian bundle realization of Section 3,

$$\text{Der}(\mathcal{A}) \cong \Gamma(TM), \quad \mathcal{N} \cong \Gamma(N), \quad \Omega_{\mathfrak{R}}^1 \cong \Gamma(T^*M).$$

The connection ∇^D becomes the Levi-Civita connection of h , $\nabla^{\mathcal{N}}$ becomes a metric connection on N , and K becomes a section of $\text{Sym}^2 T^*M \otimes N$. The intrinsic connection $\nabla_{\Psi}^{\oplus}$ thereby becomes the block metric connection on $TM \oplus N$, and its three curvature blocks become the corresponding tangential, mixed, and internal bundle maps.

Theorem A.6 (Compatibility of connection and curvature with realization). *Every Riemannian bundle realization carries the intrinsic block connection at Ψ and the curvature blocks $B_T(\Psi)$, $B_K(\Psi)$, $B_{\perp}(\Psi)$ to their corresponding connection and curvature components on $TM \oplus N$.*

Proof. The module identifications preserve direct sums, metric pairings, adjoints, contractions, covariant derivatives, and commutators of connections. They therefore carry each block of $\nabla_{\Psi}^{\oplus}$ to its bundle counterpart. Applying the curvature commutator before or after these identifications gives the same block matrix. $\square$

A.6. Internal gauge covariance. Let $u \in \mathcal{G}_{\mathcal{N}_g}^s$ preserve the regularity class of the configuration, and put

$$\tilde{u} = 1 \oplus u$$

on $\text{Der}(\mathcal{A}) \oplus \mathcal{N}$. The transformed configuration is

$$u \cdot \Psi = (u \nabla^{\mathcal{N}} u^{-1}, uK).$$

Conjugation of $\nabla^{\mathcal{N}}$ as a differential operator includes the usual inhomogeneous derivative term when u is nonconstant. The assembled connection and its curvature satisfy

$$\nabla_{u \cdot \Psi, X}^{\oplus} = \tilde{u} \nabla_{\Psi, X}^{\oplus} \tilde{u}^{-1}, \quad \mathcal{F}_{u \cdot \Psi}^{\oplus}(X, Y) = \tilde{u} \mathcal{F}_{\Psi}^{\oplus}(X, Y) \tilde{u}^{-1}.$$

Consequently,

$$B_T(u \cdot \Psi) = B_T(\Psi), \quad B_K(u \cdot \Psi) = u B_K(\Psi), \quad B_{\perp}(u \cdot \Psi) = u B_{\perp}(\Psi) u^{-1}.$$

Lemma A.7 (Gauge covariance of the curvature blocks). *Under the regularity-preserving internal gauge action, the tangential block is invariant, the mixed block transforms in the internal module, and the internal block transforms by conjugation.*

Proof. The displayed connection transformation is conjugation by $\tilde{u}$, so its curvature is conjugated by the same automorphism. Projection to the three blocks gives the stated transformations. $\square$

A.7. Orthogonal metric on configuration variations. Fix an active label λ and let τ_λ be its positive integration functional. Label subscripts on the variation modules and their pairings are suppressed throughout this subsection. The derivation metric induces the dual metric on $\Omega_{\mathfrak{A}}^1$, and the internal metric supplies adjoints and the finite-projective trace on $\text{End}_A(\mathcal{N})$. The connection-variation module

$$\mathcal{V}_\nabla = \Omega_{\mathfrak{A}}^1 \otimes_A \mathfrak{so}(\mathcal{N}_g)$$

therefore carries the pairing

$$G_\nabla(\alpha, \beta) = \tau_\lambda \left(g_D^{ij} \text{tr}_{\mathcal{N}}(\alpha_i^\dagger \beta_j) \right).$$

Here $\text{tr}_{\mathcal{N}}$ is the unnormalized finite-projective trace. The internal metric and this trace determine the Hilbert–Schmidt pairing $\text{tr}_{\mathcal{N}}(A^\dagger B)$.

Put $D = \Omega_{\mathfrak{A}}^1$. The flip on $D \otimes D$ is unitary, and

$$P_+ = \frac{1 + \varsigma}{2}$$

is the self-adjoint symmetrization projection. Restricting the tensor-product metric to its range and tensoring with $\mathcal{N}_g$ gives

$$G_K(K, L) = \tau_\lambda \left(g_D^{ia} g_D^{jb} g_{\mathcal{N}}(K_{ij}, L_{ab}) \right).$$

This construction applies in every positive finite internal rank and fixes the normalization of G_K .

The metric modules generated from the derivation and internal modules by duals, tensor products, source-determined self-adjoint retracts, and finite direct sums inherit metric maps recursively. Write

$$G_X^\flat : X \longrightarrow X^\vee$$

for the metric map on such an object. The pairings fixed by the source give these maps on the generating modules, while duality, tensor products, restriction to a self-adjoint retract, and finite direct sums determine them on the resulting modules.

For a direct sum, the canonical duality is

$$(V \oplus W)^\vee \simeq V^\vee \oplus W^\vee, \quad \text{ev}_{V \oplus W}((\phi, \psi), (v, w)) = \phi(v) + \psi(w).$$

It follows that

$$G_{V \oplus W}^\flat = G_V^\flat \oplus G_W^\flat.$$

For the canonical inclusions $i_V(v) = (v, 0)$ and $i_W(w) = (0, w)$,

$$i_V^\vee G_{V \oplus W}^\flat i_V = G_V^\flat, \quad i_W^\vee G_{V \oplus W}^\flat i_W = G_W^\flat, \quad i_V^\vee G_{V \oplus W}^\flat i_W = 0.$$

Theorem A.8 (Orthogonal direct-sum metric on the configuration space). *The fixed pairings G_∇ and G_K determine*

$$G_{\mathfrak{A}} = G_\nabla \oplus G_K, \quad G_{\mathfrak{A}}^\flat = G_\nabla^\flat \oplus G_K^\flat.$$

On the metric cotangent range $\text{ran } G_{\mathfrak{R}}^b$, the inverse is

$$G_{\mathfrak{R}}^{\sharp} = G_{\nabla}^{\sharp} \oplus G_K^{\sharp}.$$

The canonical inclusions obey

$$i_{\nabla}^{\dagger} i_{\nabla} = 1, \quad i_K^{\dagger} i_K = 1, \quad i_{\nabla}^{\dagger} i_K = 0.$$

Among metrics natural under source automorphisms and restricting to G_{∇} and G_K on the two summands, $G_{\mathfrak{R}}$ is unique.

Proof. Canonical direct-sum duality gives the displayed metric map and makes the two inclusions isometric with orthogonal images. The inverse on its range is the direct sum of the two inverse metric maps.

The metric automorphism $u = -\text{id}_{\mathcal{N}}$ centralizes the declared internal gauge group and preserves the source module types and their structure maps. It acts trivially by conjugation on a connection variation and changes the sign of a shape variation. A mixed pairing B natural under source automorphisms must therefore satisfy

$$B(\alpha, K) = B(u\alpha u^{-1}, uK) = B(\alpha, -K) = -B(\alpha, K),$$

and hence vanishes. At internal rank one the connection-variation module is already zero. The two prescribed diagonal restrictions and the vanishing mixed block determine every metric in the stated competitor class. $\square$

Both parts of the competitor class are essential. Naturality alone permits the weighted metrics

$$G^{(c_{\nabla})} = c_{\nabla} G_{\nabla} \oplus G_K, \quad c_{\nabla} > 0,$$

while the requirement that the connection inclusion remain isometric fixes $c_{\nabla} = 1$. The same action of $-\text{id}_{\mathcal{N}}$ eliminates every source-natural intertwiner from $\mathcal{V}_{\nabla}$ to $\mathcal{V}_K$, and therefore eliminates natural mixed terms.

The Hilbert–Schmidt pullback along the block-connection variation

$$(\alpha, \dot{K}) \mapsto \begin{pmatrix} 0 & -\dot{K}^{\dagger} \\ \dot{K} & \alpha \end{pmatrix}$$

is instead

$$G_{\text{HS}} = G_{\nabla} \oplus 2G_K.$$

The two rectangular blocks have equal trace and represent the same shape variation, so the pullback counts both $\dot{K}$ and the adjoint block required by metric compatibility. The coefficient two is likewise the appropriate multiplicity in the Hilbert–Schmidt norm of the full curvature endomorphism.

For the normalization used later in the rank-two torus models, let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \text{tr}(J^* J) = 2, \quad \hat{J} = \frac{J}{\sqrt{2}}.$$

Then $\hat{J}$ is the unit generator for the connection summand. An orthonormal one-form basis tensored with $\hat{J}$ makes the connection metric and its metric inverse the identity, so a pure connection momentum has kinetic energy $\|p\|^2/2$.

The coefficient-one direct-sum metric thus measures the two independent configuration variations, while the coefficient-two Hilbert–Schmidt pullback measures their appearance in the assembled block connection.

APPENDIX B. LOCAL CURVATURE-ENERGY FORMS AND QUASIFREE ADMISSIBILITY

The assembled curvature and configuration metric of Section 4 give a nonnegative density and hence compatible local and finite-label energies. Their second differentials form a configuration-indexed weak Hessian field, and stable critical members of this field admit self-adjoint stiffness operators and nonempty families of quasifree states. The weak statement holds throughout the Sobolev source class of Section 5; the operator and state statements carry the additional criticality, quotient, closure, and label hypotheses stated below.

B.1. From assembled curvature to positive energy density. For a configuration Ψ , let

$$\mathcal{F}_\Psi^\oplus = \begin{pmatrix} B_T(\Psi) & -B_K(\Psi)^\dagger \\ B_K(\Psi) & B_\perp(\Psi) \end{pmatrix}$$

be the curvature block matrix from Appendix A. The metric pairings determined by the source induce Hilbert–Schmidt norms on its three components.

Lemma B.1 (Multiplicity of the mixed curvature block). *The squared Hilbert–Schmidt norm of the assembled curvature is*

$$\|\mathcal{F}_\Psi^\oplus\|^2 = \|B_T(\Psi)\|^2 + 2\|B_K(\Psi)\|^2 + \|B_\perp(\Psi)\|^2.$$

Proof. The mixed block occurs as $B_K(\Psi)$ and as $-B_K(\Psi)^\dagger$. Since a block and its adjoint have the same Hilbert–Schmidt norm, their combined contribution is

$$\|B_K(\Psi)\|^2 + \|B_K(\Psi)^\dagger\|^2 = 2\|B_K(\Psi)\|^2.$$

□

Define the curvature-energy density by

$$\phi_\Re(\Psi) = \frac{1}{2}\|\mathcal{F}_\Psi^\oplus\|^2 = \frac{1}{2}(\|B_T(\Psi)\|^2 + 2\|B_K(\Psi)\|^2 + \|B_\perp(\Psi)\|^2).$$

Lemma B.2 (Positivity and zero set of the curvature-energy density). *The density $\phi_\Re(\Psi)$ is nonnegative and vanishes exactly when $\mathcal{F}_\Psi^\oplus$ vanishes in the Hilbert–Schmidt norm.*

Proof. Each summand is a squared norm with positive coefficient. Their sum vanishes exactly when $B_T(\Psi) = B_K(\Psi) = B_\perp(\Psi) = 0$. □

B.2. Local and finite-label curvature energies. For a local or finite cylindrical label $\lambda \in \Lambda$, let τ_λ be its positive integration functional and let $\phi_{\Re,\lambda}$ be the restriction of the curvature density to the corresponding configuration and coefficient data. The labelwise curvature energy is

$$\Phi_{\Re,\lambda}(\Psi_\lambda) = \tau_\lambda(\phi_{\Re,\lambda}(\Psi_\lambda)).$$

A Riemannian local label integrates the density over its region, while a finite cylindrical label applies its finite positive integration datum to the restricted density.

Lemma B.3 (Positivity of the labelwise curvature energy). *For every local or finite cylindrical label, $\Phi_{\Re,\lambda}(\Psi_\lambda) \geq 0$. The functional is determined by the restricted curvature expression and the integration datum of λ .*

Proof. The curvature density is nonnegative, and τ_λ is positive. Their composition gives the assertion. □

B.3. Joint source restriction. For $\lambda \subseteq \mu$, let $r_{\lambda\mu}$ restrict the source coefficients, density support, configuration slots, and integration data. On an integrated expression define

$$r_{\lambda\mu}^{\mathfrak{R}}[\tau_\mu(f_\mu)] = \tau_\lambda(r_{\lambda\mu}f_\mu).$$

Lemma B.4 (Compatibility under joint source restriction). *For every fine configuration Ψ_μ ,*

$$r_{\lambda\mu}^{\mathfrak{R}}[\Phi_{\mathfrak{R},\mu}(\Psi_\mu)] = \Phi_{\mathfrak{R},\lambda}(r_{\lambda\mu}\Psi_\mu).$$

The joint restrictions compose on every three-label chain.

Proof. Connection assembly, curvature, adjoints, contractions, and support restriction are natural under the source restriction maps. Restricting the fine density therefore gives the coarse density with its background terms, and applying the coarse integration functional gives the displayed identity. Composition follows from the corresponding composition laws for coefficients, support, configuration slots, and integration. $\square$

This contravariant restriction acts on the curvature expression and its integration datum. The response theorem uses in addition the covariant metric-dual inclusion of finite phase variables.

B.4. Gauge invariance and Riemannian realization. Let $u \in \mathcal{G}_{\mathcal{N}_g}^s$ be an internal metric-gauge transformation preserving the required Sobolev spaces. The curvature components satisfy

$$B_T(u \cdot \Psi) = B_T(\Psi), \quad B_K(u \cdot \Psi) = u B_K(\Psi), \quad B_\perp(u \cdot \Psi) = u B_\perp(\Psi) u^{-1}.$$

Lemma B.5 (Gauge invariance of curvature energy). *If u preserves the label and the Sobolev branch, then*

$$\Phi_{\mathfrak{R},\lambda}(u \cdot \Psi) = \Phi_{\mathfrak{R},\lambda}(\Psi).$$

Proof. The tangent block is fixed, and the mixed and internal blocks transform by metric isometries and conjugation. Their norms and the source integration functional are therefore preserved. $\square$

Under a Riemannian bundle realization, the derivation and internal pairings become the metrics on TM and N . If a local label is represented by a relatively compact region $U \subset M$, its integration functional becomes integration against dV_h , and

$$\Phi_U(\Psi) = \frac{1}{2} \int_U (\|B_T(\Psi)\|^2 + 2\|B_K(\Psi)\|^2 + \|B_\perp(\Psi)\|^2) dV_h.$$

Theorem B.6 (Curvature-energy realization). *A Riemannian bundle realization sends the compatible family $\Phi_{\mathfrak{R},\lambda}$ to the integrated squared-curvature functionals on the corresponding local regions and to the positive functionals of the restricted curvature density on finite cylindrical labels.*

Proof. Appendix A identifies the intrinsic curvature components with those of the metric connection on $TM \oplus N$. It also identifies the source pairings and local integration functional with the Riemannian pairings and volume measure. Substitution in the intrinsic density gives the displayed formula, while joint source restriction gives the finite-label statement. $\square$

B.5. Curvature Hessians and positive stiffness. On a relatively compact window U , retain the integer $s > \dim(U)/2 + 2$, the affine H^s configuration chart, and the full-trace-jet variation space $\mathcal{V}_0^s(U)$ of Section 5. For $v, w \in \mathcal{V}_0^s(U)$, define

$$H_{\mathfrak{R},U,\Psi}(v, w) = D^2\Phi_{\mathfrak{R},U}(\Psi)[v, w].$$

Theorem B.7 (Configuration-indexed weak curvature Hessian). *For every admitted source, window, and configuration, $H_{\mathfrak{R},U,\Psi}$ is a continuous symmetric bilinear form in the H_0^s topology. It is natural under source isomorphisms and admissible finite internal gauge transformations. For a supported inclusion $U \subseteq V$ with compatible backgrounds and zero extension E_{VU} ,*

$$H_{\mathfrak{R},V,\Psi_V}(E_{VU}v, E_{VU}w) = H_{\mathfrak{R},U,\Psi_U}(v, w).$$

The response is differentiable from H^s to $H^{s-2} \subset L^2$, and its weak linearization satisfies

$$H_{\mathfrak{R},U,\Psi}(v, w) = G_{\mathfrak{R},U}(D\mathcal{Y}_{\mathfrak{R},U}(\Psi)[v], w).$$

Proof. For the assembled connection increment $\mathcal{A}(v)$, put $L_{\Psi,v} = D_{\nabla_{\Psi}^{\oplus}}\mathcal{A}(v)$. The affine curvature identity gives

$$\mathcal{F}^{\oplus}(\Psi + v) = \mathcal{F}^{\oplus}(\Psi) + L_{\Psi,v} + \frac{1}{2}[\mathcal{A}(v) \wedge \mathcal{A}(v)].$$

Sobolev multiplication makes this a continuous polynomial into $H^{s-1} \subset L^2$. Differentiating the squared norm twice yields

$$\begin{aligned} H_{\mathfrak{R},U,\Psi}(v, w) &= \langle L_{\Psi,v}, L_{\Psi,w} \rangle_{L^2} \\ &\quad + \langle \mathcal{F}^{\oplus}(\Psi), [\mathcal{A}(v) \wedge \mathcal{A}(w)] \rangle_{L^2}. \end{aligned}$$

Both terms are symmetric and continuous in the H^s topology. Energy naturality and the second-order chain rule give source and finite-gauge naturality, since their configuration maps are affine. The same formula with support-compatible increments proves the zero-extension identity.

Appendix C writes the response as a fixed bounded block projection of $D_{\nabla_{\Psi}^{\oplus}}^{\dagger}\mathcal{F}_{\Psi}^{\oplus}$, including the symmetric projection and coefficient two in the shape component. This is a continuous polynomial from H^s to H^{s-2} , so it is Fréchet differentiable. Differentiating $D\Phi_U(\Psi)[w] = G_U(\mathcal{Y}_U(\Psi), w)$ in the direction v gives the final identity. It is an identity after pairing an H^{s-2} response derivative with an H_0^s test variation. $\square$

The weak Hessian is gauge covariant at every configuration; descent through infinitesimal gauge directions additionally requires criticality. Let $\xi_{\Psi}^{\#}$ denote the fundamental vector field of an admitted H^{s+1} gauge generator whose value and derivative preserve $\mathcal{V}_0^s(U)$.

Lemma B.8 (Gauge defect and critical descent). *For every admitted variation v ,*

$$H_{\mathfrak{R},\Psi}(v, \xi_{\Psi}^{\#}) + D\Phi_{\mathfrak{R}}(\Psi)[D_v\xi_{\Psi}^{\#}] = 0.$$

If $D\Phi_{\mathfrak{R}}(\Psi) = 0$ on the variation domain, the Hessian annihilates the H^s -closure of the infinitesimal gauge directions and therefore descends to the corresponding Sobolev quotient.

Proof. Differentiating finite gauge invariance along the one-parameter subgroup generated by ξ gives $D\Phi(\Psi)[\xi_\Psi^\#] = 0$. Differentiation in the direction v gives the displayed identity. At a critical configuration its second term vanishes. Symmetry and continuity then place the closed gauge subspace in the radical of the Hessian, which makes the quotient form independent of representatives. $\square$

Write $\mathfrak{g}_\Psi$ for the admitted infinitesimal gauge directions and

$$\mathcal{V}^s(\mathfrak{R}, \Psi)/\overline{\mathfrak{g}_\Psi} = \mathcal{V}_0^s/\overline{\mathfrak{g}_\Psi}^{H^s}.$$

The hard gauge quotient is obtained by taking the G -Hilbert completion of the variation space and then dividing by the G -closure of the same gauge directions.

Theorem B.9 (Self-adjoint curvature stiffness on the stable branch). *Let Ψ be critical and suppose that the canonical map from $\mathcal{V}^s(\mathfrak{R}, \Psi)/\overline{\mathfrak{g}_\Psi}$ into the hard gauge quotient $\mathcal{H}_{\mathfrak{R}, \Psi}$ is injective with dense image. Suppose further that the descended weak Hessian closes to a densely defined closed nonnegative form $Q_{\mathfrak{R}, \Psi}$ on that quotient and that the weak quotient is a form core. Then there is a unique nonnegative self-adjoint operator $L_{\mathfrak{R}, \Psi}$ satisfying*

$$D(Q_{\mathfrak{R}, \Psi}) = D(L_{\mathfrak{R}, \Psi}^{1/2}), \quad Q_{\mathfrak{R}, \Psi}(v, w) = \langle L_{\mathfrak{R}, \Psi}^{1/2} v, L_{\mathfrak{R}, \Psi}^{1/2} w \rangle.$$

Source and gauge isomorphisms transport the form, operator, kernel, positive hard quotient, and spectral calculus unitarily.

The residual zero-stiffness space $Z_{\mathfrak{R}, \Psi} = \ker L_{\mathfrak{R}, \Psi}$ remains part of the hard data, and the inverse stiffness is defined on

$$\mathcal{H}_{\mathfrak{R}, \Psi}^+ = Z_{\mathfrak{R}, \Psi}^\perp, \quad L_{\mathfrak{R}, \Psi}^+ = L_{\mathfrak{R}, \Psi}|_{Z_{\mathfrak{R}, \Psi}^\perp}.$$

A configuration with a negative Hessian direction does not satisfy the nonnegative hard-realization hypothesis.

Proof. The first representation theorem for densely defined closed nonnegative forms gives the operator and the equality of form domains [41, Theorem 10.7 and Corollary 10.8]. If a unitary U transports the closed form, the operator-domain characterization of the representation theorem gives $L'U = UL$. The kernel, its orthogonal complement, and the spectral calculus then transport by the spectral theorem. The kernel identity

$$\ker L_{\mathfrak{R}, \Psi} = \{v \in D(Q_{\mathfrak{R}, \Psi}) : Q_{\mathfrak{R}, \Psi}(v, v) = 0\}$$

keeps residual zero stiffness distinct from the gauge space removed before the hard form was defined. $\square$

The hypotheses of Theorem B.9 define a possibly empty family over a fixed source. Its configuration index is essential even when this family is nonempty.

Lemma B.10 (Configuration dependence of curvature stiffness). *One flat internal-rank-one source on the three-torus has two critical configurations satisfying the stable hard hypotheses whose curvature Hessians are different. Consequently its configuration-indexed Hessian field cannot be represented by one operator $L_{\mathfrak{R}}$.*

Proof. Let e_1, e_2, e_3 be the parallel tangent frame and n the parallel internal unit. Define constant symmetric tensors by

$$K_0(e_1, e_1) = n, \quad \kappa(e_2, e_2) = n,$$

with all other components zero, and let A_1 and B_2 be their skew-adjoint off-diagonal connection increments in the e_1 and e_2 directions. They rotate the (e_1, n) and (e_2, n) planes, so $[A_1, B_2]e_1 \neq 0$.

For $\Psi_t = (d, tK_0)$, the assembled connection has the single constant component $tA_1 dx^1$. Its curvature vanishes, and hence every Ψ_t is critical. For the constant variation $v = (0, \kappa)$, the Hessian formula of Theorem B.7 gives

$$H_{\mathfrak{R}, \Psi_t}(v, v) = t^2 \|[A_1, B_2], dx^1 \wedge dx^2\|_{L^2}^2.$$

The right side is zero at $t = 0$ and positive at $t = 1$.

To verify the hard hypotheses, put

$$\mathcal{V} = L^2(\text{Sym}^2 T^* \mathbb{T}^3 \otimes \mathbb{R}n), \quad \mathcal{D}_t = D_{\nabla_{\Psi_t}^\oplus} \mathcal{A}$$

on smooth symmetric tensors. Fourier transformation gives

$$\widehat{\mathcal{D}_t w}(k) = M_t(k) \widehat{w}(k), \quad M_t(k) = i\sigma_K(k) + tB,$$

where $\sigma_K(k)$ is the Codazzi symbol and B is the fixed commutator with A_1 . The maximal domain

$$D(\overline{\mathcal{D}_t}) = \left\{ w \in \mathcal{V} : \sum_{k \in \mathbb{Z}^3} \|M_t(k) \widehat{w}(k)\|^2 < \infty \right\}$$

is closed by coordinatewise convergence of the finite-dimensional Fourier blocks. Finite Fourier sums converge in its graph norm, so they are a core. It follows that

$$Q_t[w] = \|\overline{\mathcal{D}_t} w\|^2, \quad L_t = \overline{\mathcal{D}_t}^* \overline{\mathcal{D}_t}$$

give the closed nonnegative form and its self-adjoint representative, with retained zero sector

$$Z_t = \widehat{\bigoplus_{k \in \mathbb{Z}^3} \ker M_t(k)}.$$

The rank-one internal gauge group is trivial, so the weak and hard gauge quotients are unchanged and finite Fourier sums give the required Sobolev form core. The positive quotient is nonzero. Indeed, for $k \neq 0$ one may choose a real coefficient w with $\sigma_K(k)w \neq 0$; the imaginary term $i\sigma_K(k)w$ cannot cancel the real term tBw .

Let T_j be the full real Fourier range of the source-Laplacian spectral projection $\mathbf{1}_{[0, j^2]}(\Delta)$. This directed exhaustive family is independent of t . If P_t^0 and P_t^+ are the blockwise kernel and positive projections of L_t , then

$$S_{t,j}^0 = P_t^0 T_j, \quad S_{t,j} = P_t^+ T_j$$

exhaust Z_t and $Z_t^\perp$, respectively. Each $S_{t,j}$ is a finite sum of complete positive spectral spaces of $M_t(k)^* M_t(k)$, and consequently

$$S_{t,j} \subset D((L_t^+)^{1/2}) \cap D((L_t^+)^{-1/2})$$

and $S_{t,j}$ is already its own minimal reducing envelope. Thus Ψ_0 and Ψ_1 satisfy every stable hard and label condition with the same original source-label family, while their exact Hessians differ on v . $\square$

B.6. Quasifree admissibility and the state fibration. Fix a critical pair satisfying Theorem B.9 and abbreviate $L = L_{\mathfrak{R}, \Psi}^+$ on $\mathcal{H}_{\mathfrak{R}, \Psi}^+$. Suppose the source supplies a configuration-independent directed exhaustive family of original labels. Let S_i be their positive images, and assume that every seed lies in

$$\mathcal{R}_L = D(L^{1/2}) \cap D(L^{-1/2}).$$

With $b_L = L(1 + L)^{-1}$, define the minimal stiffness-reducing envelope

$$\widehat{S}_i = \overline{\text{span}}\{p(b_L)s : p \in \mathbb{R}[t], s \in S_i\}.$$

Put $K_i = \widehat{S}_i$, $P_i = K_i \oplus K_i$, and

$$P_i^\circ = (K_i \cap D(L^{1/2}) \cap D(L^{-1/2}))^{\oplus 2}.$$

The original labels retain their support meaning inside these reducing envelopes.

An operator-regular covariance on P_i is a densely defined closed nonnegative symmetric form γ_i for which P_i° is a form core. Let C_i be its nonnegative self-adjoint representative and let Γ_i be its restriction to the regular core. It is a quasifree covariance when

$$\Gamma_i + \frac{i}{2}\sigma_i \geq 0$$

on the complexified core. For $j_i q = (q, 0)$, define its configuration Cameron–Martin domain and form by

$$\begin{aligned} D(\text{CM}_{\gamma_i}) &= \{q \in K_i : j_i q \in D(C_i^{-1/2})\}, \\ \text{CM}_{\gamma_i}(q, r) &= \langle C_i^{-1/2} j_i q, C_i^{-1/2} j_i r \rangle. \end{aligned}$$

The uncertainty inequality makes C_i injective. A compatible family of operator-regular quasifree covariances is admissible for the curvature stiffness when, on every envelope,

$$D(\text{CM}_{\gamma_i}) = D(Q_{\mathfrak{R}, \Psi}) \cap K_i, \quad \text{CM}_{\gamma_i} = Q_{\mathfrak{R}, \Psi}|_{K_i}.$$

The definition also admits nonzero cross covariance between configuration and momentum, since it uses the inverse of the full covariance form.

Indeed, positivity of $\Gamma_i + i\sigma_i/2$ gives $|\sigma_i(z, w)|^2 \leq 4\gamma_i[z]\gamma_i[w]$. A vector in $\ker C_i$ is therefore symplectically orthogonal to the regular core and hence to P_i ; the nondegeneracy of the cotangent symplectic form makes that vector zero.

Theorem B.11 (Nonempty quasifree-state fibration). *For every critical source-configuration pair satisfying the hard-realization, label, and common-domain hypotheses above, the admissible quasifree-state set is nonempty. For every $c \geq \frac{1}{4}$, let $\gamma_{\mathfrak{R}, \Psi}^{(c)}$ be the closed full-phase covariance form represented by*

$$C_{\mathfrak{R}, \Psi}^{(c)} = \text{diag}(L^{-1}, cL)$$

on each reducing envelope, and denote its regular-core restriction by $\Gamma_{\mathfrak{R}, \Psi}^{(c)}$. These covariances satisfy the uncertainty inequality, restrict compatibly, and have configuration Cameron–Martin form $Q_{\mathfrak{R}, \Psi}$.

Source and gauge isomorphisms intertwining the stiffness operators induce Weyl-state pushforwards between these sets. The sets and pushforwards form a covariant functor on the groupoid of such critical pairs, and its category of elements is a discrete fibration and opfibration with nonempty fibers.

Proof. By the spectral theorem, L is multiplication by a positive spectral variable ℓ on the positive hard quotient. The Hermitian form $\Gamma^{(c)} + i\sigma/2$ decomposes into

$$\begin{pmatrix} \ell^{-1} & i/2 \\ -i/2 & c\ell \end{pmatrix},$$

whose determinant is $c - \frac{1}{4} \geq 0$ and whose upper-left entry is positive. Hence the uncertainty inequality holds. The form domain is

$$D(L^{-1/2}) \oplus D(L^{1/2}),$$

and spectral truncations show that P_i° is a form core. The inverse full covariance operator is

$$(C^{(c)})^{-1} = \text{diag}(L, c^{-1}L^{-1}),$$

so its pullback along $q \mapsto (q, 0)$ has domain $D(L^{1/2})$ and form

$$\langle L^{1/2}q, L^{1/2}r \rangle = Q_{\mathfrak{R}, \Psi}(q, r).$$

The standard Weyl and quasifree-state theorem therefore gives a state with Gaussian characteristic functional

$$\omega^{(c)}(W(z)) = \exp \left[-\frac{1}{2} \Gamma^{(c)}(z, z) \right]$$

and its GNS representation [8, Section 5.2.4].

Each $\hat{S}_i$ reduces L and its spectral calculus, which gives exact restriction of L , L^{-1} , and the covariance forms along the directed envelope system. If a unitary U intertwines two stiffness operators, $U \oplus U$ preserves the phase symplectic forms, transports the inverse covariance domains and Cameron–Martin forms, and induces the corresponding Weyl isomorphism. Pushforward of states respects identities and composition. Its category of elements is consequently a discrete opfibration, and the invertibility of every base arrow makes it also a discrete fibration. The displayed family supplies an object in every fiber. $\square$

The block covariances establish nonemptiness of an independently defined competitor class. They do not exhaust the admissible cross-correlated covariances, and the construction supplies no value of c , critical configuration, state, purity condition, GNS representation, or vacuum. A source whose curvature Hessians do not satisfy the stable hard hypotheses has an empty strong fiber while retaining its full weak Hessian field.

B.7. Local scope and additional global data. The source therefore determines compatible local curvature energies and the weak Hessian field over all admissible configurations. Stable critical configurations carry configuration-indexed self-adjoint stiffness operators and nonempty fibers of compatible quasifree states. A global total energy, a selected state, a renormalized completion, and a nonlinear measure require additional data or construction.

APPENDIX C. ANALYTIC AND ALGEBRAIC FOUNDATIONS OF THE RESPONSE

The response theorem of Section 5 rests on two analytic stages. On a relatively compact region, Green’s identity turns the differential of the curvature energy into a bulk covector and a boundary current, and the configuration metric gives the bulk covector its unique Riesz representative. The same curvature-energy derivatives then determine a Hamiltonian derivation on cotangent phase space and a

star derivation from the local polynomial CCR algebra. This appendix establishes both stages and their compatibility under supported inclusions of regions. The finite-label operator realization is treated separately in Appendix D.

C.1. Sobolev Domains and the Green Boundary Current. Let $U \Subset M$ be a relatively compact domain with smooth boundary, and fix an integer

$$s > \frac{\dim U}{2} + 2.$$

In an affine chart centered at a background configuration Ψ_U^0 , set

$$\begin{aligned} \mathcal{C}^s(U) &= \Psi_U^0 + H^s(T^*U \otimes \mathfrak{so}(N, g_N)) \\ &\quad \oplus H^s(\text{Sym}^2 T^*U \otimes N), \\ \mathcal{V}_0^s(U) &= H_0^s(T^*U \otimes \mathfrak{so}(N, g_N)) \\ &\quad \oplus H_0^s(\text{Sym}^2 T^*U \otimes N). \end{aligned}$$

Here $H_0^s(U)$ is the H^s -closure of compactly supported smooth sections. The boundary traces of all derivatives of integral order below $s - 1/2$ vanish on this space. In particular, the tangential trace that occurs in the first-order Green formula vanishes [33, Chapter 1, Theorems 11.4–11.5].

For a variation $v = (\alpha, \dot{K})$, let $\mathcal{A}(v)$ be the block-valued one-form by which v changes the assembled metric connection,

$$\mathcal{A}(v)_X = \begin{pmatrix} 0 & -\dot{K}_X^\dagger \\ \dot{K}_X & \alpha_X \end{pmatrix}.$$

Writing $D_\Psi = D_{\nabla_\Psi^\oplus}$, the affine curvature identity is

$$D\mathcal{F}_\Psi^\oplus[v] = D_\Psi \mathcal{A}(v).$$

Let $D_{\Psi, \min}$ be the graph closure of D_Ψ on compactly supported smooth block-valued one-forms, and let $D_{\Psi, \max}$ be its distributional maximal realization. The symbol $\dagger$ denotes the formal Green adjoint, whereas $*$ denotes the Hilbert-space adjoint. With the corresponding minimal and maximal realizations of the formal adjoint,

$$D_{\Psi, \min}^* = D_{\Psi, \max}^\dagger, \quad D_{\Psi, \max}^* = D_{\Psi, \min}^\dagger.$$

Indeed, testing against the compactly supported smooth core identifies the Hilbert adjoint with the distributional realization of the formal adjoint; interchanging the minimal and maximal domains gives the second identity.

Denote by γ_t the tangential trace of a block-valued one-form and by γ_n the conormal trace of a block-valued two-form. Their boundary trace-duality pairing is written $\langle \cdot, \cdot \rangle_{\partial U}$.

Lemma C.1 (Green identity and first variation). *For $\Psi \in \mathcal{C}^s(U)$ and every H^s variation v ,*

$$\begin{aligned} D\Phi_U(\Psi)[v] &= \langle D_{\Psi, \max} \mathcal{A}(v), \mathcal{F}_\Psi^\oplus \rangle_{L^2} \\ &= \langle \mathcal{A}(v), D_{\Psi, \max}^\dagger \mathcal{F}_\Psi^\oplus \rangle_{L^2} + \mathcal{B}_{\partial U}(v), \end{aligned}$$

where

$$\mathcal{B}_{\partial U}(v) = \langle \gamma_t \mathcal{A}(v), \gamma_n \mathcal{F}_\Psi^\oplus \rangle_{\partial U}.$$

The boundary current vanishes for $v \in \mathcal{V}_0^s(U)$.

Proof. Sobolev multiplication makes assembly and curvature continuous polynomial maps from H^s to H^{s-1} [3, Theorem 6.1 and Corollary 6.3]; the bundle-section norms and local differential-operator framework are treated in [4, Sections 9–10]. The energy Φ_U is therefore continuously Fréchet differentiable. Differentiating its squared curvature norm and applying the graph Green identity gives the displayed formula. The curvature lies in H^{s-1} , and its maximal formal-adjoint derivative lies in $H^{s-2} \subset L^2$, so the pairings extend from the smooth core by graph and trace continuity. The full H_0^s trace condition annihilates $\gamma_t \mathcal{A}(v)$. $\square$

Thus H_0^s variations determine the bulk covector used by the local response law. General H^s variations determine the same bulk covector together with the boundary current $\mathcal{B}_{\partial U}$.

C.2. Metric-Dual Response and Supported Inclusions. Put

$$Z_\Psi = D_{\Psi, \max}^\dagger \mathcal{F}_\Psi^\oplus.$$

Write $E_\nabla \in H^{s-2}(T^*U \otimes \mathfrak{so}(N, g_N))$ for its internal-connection block and define its raw lower-left mixed tensor by

$$E_K^0(X, Y) = \text{pr}_N(Z_\Psi(X)(Y, 0)) \in H^{s-2}(T^*U \otimes T^*U \otimes N).$$

The component admitted by the symmetric K -variation space is

$$E_K^+ = P_+ E_K^0, \quad P_+ = \frac{1}{2}(1 + \varsigma).$$

The projector is a bounded order-zero bundle map, and hence

$$E_K^+ \in H^{s-2}(\text{Sym}^2 T^*U \otimes N) \subset L^2.$$

Theorem C.2 (Local metric-dual response). *On the Hilbert completion of $\mathcal{V}_0^s(U)$ in the configuration metric, the bulk differential has the unique Riesz representative*

$$\mathcal{Y}_\Psi = (\mathcal{Y}_{\nabla, \Psi}, \mathcal{Y}_{K, \Psi}) \in H^{s-2} \subset L^2,$$

characterized by

$$D\Phi_U(\Psi)[\alpha, \dot{K}] = G_\nabla(\mathcal{Y}_{\nabla, \Psi}, \alpha) + G_K(\mathcal{Y}_{K, \Psi}, \dot{K}).$$

Its two components are

$$\mathcal{Y}_{\nabla, \Psi} = E_\nabla, \quad \mathcal{Y}_{K, \Psi} = 2E_K^+ = 2P_+ E_K^0.$$

Proof. The assembled trace contains the lower-left curvature block and its adjoint upper-right block. Their real Hilbert–Schmidt pairings with $\dot{K}$ are equal, and symmetry of $\dot{K}$ gives

$$\langle E_K^0, \dot{K} \rangle_{L^2} = \langle P_+ E_K^0, \dot{K} \rangle_{L^2}.$$

The configuration metric contains one independent symmetric K -variation summand, so the two off-diagonal contributions give the factor two. Compactly supported smooth variations are dense in the L^2 completion of $\mathcal{V}_0^s(U)$. Riesz representation gives existence, and testing the two metric summands separately gives uniqueness. $\square$

This uniqueness is relative to the configuration metric. The metric itself is the source-determined orthogonal biproduct of the connection and K -variation metrics. The central internal inversion $\text{id}_{TM} \oplus (-\text{id}_N)$ fixes connection variations and

reverses K -variations, so every natural off-diagonal pairing vanishes. The orthogonality of the two response components therefore belongs to the source metric before Riesz representation is applied.

For an admitted internal metric-gauge transformation u of H^{s+1} regularity that preserves these Sobolev spaces, covariance of assembly, curvature, the maximal Green adjoint, and the configuration metric gives

$$\mathcal{Y}_\nabla(u \cdot \Psi) = u\mathcal{Y}_\nabla(\Psi)u^{-1}, \quad \mathcal{Y}_K(u \cdot \Psi) = u\mathcal{Y}_K(\Psi).$$

The projector P_+ acts only on the tangential covector arguments and commutes with the internal action.

Let $U \Subset V$ be an admitted supported inclusion with compatible backgrounds, configurations, coefficients, and bundle data. Zero extension defines

$$E_{VU} : \mathcal{V}_0^s(U) \longrightarrow \mathcal{V}_0^s(V).$$

Its pullback on continuous covectors is

$$(E_{VU}^* \ell)(v) = \ell(E_{VU} v),$$

whereas E_{VU}^{*,L^2} denotes its Hilbert adjoint on the completed variation spaces. If $\Psi_U = \Psi_V|_U$, then

$$D\Phi_U(\Psi_U) = E_{VU}^* D\Phi_V(\Psi_V), \quad \mathcal{Y}_U(\Psi_U) = E_{VU}^{*,L^2} \mathcal{Y}_V(\Psi_V).$$

Block extraction and P_+ commute with both supported maps. The construction supplies these zero-extension arrows; ordinary restriction of an arbitrary element of $H_0^s(V)$ need not lie in $H_0^s(U)$.

At internal rank one, $\mathfrak{so}(N, g_N) = 0$, so the connection-variation space and its gauge response disappear while the shape response remains.

C.3. Hamiltonian Representative of the Phase Response. Use the cotangent symplectic form

$$\sigma((q, \eta), (Q, \Pi)) = \langle \Pi, q \rangle - \langle \eta, Q \rangle.$$

For an affine-linear configuration observable $Q(\ell)$ and momentum observable $P(v) = \langle \pi, v \rangle$, the phase response is

$$\delta_{\mathfrak{R}} Q(\ell) = \ell(G^\sharp \pi), \quad \delta_{\mathfrak{R}} P(v) = D\Phi(v) = G(\mathcal{Y}, v).$$

It has the phase-space vector field

$$(G^\sharp \pi, G^\flat \mathcal{Y}).$$

For $H_{a,b} = aT_{\text{kin}} + b\Phi$, direct contraction with the symplectic form gives

$$X_{H_{a,b}} = (aG^\sharp \pi, -bG^\flat \mathcal{Y}).$$

The declared phase-polynomial algebra carries its positive-degree augmentation. Passing to its augmentation ideal removes additive constants before representatives are compared.

Lemma C.3 (Hamiltonian representative in the kinetic-curvature family). *On every connected component of the label category containing a label with $D\Phi \neq 0$, the unique member of $H_{a,b} = aT_{\text{kin}} + b\Phi$ whose Hamiltonian derivation equals the phase response is*

$$H_R = T_{\text{kin}} - \Phi.$$

On an isolated component where $D\Phi \equiv 0$ at every label, equality of derivations fixes the kinetic coefficient, while the typing datum ϑ separately prescribes the coefficient ratio $[1 : -1]$.

Proof. Equality on configuration generators forces $a = 1$. Equality on a label with nonzero curvature differential then forces $b = -1$. Naturality carries these coefficients across the connected label component. If the curvature differential vanishes at every label of an isolated component, the momentum-generator equation contains no term in b , and the typing datum supplies the declared ratio there. The augmentation has already removed the independent additive constant. $\square$

Admitting the source volume as a third generator enlarges the family to

$$H_{R,c} = T_{\text{kin}} - \Phi + c\mathcal{V}.$$

The volume is fixed under variations of $(\nabla^{\mathcal{N}}, K)$, so $D\mathcal{V} = 0$ and every value of c gives the same phase response. Its value can nevertheless alter the constraint shell, the Master operator, and the associated spectral data. The response derivation therefore leaves this central coefficient undetermined.

C.4. Local Response Algebra and CCR Derivation. Let $\mathcal{Q}(U)$ be the smooth affine configuration core and let $\mathcal{V}_c(U)$ be its real compactly supported variation space. Write $\text{Cyl}(\mathcal{Q}(U))$ for the unital commutative star algebra of cylindrical polynomial functions on $\mathcal{Q}(U)$. Its involution is complex conjugation. Define $\mathcal{C}_\Phi(U)$ as the smallest unital commutative star subalgebra of the complex-valued functions on $\mathcal{Q}(U)$ that contains $\text{Cyl}(\mathcal{Q}(U))$, contains the curvature-energy derivative functions

$$q \longmapsto D^{k+1}\Phi_U(q)[v_0, \dots, v_k],$$

and is closed under pointwise products and directional differentiation along $\mathcal{V}_c(U)$. Functions agreeing pointwise on $\mathcal{Q}(U)$ are identified. Since the curvature energy has finite polynomial degree, only finitely many derivatives of each displayed generator are nonzero.

The complexified abelian Lie algebra $\mathcal{V}_c(U)_{\mathbb{C}}$ acts on $\mathcal{C}_\Phi(U)$ by its commuting directional derivatives. Let $U(\mathcal{V}_c(U)_{\mathbb{C}})$ denote its universal enveloping algebra and define the algebraic crossed product

$$\mathcal{B}_\Phi(U) = \mathcal{C}_\Phi(U) \rtimes U(\mathcal{V}_c(U)_{\mathbb{C}}).$$

Equivalently, $\mathcal{B}_\Phi(U)$ is generated by $\mathcal{C}_\Phi(U)$ and real momentum symbols $P_U(v)$, subject to

$$P_U(v)F - FP_U(v) = -i\partial_v F, \quad [P_U(v), P_U(w)] = 0.$$

Its involution is determined by $F^* = \overline{F}$ and $P_U(v)^* = P_U(v)$ for real v .

Lemma C.4 (PBW decomposition and faithful CCR embedding). *As a vector space,*

$$\mathcal{B}_\Phi(U) \cong \mathcal{C}_\Phi(U) \otimes \text{Sym}(\mathcal{V}_c(U)_{\mathbb{C}}).$$

The coefficient algebra and the polynomial momentum algebra embed injectively. The polynomial CCR star algebra $\mathcal{A}_{\text{CCR}}(U)$, generated by supported affine configuration observables $Q_U(\ell)$ and momenta $P_U(v)$, embeds faithfully by

$$Q_U(\ell) \longmapsto (q \longmapsto \ell(q)), \quad P_U(v) \longmapsto P_U(v).$$

Proof. The commuting directional derivatives make $\mathcal{C}_\Phi(U)$ a module algebra over the cocommutative enveloping algebra. Its smash-product multiplication is associative by the module-algebra identity and has the displayed tensor product as its underlying vector space. This gives the PBW filtration and the first two injections. The cross relation gives

$$[Q_U(\ell), P_U(v)] = i\ell(v)1,$$

while the configuration and momentum generators commute separately. The literal cylindrical functions are faithful on the smooth core, so the CCR map is injective. $\square$

Theorem C.5 (Response derivation on the local CCR algebra). *The generator assignment*

$$\Delta_U Q_U(\ell) = P_U(G_U^\sharp \ell), \quad \Delta_U P_U(v) = D\Phi_U(v)$$

descends to a star derivation

$$\Delta_U : \mathcal{A}_{\text{CCR}}(U) \longrightarrow \mathcal{B}_\Phi(U).$$

Proof. Extend the assignment by linearity, the Leibniz rule, and the star rule. Its residual on the configuration commutator is

$$-i m(G_U^\sharp \ell) + i \ell(G_U^\sharp m) = 0$$

by symmetry of the inverse Riesz map. Configuration functions commute, so the mixed CCR residual also vanishes. The momentum-commutator residual is

$$iD^2\Phi_U(w, v) - iD^2\Phi_U(v, w) = 0$$

by symmetry of the second Fréchet differential. Reality of the metric and curvature energy gives star compatibility. The assignment therefore descends through the CCR ideal. $\square$

The domain of Δ_U is the polynomial CCR algebra, and its codomain contains the curvature-energy derivative functions required by the assignment. For the one-dimensional quadratic energy $\Phi(q) = \lambda q^2/2$, the function $D\Phi(q)[1] = \lambda q$ is the existing configuration observable λQ in $\mathcal{C}_\Phi(U)$. Pointwise identification therefore creates no second symbol for the same observable.

C.5. Naturality Under Buffered Inclusions. Let $U \Subset V$ be a buffered inclusion with the required support collar, compatible background configurations, source coefficients, and bundle data. Zero extension gives

$$E_{VU} : \mathcal{V}_c(U) \longrightarrow \mathcal{V}_c(V)$$

and the corresponding extension of supported cylindrical covectors. A literal local functional on U extends to V by the same finite-jet differential expression with zero-extended smearings. This defines

$$J_{VU} : \mathcal{C}_\Phi(U) \longrightarrow \mathcal{C}_\Phi(V)$$

and gives

$$J_{VU}(FG) = J_{VU}(F)J_{VU}(G),$$

$$J_{VU}(\partial_w F) = \partial_{E_{VU}w} J_{VU}(F),$$

$$J_{VU}(D\Phi_U(v)) = D\Phi_V(E_{VU}v).$$

Finite-jet locality and the support collar establish these identities on the smooth core. The map is injective because every smooth configuration on U extends across

the compact containment to a smooth configuration on V ; a nonzero literal functional on the smaller core consequently remains nonzero after extension.

The crossed-product universal property extends J_{VU} and E_{VU} to an injective star homomorphism

$$K_{VU} : \mathcal{B}_\Phi(U) \longrightarrow \mathcal{B}_\Phi(V).$$

Its injectivity follows from the PBW decomposition and the injectivity of J_{VU} and E_{VU} . Denote the induced polynomial CCR embedding by ι_{VU} .

Theorem C.6 (Naturality under buffered inclusions). *The local response derivations satisfy*

$$K_{VU}\Delta_U = \Delta_V\iota_{VU}.$$

The embeddings compose along buffered inclusion chains and intertwine the admitted internal metric-gauge transformations.

Proof. Locality of the inverse Riesz map proves the identity on configuration generators. The third extension identity proves it on momentum generators, and the Leibniz rule then proves it on the polynomial CCR algebra. Composition follows from composition of zero extensions and the same finite-jet expressions. Metric gauge transformations preserve the configuration metric, curvature energy, point-wise products, directional derivatives, support, and the CCR pairing, so they intertwine all four maps. $\square$

The compatible supported-region system therefore determines the algebraic colimit derivation

$$\Delta_{\mathfrak{R}} : \mathcal{A}_c(\mathfrak{R}) \longrightarrow \mathcal{B}_c(\mathfrak{R}).$$

This is the local-field algebraic endpoint. Appendix D separately constructs the finite-label quotients, represented common cores, and exact operator commutators.

APPENDIX D. PHASE-SPACE RESTRICTIONS, FINITE-LABEL QUANTIZATION, AND OPERATOR DOMAINS

The phase-space and quantum expressions of Section 7 pass from finite-label phase restriction to the exact Weyl-ordered commutator representation and then to bounded commutator domains relative to a fixed represented algebra and common core. This appendix also proves transport under reducing refinements and, after an invariant quasifree state has been chosen from the admissible fiber of Appendix B, the unitary implementation of the internal gauge action. The common global operator data used for the maximal constraint and Master completion enter separately in Appendix E.

D.1. Affine phase space and finite-label restriction. For a finite label λ , distinguish the affine cotangent phase space

$$T_G^* \mathcal{C}_\lambda^s(\mathfrak{R}) = \mathcal{C}_\lambda^s(\mathfrak{R}) \times T_{G,\lambda}^* \mathcal{C}^s(\mathfrak{R})$$

from its translation symplectic space

$$\mathcal{P}_\lambda^s(\mathfrak{R}) = T\mathcal{C}_\lambda^s(\mathfrak{R}) \oplus T_{G,\lambda}^* \mathcal{C}^s(\mathfrak{R}).$$

The curvature energy is a function of the affine configuration Ψ , while the Weyl generators are indexed by vectors in $\mathcal{P}_\lambda^s(\mathfrak{R})$.

Choose an affine origin Ψ_λ^0 and write $q = \Psi - \Psi_\lambda^0$. If $\Psi_\lambda^1 = \Psi_\lambda^0 + a$, then the new coordinate is $q' = q - a$. For a configuration function F , define

$$(T_a F)(q') = F(q' + a).$$

This translation carries cylindrical polynomials and every curvature-energy derivative function in the first chart to the corresponding function in the second. It fixes the variation space and satisfies

$$\partial_v T_a F = T_a \partial_v F.$$

Hence it extends uniquely through the crossed-product universal property, fixing each momentum symbol $P(v)$, and gives a canonical $*$ -isomorphism between the two chart presentations of $\mathcal{B}_\Phi$. The defining relations then give

$$T_a \Delta^{(0)} = \Delta^{(1)} T_a$$

on the polynomial CCR algebra. The affine origin is therefore absent from the isomorphism class of the response derivation.

For every finite source label λ , let $\mathcal{E}_\Phi(\lambda)$ be the unital $*$ -algebra generated by its cylindrical configuration expressions and curvature-energy derivative expressions, closed under componentwise products and directional differentiation. For $\nu \subseteq \lambda$, let

$$R_{\nu\lambda}^0 : \mathcal{E}_\Phi(\lambda) \longrightarrow \mathcal{E}_\Phi(\nu)$$

restrict coefficients, density support, configuration and variation slots, and the integration functional together. Evaluation at λ alone is not a faithful equality test for this projective family. Two integrated expressions can cancel at that label and separate after restriction when a coarser label removes one of the contributing integration components. Accordingly, define the joint downstream-evaluation kernel

$$\mathcal{N}_\Phi(\lambda) = \bigcap_{\nu \subseteq \lambda} \ker(\text{ev}_\nu R_{\nu\lambda}^0).$$

The composition law for R^0 sends $\mathcal{N}_\Phi(\mu)$ into $\mathcal{N}_\Phi(\lambda)$ whenever $\lambda \subseteq \mu$, since for every $\rho \subseteq \lambda$,

$$\text{ev}_\rho R_{\rho\lambda}^0 R_{\lambda\mu}^0 e = \text{ev}_\rho R_{\rho\mu}^0 e = 0 \quad (e \in \mathcal{N}_\Phi(\mu)).$$

The same identity after directional differentiation gives derivative stability. Hence products, involution, directional differentiation, and source restriction descend to the quotient family.

Set

$$\mathcal{C}_\Phi^\sharp(\lambda) = \mathcal{E}_\Phi(\lambda) / \mathcal{N}_\Phi(\lambda), \quad \mathcal{B}_\Phi(\lambda) = \mathcal{C}_\Phi^\sharp(\lambda) \rtimes U(V_{\lambda, \mathbb{C}}).$$

The quotient identifies two expressions precisely when their evaluations agree after restriction to every sublabel. Its evaluation map at λ may therefore have a kernel, while an identity caused only by a cancellation in one integration component does not become an identity throughout the projective family.

The finite-label response derivation satisfies

$$\Delta_\lambda Q_\lambda(\ell) = P_\lambda(G_\lambda^\sharp \ell), \quad \Delta_\lambda P_\lambda(v) = D\Phi_\lambda(v).$$

The metric and Hessian symmetry identities prove descent through the polynomial CCR relations exactly as on finite labels.

For finite labels $\lambda \subset \mu$, let $j_{\mu\lambda} : V_\lambda \rightarrow V_\mu$ be the tangent inclusion with $G_\lambda = j_{\mu\lambda}^* G_\mu j_{\mu\lambda}$. Define

$$L_{\mu\lambda} = G_\mu^\flat j_{\mu\lambda} G_\lambda^\sharp.$$

Then

$$G_\mu^\# L_{\mu\lambda} = j_{\mu\lambda} G_\lambda^\#, \quad j_{\mu\lambda}^* L_{\mu\lambda} = I,$$

and the lifts compose on label chains. Hence

$$I_{\mu\lambda}(v, \ell) = (j_{\mu\lambda}v, L_{\mu\lambda}\ell)$$

is a symplectic phase inclusion.

Joint source restriction acts on an integrated source expression by

$$R_{\lambda\mu}(\tau_\mu(f_\mu)) = \tau_\lambda(r_{\lambda\mu}f_\mu).$$

It restricts the coefficient, density, support, configuration slots, variation slots, and integration functional together. In particular,

$$R_{\lambda\mu}(D^k \Phi_\mu[jv_1, \dots, jv_k]) = D^k \Phi_\lambda[v_1, \dots, v_k].$$

Because the restriction acts on the quotient source expression, it defines

$$R_{\lambda\mu} : \mathcal{C}_\Phi^\natural(\mu) \longrightarrow \mathcal{C}_\Phi^\natural(\lambda).$$

Let $K_{\lambda\mu}$ extend it to the tangent crossed subalgebra generated by $\mathcal{C}_\Phi^\natural(\mu)$ and the momenta $P_\mu(j_{\mu\lambda}v)$, through

$$K_{\lambda\mu}P_\mu(j_{\mu\lambda}v) = P_\lambda(v).$$

No value is assigned to an arbitrary fine momentum outside the included tangent directions.

Theorem D.1 (Compatibility of phase inclusions, restrictions, and response derivations). *The phase inclusions and response restrictions obey*

$$K_{\lambda\mu}\Delta_\mu I_{\mu\lambda} = \Delta_\lambda.$$

Proof. The metric-dual identity proves the formula on Q . Joint source restriction of $D\Phi$ in the natural quotient family $\mathcal{C}_\Phi^\natural$ proves it on P . Derivation gives the result on the generated polynomial CCR algebra. Products and involution descend componentwise, while the composition laws of j , L , and R give coherence on three-label chains. $\square$

For the nonorthogonal metric $G_\mu = \begin{pmatrix} 1 & c \\ c & 1 \end{pmatrix}$, a coordinate covector inclusion sends $(1, 0)$ under $G_\mu^\#$ outside the coarse tangent line. The metric-dual lift sends 1 to $(1, c)$, whose sharp image is the coarse tangent vector. Likewise, retaining a fine integration functional while deleting a density component fails to restrict the curvature-energy variation; simultaneous restriction removes both together.

D.2. Exact finite-label commutator representation. Represent Q by multiplication and $P(v)$ by $-i\partial_v$ on the finite polynomial/Schwartz core. Let $C_{a,b,\lambda} = \text{Op}_W(aT_{\text{kin}} + b\Phi)$.

Theorem D.2 (Exact finite-label response). *On the common core,*

$$\begin{aligned} i[C_{a,b}, Q(\ell)] &= a \text{Op}_W[\ell(G^\#\pi)], \\ i[C_{a,b}, P(v)] &= -b \text{Op}_W[D\Phi(v)]. \end{aligned}$$

For $(a, b) = (1, -1)$,

$$\pi_\lambda(\Delta_\lambda A) = i[C_\lambda, \pi_\lambda(A)]$$

for every polynomial CCR element A .

Proof. The potential commutes with Q , and the kinetic term commutes with P . The canonical commutation relations give the two leading terms. Every higher Moyal term contains at least two derivatives of the linear operand and vanishes. Thus the identities hold exactly at finite $\hbar$ on the stated core. Equality on generators extends by derivation. $\square$

The same calculation gives the connection and shape momentum identities in Section 7.4. A configuration-only polynomial F obeys

$$i[C_{a,b}, \text{Op}_W(F)] = a \text{Op}_W(DF[G^\sharp \pi]).$$

This commutator represents configuration variation generated by the kinetic term; the curvature-energy response is represented by the momentum commutator involving $D\Phi$.

D.2.1. Positive-realization boundary. The Hilbert-space representation above uses the positive base, internal, curvature, and representation pairings of the declared realization class. Each positivity assumption has an independent sharp witness. The positive base pairing $x^2 + y^2 + z^2$ becomes the Lorentzian pairing $-x^2 + y^2 + z^2$, which takes the value $-u^2$ on $(u, 0, 0)$. The rotation generator preserves the positive internal metric $\text{diag}(1, 1)$, whereas the boost generator preserves $\text{diag}(1, -1)$, for which $(0, t)$ has contraction $-t^2$. Replacing the positive curvature pairing

$$\frac{x^2}{2} + y^2 + \frac{z^2}{2}$$

by

$$\frac{x^2}{2} + y^2 - \frac{z^2}{2}$$

gives the value $-t^2/2$ on the last component. For a signed metric $\text{diag}(1, \varepsilon)$, metricity forces the off-diagonal block $-\varepsilon K^\dagger$, so the connection algebra changes with the signature. An indefinite representation pairing $u^2 - v^2$ requires a separately specified fundamental symmetry or positive majorant before it defines a Hilbert norm. Lorentzian, indefinite-internal, indefinite-curvature, and Krein-space variants therefore require the corresponding additional signature data.

D.3. Common-core and bounded commutator domains. Fix a finite regular label and abbreviate $(\mathcal{H}, \mathcal{D}, C) = (\mathcal{H}_\lambda, \mathcal{D}_\lambda, C_\lambda)$. Fix independently a represented unital C^* -algebra

$$\mathfrak{B}_\lambda \subseteq \mathcal{B}(\mathcal{H})$$

before imposing the bounded-commutator condition. Let $\text{End}^\sharp(\mathcal{D})$ consist of the linear maps $A : \mathcal{D} \rightarrow \mathcal{D}$ admitting a core adjoint $A^\sharp : \mathcal{D} \rightarrow \mathcal{D}$:

$$\langle A\xi, \eta \rangle = \langle \xi, A^\sharp \eta \rangle \quad (\xi, \eta \in \mathcal{D}).$$

Density makes $A^\sharp$ unique, and

$$(AB)^\sharp = B^\sharp A^\sharp, \quad (A^\sharp)^\sharp = A.$$

Thus $\text{End}^\sharp(\mathcal{D})$ is a $*$ -algebra on the common core.

Within this specified represented algebra set

$$\mathfrak{A}_\lambda^1 = \{R \in \mathfrak{B}_\lambda : R\mathcal{D}, R^*\mathcal{D} \subset \mathcal{D}, \exists B_R \in \mathcal{B}(\mathcal{H}_\lambda), B_R|_{\mathcal{D}} = i[C, R]|_{\mathcal{D}}\}.$$

For $R \in \mathfrak{A}_\lambda^1$, write $\delta_\lambda^v(R)$ for the unique bounded extension. Let

$$\mathfrak{P}_\lambda = \pi_\lambda(\mathcal{A}_{\text{CCR}}(\lambda)), \quad \mathfrak{J}_\lambda = \text{Alg}_\mathcal{D}^*(\mathfrak{P}_\lambda \cup \mathfrak{A}_\lambda^1).$$

Theorem D.3 (Common-core commutator derivation). *The algebra $\mathfrak{A}_\lambda^1$ is a unital $*$ -subalgebra of $\mathfrak{B}_\lambda$, and*

$$\delta_\lambda^v : \mathfrak{A}_\lambda^1 \longrightarrow \mathcal{B}(\mathcal{H})$$

is a $$ -derivation into the bounded-operator bimodule. The map*

$$\partial_\lambda(A) = i[C, A] : \mathfrak{I}_\lambda \longrightarrow \text{End}^\sharp(\mathcal{D})$$

is an algebraic $$ -derivation, with*

$$\partial_\lambda(\pi_\lambda A) = \pi_\lambda(\Delta_\lambda A), \quad \overline{\partial_\lambda(R)} = \delta_\lambda^v(R).$$

The domain $\mathfrak{A}_\lambda^1$ is maximal among the $$ -subalgebras of $\mathfrak{B}_\lambda$ satisfying these core and bounded-commutator conditions.*

Proof. The identity belongs to $\mathfrak{A}_\lambda^1$. Core and adjoint-core invariance persist under sums, products, and involution. On $\mathcal{D}$,

$$i[C, RS] = i[C, R]S + Ri[C, S].$$

The right side extends to $\delta_\lambda^v(R)S + R\delta_\lambda^v(S)$. Symmetry of C gives

$$\langle i[C, R^*]\xi, \eta \rangle = \langle \xi, i[C, R]\eta \rangle,$$

so $i[C, R^*]$ extends to $\delta_\lambda^v(R)^*$. These identities prove the bounded $*$ -derivation statement. They also hold for every word in $\mathfrak{I}_\lambda$, where all factors preserve the core, and prove the algebraic statement. The exact finite-label response theorem gives the polynomial restriction. If $\pi_\lambda(A) = 0$, that identity gives $\pi_\lambda(\Delta_\lambda A) = i[C_\lambda, 0] = 0$, so the restriction descends through the represented polynomial quotient. Two bounded extensions agreeing on the dense core coincide, which proves uniqueness. Any other $*$ -subalgebra with the same three defining properties lies elementwise in $\mathfrak{A}_\lambda^1$, which proves maximality relative to the specified represented algebra and core. $\square$

The maximality is relative to $\mathfrak{B}_\lambda$ and $\mathcal{D}$. Extension to a larger generated algebra or to a graph-closed derivation requires additional domain hypotheses. The codomain $\text{End}^\sharp(\mathcal{D})$ remains necessary because a bounded first commutator need not have a bounded second commutator, and ∂_λ need not map $\mathfrak{I}_\lambda$ into itself.

D.3.1. Polynomial overlap and finite-rank witnesses. Exact polynomial commutators and bounded commutators meet on the common core, but their represented operator classes have only scalar polynomial overlap. Separately admitted finite-rank core operators supply nonzero bounded commutators outside that overlap.

Let N be the finite-label number operator and $\mathcal{D}_{\text{fin}}$ the algebraic finite-particle core. A represented polynomial A of degree d satisfies

$$\|A\psi\| \leq c_A \|(N+1)^{d/2}\psi\| \quad (\psi \in \mathcal{D}_{\text{fin}}).$$

For a coherent vector $\Omega_z = W(z)\Omega$ and $P_n = \mathbf{1}_{[0,n]}(N)$, factorial decay of the number coefficients gives

$$P_n \Omega_z \longrightarrow \Omega_z, \quad AP_n \Omega_z \longrightarrow A\Omega_z.$$

Consequently every bounded extension of $A|_{\mathcal{D}_{\text{fin}}}$ agrees with A on coherent vectors. The regular finite-dimensional CCR representation is a Stone–von Neumann amplification of the Schrödinger representation, so the same graph-core argument applies in every nonzero multiplicity vector.

Lemma D.4 (Polynomial–bounded overlap). *For the represented scalar polynomial CCR algebra on a finite nondegenerate phase label,*

$$\mathfrak{P}_\lambda \cap \mathcal{B}(\mathcal{H}_\lambda) = \mathbb{C}1, \quad \mathfrak{P}_\lambda \cap \mathfrak{A}_\lambda^1 = \mathbb{C}1.$$

Proof. Write the Weyl symbol of a nonconstant polynomial operator as $p = p_d + \dots + p_0$, with $p_d \neq 0$ homogeneous of degree $d > 0$. Choose z with $p_d(z) \neq 0$. Weyl covariance makes

$$\langle W(tz)\Omega, \text{Op}_W(p)W(tz)\Omega \rangle$$

a polynomial in t with leading term $t^d p_d(z)$. The graph-core limit above makes this matrix element equal to that of any proposed bounded extension, whereas its absolute value would then be bounded by the operator norm. This contradiction removes the top degree; descending induction removes every positive degree. Constants lie in both algebras, which proves both equalities. $\square$

Since the polynomial–bounded overlap is $\mathbb{C}1$, its constraint commutator vanishes. Independently specified finite-rank operators provide nonzero bounded commutators. For $\xi, \eta \in \mathcal{D}_\lambda$, let $E_{\xi, \eta} = |\xi\rangle\langle\eta|$. If this operator is independently admitted to $\mathfrak{B}_\lambda$, then it and its adjoint preserve the core, and

$$i[C_\lambda, E_{\xi, \eta}] = i(|C_\lambda \xi\rangle\langle\eta| - |\xi\rangle\langle C_\lambda \eta|).$$

Hence $E_{\xi, \eta} \in \mathfrak{A}_\lambda^1$. For a unit vector ξ ,

$$\|[C_\lambda, P_\xi]\| = \|(1 - P_\xi)C_\lambda \xi\|.$$

On the untruncated three-mode torus Fock label, include the vacuum projection P_Ω in the specified represented algebra. Then

$$C_F = d\Gamma(\omega) - \frac{a}{4}(c_{-1}c_0 + c_0c_{+1}), \quad P_\Omega = |\Omega\rangle\langle\Omega|.$$

This inclusion is fixed before the commutator is evaluated and is independent of the response calculation. The free term annihilates Ω , while

$$C_F \Omega = -\frac{a}{4}(a_{-1}^* a_0^* \Omega + a_0^* a_{+1}^* \Omega).$$

The two vectors in parentheses are orthonormal, so

$$\|[C_F, P_\Omega]\| = \frac{|a|}{2\sqrt{2}}.$$

The calculation uses the full finite-mode Fock space without an occupation cutoff and witnesses a nonzero, noncentral bounded commutator in the stated Fock representation. The representation and its vacuum remain additional choices. The calculation does not establish that such projections belong to generated bounded algebras on arbitrary labels.

D.4. Reducing refinements and operator-domain boundaries. The bivariant finite-label theorem above concerns symplectic phase inclusion and contravariant restriction of source expressions. Hilbert-space transport requires the stronger condition that the relevant subspaces reduce the constraint. For this comparison, import the separately specified common-core constraint of Appendix E, let $T = C_0^*$ be its maximal adjoint, and let P_λ project onto a reducing subspace. Call the reducing projection core preserving when $P_\lambda \mathcal{D}_0 \subset \mathcal{D}_0$, and set

$$\widehat{\mathcal{D}}_\lambda = P_\lambda \mathcal{D}_0, \quad \widehat{C}_\lambda = T|_{\widehat{\mathcal{D}}_\lambda}.$$

Reduction gives

$$\widehat{C}_\lambda(P_\lambda \xi) = P_\lambda C_0 \xi \quad (\xi \in \mathcal{D}_0).$$

The projected core is dense and invariant. Assume further that the label's polynomial core equals $P_\lambda \mathcal{D}_0$. Its represented polynomial constraint is then $\widehat{C}_\lambda$ on that core.

For nested labels satisfying these reducing and core conditions, let $J_{\mu\lambda}$ be the Hilbert isometry. A compatible reducing refinement also carries the phase inclusion $I_{\mu\lambda}$, the response restriction $K_{\lambda\mu}$, and bounded operators R_λ and R_μ satisfying

$$\begin{aligned} C_\mu J_{\mu\lambda} &= J_{\mu\lambda} C_\lambda, & R_\mu J_{\mu\lambda} &= J_{\mu\lambda} R_\lambda, \\ R_\mu^* J_{\mu\lambda} &= J_{\mu\lambda} R_\lambda^*, & K_{\lambda\mu} \Delta_\mu I_{\mu\lambda} &= \Delta_\lambda. \end{aligned}$$

Theorem D.5 (Compatibility of bounded commutators under reducing refinements). *Every compatible reducing refinement satisfies*

$$\delta_\mu^v(R_\mu) J_{\mu\lambda} = J_{\mu\lambda} \delta_\lambda^v(R_\lambda).$$

Identity refinements and composites remain compatible.

Proof. For $\xi \in \mathcal{D}_\lambda$, the three intertwiners give

$$\begin{aligned} i[C_\mu, R_\mu] J_{\mu\lambda} \xi &= iC_\mu J_{\mu\lambda} R_\lambda \xi - iR_\mu J_{\mu\lambda} C_\lambda \xi \\ &= J_{\mu\lambda} i[C_\lambda, R_\lambda] \xi. \end{aligned}$$

Both sides extend boundedly and the coarse core is dense, so the equality extends uniquely. The finite-label inclusion–restriction identity supplies the polynomial identity. Identities and compositions follow from composition of I, K, J and the operator intertwiners; the adjoint equation preserves the involution. $\square$

Nonidentity refinements satisfying these hypotheses exist. Let $F = \{-e_1, 0, e_1\}$ and adjoin an interaction-isolated mode k_* to obtain F' . On the untruncated Fock spaces,

$$\mathcal{H}_{F'} \cong \mathcal{H}_F \otimes \mathcal{F}_{k_*}, \quad C_{F'} = C_F \otimes 1 + 1 \otimes \omega_* N_*.$$

The vacuum isometry $J_{F'F} \psi = \psi \otimes \Omega_*$ reduces the constraint and maps the finite-particle core exactly into the fine core. The projections

$$R_F = P_{\Omega_F}, \quad R_{F'} = P_{\Omega_F} \otimes P_{\Omega_*}$$

obey the operator and adjoint intertwiners, while their coarse bounded commutator has norm $|a|/(2\sqrt{2})$. Thus a strict compatible refinement transports a nonzero, non-central bounded commutator. This example establishes covariance for refinements satisfying the displayed reducing, core, phase, restriction, and operator-intertwining hypotheses. Arbitrary projective restrictions need not satisfy these hypotheses.

D.4.1. Independent domain failures. Each condition above excludes an independent failure. On $L^2(\mathbb{R})$ with $\mathcal{D} = \mathcal{S}(\mathbb{R})$ and $C = -i d/dx$, choose unit vectors $u \in \mathcal{D}$ and $v \notin \mathcal{D}$. The rank-one bounded operator $R = |v\rangle\langle u|$ does not preserve the core, while $R^* = |u\rangle\langle v|$ preserves it but has an adjoint that does not. The unitary multiplier $U\psi = e^{ix^2}\psi$ and its adjoint preserve Schwartz space, but

$$i[C, U]\psi = 2ixe^{ix^2}\psi$$

is unbounded. Hence boundedness and the two core-invariance conditions remain logically distinct, and no commutator derivation extends to all bounded operators.

Compression also differs from reduction. For

$$C = \begin{pmatrix} 1 & 0 & \varepsilon \\ 0 & 2 & \zeta \\ \varepsilon & \zeta & 3 \end{pmatrix}, \quad J(x, y) = (x, y, 0),$$

with $(\varepsilon, \zeta) \neq (0, 0)$, one has

$$(1 - JJ^*)CJ \neq 0, \quad J^*C^2J - (J^*CJ)^2 = \begin{pmatrix} \varepsilon^2 & \varepsilon\zeta \\ \varepsilon\zeta & \zeta^2 \end{pmatrix}.$$

Thus symmetric compression neither intertwines the constraint nor transports its Master square. Finally, every finite-dimensional compression of a defined commutator is a bounded matrix, even when the untruncated commutator is unbounded; finite cutoffs therefore cannot establish membership in $\mathfrak{A}_\lambda^1$.

D.5. Gauge action in an admitted quasifree representation. The state input for this subsection is the quasifree-state fibration of Appendix B. Fix a critical pair $(\mathfrak{R}, \Psi)$ satisfying its hard-realization, gauge-quotient, label, and common-domain hypotheses, and write

$$L = L_{\mathfrak{R}, \Psi}^+$$

on the positive hard quotient. For every original source label, let K_λ be its minimal stiffness-reducing envelope and put

$$\mathcal{P}_\lambda^+ = K_\lambda \oplus K_\lambda.$$

Choose a compatible operator-regular full-phase covariance family $\gamma = \{\gamma_\lambda\}$ from the nonempty admitted fiber, and let Γ_λ denote its restriction to the regular phase core. Its quasifree positivity condition is

$$\Gamma_\lambda + \frac{i}{2}\sigma_{\mathfrak{R}, \lambda} \geq 0$$

on the complexified regular phase core. Admissibility means that the configuration pullback of the inverse full covariance form has the same domain and values as $Q_{\mathfrak{R}, \Psi}$ on every K_λ .

This condition admits configuration–momentum cross covariance. The block forms represented by

$$C_{\mathfrak{R}, \Psi}^{(c)} = \text{diag}(L^{-1}, cL), \quad c \geq \frac{1}{4},$$

establish nonemptiness of the independently defined fiber and do not exhaust it. The source determines the fibration over stable critical configurations. The critical configuration indexing the fiber fixed above, together with its covariance, state, GNS representation, purity condition, and vacuum, remains unselected. A source with no configuration satisfying the stable hard hypotheses has an empty strong fiber while retaining its weak Hessian field.

D.5.1. Gauge covariance of the response. The internal metric gauge group acts on the configuration space, the assembled connection, and its curvature components. Its cotangent lift is defined independently of the covariance chosen above and acts by compatible symplectic transformations of the local and cylindrical phase spaces.

The connection component of the curvature variation satisfies

$$D_\nabla \Phi_U(\alpha) = G_\nabla(\mathcal{Y}_\nabla, \alpha).$$

Thus $\mathcal{Y}_\nabla$ is the metric dual of curvature-energy variation in the connection direction, whereas the Gauss generator $J(A)$ is the infinitesimal generator of an implemented gauge transformation. The shape response $\mathcal{Y}_K$ belongs to the orthogonal shape summand and acquires its gravitational interpretation on the positive-Riemannian oriented rank-one Gauss–Codazzi branch.

Lemma D.6 (Gauge covariance of the field response). *For every $u \in \mathcal{G}_{\mathcal{N}_g}^s$ preserving the configuration domain and the source pairings,*

$$\mathcal{Y}_\nabla(u \cdot \Psi) = u \cdot \mathcal{Y}_\nabla(\Psi), \quad \mathcal{Y}_K(u \cdot \Psi) = u \cdot \mathcal{Y}_K(\Psi).$$

At oriented normal rank one, the internal connection variation is trivial and $\mathcal{Y}_\nabla = 0$, while $\mathcal{Y}_K$ can remain nonzero.

Proof. Metric gauge transformations intertwine the assembled connection, curvature, curvature energy, and source motion metric. Differentiating the invariant curvature energy and using uniqueness of its Riesz representative gives both identities. The oriented rank-one metric gauge algebra is zero, so its connection-response summand vanishes. $\square$

D.5.2. *Symplectic Weyl action.* For an internal metric gauge transformation u preserving the regularity class and the projectively compatible label system, write

$$\alpha_{u,\lambda} : \mathcal{P}_\lambda^s(\mathfrak{R}) \rightarrow \mathcal{P}_{u\lambda}^s(\mathfrak{R})$$

for the induced labeled phase transformation. Write $W_\lambda(x)$ for the Weyl generator in $\mathcal{W}_\lambda(\mathfrak{R})$.

Lemma D.7 (Symplectic local Weyl action).

For every such u and every local or cylindrical label λ ,

$$\sigma_{\mathfrak{R}}(\alpha_{u,\lambda}x, \alpha_{u,\lambda}y) = \sigma_{\mathfrak{R}}(x, y).$$

Consequently u induces a C^ -isomorphism*

$$\alpha_{u,\lambda}^{\mathcal{W}} : \mathcal{W}_\lambda(\mathfrak{R}) \rightarrow \mathcal{W}_{u\lambda}(\mathfrak{R}), \quad \alpha_{u,\lambda}^{\mathcal{W}}(W_\lambda(x)) = W_{u\lambda}(\alpha_{u,\lambda}x),$$

compatible with the projective restriction maps. If $u\lambda = \lambda$, this is an automorphism of $\mathcal{W}_\lambda(\mathfrak{R})$.

Proof. The internal metric gauge action preserves $g_{\mathcal{N}}$, the source pairings, and the projectively compatible label restrictions. Its cotangent lift on each labeled phase module therefore preserves the cotangent symplectic form. The Weyl product on $\mathcal{W}_\lambda(\mathfrak{R})$ is defined only from $\sigma_{\mathfrak{R}}$, so the displayed assignment preserves the involution and the Weyl multiplication law. Compatibility with restrictions follows from the assumed preservation of the label system. $\square$

D.5.3. *Admitted state and implementable subgroup.* Let $\mathcal{G}_0 \subseteq \mathcal{G}_{\mathcal{N}_g}^s$ be the subgroup preserving the stable critical source–configuration pair fixed above, its regularity and quotient data, the label system and minimal reducing envelopes, the symplectic phase forms, and the chosen compatible covariance family. The phase transformation then restricts to

$$\alpha_{u,\lambda}^+ : \mathcal{P}_\lambda^+ \longrightarrow \mathcal{P}_{u\lambda}^+.$$

Write $\mathcal{W}_\lambda^+$ for the Weyl algebra of $\mathcal{P}_\lambda^+$ and $W_\lambda^+(x)$ for its Weyl generator. Preservation of the chosen covariance means

$$\Gamma_{u\lambda}(\alpha_{u,\lambda}^+x, \alpha_{u,\lambda}^+y) = \Gamma_\lambda(x, y)$$

for all regular-core vectors x, y . Let $(\rho_\Gamma, \mathcal{H}_\Gamma, \Omega_\Gamma)$ be the GNS representation of the resulting compatible quasifree state. The invariant covariance family and state are additional representation data.

D.5.4. Unitary implementation.

Theorem D.8 (Internal gauge implementation).

The subgroup $\mathcal{G}_0$ is implemented by unitaries $U(u)$ on $\mathcal{H}_\Gamma$ satisfying

$$U(u)\Omega_\Gamma = \Omega_\Gamma, \quad U(u)\rho_\Gamma(W_\lambda^+(x))U(u)^{-1} = \rho_\Gamma(W_{u\lambda}^+(\alpha_{u,\lambda}^+x)).$$

The operators $U(u)$ form a genuine unitary representation.

Proof. Invariance of the covariance family makes the compatible quasifree state invariant under the restricted local Weyl isomorphisms. The GNS covariance construction gives a unitary $U(u)$ fixing Ω_Γ and implementing these isomorphisms on the cyclic subspace. The restricted maps form a genuine action compatible with restrictions, and the cyclic vector is fixed. Therefore $U(uv) = U(u)U(v)$ on the cyclic subspace and hence on $\mathcal{H}_\Gamma$. No projective phase remains. $\square$

D.5.5. *Infinitesimal generators.* Let A be an element of the Lie algebra of $\mathcal{G}_0$, and let $u_t = \exp(tA)$. If $t \mapsto U(u_t)$ is strongly continuous, Stone's theorem gives a self-adjoint generator $J(A)$ by

$$U(u_t) = e^{itJ(A)}.$$

These are the internal Gauss generators of the implemented subgroup.

Lemma D.9 (Generator closure). *On a common invariant smooth cylindrical core of differentiable vectors for the unitary representation,*

$$[J(A), J(B)] = iJ([A, B])$$

for all A, B in the Lie algebra of $\mathcal{G}_0$ for which the commutator is defined on the core.

Proof. The unitary implementation is a representation of $\mathcal{G}_0$. Differentiating the group commutator identity on the common invariant core gives the displayed Lie-algebra relation. $\square$

D.5.6. *Invariant sector.* The internal gauge-invariant Hilbert subspace is

$$\mathcal{H}_\Gamma^{\mathcal{G}_0} = \{\psi \in \mathcal{H}_\Gamma : U(u)\psi = \psi \quad (\forall u \in \mathcal{G}_0)\}.$$

For the identity component generated by strongly continuous one-parameter subgroups, the fixed-vector subspace is the joint kernel of the corresponding internal Gauss generators:

$$\mathcal{H}_\Gamma^{\mathcal{G}_0^\circ} = \bigcap_A \ker J(A).$$

If $\mathcal{G}_0$ is disconnected and $\mathcal{R}$ is a set of representatives of $\mathcal{G}_0/\mathcal{G}_0^\circ$, then

$$\mathcal{H}_\Gamma^{\mathcal{G}_0} = \mathcal{H}_\Gamma^{\mathcal{G}_0^\circ} \cap \bigcap_{u \in \mathcal{R}} \ker(U(u) - 1).$$

Theorem D.10 (Internal gauge-invariant subspace).

The fixed-vector subspace $\mathcal{H}_\Gamma^{\mathcal{G}_0}$ is the internal gauge-invariant subspace of the GNS Hilbert space. Every bounded observable in a represented local or cylindrical Weyl algebra that is invariant under the implemented internal gauge action preserves this subspace.

Proof. The fixed-vector definition is closed under the unitary representation. If a bounded observable O is gauge-invariant, then $U(u)OU(u)^{-1} = O$ for every $u \in \mathcal{G}_0$. Hence $O\psi$ is fixed whenever ψ is fixed. $\square$

Lemma D.11 (Static internal Ward identity). *Let O be a gauge-invariant bounded observable in a represented local or cylindrical Weyl algebra. For every normal state ω on the represented bounded algebra and every $u \in \mathcal{G}_0$,*

$$\omega(U(u)^{-1}OU(u)) = \omega(O).$$

Equivalently, gauge-related inputs have the same value on gauge-invariant bounded observables. If A lies in the Lie algebra of a strongly continuous implemented one-parameter subgroup, then on the domain of the Gauss generator $J(A)$,

$$[J(A), O]\psi = 0.$$

Consequently $\omega_\psi([J(A), O]) = 0$ for every vector state whose vector lies in this common invariant domain. This is a kinematical gauge identity. A real-time Ward identity for H_R would require an additional temporal dynamics.

Proof. Gauge invariance gives $U(u)OU(u)^{-1} = O$, and the group-level expectation identity follows by applying ω . For $u_t = \exp(tA)$, the strong identity $U(u_t)OU(u_t)^{-1} = O$ implies that O preserves the domain of $J(A)$. Differentiating on that domain gives $i[J(A), O]\psi = 0$. The vector-state identity is the corresponding matrix element statement. $\square$

D.5.7. Rank-one collapse. When the internal metric module has rank one and the metric is oriented, the internal gauge algebra is trivial. In that case $\mathcal{G}_0$ contributes no nonzero internal Gauss generators, and the internal gauge-invariant sector is the entire GNS Hilbert space.

Remark D.12. *At oriented internal rank one, the gravitational quantities arise from curvature contractions, while internal gauge reduction is trivial.*

D.6. Constraint, optional response stationarity, and the global boundary. The derivation Δ_U defines the field response independently of a stationarity equation. A finite-label response operator R_λ gives the optional stationarity form

$$q_{Y,\lambda}[\psi] = \|R_\lambda\psi\|^2.$$

Combining it with the signed constraint gives

$$q_{HY,\lambda}[\psi] = \|C_\lambda\psi\|^2 + \|R_\lambda\psi\|^2.$$

For the rank-one model

$$C = \frac{1}{2}(P^2 - \lambda Q^2), \quad R = \lambda Q,$$

let $\epsilon^2 = q_{HY}[\psi]$ for normalized Schwartz ψ , and set $x = \|Q\psi\|$, $y = \|P\psi\|$. Then

$$x \leq \frac{\epsilon}{\lambda}, \quad y^2 \leq 2\epsilon + \frac{\epsilon^2}{\lambda}, \quad xy \geq \frac{1}{2}.$$

Consequently

$$\frac{1}{2} \leq \frac{\epsilon}{\lambda} \sqrt{2\epsilon + \frac{\epsilon^2}{\lambda}}.$$

The right side increases continuously from zero, so it equals $1/2$ at a unique $\epsilon_0 > 0$. Hence

$$q_{HY}[\psi] \geq \epsilon_0^2 \|\psi\|^2$$

on the closed row domain. In contrast, C is metaplectically equivalent to a nonzero multiple of the dilation generator and $0 \in \sigma(C^2)$. Constraint zero support can therefore coexist with nonzero response, while joint constraint–stationarity support can be empty.

The construction has now reached three distinct levels. The supported local system determines an algebraic response derivation, every finite regular label represents its pullback by an exact commutator on the specified core, and compatible reducing refinements transport the bounded commutator domain under the displayed intertwining hypotheses. A common infinite-mode representation, invariant core, unbounded constraint, and composite Master operator require the additional compatible operator data introduced in Appendix E. Bounded Master functions have their own restriction theorem under reducing hypotheses. An unbounded response operator and an invariant quasifree state family require their respective domain and representation data.

APPENDIX E. ADJOINT CONSTRAINTS AND MASTER-OPERATOR COMPLETION

The constraint completion of Section 7.4 has two operator-theoretic branches. On the first branch, a Hilbert space, a common invariant core, and a symmetric constraint on that core have been specified in addition to the labelwise representations. Its Hilbert adjoint determines a positive Master operator and, through minimal reducing envelopes, a compatible bounded spectral calculus. On the fixed-holonomy torus branch, the exact finite-label constraints have unique self-adjoint closures but do not define one operator on the admitted vacuum-containing cylinder core. Their bounded Master calculi are instead compared through a common spectral coordinate.

E.1. Common symmetric core and weak constraint. Suppose the labelwise represented constraints are carried by a common Hilbert space $\mathcal{H}_C$ and define a densely defined symmetric operator on an invariant smooth cylindrical core

$$C_0 : \mathcal{D}_0 \subset \mathcal{H}_C \longrightarrow \mathcal{H}_C.$$

The common carrier, core, and compatible operator action are additional analytic data beyond the finite-label commutator identities. Once they are specified, the symmetrically ordered kinetic representation of Section 7.4 and the represented curvature-energy form fix

$$C_0 = T_{\text{kin},0} - \Phi_0$$

on $\mathcal{D}_0$.

Since C_0 is symmetric and densely defined, it is closable. Its Hilbert adjoint is

$$H_* = C_0^*,$$

with graph domain

$$\mathcal{D}(H_*) = \left\{ \psi \in \mathcal{H}_C : \sup_{\phi \in \mathcal{D}_0 \setminus \{0\}} \frac{|\langle C_0 \phi, \psi \rangle|}{\|\phi\|_{\mathcal{H}_C}} < \infty \right\}.$$

Lemma E.1 (Weak constraint and adjoint graph domain). *The operator H_* is closed and densely defined, and*

$$H_*\psi = 0 \iff \langle C_0\phi, \psi \rangle = 0 \quad (\forall \phi \in \mathcal{D}_0).$$

Thus $\ker H_$ is the Hilbert-space weak solution space of the symmetric core constraint. The adjoint need not itself be symmetric or self-adjoint.*

Proof. The adjoint of a closable densely defined operator is closed and densely defined. By the definition of the Hilbert adjoint,

$$\langle C_0\phi, \psi \rangle = \langle \phi, H_*\psi \rangle \quad (\phi \in \mathcal{D}_0, \psi \in \mathcal{D}(H_*)).$$

The displayed weak equations are therefore equivalent to $H_*\psi = 0$, since $\mathcal{D}_0$ is dense. $\square$

E.2. Positive Master operator and bounded calculus. Define

$$M_* = H_*^* H_*.$$

Its associated quadratic form is

$$\mathfrak{m}_*[\psi] = \|H_*\psi\|^2, \quad \mathcal{D}(\mathfrak{m}_*) = \mathcal{D}(H_*).$$

Theorem E.2 (Master operator and weak zero sector). *The operator M_* is non-negative and self-adjoint, and*

$$\mathfrak{m}_*[\psi] = 0 \iff H_*\psi = 0.$$

Consequently

$$\ker M_* = \ker H_*.$$

Proof. The standard theorem for A^*A , applied to the closed densely defined operator $A = H_*$, gives self-adjointness and nonnegativity [41, Proposition 3.18(ii)]. Its closed form is $\|H_*\psi\|^2$, which vanishes exactly on $\ker H_*$. $\square$

The bounded Master coordinate and the zero-resolvent filter are

$$\mathfrak{b}_* = \frac{M_*}{1 + M_*}, \quad E_\epsilon = \frac{\epsilon}{M_* + \epsilon} \quad (\epsilon > 0).$$

Lemma E.3 (Bounded Master functions). *The operators $\mathfrak{b}_*$ and E_ϵ are bounded and positive. They satisfy*

$$\mathfrak{b}_*|_{\ker H_*} = 0, \quad E_\epsilon|_{\ker H_*} = 1.$$

Proof. The spectrum of M_* lies in $[0, \infty)$, and the functions

$$\mu \mapsto \frac{\mu}{1 + \mu}, \quad \mu \mapsto \frac{\epsilon}{\mu + \epsilon}$$

are bounded and nonnegative there. Functional calculus gives the operators, and their values at $\mu = 0$ give the stated restrictions. $\square$

Theorem E.4 (Completion from the adjoint graph domain). *For the specified Hilbert space and common symmetric core operator, the adjoint graph domain determines H_* , M_* , and their bounded functional calculus. This construction requires no self-adjoint extension of C_0 . If a self-adjoint extension is chosen, its domain and its extension-dependent square are additional operator data and are not identified here with M_* .*

Proof. The graph-domain definition fixes C_0^* ; the positive product and bounded functional calculus then fix the remaining objects. A self-adjoint extension adds a domain lying between the core operator and its adjoint, and its square belongs to that separately selected realization. $\square$

E.3. Reducing subspaces and constraint-reducing envelopes. Raw local or cylindrical labels need not reduce a common unbounded constraint. When the common operator above exists, its bounded transform constructs a reducing carrier from each original label without changing the support type of the label's observables. Put

$$T = H_*, \quad z_T = T(1 + T^*T)^{-1/2}.$$

For a closed seed subspace $S_\lambda \subset \mathcal{H}_C$ generated by the represented cylindrical vectors of a label λ , define

$$\widehat{S}_\lambda = \overline{\text{span}} \{w(z_T, z_T^*)\xi : \xi \in S_\lambda, w \in \{z_T, z_T^*\}^{<\omega}\}.$$

Theorem E.5 (Minimal constraint-reducing envelopes). *The space $\widehat{S}_\lambda$ is the unique smallest closed subspace containing S_λ that reduces z_T and z_T^* . It also reduces T , T^* , $M_* = T^*T$, their operator and form domains, and every bounded Borel function of M_* .*

If the seeds are directed by label inclusion, their envelopes are directed. If the seed union is dense in $\mathcal{H}_C$, the envelope union is dense as well. Every unitary intertwining T and the seed family transports the corresponding envelopes.

Proof. The defining span is invariant under z_T and z_T^* , hence its closure reduces both. Every closed reducing subspace containing the seed contains all the displayed words, proving minimality and uniqueness. For a closed densely defined operator,

$$1 - z_T^*z_T = (1 + T^*T)^{-1}, \quad T = z_T(1 - z_T^*z_T)^{-1/2}$$

on the spectral domain of the unbounded factor. A projection commuting with z_T and z_T^* therefore preserves the spectral domains and reduces T , T^* , and T^*T . Directedness, density, and unitary transport follow directly from the defining closed spans. $\square$

For $\lambda \subseteq \lambda'$, let

$$J_{\lambda'\lambda} : \widehat{S}_\lambda \longrightarrow \widehat{S}_{\lambda'}$$

be the isometric inclusion, and let T_λ and $M_{*,\lambda}$ denote the restrictions of T and M_* . The smaller envelope reduces the restriction to the larger one.

Lemma E.6 (Reducing transport of the Master calculus). *For every nested pair of minimal constraint-reducing envelopes,*

$$T_{\lambda'}J_{\lambda'\lambda} = J_{\lambda'\lambda}T_\lambda, \quad M_{*,\lambda'}J_{\lambda'\lambda} = J_{\lambda'\lambda}M_{*,\lambda}$$

on their corresponding domains, and every bounded Borel function satisfies

$$f(M_{*,\lambda'})J_{\lambda'\lambda} = J_{\lambda'\lambda}f(M_{*,\lambda}).$$

In particular, the identity holds for $\mathfrak{b}_$, E_ϵ , resolvents, and the zero spectral projection.*

Proof. The envelope projections commute with T , T^* , and M_* , including their domains. The restrictions therefore form orthogonal direct summands. The spectral measure of a reduced self-adjoint operator splits along the same projection, and bounded Borel functional calculus gives the final identity [41, Proposition 5.26]. $\square$

Separately defined label core operators admit the same conclusion only when their domains split exactly. Let $J : \mathcal{H}_{C,\lambda} \rightarrow \mathcal{H}_{C,\lambda'}$ be isometric and write $P = JJ^*$. An exact reducing core inclusion satisfies

$$\begin{aligned} P\mathcal{D}_{0,\lambda'} &= J\mathcal{D}_{0,\lambda}, & P\mathcal{D}_{0,\lambda'} &\subseteq \mathcal{D}_{0,\lambda'}, \\ C_{0,\lambda'}J &= JC_{0,\lambda}, & PC_{0,\lambda'}\xi &= C_{0,\lambda'}P\xi \quad (\xi \in \mathcal{D}_{0,\lambda'}). \end{aligned}$$

These identities give the literal direct sum

$$C_{0,\lambda'} = JC_{0,\lambda}J^* \oplus C_{0,\lambda'}|_{(1-P)\mathcal{D}_{0,\lambda'}}.$$

The adjoint, positive product, and bounded Master calculus then split as above [41, Definition 1.8 and Proposition 1.15]. Mere compression does not suffice: an off-diagonal core coupling K leaves the compressed linear operator unchanged but contributes the positive defect K^*K to the compressed square.

E.4. Normal states on the zero eigenspace. For a positive normal functional ω_λ on a represented von Neumann algebra containing $W^*(M_{*,\lambda})$, define

$$\omega_{\lambda,\epsilon}(A) = \frac{\omega_\lambda(E_{\epsilon,\lambda}AE_{\epsilon,\lambda})}{\omega_\lambda(E_{\epsilon,\lambda}^2)}$$

whenever the denominator is nonzero.

Lemma E.7 (Normalization by the Master filter). *If $\omega_\lambda(E_{\epsilon,\lambda}^2) > 0$, then $\omega_{\lambda,\epsilon}$ is a positive normalized functional.*

Proof. For $A \geq 0$, the filtered operator is

$$E_{\epsilon,\lambda}AE_{\epsilon,\lambda} = (A^{1/2}E_{\epsilon,\lambda})^*(A^{1/2}E_{\epsilon,\lambda}),$$

so positivity follows from positivity of ω_λ . Substitution of $A = 1$ gives normalization. $\square$

Define the zero-window mass by

$$\mu_0(\omega_\lambda) = \lim_{\epsilon \downarrow 0} \omega_\lambda(E_{\epsilon,\lambda}^2).$$

The limit exists because the bounded positive operators $E_{\epsilon,\lambda}^2$ decrease as $\epsilon \downarrow 0$. They converge strongly to $P_{\ker M_{*,\lambda}}$, and normality gives

$$\mu_0(\omega_\lambda) = \omega_\lambda(P_{\ker M_{*,\lambda}}).$$

To state compatibility, let $\mathcal{A}_\lambda$ be represented von Neumann algebras containing the corresponding Master calculi. For $\lambda \subseteq \lambda'$, suppose there is a normal unital $*$ -homomorphism

$$r_{\lambda'\lambda} : \mathcal{A}_{\lambda'} \longrightarrow \mathcal{A}_\lambda$$

such that

$$r_{\lambda'\lambda}(f(M_{*,\lambda'})) = f(M_{*,\lambda})$$

for every bounded Borel f , and suppose

$$\omega_{\lambda'} = \omega_\lambda \circ r_{\lambda'\lambda}.$$

Theorem E.8 (Compatible normal kernel limits). *If $\mu_0(\omega_\lambda) > 0$ at every label, then $\omega_{\lambda,\epsilon}$ converges pointwise on $\mathcal{A}_\lambda$ to*

$$\omega_{\lambda,0}(A) = \frac{\omega_\lambda(P_{\ker M_{*,\lambda}} A P_{\ker M_{*,\lambda}})}{\mu_0(\omega_\lambda)}.$$

The limit is a positive normalized functional satisfying

$$\omega_{\lambda,0}(\mathfrak{b}_{*,\lambda}) = 0, \quad \omega_{\lambda',0} = \omega_{\lambda,0} \circ r_{\lambda'\lambda}.$$

Proof. Strong convergence of the uniformly bounded filters gives

$$E_{\epsilon,\lambda} A E_{\epsilon,\lambda} \longrightarrow P_{\ker M_{*,\lambda}} A P_{\ker M_{*,\lambda}}$$

in the weak operator topology. Normality carries this limit through ω_λ , while the denominator converges to the strictly positive zero-window mass. The basic Master coordinate vanishes on the kernel. Normality of $r_{\lambda'\lambda}$ and preservation of bounded Borel calculus give

$$r_{\lambda'\lambda}(P_{\ker M_{*,\lambda'}}) = P_{\ker M_{*,\lambda}},$$

and the displayed compatibility of the reference functionals then gives the compatibility of the limits. $\square$

This construction is the normal zero-eigenvalue branch. Strong convergence to a kernel projection does not control a singular functional. A singular functional may assign value one to every E_ϵ^2 even when the kernel projection is zero.

E.5. Continuous spectral zero and singular extensions. Let $M \geq 0$, set

$$b = M(1 + M)^{-1}, \quad A = C^*(1, b).$$

If $0 \in \sigma(M)$, continuous functional calculus identifies A with the continuous functions on $\sigma(b)$ and gives the character

$$\delta_0(f(b)) = f(0).$$

The filter is the continuous function

$$E_\epsilon = e_\epsilon(b), \quad e_\epsilon(t) = \frac{\epsilon(1-t)}{\epsilon + (1-\epsilon)t},$$

so

$$\delta_0(b) = 0, \quad \delta_0(E_\epsilon) = \delta_0(E_\epsilon^2) = 1.$$

Theorem E.9 (Extension of the zero character). *The character δ_0 extends to a positive unital functional on every unital C^* -algebra containing A , including $W^*(M)$. If $\ker M = \{0\}$, every extension to $W^*(M)$ has zero normal part and assigns zero to $P_{\ker M}$.*

Proof. The state-extension theorem gives a positive unital extension [7, Proposition 2.3.24]. Suppose $\ker M = \{0\}$. The represented spectral projection of b at zero then vanishes. Every positive extension has a unique positive normal-singular decomposition $\varphi = \varphi_n + \varphi_s$ [47, Chapter III, Definition 2.13, Theorem 2.14, Eq. (10), and Definition 2.15]. Since $b \geq 0$ and $\varphi(b) = 0$, its normal part satisfies $\varphi_n(b) = 0$. The spectral density of that normal part vanishes where $b > 0$, and the zero spectral projection supplies no remaining support. Hence $\varphi_n = 0$. The actual kernel projection is the zero operator and receives value zero. $\square$

The extension is a state; no multiplicative extension of the character to the containing algebra is used. The represented bounded-Borel calculus quotients functions by equality with respect to the maximal spectral measure. When zero lies in continuous spectrum, $\mathbf{1}_{\{0\}}(M) = 0$. Literal point evaluation would send this zero operator to one and therefore does not descend to the represented Borel quotient. The state-extension theorem preserves the continuous character without evaluating represented Borel classes at a point.

If $M = mI$ with $m > 0$, then zero is absent from $\sigma(M)$ and $b = m/(1+m) > 0$. Every state has positive b -expectation, so there is no state supported at spectral zero. In the normal filtering construction, every denominator is positive at fixed ϵ , but $\mu_0 = 0$ and the filtered functional remains the original reference functional. Positive zero-window mass is therefore the load-bearing hypothesis of the normal branch.

E.6. Fixed-holonomy torus spectrum. Consider the rank-two normal connection on T^3 from Section 9. The unit internal generator is

$$\hat{J} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \|\hat{J}\|_{\text{HS}} = 1.$$

Large $SO(2)$ gauge transformations

$$u_k(x) = \exp((k_1 x^1 + k_2 x^2 + k_3 x^3)J), \quad k \in \mathbb{Z}^3,$$

identify the harmonic coefficients by $\alpha_i \sim \alpha_i + k_i$. Fixing $[\alpha] \in (\mathbb{R}/\mathbb{Z})^3$ leaves the coexact configuration space

$$\mathfrak{h}_+ = d^* \Omega^2(T^3; \mathbb{R}\hat{J}).$$

The fixed holonomy class removes the harmonic coefficients from the Schrödinger configuration coordinates.

For $n \geq 1$, let

$$F_n = \{k \in \mathbb{Z}^3 : |k_i| \leq n\}$$

and let E_n be the real coexact span of the nonzero modes in F_n . These nested labels exhaust the finite-Fourier core of $\mathfrak{h}_+$. The operator

$$L_n = d^*d = -\Delta_H$$

is positive on E_n , and every E_n contains the background mode

$$A_b = a \sin x^1 dx^2 J = \sqrt{2}a \sin x^1 dx^2 \hat{J}.$$

The scalar-measure assertion below uses the chosen pure quasifree covariance on the positive spectral subspace,

$$C_{qq,n} = L_n^{-1}, \quad C_{pp,n} = L_n/4, \quad C_{qp,n} = 0.$$

The connection component of the source metric in Theorem A.8 makes the inverse kinetic metric the identity in an orthonormal $\hat{J}$ -basis. Abelian curvature gives

$$F(A + q) = F(A) + dq, \quad d^*F(A) = A_b, \quad L_n A_b = A_b.$$

Consequently the exact signed constraint on T^*E_n is

$$\begin{aligned} H_n(q, p) &= \frac{1}{2} \|p\|^2 - \frac{1}{2} \|F(A) + dq\|^2 \\ &= \frac{1}{2} \|p\|^2 - \frac{1}{2} \langle q + A_b, L_n(q + A_b) \rangle. \end{aligned}$$

This form retains the affine term $-\langle A_b, q \rangle$ and the scalar term $-\|F(A)\|^2/2$.

Let $C_{0,n} = \text{Op}_W(H_n)$ on the finite-dimensional Schwartz core. In a real orthonormal eigenbasis $L_n u_j = \lambda_j u_j$, with $\lambda_j \geq 1$, translation by A_b gives the unitarily equivalent sum of inverted oscillators

$$\tilde{C}_{0,n} = \sum_{j=1}^{N_n} \frac{1}{2} (P_j^2 - \lambda_j Q_j^2).$$

Theorem E.10 (Fixed-label Master spectrum). *The operator $C_{0,n}$ is essentially self-adjoint for every $n \geq 1$. Its closure satisfies $C_n = \overline{C_{0,n}} = C_{0,n}^*$, and $M_n = C_n^2$ satisfies*

$$\sigma(C_n) = \sigma_{\text{ess}}(C_n) = \mathbb{R}$$

with purely absolutely continuous spectral type, and

$$\sigma(M_n) = \sigma_{\text{ess}}(M_n) = [0, \infty), \quad \ker M_n = \{0\}.$$

The Gaussian cyclic vector determined by the chosen covariance has scalar Master measure equivalent to Lebesgue measure on $(0, \infty)$.

Proof. Put $\omega_j = \sqrt{\lambda_j}$ and introduce the symplectic coordinates

$$X_j = \frac{P_j + \omega_j Q_j}{\sqrt{2\omega_j}}, \quad Y_j = \frac{P_j - \omega_j Q_j}{\sqrt{2\omega_j}}.$$

The identity $[X_j, Y_k] = i\delta_{jk}$ shows that the change is symplectic. It therefore has a metaplectic implementer that preserves Schwartz space and intertwines Weyl quantization [17, Lemma 2.14 and Theorem 2.15]. Direct calculation gives

$$\frac{1}{2}(P_j^2 - \lambda_j Q_j^2) = \frac{\omega_j}{2}(X_j Y_j + Y_j X_j) = \omega_j D_j.$$

In the X -representation, $D = -i(x\partial_x + 1/2)$. On either half-line the unitary map

$$(U_{\pm} f)(y) = e^{y/2} f(\pm e^y)$$

sends D to $-i\partial_y$. Fourier transform therefore sends the finite sum to multiplication by the nonzero linear form $s \mapsto \sum_j \omega_j s_j$. This proves essential self-adjointness on the transported Schwartz core, pure absolute continuity, and full real essential spectrum. Squaring gives the stated Master spectrum and zero kernel.

Translation and the metaplectic map send a selected nonzero Gaussian factor to

$$g(x) = c \exp(-\gamma x^2 + \eta x + \zeta), \quad \text{Re } \gamma > 0.$$

Its half-line Mellin transforms

$$(\mathcal{M}_{\pm} g)(s) = (2\pi)^{-1/2} \int_0^{\infty} x^{-1/2-is} g(\pm x) dx$$

are holomorphic for $\text{Im } s > -1/2$. Mellin–Plancherel injectivity follows by applying Fourier–Plancherel to the logarithmic maps $U_{\pm}$ above [41, Theorem C.4]. Each transform is therefore a nonzero holomorphic function, so its real zeros are discrete. The sum of their squared moduli is positive almost everywhere. The remaining dilation factors contribute nonzero absolutely continuous measures, and convolution preserves a positive almost-everywhere density. Squaring the constraint coordinate gives a Master density equivalent to Lebesgue measure on $(0, \infty)$ and creates no atom at zero. $\square$

E.7. Failure of a common vacuum-core constraint. In the Fock coordinates associated with the chosen compatible quasifree covariance,

$$Q_j = \lambda_j^{-1/2}(a_j + a_j^\dagger), \quad P_j = \frac{\sqrt{\lambda_j}}{2i}(a_j - a_j^\dagger).$$

For the normalized two-particle vector $|2_j\rangle = (a_j^\dagger)^2\Omega/\sqrt{2}$, the Weyl-ordered summand obeys

$$\frac{1}{2}(P_j^2 - \lambda_j Q_j^2)\Omega = \left(\frac{\lambda_j}{8} - \frac{1}{2}\right)\Omega - \frac{\sqrt{2}(\lambda_j + 4)}{8}|2_j\rangle.$$

The two-particle vectors are orthonormal, so every scalar subtraction c_n leaves

$$\|(C_{0,n} - c_n)\Omega\|^2 \geq \sum_{j \in E_n} \frac{(\lambda_j + 4)^2}{32}.$$

The exhaustive torus Hodge spectrum makes this sum diverge. The affine background contributes only finitely supported one-particle terms and cannot cancel the two-particle component. Hence the exact label operators do not define a common operator on the vacuum-containing finite-particle cylinder core.

This obstruction does not contradict the reducing-envelope theorem. That theorem begins with an independently specified common closed operator; the fixed-holonomy torus labels fail precisely that antecedent on the stated Fock core.

E.8. Spectral-coordinate comparison maps. For the fixed-holonomy labels set

$$b_n = M_n(1 + M_n)^{-1}, \quad A_n = C^*(1, b_n), \quad \mathcal{B}_n = W^*(M_n) = W^*(b_n).$$

Theorem E.10 gives faithful functional-calculus isomorphisms

$$C([0, 1]) \longrightarrow A_n, \quad L^\infty([0, 1], m) \longrightarrow \mathcal{B}_n, \quad f \longmapsto f(b_n),$$

where m is Lebesgue measure and the second map acts on represented equivalence classes. Denote the second isomorphism by ρ_n . All labels therefore carry the same represented spectral coordinate and the same Lebesgue null ideal.

For $n \leq N$, define

$$r_{Nn} = \rho_n \rho_N^{-1} : \mathcal{B}_N \longrightarrow \mathcal{B}_n.$$

Theorem E.11 (Comparison of the bounded Master calculi). *The map r_{Nn} is the unique normal unital $*$ -isomorphism satisfying*

$$r_{Nn}(b_N) = b_n.$$

These maps obey

$$r_{nn} = \text{id}, \quad r_{Ln} = r_{Nn} \circ r_{LN} \quad (n \leq N \leq L).$$

The continuous zero characters on A_n admit a nonempty compatible family of singular extensions ω_n to $\mathcal{B}_n$ satisfying

$$\omega_N = \omega_n \circ r_{Nn}, \quad \omega_n(b_n) = 0,$$

$$\omega_n(E_{\epsilon,n}) = \omega_n(E_{\epsilon,n}^2) = 1, \quad \omega_n(P_{\ker M_n}) = 0.$$

Proof. The maps ρ_n give existence. A normal unital map carrying b_N to b_n agrees with r_{Nn} on polynomials and then on continuous functions by norm approximation. Normality carries monotone continuous approximants to the Borel spectral projections, and the monotone class theorem extends the equality to every represented bounded Borel function. Since b_N generates $\mathcal{B}_N$, the map is unique. Uniqueness gives the identity and composition laws.

For $k \geq 1$, define a state on the common spectral-coordinate algebra by

$$\Phi_k(f) = k \int_0^{1/k} f(t) dt.$$

A free-ultrafilter limit $\Phi(f) = \lim_{k \rightarrow \mathcal{U}} \Phi_k(f)$ is a state on $L^\infty([0, 1], m)$. Uniform continuity gives $\Phi(f) = f(0)$ for continuous f , so Φ extends the zero character without evaluating a Borel equivalence class at a point. Set

$$\omega_n = \Phi \rho_n^{-1}.$$

The identity $r_{Nn} \rho_N = \rho_n$ gives compatibility. Evaluation of the continuous filter function at zero gives the displayed values. Theorem E.10 makes the zero projection vanish, and positivity of b_n almost everywhere forces the normal part of ω_n to vanish. $\square$

If a common operator and reducing Hilbert restrictions exist on another branch, compression of bounded functional calculus gives a normal unital map preserving b . Whenever that branch has the same represented spectral coordinate, uniqueness identifies its Hilbert-induced map with the map above. In the present torus model, r_{Nn} is supplied directly by the shared spectral coordinate because the common vacuum-core operator is obstructed.

The fixed law and label family determine M_n, b_n , the continuous zero characters on A_n , and the comparison maps. Choosing Φ determines an extension to $\mathcal{B}_n$. Different extension choices can give different compatible singular families.

E.9. Robustness under a fixed positive kinetic coefficient. The source metric fixes the connection coefficient at one. To test the spectral conclusion independently of that normalization, fix $c_\nabla > 0$ and set

$$H_n^{(c_\nabla)}(q, p) = \frac{1}{2c_\nabla} \|p\|^2 - \frac{1}{2} \langle q + A_b, L_n(q + A_b) \rangle.$$

Let $C_n^{(c_\nabla)}$ be the closure of its Weyl quantization and define

$$M_n^{(c_\nabla)} = (C_n^{(c_\nabla)})^2, \quad b_n^{(c_\nabla)} = M_n^{(c_\nabla)}(1 + M_n^{(c_\nabla)})^{-1}, \\ E_{\epsilon, n}^{(c_\nabla)} = \epsilon(M_n^{(c_\nabla)} + \epsilon)^{-1}.$$

Corollary E.12 (Stability under positive kinetic rescaling). *For each fixed $c_\nabla > 0$, the operators $C_n^{(c_\nabla)}$ and $M_n^{(c_\nabla)}$ have the same spectral conclusions and zero kernel as the source-normalized operators, and the scalar Master measure of the Gaussian vector remains equivalent to Lebesgue measure. The pairs*

$$(W^*(M_n^{(c_\nabla)}), b_n^{(c_\nabla)})$$

*carry unique normal unital $b^{(c_\nabla)}$ -preserving *-isomorphisms $r_{Nn}^{(c_\nabla)}$ and compatible singular states $\omega_n^{(c_\nabla)}$ satisfying*

$$\omega_n^{(c_\nabla)}(b_n^{(c_\nabla)}) = 0,$$

$$\omega_n^{(c_\nabla)}(E_{\epsilon,n}^{(c_\nabla)}) = \omega_n^{(c_\nabla)}((E_{\epsilon,n}^{(c_\nabla)})^2) = 1.$$

The obstruction to a common operator on the vacuum-containing core also persists. The result gives one comparison family for each fixed positive coefficient; it supplies no map between distinct coefficients.

Proof. The symplectic scaling

$$Q'_j = \sqrt{c_\nabla} Q_j, \quad P'_j = P_j / \sqrt{c_\nabla}$$

changes the dilation frequency to $\omega_j^{(c_\nabla)} = \sqrt{\lambda_j / c_\nabla}$ and leaves the metaplectic and Mellin arguments unchanged. In the chosen quasifree coordinates the two-particle component becomes

$$-\frac{\sqrt{2}(\lambda_j + 4c_\nabla)}{8c_\nabla} |2_j\rangle,$$

whose squared sum diverges for every fixed positive c_∇ . For that coefficient, functional calculus gives faithful maps

$$\rho_n^{(c_\nabla)} : L^\infty([0, 1], m) \longrightarrow W^*(M_n^{(c_\nabla)}).$$

The formulas

$$r_{Nn}^{(c_\nabla)} = \rho_n^{(c_\nabla)}(\rho_N^{(c_\nabla)})^{-1}, \quad \omega_n^{(c_\nabla)} = \Phi(\rho_n^{(c_\nabla)})^{-1}$$

define the comparison maps and compatible states at the fixed coefficient. $\square$

E.10. Completion data and further structures. The common-core branch begins with a specified symmetric operator and determines its maximal adjoint, positive Master operator, bounded functional calculus, and minimal constraint-reducing envelopes. A compatible normal reference family with positive zero-window mass then determines the corresponding normal kernel-compressed limit family. The fixed-holonomy torus branch determines exact finite-label Master operators, continuous zero characters, a common spectral coordinate, and the associated comparison maps. Extension of those characters to compatible singular states remains a choice of state on the represented spectral algebra.

Appendix D separately proves that imposing response stationarity together with the signed constraint can remove the continuous zero-support sector. That joint equation is distinct from the Master completion here.

A real-time Hamiltonian and temporal development determine evolution. Boundary conditions may select a self-adjoint extension. An infinite-mode constraint or energy requires a common carrier, domain, and convergence or renormalization prescription, while a state on a larger represented observable algebra requires an additional extension. A normalizable zero vector exists only when the resulting Master operator has a zero eigenvalue; continuous spectral support at zero alone supplies the singular alternative proved above.

APPENDIX F. RANK-ONE CURVATURE OBSERVABLES AND LOCAL COMPLETE OBSERVABLES

This appendix derives the scalar and one-form contractions of the assembled curvature on the oriented positive-Riemannian rank-one branch. It then establishes the bounded functional calculus generated by those contractions, their behavior under Master-operator filtering and label comparison, and the local complete-observable construction on the regular Hamiltonian locus.

F.1. Rank-One Curvature Observables.

Section 6 defines the rank-one curvature-contraction map from the tangent and mixed components of the assembled curvature. Assume here that the base and internal metrics are positive Riemannian and that the internal line is oriented. The contractions are

$$\mathfrak{s} : \mathcal{B}_T^{\mathfrak{R}} \rightarrow \mathcal{A}, \quad \mathfrak{v} : \mathcal{B}_K^{\mathfrak{R}} \rightarrow \Omega_{\mathfrak{R}}^1,$$

and

$$\mathcal{R}_1(B_T, B_K, B_{\perp}) = (\mathfrak{s}(B_T), \mathfrak{v}(B_K)).$$

The first component is the scalar Gauss contraction, and the second is the one-form Codazzi contraction.

Lemma F.1 (Rank-one Gauss–Codazzi formula). *Suppose the split realization has positive Riemannian metrics, an orientation, and internal rank one. Write $K = k \otimes n$ for a local oriented unit section n . Then*

$$\mathfrak{s}(B_T) = \text{Scal}_h - (\text{tr}_h k)^2 + \|k\|_h^2, \quad \mathfrak{v}(B_K) = \text{div}_h k - d(\text{tr}_h k).$$

These are the contracted Gauss and Codazzi expressions of the positive-Riemannian split realization.

Proof. Let e_i be a local h -orthonormal frame. In rank one,

$$K_{e_i} e_j = k_{ij} n, \quad K_{e_i}^{\dagger} n = k_i^{\sharp}.$$

The tangent curvature block is

$$B_T(e_i, e_j) e_j = R^h(e_i, e_j) e_j - k_{jj} k_i^{\sharp} + k_{ij} k_j^{\sharp}.$$

Tracing with $h(\cdot, e_i)$ over i, j gives

$$\sum_{i,j} h(B_T(e_i, e_j) e_j, e_i) = \text{Scal}_h - (\text{tr}_h k)^2 + \|k\|_h^2.$$

For the mixed component,

$$B_K(e_i, e_j) e_i = (\nabla_i k)_{ji} n - (\nabla_j k)_{ii} n.$$

Contracting the normal coefficient over i gives

$$\sum_i g_N(B_K(e_i, e_j) e_i, n) = \nabla^i k_{ij} - \nabla_j (\text{tr}_h k),$$

which is the displayed one-form expression. $\square$

The tangent and mixed curvature components, the source metrics, and the oriented line identification therefore determine $\mathcal{R}_1$. The constraint operator enters only when a state is filtered before evaluating an observable already generated from these curvature contractions.

At a represented rank-one label λ , suppose the components of $\mathcal{R}_{1,\lambda}$ form the declared commuting self-adjoint curvature coordinates. Their source-generated bounded observables are

$$A_{\lambda} = F_{\lambda}(\mathcal{R}_{1,\lambda}), \quad F_{\lambda} \in C_b(\text{Spec } \mathcal{R}_{1,\lambda}).$$

This is the bounded joint continuous functional calculus declared by the rank-one readout recipe. A represented von Neumann algebra may separately admit bounded Borel functions of the same coordinates. Such functions are represented attachments to the source-generated family and require that additional declaration.

F.2. Master-Filtered Evaluations. Let $M_{*,\lambda}$ be the positive Master operator of Appendix E at a finite, local, or cylindrical label, and let ω_λ be a normal state on a represented von Neumann algebra containing its bounded spectral calculus and the observable A_λ . For $\epsilon > 0$, set

$$E_{\epsilon,\lambda} = \frac{\epsilon}{M_{*,\lambda} + \epsilon}.$$

Whenever $\omega_\lambda(E_{\epsilon,\lambda}^2) > 0$, the filtered evaluation is

$$\omega_{\lambda,\epsilon}(A_\lambda) = \frac{\omega_\lambda(E_{\epsilon,\lambda} A_\lambda E_{\epsilon,\lambda})}{\omega_\lambda(E_{\epsilon,\lambda}^2)}.$$

Write $P^{M_{*,\lambda}}$ for the spectral measure of $M_{*,\lambda}$ and

$$\nu_\lambda(B) = \omega_\lambda(P^{M_{*,\lambda}}(B)).$$

The scalar Master moment of the state is

$$\omega_\lambda[M_{*,\lambda}] = \int_{[0,\infty)} \mu \, d\nu_\lambda(\mu) \in [0, \infty].$$

The filtered Master evaluation is defined entirely through bounded spectral calculus by

$$\omega_{\lambda,\epsilon}[M_{*,\lambda}] = \frac{\omega_\lambda(M_{*,\lambda} E_{\epsilon,\lambda}^2)}{\omega_\lambda(E_{\epsilon,\lambda}^2)}.$$

Indeed, $M_{*,\lambda} E_{\epsilon,\lambda}^2$ is bounded and positive.

Lemma F.2 (Master-filter bounds). *For $\mu \geq 0$ and*

$$w_\epsilon(\mu) = \frac{\epsilon}{\mu + \epsilon},$$

one has

$$1 - w_\epsilon(\mu) \leq \frac{\mu}{\epsilon}, \quad 1 - w_\epsilon(\mu)^2 \leq \frac{2\mu}{\epsilon}, \quad \mu w_\epsilon(\mu)^2 \leq \frac{\epsilon}{4}.$$

Proof. The first two inequalities follow from

$$1 - w_\epsilon(\mu) = \frac{\mu}{\mu + \epsilon} \leq \frac{\mu}{\epsilon}$$

and

$$1 - w_\epsilon(\mu)^2 = (1 - w_\epsilon(\mu))(1 + w_\epsilon(\mu)) \leq 2\frac{\mu}{\epsilon}.$$

For the third, write $t = \mu/\epsilon$. Then

$$\mu w_\epsilon(\mu)^2 = \epsilon \frac{t}{(1+t)^2} \leq \frac{\epsilon}{4},$$

because $4t \leq (1+t)^2$. □

It follows immediately that

$$\omega_{\lambda,\epsilon}[M_{*,\lambda}] \leq \frac{\epsilon}{4\omega_\lambda(E_{\epsilon,\lambda}^2)}.$$

Lemma F.3 (Stability of bounded evaluations under Master filtering). *Let $\|A_\lambda\| \leq R$, and suppose*

$$\omega_\lambda[M_{*,\lambda}] \leq \delta\epsilon, \quad 0 < \delta < \frac{1}{2}.$$

Then

$$\omega_\lambda(E_{\epsilon,\lambda}^2) \geq 1 - 2\delta$$

and

$$|\omega_{\lambda,\epsilon}(A_\lambda) - \omega_\lambda(A_\lambda)| \leq 2R \left(\sqrt{\delta} + \frac{2\delta}{1 + \sqrt{1 - 2\delta}} \right).$$

Proof. Let $(\pi_\lambda, \mathcal{H}_{\omega_\lambda}, \Omega_\lambda)$ be the GNS representation of the bounded represented algebra, and suppress π_λ on its bounded elements. Spectral integration of the first two filter bounds gives

$$\|(1 - E_{\epsilon,\lambda})\Omega_\lambda\|^2 \leq \delta, \quad \omega_\lambda(E_{\epsilon,\lambda}^2) \geq 1 - 2\delta.$$

Set

$$\Omega_{\lambda,\epsilon} = \frac{E_{\epsilon,\lambda}\Omega_\lambda}{\|E_{\epsilon,\lambda}\Omega_\lambda\|}.$$

Since

$$1 - \|E_{\epsilon,\lambda}\Omega_\lambda\| \leq \frac{2\delta}{1 + \sqrt{1 - 2\delta}},$$

the triangle inequality gives

$$\|\Omega_{\lambda,\epsilon} - \Omega_\lambda\| \leq \sqrt{\delta} + \frac{2\delta}{1 + \sqrt{1 - 2\delta}}.$$

The vector-state estimate

$$|\langle \Omega_{\lambda,\epsilon}, A_\lambda \Omega_{\lambda,\epsilon} \rangle - \langle \Omega_\lambda, A_\lambda \Omega_\lambda \rangle| \leq 2R \|\Omega_{\lambda,\epsilon} - \Omega_\lambda\|$$

proves the result. $\square$

The filter scale controls two competing quantities. Decreasing ϵ improves the bound on the filtered Master expectation. The surviving mass in the denominator controls stability of the normalized evaluation. The moment hypothesis relates these requirements at the chosen scale.

The finite-scale construction applies whenever its denominator is positive, on either spectral branch of Appendix E. A normal zero-width limit additionally requires positive zero-window mass. The fixed-holonomy torus labels instead have continuous spectral support at zero and zero kernel; their zero-supported extensions are singular, so they do not produce the normal kernel-compressed limit used below.

F.3. Compatibility and Exact Tensor Cancellation. Let $\mathcal{A}_\lambda^{\text{vN}}$ be represented von Neumann algebras containing both the bounded Master calculus and the rank-one curvature observable. Given $\lambda \subseteq \lambda'$, suppose there is a map

$$r_{\lambda'\lambda} : \mathcal{A}_{\lambda'}^{\text{vN}} \longrightarrow \mathcal{A}_\lambda^{\text{vN}}$$

that is a normal unital $*$ -homomorphism and satisfies

$$r_{\lambda'\lambda}(f(M_{*,\lambda'})) = f(M_{*,\lambda})$$

for every bounded Borel function f ,

$$r_{\lambda'\lambda}(A_{\lambda'}) = A_\lambda, \quad \omega_{\lambda'} = \omega_\lambda \circ r_{\lambda'\lambda}.$$

Theorem F.4 (Compatibility of Master-filtered curvature evaluations). *If the two filter denominators are positive, then*

$$\omega_{\lambda',\epsilon}(A_{\lambda'}) = \omega_{\lambda,\epsilon}(A_{\lambda}).$$

If the reference states have positive zero-window mass, the normal kernel-supported limits of Appendix E also satisfy

$$\omega_{\lambda',0}(A_{\lambda'}) = \omega_{\lambda,0}(A_{\lambda}).$$

Proof. Preservation of bounded Borel calculus gives

$$r_{\lambda'\lambda}(E_{\epsilon,\lambda'}) = E_{\epsilon,\lambda}.$$

Applying the compatible states to the numerator and denominator of the filtered evaluation gives the finite-scale identity. Positive zero-window mass gives the point-wise normal limits at both labels, and passing to the limit proves the second identity. $\square$

Exact tensor cancellation is a separate sufficient condition. Suppose an extension is represented by

$$\begin{aligned} \mathcal{H}_{\lambda'} &= \mathcal{H}_{\lambda} \otimes \mathcal{H}_z, & A_{\lambda'} &= A_{\lambda} \otimes 1, \\ \omega_{\lambda'} &= \omega_{\lambda} \otimes \omega_z, & E_{\epsilon,\lambda'} &= E_{\epsilon,\lambda} \otimes E_{\epsilon,z}. \end{aligned}$$

The last identity is an explicit hypothesis on the Master filters rather than a consequence of tensor factorization alone.

Lemma F.5 (Exact cancellation under a factorized Master filter). *If $\omega_z(E_{\epsilon,z}^2) > 0$, then*

$$\omega_{\lambda',\epsilon}(A_{\lambda'}) = \omega_{\lambda,\epsilon}(A_{\lambda}).$$

Proof. Tensor factorization gives

$$\omega_{\lambda'}(E_{\epsilon,\lambda'} A_{\lambda'} E_{\epsilon,\lambda'}) = \omega_{\lambda}(E_{\epsilon,\lambda} A_{\lambda} E_{\epsilon,\lambda}) \omega_z(E_{\epsilon,z}^2)$$

and

$$\omega_{\lambda'}(E_{\epsilon,\lambda'}^2) = \omega_{\lambda}(E_{\epsilon,\lambda}^2) \omega_z(E_{\epsilon,z}^2).$$

The common positive factor cancels. $\square$

F.4. Finite-Label Norm Approximation. Let A be a bounded rank-one curvature observable, let A_F be a bounded observable at finite label F , and let $\iota_F(A_F)$ denote its image in the limiting bounded algebra.

Lemma F.6 (Finite-label error estimate). *For states ω and ω_F ,*

$$\begin{aligned} |\omega(A) - \omega_{F,\epsilon}(A_F)| &\leq \|A - \iota_F(A_F)\| \\ &\quad + |\omega(\iota_F(A_F)) - \omega_F(A_F)| \\ &\quad + |\omega_F(A_F) - \omega_{F,\epsilon}(A_F)|. \end{aligned}$$

If $\|A_F\| \leq R$ and

$$\omega_F[M_{*,F}] \leq \delta_F \epsilon, \quad 0 < \delta_F < \frac{1}{2},$$

then the final term is at most

$$2R \left(\sqrt{\delta_F} + \frac{2\delta_F}{1 + \sqrt{1 - 2\delta_F}} \right).$$

Proof. Insert $\omega(\iota_F(A_F))$ and $\omega_F(A_F)$ between the two endpoint evaluations and apply the triangle inequality. The first term is bounded by the norm error because ω is a state. The stated bound on the final term is the Master-filter stability lemma. $\square$

Consequently,

$$\omega_{F,\epsilon}(A_F) \longrightarrow \omega(A)$$

whenever the norm-approximation error, the finite-label comparison error, and the filtering error tend to zero along the chosen labels and scales. Under the scalar moment hypothesis, it is enough for the first two errors and δ_F to tend to zero while the observables remain uniformly bounded.

F.5. Complete Observables on the Regular Hamiltonian Locus. Let $\mathcal{P}$ be a finite-dimensional cylindrical phase space, or a smooth local phase chart on which the signed constraint H_R has a smooth local Hamiltonian flow φ_u . Write X_{H_R} for its Hamiltonian vector field. A local clock transverse to the flow is a smooth function T satisfying

$$X_{H_R}(T)(z_0) \neq 0$$

at the point under consideration. On a finite-dimensional cylindrical phase space, the flow-box theorem supplies such a function at every point where $X_{H_R} \neq 0$. On a more general local phase chart, the smooth flow and transverse clock are the stated local hypotheses.

Theorem F.7 (Local complete observables). *Let T be transverse to X_{H_R} at z_0 , and let τ be close to $T(z_0)$. After restricting to a neighborhood of z_0 , there is a unique smooth function $s(z, \tau)$ near zero such that*

$$T(\varphi_{s(z, \tau)} z) = \tau.$$

For every smooth function f on the flow tube,

$$O_{f,T}(\tau)(z) = f(\varphi_{s(z, \tau)} z)$$

satisfies

$$X_{H_R}(O_{f,T}(\tau)) = 0.$$

It also recovers the slice observable,

$$T(z) = \tau \implies O_{f,T}(\tau)(z) = f(z).$$

If f is bounded on the flow tube, then $O_{f,T}(\tau)$ is bounded on its domain.

Proof. Apply the implicit-function theorem to

$$G(s, z, \tau) = T(\varphi_s z) - \tau.$$

At $(0, z_0, T(z_0))$, its derivative in s is $X_{H_R}(T)(z_0)$, which is nonzero. This gives the unique smooth hitting time $s(z, \tau)$. The flow law and uniqueness imply

$$s(\varphi_u z, \tau) = s(z, \tau) - u$$

whenever both sides are defined. Hence

$$O_{f,T}(\tau)(\varphi_u z) = f(\varphi_{s(\varphi_u z, \tau)} \varphi_u z) = f(\varphi_{s(z, \tau)} z),$$

and differentiation at $u = 0$ gives $X_{H_R}(O_{f,T}(\tau)) = 0$. If $T(z) = \tau$, uniqueness gives $s(z, \tau) = 0$, which proves slice recovery. Boundedness follows directly from the definition. $\square$

For the deparametrized local constraint

$$C = P + \frac{1}{2}p^2$$

in canonical coordinates $(T, x; P, p)$, this construction gives [14]

$$O_x(\tau) = x + (\tau - T)p, \quad O_p(\tau) = p.$$

Indeed,

$$\{T, C\} = 1, \quad \{x, C\} = p, \quad \{p, C\} = 0,$$

and therefore

$$\{O_x(\tau), C\} = 0, \quad \{O_p(\tau), C\} = 0, \quad \{O_x(\tau), O_p(\tau)\} = 1.$$

The theorem supplies bounded classical constraint-invariant observables when the chosen slice function is bounded on the local flow tube. A quantum version begins after a representation and a quantization map for these classical functions have been supplied.

The oriented rank-one branch thus carries a source-generated continuous calculus of Gauss–Codazzi curvature observables. Master filtering evaluates those observables at finite scale, compatible maps preserve the evaluations, and positive zero-window mass yields their normal kernel-supported limits. Local clocks independently produce classical complete observables on the regular Hamiltonian locus.

APPENDIX G. CHARGE SECTORS, COVARIANT TRANSITIONS, AND BOUNDED OBSERVABLES

Section 8 distinguishes the bounded operations generated on represented components from the covariant morphisms between charge sectors. This appendix constructs the completed transition category and its generated endomorphism algebras, defines the bounded-law groupoid independently of the source assignment, and proves the functorial source determination used in Section 8.3. States evaluate these structures after they have been fixed.

G.1. The Bounded Transition Category. Fix a transition index U . The internal metric-gauge data supply an additive real metric $*$ -category $\mathbf{C}_U$ and a faithful finite-projective metric probe

$$\omega_U : \mathbf{C}_U \longrightarrow \mathbf{Proj}_g(A_U).$$

For $T \in \mathbf{C}_U(i, j)$, define

$$\|T\|_{U, \omega} = \|\omega_U(T)\|.$$

Faithfulness makes this a norm. The metric $*$ -structure and the operator norm of the probe give

$$\|T^*T\|_{U, \omega} = \|T\|_{U, \omega}^2, \quad \|ST\|_{U, \omega} \leq \|S\|_{U, \omega} \|T\|_{U, \omega}.$$

Completing every Hom space in this norm therefore extends adjoint and composition continuously and gives a real metric C^* -category $\widehat{\mathbf{C}}_U$.

For $U \leq V$, the source restriction functor $s_{VU} : \mathbf{C}_V \rightarrow \mathbf{C}_U$ is contractive in the probe norms. It extends uniquely to a bounded $*$ -functor

$$\widehat{s}_{VU} : \widehat{\mathbf{C}}_V \longrightarrow \widehat{\mathbf{C}}_U.$$

The source comparison maps extend with it, so the completed categories form the coherent projective diagram

$$\widehat{\mathbf{C}}_\bullet = \{\widehat{\mathbf{C}}_U, \widehat{s}_{VU}\}_{U \leq V}.$$

A charge sector is an object i of $\widehat{\mathbf{C}}_U$. Its identity 1_i is the unit of the real C^* -algebra

$$\text{End}(i) = \text{End}_{\widehat{\mathbf{C}}_U}(i).$$

The construction therefore treats charge sectors intrinsically as category objects. A common linking representation that realizes their identities as projections would be additional data.

G.2. Source-Supplied Transitions and Generated Endomorphism Algebras. Let

$$P_U^{VT}(i, j) \subseteq \text{Hom}_{\mathbf{C}_U}(i, j)$$

be the original source-supplied transitions from i to j . A transition $T : i \rightarrow j$ remains a morphism after completion, while its positive products have the distinct endomorphism types

$$T^*T \in \text{End}(i), \quad TT^* \in \text{End}(j).$$

At an object i , set

$$\begin{aligned} \Sigma_{U,i}^{VT} &= \{T^*T : \exists j, T \in P_U^{VT}(i, j)\} \\ &\cup \{TT^* : \exists j, T \in P_U^{VT}(j, i)\}. \end{aligned}$$

The source-generated transition algebra is represented by the faithful unital inclusion

$$\iota_{U,i} : \mathfrak{V}_{U,i}^0 \hookrightarrow \text{End}(i), \quad \iota_{U,i}(\mathfrak{V}_{U,i}^0) = C_{\text{End}(i)}^*(\Sigma_{U,i}^{VT}).$$

Thus the transition itself and its two positive products occupy different parts of the bounded law. Completion can add morphisms to a Hom space, but it does not add parameters to the fixed family $P_U^{VT}(i, j)$. Applying the constraint derivation to T^*T or TT^* requires a separate hypothesis placing that endomorphism in the derivation domain.

Lemma G.1 (Strict inclusion on the weight-one real torus object). *For the weight-one real $SO(2)$ object X in the admitted torus source,*

$$\iota_{U,X}(\mathfrak{V}_{U,X}^0) \subsetneq \text{End}_{\widehat{\mathbf{C}}_U}(X).$$

Proof. The object is $X = \mathcal{A}_U \otimes_{\mathbb{R}} \mathbb{R}^2$ with the standard $SO(2)$ action. The constant equivariant endomorphism

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^* = -J,$$

belongs to $\text{End}_{\widehat{\mathbf{C}}_U}(X)$. At every point ξ of the compact spectrum of $\mathcal{A}_U$, the commutant of the weight-one real fiber is

$$\text{End}_{SO(2)}(X_\xi) = \mathbb{R}[J].$$

Its self-adjoint part is $\mathbb{R}1_X$. Every generator in $\Sigma_{U,X}^{VT}$ is positive, self-adjoint, and equivariant, so its evaluation at ξ is scalar. Fiber evaluation therefore carries the entire generated algebra $\iota_{U,X}(\mathfrak{V}_{U,X}^0)$ into $\mathbb{R}1_X$, whereas it carries J to the nonscalar matrix J . Hence J does not lie in the range of $\iota_{U,X}$, which proves the strict inclusion. $\square$

G.3. Restrictions and Residual Source Symmetry. For $U \leq V$, functoriality of adjoint and composition gives

$$\begin{aligned}\widehat{s}_{VU}(T^*T) &= s_{VU}(T)^* s_{VU}(T), \\ \widehat{s}_{VU}(TT^*) &= s_{VU}(T) s_{VU}(T)^*.\end{aligned}$$

The restriction functor consequently induces a unital $*$ -homomorphism

$$\beta_{VU,i} : \mathfrak{V}_{V,i}^0 \longrightarrow \mathfrak{V}_{U,\widehat{s}_{VU}i}^0$$

which satisfies the inclusion square

$$\bar{s}_{VU,i} \iota_{V,i} = \iota_{U,\widehat{s}_{VU}i} \beta_{VU,i}.$$

Here $\bar{s}_{VU,i}$ denotes the induced map on the endomorphism algebra. The comparison maps for three nested indices give the corresponding coherence for the $\beta_{VU,i}$.

The residual source-symmetry group D_G acts by metric $*$ -functors. It transports objects, morphisms, adjoints, positive products, and the included generated algebras. If a subgroup fixes an object i , its invariant endomorphisms form the fixed-point C^* -subalgebra of $\text{End}(i)$, and its invariant generated observables form the corresponding fixed-point subalgebra of $\iota_{U,i}(\mathfrak{V}_{U,i}^0)$. This equivariance selects no preferred representative of a symmetry orbit.

G.4. State Evaluation and External Tensor Factors. Let $\Omega = (\omega_x)_x$ and $\Omega' = (\omega'_x)_x$ be compatible families of positive functionals on the same bounded-law object. They are equivalent on its generated observables when their restrictions agree on every common represented and local-operation algebra and, at every transition object, after the faithful inclusion

$$\Omega \sim_{\mathfrak{V}} \Omega' \iff \begin{aligned} \omega_x|_{\mathfrak{V}_x^0} &= \omega'_x|_{\mathfrak{V}_x^0} & (x \in \mathcal{I}^{(0)} \sqcup \mathcal{I}^{(1)}), \\ \omega_{U,i} \circ \iota_{U,i} &= \omega'_{U,i} \circ \iota_{U,i} & ((U,i) \in \mathcal{I}^{(2)}). \end{aligned}$$

This is an equivalence of functional families. It does not select or classify sector labels, reference states, or vacua.

The same distinction is visible when an external tensor factor is present. For a state ω on a component algebra, states η and η' on an external algebra, and $a \in \mathfrak{V}_x^0$, the product states on the spatial tensor product satisfy

$$(\omega \otimes \eta)(a \otimes I) = \omega(a) = (\omega \otimes \eta')(a \otimes I).$$

The two product states therefore have the same restriction to $\mathfrak{V}_x^0 \otimes I$. More generally, any two states with equal restrictions to the declared algebra are indistinguishable by its elements. Declaring an additional operator $I \otimes b$ as a bounded generator enlarges the bounded-law data.

G.5. The Independently Defined Bounded-Law Groupoid. Let $\mathbf{L}$ be the groupoid whose objects are the common field-law data L of Section 8.3 and whose arrows are their role-preserving isomorphisms. The bounded-law target is defined before the source functor. Write J for the five recipe-row labels and $\mathcal{D}$ for their family of parameter sets and output maps. An object of $\mathbf{L}^B$ is then specified by

$$J = \{0, 1, 2, 3, 4\}, \quad \mathcal{D} = ((P^r, \sigma^r))_{r \in J},$$

$$\mathcal{T} = (L_0, \mathcal{I}^0, \mathcal{D}, \mathfrak{V}^0, \widehat{\mathbf{C}}_\bullet, \iota, \mathcal{M})$$

satisfying the following conditions.

The component index is the tagged disjoint union

$$\mathcal{I}^0 = \{0\} \times \mathcal{I}^{(0)} \sqcup \{1\} \times \mathcal{I}^{(1)} \sqcup \{2\} \times \mathcal{I}^{(2)}.$$

The tags are part of the component types, and a transition component is a pair (U, i) formed from an index and an object of the category at that index. Each recipe row has a typed parameter set P_x^r , a presence predicate, and an output map $\sigma_x^r : P_x^r \rightarrow \mathfrak{V}_x^0$ wherever that row is present. V0 contains the represented units. VM contains the declared bounded continuous functions of the Master operator. VR contains the declared bounded joint continuous functions of the rank-one curvature contractions. VH contains the declared bounded cylinder operations. VT contains the two positive outputs v_T^- and v_T^+ indexed by an original source-supplied transition $T : i \rightarrow j$, with

$$\iota_{U,i}(v_T^-) = T^*T, \quad \iota_{U,i}(v_T^+) = TT^*.$$

The presence predicates retain the domains stated in Section 8.1. In particular, VR and VH need not be present at every component.

At every represented or local-operation component, $\mathfrak{V}_x^0$ is the smallest unital C^* -subalgebra of its specified ambient algebra containing all declared outputs present at x . At every transition component (U, i) , $\mathfrak{V}_{U,i}^0$ is a unital real C^* -algebra with a faithful inclusion into $\text{End}_{\widehat{\mathcal{C}}_U}(i)$, and its image is exactly the unital C^* -algebra generated by the positive products of the original transitions. Morphisms introduced by completion are not new VT parameters. The transition entry is a projective diagram of small real metric C^* -categories and bounded $*$ -functors with specified unitary restriction comparisons satisfying their unit and associativity coherence. The family $\mathcal{M}$ records only the component maps and restriction comparisons that are actually defined, including the maps $\beta_{VU,i}$ and their inclusion squares.

An arrow $\mathcal{T} \rightarrow \mathcal{T}'$ is a tuple

$$\Theta = (\ell, \kappa, \{p_x^r\}_{r,x}, \{\beta_x\}_x, \Gamma).$$

Here $\ell : L_0 \rightarrow L'_0$ is a field-law isomorphism, $\kappa : \mathcal{I}^0 \rightarrow \mathcal{I}'^0$ is a strict type-preserving index isomorphism, each p_x^r is a parameter bijection preserving its row and presence domain, each β_x is a unital $*$ -isomorphism of the corresponding component algebras, and Γ is an isomorphism of the projective completed transition diagrams. For every non-VT parameter q , these maps satisfy

$$\beta_x \sigma_x^r(q) = \sigma_{\kappa(x)}^r(p_x^r q).$$

For VT, p^{VT} sends $(T, \pm)$ to $(\Gamma(T), \pm)$, and the positive-product equations commute with β , Γ , and the inclusions. The restriction comparisons obey their projective coherence and

$$\Gamma_{U,i} \iota_{U,i} = \iota'_{\kappa(U,i)} \beta_{U,i},$$

where $\Gamma_{U,i}$ is the induced endomorphism-algebra map. All recorded component maps in $\mathcal{M}$ commute with the corresponding maps in $\mathcal{M}'$.

Every tuple satisfying these conditions is admitted, whether or not it is induced by a source isomorphism. Componentwise identities, compositions, and inverses make $\mathbf{L}^B$ an independently defined groupoid. Projection to the first coordinate defines

$$U : \mathbf{L}^B \longrightarrow \mathbf{L}, \quad U(\mathcal{T}) = L_0.$$

G.6. Functorial Determination by the Source.

Theorem G.2 (Functorial source determination of the bounded-law data). *The source assignment defines a functor*

$$\mathbb{L} : \mathbf{S}^F \longrightarrow \mathbf{L}^B$$

such that

$$U\mathbb{L} = L$$

on objects and arrows.

Proof. For a source $\mathfrak{R}$, the previously determined field law supplies the first coordinate $L(\mathfrak{R})$. The source-typed represented, local-operation, and transition components give the tagged index $\mathcal{I}_{\mathfrak{R}}^0$. The five declared recipe rows give their parameter sets, presence predicates, and output maps. Their unital norm-closed hulls give the minimal component algebras. The metric transition category, its faithful probes, and its restriction comparisons give $\widehat{\mathbf{C}}_\bullet$; the positive products of the original transitions give the faithful inclusions ι . The available component maps and comparisons give $\mathcal{M}$. The construction above verifies every object axiom of the independently defined target, so $\mathbb{L}(\mathfrak{R})$ is an object of $\mathbf{L}^B$.

Let $f : \mathfrak{R} \rightarrow \mathfrak{R}'$ be a source isomorphism. Its field-law map gives $\ell_f = L(f)$. It transports represented labels, local indices, and transition objects without changing their tags, which gives a strict typed-index isomorphism κ_f . The source isomorphism preserves every row-presence predicate and carries each declared parameter and output to its counterpart. The resulting ambient algebra maps restrict, by minimality, to component-algebra isomorphisms $\beta_{f,x}$.

On the transition categories, the source isomorphism gives a metric $*$ -functor on each uncompleted Hom space. Compatibility of the faithful probes makes it isometric in the probe norm, so it extends uniquely to an isomorphism $\Gamma_{f,U}$ of the completed categories. These extensions intertwine restriction functors and their comparison maps. They carry original transitions to original transitions and satisfy

$$\Gamma_{f,U,i} \iota_{U,i} = \iota'_{fU,\Gamma_f i} \beta_{f,U,i},$$

together with the two transported positive-product equations. Thus

$$\mathbb{L}(f) = (L(f), \kappa_f, \{p_{f,x}^r\}, \{\beta_{f,x}\}, \Gamma_f)$$

is an arrow in $\mathbf{L}^B$.

For the identity source arrow, every component of this tuple is an identity. For composable source isomorphisms f and g , the field-law maps, strict index maps, rowwise parameter maps, component-algebra maps, and completed transition functors satisfy their respective composition laws. Consequently

$$\mathbb{L}(1_{\mathfrak{R}}) = 1_{\mathbb{L}(\mathfrak{R})}, \quad \mathbb{L}(gf) = \mathbb{L}(g)\mathbb{L}(f).$$

The first coordinate of both the object and arrow assignments is L , so projection gives $U\mathbb{L} = L$ strictly. $\square$

G.7. The Bounded-Law Endpoint. Declared continuous bounded operations generate the component C^* -algebras by unital norm closure. A bounded Borel calculus, including spectral projections in a specified representation, is a separate represented attachment unless it is already supplied by a declared recipe. Closed quadratic forms retain their form domains and are not elements of a component C^* -algebra merely because bounded functions of an associated operator are present.

The VH row records declared bounded cylinder operations. The VT row records the positive source and target products of original source-supplied transitions, while the transitions themselves remain morphisms of $\widehat{\mathbf{C}}_\bullet$. No VH generator is declared on the fixed-holonomy torus components considered in this paper. Adding a new bounded cylinder operation, spectral projection, or external tensor observable changes the declared bounded-law data.

The source therefore determines the generated component algebras, completed transition category, and their typed relations before any state, representation choice, extension parameter, detector model, or temporal development is supplied. Those further data determine evaluations or additional structures without changing which bounded operations were generated by the source.

APPENDIX H. IRREDUNDANCY, RESIDUAL SYMMETRY, AND SOURCE COMPARISONS

This appendix supplies the calculations used in Section 10. It first records the downstream roles of the seven source components and the finite incidence certificate for their declared single-deletion test. The normalizer quotient then identifies the residual internal symmetry of the source. Two finite models serve different purposes: a marked transport family isolates the assignment datum, whereas a five-dimensional preservation problem determines one complete infinitesimal stabilizer. Between them, eight pre-source comparisons test the individual roles of the conditions in the definition of a field source.

The marked transport family, five-dimensional preservation model, and pre-source comparison classes are defined in their respective subsections. The field-level source automorphism, source recovery, and source-to-field theorems are the results of Sections 10 and 11.

H.1. Roles and Single-Deletion Incidence. Order the seven components as

$$P = (\mathcal{A}, \Lambda, \text{Der}(\mathcal{A}), \mathcal{E}, \mathcal{N}_g, \mathcal{G}_{\mathcal{N}_g}, \vartheta) .$$

Their representative downstream roles are as follows.

source datum	representative determined role	principal use
$\mathcal{A}$	local quantities, coefficients, and module and tensor operations	Sections 3–8
Λ	localization, restriction, and projective labels	Sections 3, 11
$\text{Der}(\mathcal{A})$	directions of variation and the first-order calculus	Sections 3, 4.1
$\mathcal{E}$	the derivation metric, its torsion-free connection, integration, regularity, and the motion pairing	Sections 3, 4.3
$\mathcal{N}_g$	internal values, metric contractions, adjoints, and the target of the shape field	Sections 4.1–5
$\mathcal{G}_{\mathcal{N}_g}$	the declared internal metric-gauge action	Sections 3, 5
ϑ	the tangential/internal, carrier/constraint, rank/readout, visible/hidden, and law/input assignments	Sections 3, 11

The table gives one representative role for each component; later constructions may use the same component more than once.

The fixed finite test has eight formal dependence profiles $Q_1, \dots, Q_8$. With columns ordered by P , they are

$$\begin{array}{l|l} Q_1 & \mathcal{A}, \Lambda, \text{Der}(\mathcal{A}), \mathcal{E}, \mathcal{N}_g, \mathcal{G}_{\mathcal{N}_g} \\ Q_2 & \mathcal{A}, \Lambda, \text{Der}(\mathcal{A}), \mathcal{N}_g, \vartheta \\ Q_3 & \mathcal{A}, \text{Der}(\mathcal{A}), \mathcal{E}, \mathcal{N}_g, \mathcal{G}_{\mathcal{N}_g}, \vartheta \\ Q_4 & \Lambda, \text{Der}(\mathcal{A}), \mathcal{E}, \mathcal{N}_g, \vartheta \\ Q_5 & \text{Der}(\mathcal{A}), \mathcal{E}, \mathcal{N}_g, \vartheta \\ Q_6 & \mathcal{A}, \Lambda, \text{Der}(\mathcal{A}), \mathcal{E}, \vartheta \\ Q_7 & \Lambda, \text{Der}(\mathcal{A}), \mathcal{E}, \mathcal{N}_g, \mathcal{G}_{\mathcal{N}_g}, \vartheta \\ Q_8 & \mathcal{A}, \vartheta. \end{array}$$

These profiles fix the row order of the finite comparison; the preceding table records the live downstream roles. Writing $j_{ai} = 1$ when the a -th profile uses the i -th component gives

$$J_g = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

H.1.1. *The typed metric-module datum.* The fifth component of P is the pair

$$\mathcal{N}_g = (\mathcal{N}, g_{\mathcal{N}}),$$

and a source isomorphism preserves both its module and its metric. An artificial bookkeeping split of this pair gives the eight columns

$$(\mathcal{A}, \Lambda, \text{Der}(\mathcal{A}), \mathcal{E}, \mathcal{N}, g_{\mathcal{N}}, \mathcal{G}_{\mathcal{N}_g}, \vartheta)$$

and the matrix

$$J_8 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Lemma H.1 (Kernel of the artificial metric-module split). *The fifth and sixth columns of J_8 coincide, and*

$$\ker J_8 = \text{span}\{(0, 0, 0, 0, -1, 1, 0, 0)^T\}.$$

Proof. Merging the two equal columns gives J_g . If $v \in \ker J_8$, then

$$J_g(v_1, v_2, v_3, v_4, v_5 + v_6, v_7, v_8)^T = 0.$$

The first seven rows of J_g have determinant 1, so J_g is injective. Hence

$$v_1 = v_2 = v_3 = v_4 = v_7 = v_8 = 0, \quad v_5 + v_6 = 0.$$

The displayed generator lies in the kernel because the two columns are equal. $\square$

This lemma concerns the typing of one source component and the kernel of one finite incidence map. It neither removes the metric from the source datum nor identifies arbitrary positive metrics on a finite-projective module.

Theorem H.2 (Full-rank and single-deletion incidence). *Within the fixed seven-component architecture and the eight-row test above, J_g has full column rank and every column is nonzero. Every source component therefore has a determined downstream role, and deleting one component removes at least one declared incidence.*

Proof. The determinant-one minor formed by the first seven rows proves full column rank; equivalently,

$$\det(J_g^T J_g) = 15.$$

Every column contains a nonzero entry, so deleting that column removes each incidence marked by one of its nonzero entries. The role table identifies a corresponding downstream use for every component. The conclusion is the declared single-deletion statement within the fixed architecture. $\square$

If an additional datum occurs in none of the eight fixed profiles, its adjoined column is zero and the column space of J_g is unchanged. This observation concerns the fixed incidence test; Section 10 separately determines how the six stated time, signature, coupling, and lift data affect the field law.

H.2. Normalizer Quotients in Standard Real Representations. Let

$$N_G = N_{\text{Aut}(\mathcal{N}_g)}(\mathcal{G}_{\mathcal{N}_g}), \quad D_G = N_G / \mathcal{G}_{\mathcal{N}_g}.$$

For constant fibers with the standard real metric representation, the orthogonal normalizers give

G	N	$O(N)$	D_G
$O(r)$	$\mathbb{R}^r$	$O(r)$	1
$SO(r)$, $r \geq 2$	$\mathbb{R}^r$	$O(r)$	$\mathbb{Z}_2$
$SO(8)$	$\mathbb{R}^8$	$O(8)$	$\mathbb{Z}_2$
$SU(2)$	$\mathbb{R}^4 \simeq \mathbb{H}$	$O(4)$	$SO(3)$
$SU(m)$, $m \geq 3$	$\mathbb{C}_{\mathbb{R}}^m$	$O(2m)$	$U(1) \rtimes \mathbb{Z}_2$
$U(m)$	$\mathbb{C}_{\mathbb{R}}^m$	$O(2m)$	$\mathbb{Z}_2$.

Theorem H.3 (Representatives and continuous residual symmetry). *Orientation reversal represents the quotient for $SO(r)$. Right quaternionic multiplication, modulo its common center with left $SU(2)$, represents the $SO(3)$ residual of the real $SU(2)$ -module. Central complex phases and complex conjugation represent $U(1) \rtimes \mathbb{Z}_2$ for $SU(m)$, while complex conjugation represents the $\mathbb{Z}_2$ quotient for $U(m)$.*

For closed Lie subgroups $\mathcal{G}_{\mathcal{N}_g} \triangleleft N_G$, the quotient D_G is discrete exactly when

$$N_G^0 \subseteq \mathcal{G}_{\mathcal{N}_g}.$$

Equivalently,

$$\mathfrak{n}_G / \mathfrak{g}_{\mathcal{N}_g} = 0.$$

A positive-dimensional subgroup of the centralizer that survives modulo the declared gauge group gives a continuous component of D_G .

Proof. Normality of $SO(r)$ in $O(r)$ gives the determinant quotient. In the quaternionic model,

$$SO(4) \simeq (SU(2)_L \times SU(2)_R) / \{\pm(1, 1)\},$$

and $SU(2)_R$ centralizes the declared left action. Quaternionic conjugation exchanges the two factors, so an orientation-reversing class does not normalize the fixed left factor. Hence

$$N_{O(4)}(SU(2)_L) / SU(2)_L \simeq SU(2)_R / \{\pm 1\} \simeq SO(3).$$

For $m \geq 3$, the real orthogonal normalizers are

$$N_{O(2m)}(SU(m)) = U(m) \rtimes \langle c \rangle, \quad N_{O(2m)}(U(m)) = U(m) \rtimes \langle c \rangle,$$

where c is complex conjugation. Taking the respective quotients gives the two complex rows. Finally, the quotient of closed Lie groups is discrete precisely when its Lie algebra vanishes, which gives the last two conditions. $\square$

For the pairwise nonisomorphic product datum

$$\begin{aligned} G &= U(1) \times SU(2) \times SU(3), & V &= \mathbb{R}^2 \oplus \mathbb{R}^4 \oplus \mathbb{R}^6, \\ O(V) &= O(2) \times O(4) \times O(6), & D_G &= \mathbb{Z}_2 \times SO(3) \times (U(1) \rtimes \mathbb{Z}_2). \end{aligned}$$

The factors have distinct dimensions and representation types, so the normalizer preserves the three blocks and the quotient is the product of the individual rows.

For repeated isomorphic blocks, suppose the source datum admits exactly the independent block normalizers and permutations of the k_α identical copies, with no further multiplicity-space transformations. Under this hypothesis,

$$G = \prod_{\alpha} G_{\alpha}^{k_{\alpha}} \implies D_G = \left(\prod_{\alpha} D_{G_{\alpha}}^{k_{\alpha}} \right) \rtimes \prod_{\alpha} S_{k_{\alpha}}.$$

The permutation class exchanges identical summand blocks. Distinct metric or representation types remove that exchange from the source normalizer.

Declaring $U(m)$ instead of $SU(m)$, or declaring an invariant complex or quaternionic structure as part of the internal datum, absorbs the corresponding continuous centralizer. These choices define different source data and therefore different normalizer quotients.

For $SO(8)$ on its vector module, orthogonal conjugation yields the determinant class displayed in the table. Triality permutes vector and spinor representation types and requires spinor-module data, so it belongs to a different source structure from the single vector-typed module used here.

H.3. A Marked Finite Transport Family. The transport calculation is made on a family whose six scalar coordinates and one assignment coordinate are fixed before the incidence matrix is formed. Let $V = \mathbb{R}^6$, with basis $b_1, \dots, b_6$, and define the unital commutative algebra

$$B = \mathbb{R} \oplus V, \quad (r, v)(s, w) = (rs, rw + sv).$$

Thus V is the square-zero radical of B . Choose

$$a, \lambda \geq 1, \quad 1 \leq d \leq 6, \quad e > 0, \quad n \geq 2, \quad g \geq 3.$$

The algebra datum is

$$\mathcal{A}^{(a)} = \mathbb{R}^a \times B,$$

with the a semisimple factors distinguished from the nonsemisimple factor B . A localization set L_λ of cardinality λ carries each distinguished factor, giving a component-chart incidence set $I_{a,\lambda}$ of cardinality $a\lambda$.

For $v = \sum_j v_j b_j$, define

$$\delta_i(r, v) = (0, v_i b_i)$$

on B , extended by zero on $\mathbb{R}^a$. Since

$$\delta_i((r, v)(s, w)) = (0, (rw_i + sv_i)b_i) = \delta_i(r, v)(s, w) + (r, v)\delta_i(s, w),$$

each δ_i is a derivation. The declared derivation datum is the ordered subspace

$$\mathcal{D}_d = \text{span}_{\mathbb{R}}\{\delta_1, \dots, \delta_d\}.$$

The finite quadratic energy vanishes on the semisimple factors and the scalar part of B , while on V it satisfies

$$\mathcal{E}_e(b_i, b_j) = e\delta_{ij}.$$

Its restriction to the coordinate subspace selected by $\mathcal{D}_d$ has trace de . A fixed positive functional on the semisimple and scalar factors completes this finite energy datum:

$$\omega(x_1, \dots, x_a; r, v) = \sum_{i=1}^a x_i + r.$$

Let p be the identity of the first distinguished $\mathbb{R}$ -factor. The metric-module datum is

$$\mathcal{N}_n = (p\mathcal{A}^{(a)})^n, \quad g_n(u, v) = \sum_{j=1}^n u_j v_j,$$

and the gauge datum is the cyclic orthogonal subgroup

$$\mathcal{G}_g = \langle R_{2\pi/g} \rangle \leq O(n),$$

acting by rotation on the first two coordinates. Its order is g .

Finally, let $\Theta = \{\theta_0, \theta_1\}$, where the two values agree on the preceding six data and exchange the named carrier and constraint roles. Structural equivalence in this family preserves the distinguished algebra factors, chart incidence, ordered derivation and image data, quadratic energy, supported metric module, cyclic gauge subgroup, and assignment. Consequently θ_0 and θ_1 define inequivalent marked objects.

Theorem H.4 (Finite transport obstruction). *On the marked family above, the six evaluations*

$$\mathfrak{T}(a, \lambda, d, e, n, g, \theta) = (a\lambda, a, d, de, n, ng)$$

determine the six scalar incidence coordinates (a, λ, d, e, n, g) . Their multiplicative incidence matrix has rank six and has the assignment coordinate as its one-dimensional kernel. The two inequivalent assignments θ_0 and θ_1 have the same transport data.

Proof. The six entries are, respectively, the cardinality of $I_{a,\lambda}$, the number of distinguished semisimple factors, the dimension of $\mathcal{D}_d$, the trace of the energy on its selected coordinate subspace, the module rank, and the cardinality of $\{1, \dots, n\} \times \mathcal{G}_g$. They give

$$a = T_2, \quad \lambda = T_1/T_2, \quad d = T_3, \quad e = T_4/T_3, \quad n = T_5, \quad g = T_6/T_5.$$

The exponent vectors of these evaluations, in coordinates $(a, \lambda, d, e, n, g, \theta)$, form

$$J_T = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}.$$

The inverse formulas prove that J_T has rank six, and its final column vanishes. Since no evaluation contains θ ,

$$\mathfrak{T}(a, \lambda, d, e, n, g, \theta_0) = \mathfrak{T}(a, \lambda, d, e, n, g, \theta_1).$$

□

The matrix J_T is an exponent-incidence matrix for the six displayed scalar evaluations; the discrete coordinates are not being varied infinitesimally. This finite algebraic family supplies neither six complete source objects nor an object of the analytic field-source class. An assignment-sensitive seventh evaluation would be additional data and would remove this particular kernel. Section 11 instead recovers the source from indexed local algebras, quadratic forms, transition categories, and compatible assignments.

H.4. Comparison-Relative Sharpness of the Source Conditions. Number the eight conditions in the definition of a field source by P1–P8. A pre-source for the following comparisons carries the seven slots

$$(\mathcal{A}, \Lambda, \text{Der}(\mathcal{A}), \mathcal{E}, \mathcal{N}_g, \mathcal{G}_{\mathcal{N}_g}, \vartheta)$$

and the local, cylinder, and transition systems generated functorially from them, while any subset of P1–P8 may fail. Index objects and the algebra, module, or transition data assigned to them remain distinct types.

A structural isomorphism of pre-sources preserves the seven slots, index incidence, restriction functors, Dirichlet form, positive cone, metric-module and gauge data, assignments, generated transition systems, and projective topology. It therefore preserves index-fiber multiplicities, derivation/restriction commutators, symbol rank, positivity on the unit, finite duality, metric invariance, assignment bijectivity, and the kernel of cofinal restriction on objects and arrows.

The principal control is the standard flat source on T^3 , with ordinary smooth restrictions, its elliptic Dirichlet form and positive volume, the trivial oriented metric line, trivial gauge group, and the standard assignments. For P2, use the same data on a two-level star whose top is T^3 and whose leaves are a finite smooth cover. For P8, use the disjoint union $T_v^3 \sqcup T_h^3$, initially with lower-window covers on both components. These controls satisfy P1–P8.

Theorem H.5 (Admitted-source sufficiency and comparison-relative sharpness). *Every $\mathfrak{R} \in \mathbf{S}^F$ satisfies P1–P8 together with the remaining regularity and compatibility hypotheses in the definition of a field source, and Theorem 3.8 determines its field and law.*

Separately, for each $i = 1, \dots, 8$, there is a pre-source in the comparison class described below that satisfies the other seven predicates, fails P_i , and loses the named structural guarantee or acquires the named ambiguity. The eight conclusions are invariant under structural equivalence.

Proof. The first assertion is the field-source definition followed by Theorem 3.8. It uses the full admitted-source hypotheses, including the Sobolev, support, Weyl, and compatible-state conditions stated after P1–P8.

For the second assertion, the eight comparisons and their structural invariants are

	single change	lost structure or acquired ambiguity
P1	duplicate one localization-index object with the same fiber and incidence	unique recovery of the stated index object; invariant: index-fiber multiplicity
P2	replace the derivation restrictions on the two-level star by $R_i(X) = 2X _{U_i}$	restriction-natural first-order calculus; invariant: the derivation/restriction commutator
P3	replace the Dirichlet form by the zero form on $L^2(T^3)$	the nondegenerate derivation metric and its torsion-free metric connection; invariant: symbol rank
P4	replace every local integral by its negative	the positive curvature-energy and motion pairing; invariant: positivity on the unit
P5	replace the metric line by a countable algebraic direct sum	finite-projective duality; invariant: finite generation and dualizability
P6	replace the trivial gauge group by the dilation action $S_t = e^t I$	metric descent of contractions and energy; invariant: metric invariance
P7	send both named modules to the tangential label	the uniquely typed internal target of the shape field; invariant: assignment image size
P8	retain only visible-component restrictions on a two-component source	faithful recovery of a global structural arrow; invariant: the kernel pair of cofinal restriction.

For P1, adjoin a distinct index object U' with the same fiber and restriction incidence as U . Composition remains compatible, and the original cofinal fiber family remains jointly faithful, so P2–P8 retain their control values. The localization description now has a two-element fiber over the description of U . This comparison establishes the stated index-recovery ambiguity; all field-law data on the duplicated presentation remain outside its conclusion.

For P2, retain the standard local graph-dense cores, algebra restrictions $r_i(f) = f|_{U_i}$, and local derivation modules on the two-level star, but set

$$R_i(X) = 2X|_{U_i}.$$

There are no composable pairs of nonidentity arrows, and multiplication by two is continuous and injective. The lower cover therefore remains cofinal and jointly faithful. However,

$$R_i(X)(r_i f) - r_i(Xf) = (Xf)|_{U_i},$$

which is nonzero for suitable X and f . P2 alone fails.

For P3, set

$$\mathcal{E}_0(u, v) = 0, \quad \text{Dom } \mathcal{E}_0 = L^2(T^3).$$

This form is closed, Markovian, strongly local, and regular; its graph norm is the L^2 norm, so the standard smooth core remains dense. Its symbol vanishes, making the induced pairing on nonzero derivations degenerate. All other data retain their control values.

For P4, keep the positive reference measure and Dirichlet form and replace each local functional by

$$\tilde{I}_U(f) = - \int_U f \, dx.$$

All linear restriction and gluing identities remain valid, while $\tilde{I}_U(1) < 0$. A nonzero squared-curvature density consequently has negative integral, so the required positive energy and motion pairing are unavailable.

For P5, take

$$\mathcal{N}_\infty = \bigoplus_{k \geq 1} C^\infty(T^3) e_k, \quad g(x, y) = \sum_k x_k y_k,$$

with coordinatewise restrictions and the locally convex direct-sum topology. Finite cylinders retain the coordinates in $F \in \mathbb{N}$, and their transition categories have diagonal multiplier arrows with componentwise restriction and truncation. Windows and finite cylinders jointly separate all objects and arrows, so P8 remains true. A finite set of module generators has support in only finitely many coordinates, and hence cannot generate a basis vector outside that union. Thus P5 fails, together with the finite dual-basis identity it would supply.

For P6, the maps $S_t = e^t I$ form a continuous group and commute with every restriction, while

$$g(S_t v, S_t v) = e^{2t} g(v, v).$$

The metric module itself remains finite projective and all other predicates retain their control values, but contractions fail to descend through the declared action when $t \neq 0$.

For P7, keep the six nonassignment slots and map both $\text{Der}(\mathcal{A})$ and $\mathcal{N}_g$ to the tangential label. The assignment commutes with restriction but has one-element image, so it supplies no internal label for $\text{Sym}^2 \Omega^1 \otimes \mathcal{N}$. The remaining seven predicates are unchanged.

For P8, begin with

$$\mathcal{A}^{(2)} = C^\infty(T_v^3) \oplus C^\infty(T_h^3)$$

and replace the two-component localization by the star whose leaves cover only T_v^3 . Every local module and transition system is obtained by base change along

$$r_i(f_v, f_h) = f_v|_{U_i^v}.$$

The visible leaves remain a downward-cofinal family and have compatible, distinct index data, so P1 holds. The global metric-module automorphism

$$\tau(n_v, n_h) = (n_v, -n_h)$$

preserves the metric, trivial gauge group, and assignments, yet every visible restriction sends it to the identity. Thus cofinal restriction is not faithful on arrows, whereas P1–P7 retain their stated values. This completes the eight comparisons. $\square$

The comparison theorem concerns the eight fixed classes of pre-sources and the named structures in its proof. Membership in $\mathbf{S}^F$ additionally requires the regularity, support-compatibility, Weyl, and compatible-state hypotheses in the field source definition. The comparison objects are not thereby admitted to that class.

H.5. Infinitesimal Stabilizer of a Finite Source Model. Let

$$V = B \oplus W, \quad B = \mathbb{R}^3, \quad W = \mathbb{R}^2,$$

with ordered bases e_1, e_2, e_3 and f_1, f_2 . The finite source model $\mathfrak{P}$ has the following data. The base algebra is $\mathbb{R}^3$ with

$$m(e_i, e_j) = \delta_{ij}e_i, \quad \mathbf{1} = e_1 + e_2 + e_3,$$

and the product is zero if either argument lies in W . The localization atoms are e_i ; the ordered base frame is

$$d_1 = e_1 - e_2, \quad d_2 = e_2 - e_3.$$

The Dirichlet and integration data are

$$\Delta = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 2 \end{pmatrix}, \quad \ell = (1, 2, 4),$$

the internal metric is I_2 , and the connected declared gauge group is $SO(2)$, with

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The projector $P = 0_B \oplus I_W$ records the tangential/internal assignment.

A competitor $g \in GL(V)$ preserves this finite source precisely when

$$\begin{aligned} (H.1) \quad & gP = Pg, \quad g\mathbf{1} = \mathbf{1}, \quad g m(v, w) = m(gv, gw), \\ & g d_a = d_a, \quad g_B^T \Delta g_B = \Delta, \quad \ell g_B = \ell, \\ & g_W^T g_W = I_2, \quad g_W(\mathbb{R}J)g_W^{-1} = \mathbb{R}J. \end{aligned}$$

The blocks g_B and g_W are consequences of the first equation rather than restrictions on the competitor space.

Let $g(t) = I + tX + O(t^2)$, where the 25 entries of $X = (x_{ij})_{1 \leq i, j \leq 5}$ are ordered row by row, and put $(b_1, \dots, b_5) = (e_1, e_2, e_3, f_1, f_2)$. The infinitesimal normalizer condition is $[X_W, J] \in \mathbb{R}J$. The commutator has equal off-diagonal entries, whereas a multiple of J has opposite off-diagonal entries, so the multiple must vanish. Linearizing (H.1) therefore gives

$$\begin{aligned} (H.2) \quad & [X, P] = 0, \quad X\mathbf{1} = 0, \\ & X m(b_i, b_j) - m(Xb_i, b_j) - m(b_i, Xb_j) = 0 \quad (1 \leq i, j \leq 5), \\ & X d_a = 0, \quad X_B^T \Delta + \Delta X_B = 0, \quad \ell X_B = 0, \\ & X_W^T + X_W = 0, \quad [X_W, J] = 0. \end{aligned}$$

Vectorizing every nonzero scalar equation in (H.2) produces a coefficient matrix $C \in M_{86 \times 25}(\mathbb{R})$. Its rows comprise 12 role-projector equations, 5 unit equations, 39 product equations, 10 frame equations, 9 Dirichlet equations, 3 integration equations, 4 metric equations, and 4 gauge-normalizer equations. Thus (H.2) is a compact construction of every row of C .

Theorem H.6 (Infinitesimal stabilizer of the finite model). *The complete linearized preservation system (H.2) has rank 24, and*

$$\ker C = \{0_B \oplus aJ : a \in \mathbb{R}\}.$$

Its quotient by the embedded declared gauge algebra $\mathfrak{so}(2) = \mathbb{R}J$ is zero-dimensional.

Proof. The role equation first makes X block diagonal. On B , the product equation says that X_B is a derivation of $\mathbb{R}^3$. Since $e_i^2 = e_i$,

$$X_B e_i = 2e_i(X_B e_i).$$

The right side has only an e_i -component, and comparison of that component sets it to zero. Hence $X_B = 0$. The frame, Dirichlet, and integration equations then hold automatically.

The metric equation gives $X_W^T + X_W = 0$, so $X_W = aJ$. Such a matrix also commutes with J , and conversely $0_B \oplus aJ$ satisfies every equation in (H.2). The kernel is therefore one-dimensional. Since C has 25 columns, rank-nullity gives $\text{rank } C = 24$, and quotienting by the same gauge line leaves dimension zero. $\square$

Four exact nonlinear families provide complementary checks:

$$\begin{aligned} g_\sigma &= \Pi_\sigma \oplus I_W, & \sigma &\in S_3, \\ g_{\alpha,\beta} &= \text{diag}(\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2), & \alpha_i, \beta_a &> 0, \\ g_U^+ &= \begin{pmatrix} I_3 & U \\ 0 & I_2 \end{pmatrix}, & g_Z^- &= \begin{pmatrix} I_3 & 0 \\ Z & I_2 \end{pmatrix}, \\ g_A &= I_B \oplus A, & A &\in GL_2(\mathbb{R}) \setminus O(2). \end{aligned}$$

Theorem H.7 (Exact exclusions in the four declared families). *The identity is the only source-preserving locality permutation or positive coordinate rescaling. Every nonzero typed shear and every nonmetric internal map violates an equation in (H.1).*

Proof. For a locality permutation, integration preservation gives

$$\ell \Pi_\sigma = \ell.$$

The entries 1, 2, 4 are distinct, so σ is the identity. For a positive diagonal rescaling, unit preservation gives $\alpha_1 = \alpha_2 = \alpha_3 = 1$, while metric preservation gives $\beta_1^2 = \beta_2^2 = 1$; positivity then gives $\beta_1 = \beta_2 = 1$.

For the shears,

$$[g_U^+, P] = \begin{pmatrix} 0 & U \\ 0 & 0 \end{pmatrix}, \quad [g_Z^-, P] = \begin{pmatrix} 0 & 0 \\ -Z & 0 \end{pmatrix},$$

so the role equation holds exactly when the shear vanishes. Finally,

$$\|A^T A - I_2\|_F^2 > 0$$

for $A \notin O(2)$, which violates the internal metric equation. $\square$

The first theorem classifies the full infinitesimal competitor space of this finite object, while the second concerns exactly the four displayed nonlinear families. An internal reflection preserves the metric and normalizes $SO(2)$, so a disconnected class remains in the exact stabilizer modulo the connected declared gauge group. The field-level source automorphism and normalizer theorems retain the independent arguments of Section 10.

H.6. Consequences for Source Recovery. The seven source components have established roles under the fixed single-deletion test, and the residual internal symmetry is the normalizer quotient D_G . The marked transport family determines

six scalar incidence coordinates and remains insensitive to its assignment coordinate, while the eight pre-source comparisons establish the stated comparison-relative sharpness of P1–P8. The finite preservation model has only the declared connected gauge direction in its infinitesimal stabilizer.

Complete source recovery uses the indexed local algebras, quadratic forms, transition categories, and compatible assignment data of Section 11. Under its regularity and compatibility hypotheses, those data recover the source up to structural equivalence, after which the source determines the field and its law.

APPENDIX I. ANALYTIC AND CATEGORICAL FOUNDATIONS OF SOURCE RECOVERY

Let $\mathcal{B}$ be the class of indexed operational data satisfying conditions (R1)–(R9) of Section 11, specified independently of a source presentation. The recovery proceeds through the assignments

$$(I.1) \quad \begin{array}{ll} \{A_W, r_{VU}\} & \mapsto (\mathcal{A}, \Lambda), \\ \{D_W, Q_W, \tau_W, \Gamma_W, A_W^0\} & \mapsto (\mathcal{E}, \Omega_{\mathfrak{R}}^1, \text{Der}(\mathcal{A})), \\ \{\mathbf{C}_U, \omega_U, \chi_{VU}\} & \mapsto (\mathcal{N}_g, \mathcal{G}_{\mathcal{N}_g}), \\ \{\mu, \rho_\Sigma\} & \mapsto \vartheta. \end{array}$$

The local constructions are first made compatible with restriction and then assemble to an equivalence between the regular source and operational-data groupoids. A second result restricts this equivalence to the branch on which the response domains are closed form domains and lifts it after both sides are given the same compatible carrier-role point. The point belongs to this refinement as supplied data; the unpointed equivalence retains the primitive source recovery displayed in (I.1).

I.1. Local Function Algebras and Restriction Variance. Condition (R9) makes the declared completion A_W of each regular operation core A_W^0 a commutative unital real Banach algebra with the square-norm and invertibility properties stated in Section 11. For a restriction $U \leq V$, write

$$r_{VU} : A_V \longrightarrow A_U.$$

The algebra clause of (R1) makes r_{VU} a unital homomorphism and supplies the Gelfand representation used below.

Lemma I.1 (Operation-algebra representation). *Under (R9) and the algebra and separatedness clauses of (R1), every index W has a compact Hausdorff spectrum X_W and an isometric algebra isomorphism*

$$\kappa_W : A_W \xrightarrow{\cong} C(X_W; \mathbb{R}).$$

Each restriction homomorphism induces a continuous map

$$r_{VU}^\vee : X_U \longrightarrow X_V,$$

and the identities

$$r_{TW} = r_{VW} r_{TV}, \quad r_{TW}^\vee = r_{TV}^\vee r_{VW}^\vee$$

hold whenever $W \leq V \leq T$, with indices renamed according to the displayed variance.

Proof. A commutative unital real Banach algebra whose norm satisfies $\|f^2\| = \|f\|^2$ and in which $1 + f^2$ is invertible for every element is isometrically isomorphic to the real-valued continuous functions on its compact Hausdorff spectrum [32, Theorem 1.2.9, Corollary 1.2.12, and pp. 26–27]. Unital algebra homomorphisms induce continuous maps of spectra in the opposite direction [32, Corollary 1.3.21]. Functoriality of the Gelfand transform gives the composition identities, while the separatedness clause of (R1) supplies the required projective separation. $\square$

I.2. The Canonical Response-Domain Graph. For a positive functional τ_W on A_W , define

$$n_{\tau,W}(a)^2 = \tau_W(a^2), \quad N_{\tau,W} = \{a \in A_W : n_{\tau,W}(a) = 0\}.$$

Positivity applied to $\tau_W((sa + tb)^2)$ gives Cauchy–Schwarz. Moreover,

$$0 \leq \tau_W(a^2 c^2) \leq \|c\|_\infty^2 \tau_W(a^2),$$

so $N_{\tau,W}$ is an ideal. The quotient therefore carries the inner product

$$\langle [a], [b] \rangle_{\tau,W} = \tau_W(ab),$$

whose completion is denoted by $H_{\tau,W}$. Faithfulness of τ_W is unnecessary.

Definition I.2 (Graph-admitting operational data). *An indexed operational datum is graph admitting when, at every W , it has linear maps*

$$\iota_W : A_W^0 \longrightarrow D_W, \quad j_W : D_W \hookrightarrow H_{\tau,W}, \quad j_W \iota_W(a) = [a],$$

where j_W is injective, and the following conditions hold.

First, D_W is complete for

$$(I.2) \quad \|u\|_{D,W}^2 = \|j_W u\|_{H_{\tau,W}}^2 + Q_W(u),$$

and $\iota_W(A_W^0)$ is dense for this norm. Second, every restriction $V \rightarrow U$ induces bounded maps

$$r_{VU}^0 : H_{\tau,V} \longrightarrow H_{\tau,U}, \quad r_{VU}^D : D_V \longrightarrow D_U$$

satisfying

$$r_{VU}^0 j_V = j_U r_{VU}^D$$

and the identity and composition laws. The restrictions are bounded for the graph norms as well as for the Hilbert norms.

Let $\mathbf{B}^D$ be the replete groupoid of graph-admitting objects in $\mathbf{B}$, with operational isomorphisms whose existing response-domain components preserve τ, Q, j and the regular cores. Let $\mathbf{S}^D$ be the full replete subgroupoid of regular sources whose local closed Dirichlet domains have the realization in Definition I.2 and whose restrictions are bounded in both norms.

Lemma I.3 (Closed response form). *For graph-admitting data, the formula*

$$(I.3) \quad \mathfrak{q}_W(j_W u) = Q_W(u), \quad \text{Dom } \mathfrak{q}_W = j_W(D_W),$$

defines a closed nonnegative form on $H_{\tau,W}$. The core form

$$q_W^0([a]) = Q_W(\iota_W a), \quad a \in A_W^0,$$

is closable and its closure is $\mathfrak{q}_W$.

Proof. Injectivity of j_W makes (I.2) a norm. A form-norm Cauchy sequence in $j_W(D_W)$ pulls back to a graph-norm Cauchy sequence in D_W ; completeness gives its limit and proves that (I.3) is closed. The graph of q_W is the closure of the graph of q_W^0 because the regular core is graph dense. Hence the core form is closable and has the stated closure. $\square$

If

$$\tau_U((r_{VU}a)^2) \leq C_{VU}^2 \tau_V(a^2), \quad Q_U(r_{VU}^D u) \leq C_{Q,VU}^2 Q_V(u),$$

then restriction descends through the null ideals and

$$(I.4) \quad \|r_{VU}^D u\|_{D,U}^2 \leq \max(C_{VU}^2, C_{Q,VU}^2) \|u\|_{D,V}^2.$$

Thus the graph domains restrict functorially. This construction depends on the local tuple

$$(A_W^0, A_W, \tau_W, D_W, j_W, Q_W)$$

and its restriction maps.

Theorem I.4 (Graph normalization of the comparison maps). *The source and behavior comparison maps of the regular source–operational equivalence restrict to unique surjective graph isometries on $\mathbf{S}^D$ and $\mathbf{B}^D$. They preserve the positive functionals, quadratic forms, injections, regular cores, and restrictions.*

Proof. For operational data B , let $\kappa_W : A_W \rightarrow C(X_W; \mathbb{R})$ be the Gelfand isomorphism. The measure reconstructed from τ_W satisfies

$$\tau_W(a) = \int_{X_W} \kappa_W(a) d\mu_W.$$

The algebra component κ_W^{-1} of the behavior comparison therefore preserves τ_W and the Hilbert inner product. On the regular core it also preserves Q_W , since the reconstructed core energy is defined from the recorded response form. The core map is consequently a graph isometry and extends uniquely to the completed response domain. Its range is closed because its domain is complete and dense because it contains the target core, so the extension is surjective. The identity with j_W holds first on the core and then on the full graph domain by continuity.

The source comparison uses the inverse Gelfand evaluation map and the same argument. Restriction naturality and naturality under operational or source isomorphisms follow because the two candidate composites agree on every graph-dense core. Uniqueness of graph completion then makes them equal on the full domains. These extensions are the response-domain components of the original comparison maps, so each arrow carries a single map on D_W . $\square$

I.3. Positive Integration and the Generated First-Order Calculus. The positive functional τ_W on $A_W \cong C(X_W; \mathbb{R})$ has a unique finite regular Borel representing measure μ_W by the Riesz–Markov theorem. Condition (R4) relates the quadratic form to the symmetric local square field by

$$(I.5) \quad Q_W(\iota_W f) = \tau_W(\Gamma_W(f, f)) = \int_{X_W} \Gamma_W(f, f) d\mu_W.$$

The symmetric bilinearity of Γ_W gives

$$(I.6) \quad \Gamma_W(f, g) = \frac{1}{4}(\Gamma_W(f + g, f + g) - \Gamma_W(f - g, f - g)).$$

For the regular core A_U^0 , form

$$\Omega_{U,0}^1 = \text{span}\{a d_U b : a, b \in A_U^0\},$$

with pairing

$$(I.7) \quad \left\langle \sum_i a_i d_U b_i, \sum_j c_j d_U e_j \right\rangle_U = \sum_{i,j} a_i^* c_j \Gamma_U(b_i, e_j).$$

Positivity gives Cauchy–Schwarz for this pairing, so its zero-pairing set is an A_U^0 -submodule and the quotient is a pre-Hilbert module. Its completion is denoted by Ω_U^1 .

Theorem I.5 (Restriction-natural generated first-order calculus). *The formula*

$$(I.8) \quad r_{VU}^{(1)}(a d_V b) = r_{VU}(a) d_U(r_{VU} b)$$

descends to the null quotient, extends continuously to $\Omega_V^1 \rightarrow \Omega_U^1$, commutes with the derivations, and satisfies the restriction composition law. Conversely, a completed first-order calculus generated densely by $A_U^0 d_U A_U^0$ with pairing (I.7) is returned up to a restriction-natural unitary module isomorphism.

Proof. Square-field compatibility gives

$$\langle r_{VU}^{(1)} \xi, r_{VU}^{(1)} \zeta \rangle_U = r_{VU} \langle \xi, \zeta \rangle_V.$$

The represented unital algebra restriction is contractive, so restriction sends null vectors to null vectors and is bounded for the quotient norm. Continuity gives the completed map. The derivation and composition identities hold on generators and extend by density. In the reverse direction, the map sending a reconstructed symbol $a d_U b$ to the same element of the declared calculus preserves the pairing and has dense range, hence extends to the required unitary. Formula (I.8) makes these unitaries natural. $\square$

This is the canonical derivation representation in the commutative, strongly local setting of [11, Theorems 8.1–8.3]. On the nondegenerate source branch of Section 3, the declared duality

$$\text{Der}_{\mathcal{A},U} \cong \text{Hom}_{A_U}(\Omega_U^1, A_U)$$

recovers the derivation modules, and the comparisons (I.8) make this dual recovery compatible with restriction.

I.4. Internal Source Recovery and a Supplied Carrier Point. At every index U , let $\mathbf{C}_U$ be the additive, idempotent-complete, rigid symmetric metric tensor category recorded by the transition data and let

$$\omega_U : \mathbf{C}_U \longrightarrow \mathbf{Proj}_g(A_U)$$

be its faithful finite-projective metric probe. Here $\mathbf{Proj}_g(A_U)$ is the category of finite-projective metric A_U -modules and adjointable maps. For $U \leq V$, coefficient base change b_{VU} , transition restriction s_{VU} , and the strong symmetric monoidal metric comparison

$$\chi_{VU} : b_{VU} \omega_V \Longrightarrow \omega_U s_{VU}$$

satisfy the restriction cocycle.

For $f, g : E \rightarrow F$ in $\mathbf{Proj}_g(A_U)$, the finite-projective trace gives the positive Hom-module pairing

$$(I.9) \quad \langle f, g \rangle_{\text{HS}} = \text{Tr}_E(f^*g) \in A_U.$$

At a point of the spectrum this is the ordinary Hilbert–Schmidt inner product, which proves positivity and definiteness. Finite-projective trace commutes with coefficient base change, while composition, tensor product, and duality satisfy the usual trace identities. These structures distinguish the object-metric and monoidal clauses of (R6) from the Hom-module and probe faithfulness clauses of (R7).

Define $\text{Aut}_{\mathbf{I}}^{\otimes,*}(\omega)$ to consist of families $g = (g_{U,X})$ of tensor-natural metric automorphisms satisfying

$$(I.10) \quad \chi_{VU,X} b_{VU}(g_{V,X}) = g_{U,s_{VU}X} \chi_{VU,X}.$$

Theorem I.6 (Indexed metric-gauge reconstruction). *The indexed transition data reconstruct the restriction-compatible gauge datum as*

$$\mathcal{G}_B = \text{Aut}_{\mathbf{I}}^{\otimes,*}(\omega).$$

For the forward data of a regular source, the canonical action map

$$\mathcal{G}_{\mathcal{N}_g} \longrightarrow \mathcal{G}_{F(\mathfrak{R})}$$

is an isomorphism, and conjugation identifies the corresponding normalizer quotients D_G . The Serre–Swan comparison returns the primitive finite-projective metric module in the source round trip.

Proof. Equation (I.10) is closed under products and inverses, and the cocycle for χ makes the definition independent of a refinement chain. Every source gauge transformation acts naturally on the faithful generating family. Its kernel is trivial by faithfulness. At each fiber, compact Tannaka–Krein reconstruction identifies a tensor-natural metric automorphism with a unique source gauge element [26, Section 1, Theorem 20]. Naturality in coefficient multiplication and equation (I.10) make the fiber elements continuous and restriction compatible, so they assemble to a unique source gauge transformation. Conjugation gives the normalizer statement. The finite-projective module comparison is the natural Serre–Swan evaluation [42, Section 50][46, Section 3, Theorem 2]. $\square$

The primitive source recovery above uses the whole indexed probe datum. A particular carrier object requires a separate pointed refinement.

Definition I.7 (Compatible carrier-role point). *A carrier-role point consists of objects and metric comparisons*

$$x_{g,W} \in \mathbf{C}_W, \quad \xi_{VU} : s_{VU}x_{g,V} \xrightarrow{\cong} x_{g,U},$$

obeying the identity and restriction cocycles. Its probe is a positive finite-projective metric module of positive locally constant rank. Exactly one isomorphism class carries the stable type

$$t_g = (\mathbf{N}, \mathbf{C}).$$

For every point p of the local spectrum, the point is fiberwise faithful when

$$(I.11) \quad (g_p)_{x_{g,W},p} = 1 \implies g_p = 1 \quad (g_p \in (\text{Aut}_{\mathbf{I}}^{\otimes,*}(\omega_W))_p).$$

Let $\widehat{\mathbf{B}}^{D,g}$ have objects $(B, \mathfrak{d}, x_g, \xi, t_g)$, where $B \in \mathbf{B}^D$, $\mathfrak{d}$ is its graph lift, and Definition I.7 holds. Its arrows are graph-preserving operational isomorphisms together with metric point isomorphisms preserving ξ and t_g .

Let $\widehat{\mathbf{S}}^{D,g}$ have objects $(\mathfrak{R}, y_g, v, t_g)$, where $\mathfrak{R} \in \mathbf{S}^D$ and y_g is a supplied compatible point in the transition category of $F(\mathfrak{R})$. Source arrows carry metric point isomorphisms with the same coherence and tag preservation.

I.5. Natural Equivalences.

Theorem I.8 (Equivalence of regular sources and indexed data). *The forward and reconstruction functors*

$$F : \mathbf{S} \longrightarrow \mathbf{B}, \quad R : \mathbf{B} \longrightarrow \mathbf{S}$$

are quasi-inverse. There are natural isomorphisms

$$\epsilon : RF \Longrightarrow \text{Id}_{\mathbf{S}}, \quad \eta : FR \Longrightarrow \text{Id}_{\mathbf{B}},$$

and hence

$$(I.12) \quad \mathbf{S} \simeq \mathbf{B}.$$

Proof. For a source $\mathfrak{R}$, Gelfand and Riesz–Markov evaluation return the local algebras and positive integration; the unitary comparison in Theorem I.5 and its nondegenerate dual return the generated first-order and derivation modules; Serre–Swan evaluation and Theorem I.6 return the primitive internal and gauge data; and the structural labels return by identity. These components commute with restriction and define $\epsilon_{\mathfrak{R}} : R(F(\mathfrak{R})) \rightarrow \mathfrak{R}$.

For $B \in \mathbf{B}$, the corresponding evaluation maps return its operation algebras, response domains, quadratic forms, positive functionals, square fields, regular cores, generated first-order modules, transition categories, probes, χ -comparisons, Hom pairings, generating family, structural labels, assignments, and restrictions. They define $\eta_B : F(R(B)) \rightarrow B$. The componentwise naturality of Gelfand, Riesz–Markov, Serre–Swan, the first-order comparison, and the indexed tensor comparison proves naturality of both families. $\square$

Theorem I.9 (Graph-and-point refinement). *The equivalence (I.12) restricts and lifts to*

$$(I.13) \quad \widehat{\mathbf{S}}^{D,g} \simeq \widehat{\mathbf{B}}^{D,g}.$$

The response-domain components are the graph isometries of Theorem I.4, and the point components transport the supplied points through the transition-category components of the base comparisons.

Proof. Retain the graph-normalized behavior comparison $\eta : FR \Rightarrow \text{Id}$. Full faithfulness of F defines a unique natural isomorphism

$$\alpha : \text{Id} \Longrightarrow RF, \quad F(\alpha_X) = \eta_{FX}^{-1}.$$

The identities

$$\eta_{FX} F(\alpha_X) = 1_{FX}, \quad R(\eta_B) \alpha_{RB} = 1_{RB}$$

follow respectively from the definition and from naturality of η , after applying the faithful functor F . Thus (F, R, α, η) is an adjoint equivalence. Put $\epsilon = \alpha^{-1}$; then $F(\epsilon_X) = \eta_{FX}$. Theorem I.4 shows that these comparisons preserve the graph data.

For a pointed behavior (B, x_g) , define the reconstructed point by

$$(I.14) \quad y_{g,W} = (\eta_{B,W}^C)^{-1}(x_{g,W}).$$

Pulling ξ and t_g through the same transition-category comparison gives the source point restrictions and tag. Naturality of η defines the construction on arrows. In the behavior round trip, η_B^C sends (I.14) back to x_g . In the source round trip, the identity $F(\epsilon_X) = \eta_{FX}$ sends the pulled-back point to the original source point. These point comparisons and the graph-normalized base comparisons give the two natural isomorphisms in (I.13). $\square$

On the full source subgroupoid for which y_g is the transition representation of the primitive $\mathcal{N}_g$, the returned probed module is isomorphic to $\mathcal{N}_g$. This is the canonical-carrier forward subgroupoid.

Theorem I.10 (Carrier-point rank obstruction). *There is an object of $\widehat{\mathbf{B}}^{D,g}$ whose primitive internal module has rank one and whose supplied carrier-role point probes to a rank-two module.*

Proof. Use one index and

$$A^0 = A = H_\tau = D = \mathbb{R}, \quad \tau(a) = a, \quad j = 1, \quad Q = 0,$$

with primitive $\mathcal{N}_g = \mathbb{R}$, trivial admitted gauge group, transition category $\mathbf{Proj}_g(\mathbb{R})$, and identity probe. Its tensor automorphism group is trivial because the tensor-unit component is fixed and naturality with all maps $\mathbb{R} \rightarrow V$ fixes every vector. The same unpointed behavior therefore admits the uniquely tagged supplied point $x_g = \mathbb{R}^2$, which satisfies (I.11). All graph and restriction conditions hold at the singleton index. Reconstruction leaves the primitive source module at rank one, while $\omega(x_g)$ has rank two. Rank is preserved by finite-projective metric-module isomorphisms, so $\omega(x_g) \not\cong \mathcal{N}_g$. $\square$

I.6. Comparison-Relative Necessity of the Regularity Conditions. The comparison theorem is formulated in a pre-regular universe before its witnesses are chosen. An object contains every primitive component of the indexed operational datum together with an injective map $j_W : D_W \rightarrow H_{\tau,W}$, declared restriction families, and a supplied point with its comparisons and carrier-role type. Isomorphisms preserve all these components.

In this universe the exact conditions are as follows.

- (R1) Restrictions are oriented from a record to its declared restriction, preserve the cores and unital algebra products, commute with j , are continuous for the declared norms, satisfy identity and composition, and are jointly injective on every declared restriction family.
- (R2) Every D_W is complete for (I.2), restrictions are graph bounded, and Q_W is Markov under normal unit contractions.
- (R3) The image $\iota_W(A_W^0)$ is graph dense in D_W , and the cores restrict compatibly.
- (R4) The τ_W are positive and restriction bounded on their null quotients, while the Γ_W are symmetric, positive, local, polarized, restriction compatible, and satisfy (I.5).
- (R5) The pairing (I.7) has a coefficient submodule as its null space, and its restriction-natural completion is the entire declared Ω_W^1 .

- (R6) Probe objects are finite-projective positive metric modules, and object metrics, adjoints, tensor products, duals, evaluations, coevaluations, unitary strong-monoidal restriction comparisons, and the supplied point are preserved.
- (R7) Hom modules carry the positive structure (I.9), the probe is faithful on morphisms, and the separately declared observed family generates under finite sums, tensor products, duals, and idempotent summands.
- (R8) Structural labels and assignments are well typed, restriction compatible, and bijective, including the distinction between fixed laws and variable inputs and the carrier-role type.
- (R9) Every completed core is a commutative unital real function algebra with the square-norm and strict-reality properties used in Lemma I.1.

For $i = 1, \dots, 8$, let

$$(I.15) \quad \mathbf{C}_i = \{B : B \models (R_9), \quad B \models \bigwedge_{j \neq i} (R_j)\}.$$

This definition contains no reconstruction conclusion. The conclusion tested below is membership in the essential image of the enhanced forward functor in Theorem I.9.

Three controls will be used repeatedly. At a singleton index, take

$$(I.16) \quad A^0 = A = D = H_\tau = \mathbb{R}, \quad \tau(a) = a, \quad j = 1, \quad Q = \Gamma = 0, \quad \Omega^1 = 0.$$

For a real function algebra A , take the tensor control

$$(I.17) \quad \mathbf{C} = \mathbf{Proj}_g(A), \quad \omega = \text{Id}, \quad x_g = A,$$

with the pairing (I.9). The identity probe is faithful, the unit generates every finite-projective module by sums and idempotent summands, and its tensor automorphism group is trivial. Finally, use distinct tangential and internal labels with typed bijections on the carrier/constraint, rank/readout, visible/hidden, and law/input roles. These controls satisfy all conditions relevant to their respective components.

Theorem I.11 (Necessity on the fixed comparison classes). *For every $i = 1, \dots, 8$, there is an object $B_i \in \mathbf{C}_i$ that fails (Ri) and lies outside the enhanced forward essential image. Hence each of (R1)–(R8) is individually necessary for membership in that image relative to the fixed classes (I.15).*

Proof. For (R1), let $U \leq V$ with one restriction $V \rightarrow U$, and set

$$A_V^0 = A_V = D_V = H_{\tau,V} = \mathbb{R}^2, \quad A_U^0 = A_U = D_U = H_{\tau,U} = \mathbb{R},$$

with coordinatewise products,

$$r_{VU}(a, b) = a, \quad \tau_V(a, b) = a + b, \quad \tau_U(c) = c,$$

and $j = 1$, $Q = \Gamma = 0$. Use zero first-order modules and the controls (I.17). Finite dimensionality proves (R2)–(R3), positivity and $\tau_U((r(a, b))^2) \leq \tau_V((a, b)^2)$ prove (R4), and the remaining controls prove (R5)–(R9). The nonzero element $(0, 1)$ restricts to zero, so the declared restriction family is not jointly injective. Every forward object is projectively separated, an invariant under enhanced isomorphism.

For (R2), let X be the closed unit disk and take

$$A = C(X; \mathbb{R}), \quad A^0 = C^\infty(\overline{X}; \mathbb{R}), \quad H_\tau = L^2(X),$$

$$D = H^1(X) \cap C(X), \quad Q(f) = \int_X |\nabla f|^2 dx, \quad \tau(f) = \int_X f dx, \quad \Gamma(f, g) = \nabla f \cdot \nabla g.$$

The generated first-order module is $C(X; \mathbb{R}^2)$, since $v_1 dx_1 + v_2 dx_2$ represents every smooth vector field and smooth fields are uniformly dense. The other conditions follow from the singleton tensor and structural controls, smooth graph-core density, positivity, and the Sobolev chain rule for normal contractions. The radial function

$$u(r) = \log \log(e/r)$$

belongs to $H^1(X) \cap L^2(X)$ and has no continuous representative. Its truncations $u_n = \min(u, n)$ lie in D and converge to u in the graph norm. Thus D is incomplete. Closed graph domains characterize the enhanced forward image and are preserved by graph isometry.

For (R3), take

$$A = C(S^1; \mathbb{R}), \quad A^0 = C^\infty(S^1), \quad H_\tau = L^2(0, 1), \quad D = H^1(0, 1),$$

with $Q(f) = \int_0^1 |f'|^2 dt$, $\tau(f) = \int_0^1 f dt$, and $\Gamma(f, g) = f'g'$. The one-form module is $C(S^1)dt$; indeed, for $s(t) = \sin(2\pi t)$ and $c(t) = \cos(2\pi t)$,

$$\frac{c}{2\pi} ds - \frac{s}{2\pi} dc = dt.$$

The remaining conditions follow from the standard controls. The continuous trace $T(f) = f(1) - f(0)$ vanishes on A^0 , so the graph closure of the core is $H^1(S^1)$, a proper subspace of D witnessed by $f(t) = t$. Graph-core density is an invariant of the forward image.

For (R4), take one index with

$$A^0 = A = D = H_\tau = \mathbb{R}^2, \quad \tau(a, b) = a + b, \quad Q(a, b) = (a - b)^2, \quad \Gamma = 0.$$

The zero generated first-order module and the common controls establish every other condition. Normal contractions decrease Q , while for $f = (1, 0)$,

$$Q(f) = 1, \quad \tau(\Gamma(f, f)) = 0.$$

The core energy/square-field identity holds throughout the forward image and is preserved by enhanced isomorphism.

For (R5), use the analytic control (I.16) and declare instead

$$\Omega^1 = \mathbb{R}e, \quad \langle ae, be \rangle = ab, \quad d(a) = 0.$$

Every other condition is unchanged. Since $A^0 dA^0 = 0$, its completion is zero and cannot equal the declared nonzero module $\mathbb{R}e$. The generated span is preserved by operational isomorphisms.

For (R6), take two indices $U \leq V$, use the analytic and structural controls at both, and set

$$\mathbf{C}_U = \mathbf{C}_V = \mathbf{Proj}_g(\mathbb{R}), \quad \omega_U = \omega_V = \text{Id}, \quad x_{g,U} = x_{g,V} = \mathbb{R}.$$

All restriction functors are identities, while the natural probe comparison is

$$\chi_{VU,X} = 2 \mathbf{1}_X.$$

The Hom pairings, probe faithfulness, generating family, point comparison, and all analytic data satisfy (R1)–(R5) and (R7)–(R9). The comparison χ_{VU} is neither unitary nor monoidal, since

$$\chi_{VU,X \otimes Y} = 2 \mathbf{1}_{X \otimes Y}, \quad \chi_{VU,X} \otimes \chi_{VU,Y} = 4 \mathbf{1}_{X \otimes Y}.$$

Unitary strong-monoidal comparison is invariant under enhanced isomorphism.

For (R7), use one index and

$$\mathbf{C} = \mathbf{Proj}_g(\mathbb{R}) \times \mathbf{Proj}_g(\mathbb{R}), \quad \omega(X, Y) = X, \quad x_g = (\mathbb{R}, \mathbb{R}).$$

The category has componentwise object metrics and tensor structure, while its Hom pairing is

$$\langle (f_1, f_2), (g_1, g_2) \rangle = \text{tr}(f_1^* g_1) + \text{tr}(f_2^* g_2).$$

This pairing is positive definite and the supplied point generates after finite sums and independent idempotent splittings. The object-level probe data satisfy (R6). For $Y = (0, \mathbb{R})$, however,

$$h = (0, 1_{\mathbb{R}}) \neq 0, \quad \omega(h) = 0.$$

Thus the probe is not faithful; the displayed Hom pairing also lacks tensor multiplicativity, both defects belonging to (R7). Probe faithfulness is invariant under equivalence and holds in every forward object.

For (R8), use the common controls and retain every structural bijection except the law/input assignment on $\{L, I\}$, which is

$$\rho(L) = L, \quad \rho(I) = L.$$

All analytic and categorical conditions remain true, while this assignment is not injective. Image cardinality is preserved by structural isomorphism, and every forward object keeps the two roles distinct.

Each witness lies in its class $\mathbf{C}_i$ and is excluded by the stated isomorphism invariant. This proves the theorem. $\square$

The theorem concerns the fixed classes (I.15) and the enhanced forward essential image. Condition (R9) is retained in every witness. The result assigns no necessity statement to (R9) and no minimality statement to other possible recovery architectures.

APPENDIX J. EXPLICIT CURVATURE, SPECTRAL, AND OPERATOR CALCULATIONS ON THE THREE-TORUS

Section 9 presents five principal field configurations and six groups of conclusions. This appendix follows the order used there. The first three calculations exhibit simultaneous connection and shape responses at internal rank two, a one-mode rank-one gravitational response, and a two-coordinate rank-one response followed through its constraint and selected observables. The next two calculations distinguish fixed-holonomy constraint data with $K = 0$ from the global response spectrum of a flat coupled configuration with $K \neq 0$. The final calculation treats separately declared graph and finite-operator data. A decaying-harmonic connection profile supplies its stated comparison family.

J.1. Simultaneous Gauge and Shape Responses at Internal Rank Two.

Let $T^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$, $V = (2\pi)^3$, and let the oriented trivial normal plane have orthonormal frame (e_1, e_2) and $Je_1 = e_2$, with $\|J\|_{\text{HS}}^2 = 2$. On the following two-parameter family of configurations, take

$$(24) \quad A = (\alpha_1 dx^1 + \alpha_2 dx^2 + \alpha_3 dx^3 + a \sin x^1 dx^2)J, \quad K = k dx^1 \otimes dx^1 \otimes e_1.$$

Theorem J.1 (Simultaneous nonzero response components at internal rank two). *For (24), $B_T = 0$ and the positive curvature-energy functional is*

$$(25) \quad \Phi(a, k; \alpha) = V k^2 \left(\alpha_2^2 + \alpha_3^2 + \frac{a^2}{2} \right) + \frac{V}{2} a^2.$$

The restriction of the configuration-space metric to this family is $V da^2 + V dk^2$, so

$$(26) \quad \mathcal{Y}_a = a(1 + k^2), \quad \mathcal{Y}_k = 2k \left(\alpha_2^2 + \alpha_3^2 + \frac{a^2}{2} \right),$$

and

$$(27) \quad \frac{\partial^2 \Phi}{\partial a \partial k} = 2Vak.$$

The quantities $\mathcal{Y}_a$ and $\mathcal{Y}_k$ are the Riesz coordinates of the pullbacks of the ambient connection and shape responses to this two-parameter slice. At $\alpha_2 = 1/4$, $\alpha_3 = 0$, $a = k = 1$, the two response values are 2 and $9/8$. Thus one compact rank-two model has simultaneously nonzero ambient connection and shape responses and a nonzero mixed second derivative.

Proof. The abelian normal curvature is $B_\perp = a \cos x^1 dx^1 \wedge dx^2 J$. The only mixed terms are, up to two-form antisymmetry,

$$B_K(e_2, e_1)e_1 = k(\alpha_2 + a \sin x^1)e_2, \quad B_K(e_3, e_1)e_1 = k\alpha_3 e_2.$$

The single shape endomorphism has zero antisymmetrized quadratic tangent block. Substitution into the doubled mixed-block norm and use of $\int \sin x^1 dV = 0$ and $\int \sin^2 x^1 dV = \int \cos^2 x^1 dV = V/2$ gives (25). Differentiation followed by taking the metric dual with respect to the restricted configuration-space metric gives (26)–(27) and the named values. $\square$

Because the internal rank is two, the second component is a shape response. Its identification as a gravitational response belongs to the oriented rank-one class with positive Riemannian base metric in Theorem 6.3.

J.2. A One-Mode Rank-One Gravitational Response. On the same flat three-torus, replace the rank-two normal bundle by an oriented metric normal line with unit section n and trivial normal connection. Let

$$\kappa = \sin(x^2) dx^1 \otimes dx^1 \otimes n, \quad e = \frac{\kappa}{\|\kappa\|_G},$$

where G is the configuration-space metric restricted to shape variations, and set

$$K(q) = qe, \quad \lambda = 2\Phi(e) > 0.$$

The tangent and normal curvature components vanish, while the mixed curvature is linear in q . Since $G(e, e) = 1$, the restricted curvature energy and its metric-dual response are

$$\Phi(q) = \frac{\lambda}{2} q^2, \quad \mathcal{Y}_K(q) = \lambda q.$$

The response is nonzero for $q \neq 0$. The internal gauge algebra is $\mathfrak{so}(1) = 0$, and on this oriented positive-Riemannian rank-one realization the surviving shape response is the gravitational response.

Let $\mathcal{H} = L^2(\mathbb{R}, dq)$. On the Schwartz core $\mathcal{S}(\mathbb{R})$, define

$$(Q\psi)(q) = q\psi(q), \quad (P\psi)(q) = -i\psi'(q), \quad C_0 = \frac{1}{2}(P^2 - \lambda Q^2).$$

The finite-label response identity is

$$i[C_0, P] = \lambda Q$$

on this common core. A linear symplectic change takes C_0 to a positive multiple of the dilation generator, and its metaplectic implementer preserves Schwartz space. Thus C_0 is essentially self-adjoint; write $C = \overline{C_0}$.

For the simultaneous constraint-and-response statement, set

$$\mathcal{T}_0\psi = (C_0\psi, \lambda Q\psi), \quad \mathcal{T} = \overline{\mathcal{T}_0}, \quad m[\psi] = \|\mathcal{T}\psi\|^2$$

on the closed row domain $\text{Dom } \mathcal{T}$. The form m has zero kernel. By contrast, C^2 has continuous spectral support at zero. Continuous constraint support is therefore compatible with a nonzero gravitational response, although the constraint and response have no simultaneous normalizable zero vector on this label.

J.3. Curvature, Constraint, and a Bounded Commutator in a Rank-One Field. Let

$$T^3 = (\mathbb{R}/2\pi\mathbb{Z})^3, \quad h = \sum_{i=1}^3 d\theta^i \otimes d\theta^i, \quad d\mu = (2\pi)^{-3} d\theta^1 d\theta^2 d\theta^3.$$

Thus $\int_{T^3} 1 d\mu = 1$. Let $L = T^3 \times \mathbb{R}n$ be an oriented metric line with $g_L(n, n) = 1$ and trivial normal connection. Set

$$S = d\theta^2 \otimes d\theta^2 - d\theta^3 \otimes d\theta^3, \quad K_{q,x} = (qh + x \sin(\theta^1)S) \otimes n.$$

Denote this family by

$$\iota(q, x) = (\nabla^{\mathcal{N}} = d, K_{q,x}).$$

The flat base, normal line, trivial normal connection, and displayed family $K_{q,x}$ determine the curvature, energy, response, signed constraint, characteristic vector field, and Gauss–Codazzi contractions below. The Gaussian projection is supplied separately in the represented operator algebra, and the relational observable additionally requires a clock, a clock reading, and a transverse point. Write $r = x \sin \theta^1$, so the matrix of the shape tensor is $\text{diag}(q, q+r, q-r)$ in the flat orthonormal frame $e_i = \partial_{\theta^i}$.

The assembled curvature has tangent, mixed, and normal blocks

$$B_T(e_i, e_j)e_\ell = -k_{j\ell}k_i^\sharp + k_{i\ell}k_j^\sharp, \quad B_K(e_i, e_j)e_\ell = (\partial_i k_{j\ell} - \partial_j k_{i\ell})n, \quad B_\perp = 0.$$

The independent nonzero tangent values are

$$\begin{aligned} B_T(e_1, e_2)e_1 &= q(q+r)e_2, & B_T(e_1, e_2)e_2 &= -q(q+r)e_1, \\ B_T(e_1, e_3)e_1 &= q(q-r)e_3, & B_T(e_1, e_3)e_3 &= -q(q-r)e_1, \\ B_T(e_2, e_3)e_2 &= (q^2 - r^2)e_3, & B_T(e_2, e_3)e_3 &= -(q^2 - r^2)e_2, \end{aligned}$$

and the independent nonzero mixed values are

$$B_K(e_1, e_2)e_2 = x \cos \theta^1 n, \quad B_K(e_1, e_3)e_3 = -x \cos \theta^1 n.$$

All remaining values follow by two-form antisymmetry or vanish. Consequently,

$$\|B_T\|^2 = 6q^4 + 2x^4 \sin^4 \theta^1, \quad \|B_K\|^2 = 2x^2 \cos^2 \theta^1, \quad \|B_\perp\|^2 = 0.$$

The doubled mixed-block weight gives

$$\phi = 3q^4 + x^4 \sin^4 \theta^1 + 2x^2 \cos^2 \theta^1, \quad \Phi(q, x) = \int_{T^3} \phi d\mu = 3q^4 + x^2 + \frac{3}{8}x^4.$$

The restriction of the configuration-space metric to this two-parameter family is $3dq^2 + dx^2$. The restriction of the metric-dual response to its tangent plane is therefore

$$\iota^*\mathcal{Y}_K = 4q^3 \partial_q + \left(2x + \frac{3}{2}x^3\right) \partial_x.$$

These are the two tangent components of the full response along the displayed family. In canonical coordinates,

$$H_R = \frac{p_q^2}{6} + \frac{p_x^2}{2} - 3q^4 - x^2 - \frac{3}{8}x^4,$$

with characteristic equations

$$X_{H_R}q = \frac{p_q}{3}, \quad X_{H_R}x = p_x, \quad X_{H_R}p_q = 12q^3, \quad X_{H_R}p_x = 2x + \frac{3}{2}x^3.$$

Use normalized coordinates

$$Q_q = \sqrt{3}q, \quad Q_x = x, \quad P_q = \frac{p_q}{\sqrt{3}}, \quad P_x = p_x.$$

Before evaluating the constraint commutator, fix the normalized Gaussian vector and its rank-one projection

$$\gamma(Q_q, Q_x) = \pi^{-1/2} e^{-(Q_q^2 + Q_x^2)/2}, \quad P_\gamma = |\gamma\rangle\langle\gamma|.$$

The projection is a separately specified operator in the fixed represented algebra $\mathfrak{B}_{\lambda_2}$, and it generates $C^*(1, P_\gamma)$. On the Schwartz core of the two-mode representation, symmetric ordering gives

$$\widehat{C}_{\lambda_2} = \frac{1}{2}(\widehat{P}_q^2 + \widehat{P}_x^2) - \frac{1}{3}\widehat{Q}_q^4 - \widehat{Q}_x^2 - \frac{3}{8}\widehat{Q}_x^4.$$

The generator Δ_{λ_2} is implemented there by

$$\begin{aligned} i[\widehat{C}_{\lambda_2}, \widehat{Q}_q] &= \widehat{P}_q, & i[\widehat{C}_{\lambda_2}, \widehat{Q}_x] &= \widehat{P}_x, \\ i[\widehat{C}_{\lambda_2}, \widehat{P}_q] &= \frac{4}{3}\widehat{Q}_q^3, & i[\widehat{C}_{\lambda_2}, \widehat{P}_x] &= 2\widehat{Q}_x + \frac{3}{2}\widehat{Q}_x^3. \end{aligned}$$

The projection preserves the relevant core and adjoint-core domains, and its commutator extends to a bounded finite-rank operator. Hence P_γ belongs to the bounded first-commutator domain $\mathfrak{A}_{\lambda_2}^1$, relative to the represented algebra $\mathfrak{B}_{\lambda_2}$ and the displayed common core. This is a represented-domain conclusion; membership in a source-generated algebra $\mathfrak{V}_x^0$ would require P_γ to occur among its declared generators. Theorem 7.14 identifies the bounded commutator with the restriction of the common-core derivation that implements Δ_{λ_2} . Direct calculation on the full two-mode representation gives

$$\|[\widehat{C}_{\lambda_2}, P_\gamma]\| = \sqrt{\frac{475}{96}}.$$

The calculation uses the full two-mode representation without truncating occupation number, together with the closure of the displayed symmetric constraint.

The scalar Gauss and one-form Codazzi contractions on the positive-Riemannian rank-one branch are

$$\mathfrak{s}(B_T) = -6q^2 + 2x^2 \sin^2 \theta^1, \quad \mathfrak{v}(B_K) = 0.$$

The one-form contraction vanishes because its divergence index contracts no nonzero derivative of the diagonal shape tensor; the mixed component B_K nevertheless remains nonzero when $x \cos \theta^1 \neq 0$. The internal metric gauge algebra is $\mathfrak{so}(1) = 0$, so this oriented line model has no nontrivial infinitesimal internal gauge response.

On the local region $p_q \neq 0$, choose $T = q$ and a reading τ . The equation $q(\varphi_s z) = \tau$ has a unique local solution $s = s(z, \tau)$, and

$$O_x(\tau)(z) = x(\varphi_{s(z, \tau)}(z))$$

is invariant under the characteristic vector field on that chart. Its leading form is

$$s(z, \tau) = \frac{3(\tau - q)}{p_q} + O((\tau - q)^2), \quad O_x(\tau) = x + \frac{3p_x}{p_q}(\tau - q) + O((\tau - q)^2).$$

Here s is the parameter of the characteristic flow of H_R . On the fixed representation,

$$M = \widehat{C}_{\lambda_2}^2, \quad b = M(1 + M)^{-1}$$

is a nonscalar represented Master function. Separately, the bounded continuous function

$$h : \mathbb{R}^2 \longrightarrow (-1, 1), \quad h(u, \eta) = \frac{u}{1 + |u|}$$

gives the nonscalar bounded Gauss–Codazzi observable

$$h(\mathfrak{s}(B_T), \mathfrak{v}(B_K)).$$

The first element establishes nonvacuity of the represented Master-functional generator family, and the second establishes nonvacuity of the Gauss–Codazzi-functional generator family at this worked-model scope.

J.4. Constraint Spectrum and Holonomy on the Three-Torus. Let $M = T^3 = \mathbb{R}^3 / (2\pi\mathbb{Z})^3$ have its flat metric h , standard volume form, and flat Dirichlet form

$$\mathcal{E}(f) = \int_{T^3} \|df\|_h^2 dV_h.$$

Let $N = T^3 \times \mathbb{R}^2$ with the Euclidean metric, and let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

generate the standard $SO(2)$ -action. Put $\widehat{J} = J/\sqrt{2}$, so $\|\widehat{J}\|_{\text{HS}} = 1$. Choose a fixed holonomy class $[\alpha] \in (\mathbb{R}/\mathbb{Z})^3$, with at least one nonzero component, and a metric normal connection

$$\nabla^{\mathcal{N}} = d + A, \quad A = (\alpha_1 dx^1 + \alpha_2 dx^2 + \alpha_3 dx^3 + a \sin x^1 dx^2)J,$$

with $a \neq 0$. Set $K = 0$. Then

$$F^{\mathcal{N}} = dA = a \cos x^1 dx^1 \wedge dx^2 J,$$

so the curvature energy is finite and nonzero on every finite Fourier label containing this background mode. The holonomies around the coordinate circles include $\exp(2\pi\alpha_i J)$, and therefore determine nontrivial transition data. The declared generator maps contain no bounded holonomy cylinder function, so the holonomy contributes no generator to the source-generated bounded observable algebra.

The constant rotations $\exp(tJ) \subset SO(2)$ commute with the chosen connection and form a nontrivial source gauge subgroup. Together with the nontrivial fundamental group $\pi_1(T^3) \cong \mathbb{Z}^3$, the displayed holonomies supply the transition data associated with this model.

The full $SO(2)$ gauge group contains the winding transformations $u_k(x) = \exp((k_i x^i)J)$, which identify $\alpha_i \sim \alpha_i + k_i$ for $k \in \mathbb{Z}^3$. Set

$$\alpha_{\text{harm}} = \sum_{i=1}^3 \alpha_i dx^i J.$$

Flatness and harmonicity give

$$d\alpha_{\text{harm}} = 0, \quad d^* \alpha_{\text{harm}} = 0, \quad d^* d \alpha_{\text{harm}} = 0.$$

Thus the harmonic form lies in $\ker d^* d$. The holonomy class $[\alpha] \in (\mathbb{R}/\mathbb{Z})^3$ is held fixed as a parameter, while the Schrödinger representation is formed on the positive spectral subspace

$$\mathfrak{h}_+ = d^* \Omega^2(T^3; \mathbb{R} \hat{J}).$$

For $n \geq 1$, let

$$F_n = \{k \in \mathbb{Z}^3 : |k_i| \leq n\}$$

and let E_n be the real coexact Fourier span of its nonzero modes. These coexact cubic Fourier labels form a nested exhaustive family and contain the background mode

$$A_b = a \sin x^1 dx^2 J = \sqrt{2} a \sin x^1 dx^2 \hat{J}.$$

On E_n , the stiffness is $L_n = d^* d > 0$. The covariance used for the spectral-measure calculation is

$$C_{qq,n} = L_n^{-1}, \quad C_{pp,n} = L_n/4, \quad C_{qp,n} = 0.$$

It defines the particular quasifree state used in that calculation; this choice of state is additional represented data. The covariance is invariant under the constant rotations, so their source action has a quasifree unitary implementation in this representation. The configuration-space metric of Theorem 4.9 gives the signed constraint

$$H_n(q, p) = \frac{1}{2} \|p\|^2 - \frac{1}{2} \langle q + A_b, L_n(q + A_b) \rangle.$$

Expanding the translated quadratic form gives the background affine and scalar terms. Let $C_{0,n}$ be its symmetric/Weyl quantization on the Schwartz core $\mathcal{S}(E_n) \subset L^2(E_n)$, and set

$$C_n = \overline{C_{0,n}}, \quad M_n = C_n^2.$$

Theorem J.2 (Master spectrum in the compact torus model). *Fix the holonomy class $[\alpha]$. For every coexact cubic Fourier label E_n , $n \geq 1$,*

$$\sigma(M_n) = \sigma_{\text{ess}}(M_n) = [0, \infty), \quad \ker M_n = \{0\}.$$

The spectrum of C_n is $\mathbb{R}$ and is purely absolutely continuous. For the chosen cyclic vector, the scalar spectral measure of M_n is equivalent to Lebesgue measure on $(0, \infty)$. Thus $0 \in \sigma(M_n)$, while M_n has no normalizable zero eigenvector.

Set

$$b_n = M_n(1 + M_n)^{-1}, \quad A_n = C^*(1, b_n).$$

Thus A_n is the continuous functional-calculus algebra of M_n . The functional calculus gives faithful normal isomorphisms

$$\rho_n : L^\infty([0, 1], m) \xrightarrow{\cong} W^*(M_n), \quad \rho_n(f) = f(b_n),$$

where m is Lebesgue measure and the functions are represented modulo null sets. For $n \leq N$, define the spectral-coordinate comparison isomorphism

$$r_{Nn} = \rho_n \rho_N^{-1} : W^*(M_N) \xrightarrow{\cong} W^*(M_n).$$

It is the unique normal unital $*$ -isomorphism carrying b_N to b_n . The canonical character $\delta_{0,n}(f(b_n)) = f(0)$ has compatible singular extensions ω_n to $W^*(M_n)$, and

$$r_{Ln} = r_{Nn} \circ r_{LN} \quad (n \leq N \leq L),$$

and

$$\begin{aligned} \omega_N &= \omega_n \circ r_{Nn}, & \omega_n(b_n) &= 0, \\ \omega_n(E_{\epsilon,n}) &= \omega_n(E_{\epsilon,n}^2) = 1, & \omega_n(P_{\ker M_n}) &= 0. \end{aligned}$$

Proof. Appendix E translates the exact Weyl polynomial to a finite sum of dilation generators by a metaplectic transformation. The half-line logarithmic transform and Mellin transform give the full essential spectrum and the scalar spectral measure of the chosen quasifree state. The faithful maps $\rho_n : L^\infty([0, 1], m) \rightarrow W^*(M_n)$ determine the unique normal comparison isomorphisms carrying b_N to b_n , together with the compatible family of singular state extensions. $\square$

The fixed field data and represented constraints determine M_n , b_n , the character of A_n , and the spectral-coordinate comparison maps r_{Nn} . Extensions of that character to states on $W^*(M_n)$ are additional represented data and may be nonunique.

The operators C_n are not restrictions of a single operator on the vacuum-containing finite-particle cylinder domain, because the squared norm of the resulting two-particle component diverges. The cross-label represented structure is instead given by the von Neumann algebras $W^*(M_n)$, the elements b_n , and the spectral-coordinate comparison maps r_{Nn} .

J.5. Holonomy Dependence of the Curvature Hessian on the Flat Three-

Torus. We prove Theorem 9.1. Let $M = \mathbb{T}^3$ have period 2π , and let $N = \mathbb{T}^3 \times \mathbb{R}^3$ have oriented orthonormal frame (e_1, e_2, e_3) . Write (t_1, t_2, t_3) for the parallel tangent frame. For orthonormal vectors u, v , let $J_{uv}u = v$, $J_{uv}v = -u$, and let it vanish on the orthogonal complement. Put

$$J = J_{e_2 e_3}, \quad A_\theta = \theta dx^1 J, \quad K_k = k dx^1 \otimes dx^1 \otimes e_1,$$

where $\theta \in \mathbb{R}/\mathbb{Z}$ and $k \neq 0$.

All coefficients of A_θ are constant multiples of J , and $Je_1 = 0$. Every nonzero map $(K_k)_X$ is a multiple of the same rank-one map. Consequently

$$F_{A_\theta} = 0, \quad qquad D_{A_\theta} K_k = 0, \quad qquad \mathcal{F}^\oplus(A_\theta, K_k) = 0, \quad qquad \Phi(A_\theta, K_k) = 0.$$

Each member of the family is critical. On a contractible region, $u_\theta(x) = \exp(\theta x^1 J)$ gauges A_θ to zero and fixes K_k . Globally,

$$\text{Hol}_{A_\theta}(\gamma_1) = \exp(2\pi\theta J),$$

and this gauge is single valued only for integral θ .

The line $L_K = \mathbb{R}e_1$ is preserved by A_θ , and the assembled connection has the orthogonal parallel splitting

$$TM \oplus N = (TM \oplus L_K) \oplus L_K^\perp.$$

The first summand is the oriented rank-one assembled connection. Orthogonal splitting of the energy and motion metric, followed by uniqueness of the Riesz representative, gives

$$P_K \mathcal{Y}_K^N = \mathcal{Y}_K^{L_K}.$$

Section 6 identifies the right-hand side as the gravitational response. Since $\partial_k K_k = dx^1 \otimes dx^1 \otimes e_1$, the parameter k is a gravitational configuration direction. The complementary rank-three shape directions retain their shape-response designation.

To compute the Hessian on the two nontrivial J -eigenspaces, complexify $E_K = L_K^\perp$ and choose q_σ with $Jq_\sigma = \sigma i q_\sigma$, $\sigma = \pm 1$. Define

$$B = J_{t_1 e_1}, \quad M_\sigma = J_{e_1 q_\sigma}, \quad S_{\ell, \sigma} = J_{t_\ell q_\sigma}.$$

Direct action on the basis gives

$$[B, M_\sigma] = -S_{1, \sigma}, \quad [B, S_{1, \sigma}] = M_\sigma, \quad [B, S_{2, \sigma}] = [B, S_{3, \sigma}] = 0.$$

An arbitrary variation in the σ -eigenspace has coefficients

$$a = \sum_i a_i dx^i M_\sigma, \quad \kappa = \sum_{i,j} \kappa_{ij} dx^i \odot dx^j q_\sigma, \quad \kappa_{ij} = \kappa_{ji}.$$

For a Fourier mode, put $p_i = n_i + \sigma \theta_i$, $d_i = ip_i$, with $(\theta_1, \theta_2, \theta_3) = (\theta, 0, 0)$. Direct linearization of the assembled curvature gives, for $i < j$,

$$(28) \quad m_{ij} = d_i a_j - d_j a_i + k(\delta_{i1} \kappa_{j1} - \delta_{j1} \kappa_{i1}),$$

$$(29) \quad s_{ij}^1 = d_i \kappa_{j1} - d_j \kappa_{i1} - k(\delta_{i1} a_j - \delta_{j1} a_i),$$

$$(30) \quad s_{ij}^\ell = d_i \kappa_{j\ell} - d_j \kappa_{i\ell}, \quad \ell = 2, 3.$$

These twelve coefficients contain all three connection and all six symmetric shape variables. With the configuration-space and curvature norms,

$$\|(a, \kappa)\|_G^2 = 2 \sum_i |a_i|^2 + \sum_i |\kappa_{ii}|^2 + 2 \sum_{i < j} |\kappa_{ij}|^2,$$

and the flat-background Hessian is

$$H((a, \kappa), (a, \kappa)) = 2 \sum_{i < j} \left(|m_{ij}|^2 + \sum_{\ell=1}^3 |s_{ij}^\ell|^2 \right).$$

An infinitesimal internal-gauge parameter ξM_σ gives

$$a_i = -d_i \xi, \quad \kappa_{11} = k\xi, \quad \kappa_{ij} = 0 \quad ((i, j) \neq (1, 1)),$$

which vanishes under (28)–(30). Thus the gauge quotient acts on the combined connection–shape variation space.

In the input order

$$(a_1, a_2, a_3, \kappa_{11}, \kappa_{12}, \kappa_{13}, \kappa_{22}, \kappa_{23}, \kappa_{33}),$$

the vector

$$w(d, k) = \begin{pmatrix} -d_1(k^2 + d_1^2), -d_1^2 d_2, -d_1^2 d_3, 0, \\ -kd_1 d_2, -kd_1 d_3, -kd_2^2, -kd_2 d_3, -kd_3^2 \end{pmatrix}$$

also vanishes under the complete linearized curvature map. Whenever it is nonzero, it is not tangent to the gauge orbit because its κ_{11} component vanishes, whereas a nonzero gauge vector has $\kappa_{11} = k\xi \neq 0$. When $w(d, k) = 0$, the vector with $a_1 = 1$ and all other entries zero supplies a nonzero kernel class in the gauge quotient. Every Fourier sector therefore contains a non-gauge zero direction, and

$$\dim \ker L_{\theta, k} = \infty.$$

Let $\mathcal{L}_{\theta, k}$ denote the linearized curvature map on the smooth Fourier core. If $v_r \rightarrow 0$ in the kinetic norm and $\mathcal{L}_{\theta, k} v_r \rightarrow z$ in the curvature norm, integration by parts gives

$$\langle z, w \rangle = \lim_r \langle v_r, \mathcal{L}_{\theta, k}^\dagger w \rangle = 0$$

for every smooth test field w . Thus $\mathcal{L}_{\theta, k}$ is closable. Its graph closure vanishes on the closure of the infinitesimal gauge directions, and

$$Q_{\theta, k}([v]) = \|\overline{\mathcal{L}}_{\theta, k} v\|^2$$

is a densely defined closed nonnegative form on the corresponding Hilbert-space quotient. The first representation theorem gives the nonnegative self-adjoint operator $L_{\theta, k}$. Its infinite-dimensional zero eigenspace implies that the full resolvent is noncompact and $\text{Tr}(e^{-tL_{\theta, k}}) = \infty$ for every $t > 0$.

The rank-one tangential factor of K_k determines the line $\mathbb{R} dx^1$, and hence $\eta = dx^1$ up to sign. Haar averaging over translations in the orthogonal x^2, x^3 directions defines

$$P_\eta v = \int_{\mathbb{T}^2} T_y v dy.$$

The background coefficients, configuration-space metric, gauge map, and linearized curvature commute with these translations. The projector consequently descends to the Hilbert-space quotient, reduces $L_{\theta, k}$, and selects the Fourier modes $(n, 0, 0)$.

On this range set $p = n + \sigma\theta$. The nonzero part of the complete linearized map is

$$\begin{aligned} (m_{12}, s_{12}^1) &= (ipa_2 + k\kappa_{12}, -ka_2 + ip\kappa_{12}), \\ (m_{13}, s_{13}^1) &= (ipa_3 + k\kappa_{13}, -ka_3 + ip\kappa_{13}), \\ s_{12}^2 &= ip\kappa_{22}, & s_{12}^3 &= ip\kappa_{23}, \\ s_{13}^2 &= ip\kappa_{23}, & s_{13}^3 &= ip\kappa_{33}. \end{aligned}$$

All $(2, 3)$ outputs vanish. The variables (a_1, κ_{11}) do not appear and supply two zero directions, one tangent to the gauge orbit and one non-gauge. The remaining map consists of two identical blocks

$$C(p, k) = \begin{pmatrix} ip & k \\ -k & ip \end{pmatrix}, \quad C(p, k)^\dagger C(p, k) = \begin{pmatrix} p^2 + k^2 & -2ipk \\ 2ipk & p^2 + k^2 \end{pmatrix},$$

together with three tensor directions of eigenvalue $2p^2$. After the gauge-orbit zero line is quotiented and the non-gauge zero line retained, the nonzero spectrum is

$$\frac{\lambda}{m(\lambda)} \left| \begin{array}{ccc} 2p^2 & (p-k)^2 & (p+k)^2 \\ 3 & 2 & 2. \end{array} \right.$$

Additional zeros at $p = 0$ or $p = \pm k$ remain in the spectrum.

For $t > 0$, the bounded continuous function $h_t(x) = xe^{-tx}$ vanishes on the entire kernel. The trace on one J -eigenspace and one Fourier mode is

$$6p^2e^{-2tp^2} + 2(p-k)^2e^{-t(p-k)^2} + 2(p+k)^2e^{-t(p+k)^2}.$$

Summing over $n \in \mathbb{Z}$ and $\sigma = \pm 1$ is absolutely and locally uniformly convergent, since the summand is bounded on compact parameter sets by $C(1+n^2)e^{-cn^2}$. Hence $P_\eta h_t(L_{\theta,k})$ is trace class, and its trace may be differentiated termwise in θ , k , and t .

The Gaussian Poisson identity and its t -derivative give

$$\sum_{n \in \mathbb{Z}} c(n+a)^2 e^{-tc(n+a)^2} = \sum_{m \in \mathbb{Z}} \mathcal{A}_c(m, t) e^{2\pi i m a},$$

where $\mathcal{A}_c$ is the coefficient stated in Theorem 9.1. Applying this identity to the three spectral families yields the formula for $\Xi_{\theta,k}(t)$ in that theorem. Its zero-winding term is independent of θ and k . The two nonzero-winding factors are the conjugacy-invariant characters

$$\begin{aligned} \frac{1}{2} \operatorname{Tr}_{E_K} \operatorname{Hol}_{A_\theta}(\gamma_1)^m &= \cos(2\pi m \theta), \\ \frac{1}{2} \operatorname{Tr}_{\operatorname{span}\{t_1, e_1\}} \operatorname{Hol}_{TM \oplus L_K}(\gamma_1)^m &= \cos(2\pi m k). \end{aligned}$$

It remains to exclude cancellation in the complete trace. At $t = 1$ and $m \neq 0$, both $\mathcal{A}_1(m, 1)$ and $\mathcal{A}_2(m, 1)$ are negative, and direct substitution gives

$$\frac{|\mathcal{A}_1(m, 1)|}{|\mathcal{A}_2(m, 1)|} = 2\sqrt{2} \frac{\pi^2 m^2 - 1/2}{\pi^2 m^2 - 1} e^{-\pi^2 m^2/2} < \frac{3}{4}.$$

Consequently

$$6\mathcal{A}_2(m, 1) + 8\mathcal{A}_1(m, 1) \cos(2\pi m k) < 0.$$

Absolute convergence then gives

$$\Xi_{0,k}(1) - \Xi_{1/2,k}(1) = 2 \sum_{\substack{m \in \mathbb{Z} \\ m \in 2\mathbb{Z}+1}} [6\mathcal{A}_2(m, 1) + 8\mathcal{A}_1(m, 1) \cos(2\pi m k)] < 0.$$

Taking $k = 1/6$ proves the stated global distinction.

Finally, termwise differentiation gives

$$\partial_k \Xi_{\theta,k}(t) = -16\pi \sum_{m \in \mathbb{Z}} m \mathcal{A}_1(m, t) \cos(2\pi m \theta) \sin(2\pi m k).$$

At $k = 1/6, t = 1$, its first Fourier coefficient is nonzero. Uniqueness of the absolutely convergent Fourier series and continuity show that the derivative is nonzero on a nonempty open subset of the holonomy circle. Let

$$W_m^{\theta k} = 8\mathcal{A}_1(m, t) \cos(2\pi m \theta) \cos(2\pi m k),$$

the mixed holonomy contribution to the m -th Fourier coefficient. At $m = 1$ and $\theta = k = 1/6$,

$$\frac{W_m^{\theta k}}{8\mathcal{A}_1(m, t)} = \frac{1}{4}, \quad \frac{\partial_k W_m^{\theta k}}{-16\pi m \mathcal{A}_1(m, t)} = \frac{\sqrt{3}}{4}.$$

This completes the proof.

J.6. Graph-Domain Ideals and Finite Operator Families. The remaining calculations use additional graph generators or a separately declared finite operator family on fixed untruncated Fock representations. They prove the ideal, transport, kernel, lower-bound, and membership statements used in Section 9.6.

J.6.1. The fixed-label graph ideal. Fix a full finite-mode label x before occupation-number truncation, its Fock carrier $\mathcal{H}_x$, and its self-adjoint represented constraint A_x . Put

$$\delta_x(T) = i[A_x, T], \quad \|T\|_{1,x} = \|T\| + \|\delta_x(T)\|$$

whenever the commutator extends boundedly. Let Σ_x^1 be the adjoint-closed family consisting of the rank-one occupation projections, configuration multipliers with bounded commutator, bounded functions of A_x^2 , represented complete observables commuting with A_x , and compact-gauge averages with graph-norm-continuous orbits. Define

$$\begin{aligned} \mathcal{A}_{1,x} &= \overline{\text{Alg}^*(\Sigma_x^1)}^{\|\cdot\|_{1,x}}, & \mathfrak{V}_x &= C^*(\mathcal{A}_{1,x}), \\ D_x &= C^*(\mathfrak{V}_x, \delta_x(\mathcal{A}_{1,x})), & I_x &= \text{Ideal}_{D_x}\{P_\nu : \nu \in \mathbb{N}_0^{d_x}\}, \\ J_x &= \text{Ideal}_{D_x}\{\delta_x(T)^* \delta_x(T) : T \in \mathcal{A}_{1,x}\}. \end{aligned}$$

The commutator of every declared generator lies in I_x . For an occupation projection,

$$\delta_x(P_\nu) = \delta_x(P_\nu)P_\nu + P_\nu\delta_x(P_\nu) \in I_x.$$

In the constant-semidefinite Schrödinger coordinates of this finite-dimensional label, write

$$A_x = -\frac{1}{2}g_x^{rs}\partial_r\partial_s + V_x(q), \quad g_x = g_x^T \geq 0.$$

A configuration multiplier M_φ with bounded commutator satisfies

$$[A_x, M_\varphi]\psi = -(g_x\nabla\varphi) \cdot \nabla\psi - \frac{1}{2}\text{div}(g_x\nabla\varphi)\psi.$$

Localized oscillatory tests force $g_x\nabla\varphi = 0$, and the commutator vanishes. Spectral calculus gives $[A_x, f(A_x^2)] = 0$, while a represented complete observable satisfies $[A_x, R_f] = 0$ by its defining relation. If a compact gauge group strongly commutes with A_x , normalizes I_x , and has graph-continuous orbit, then

$$\delta_x\left(\int_G U_g T U_g^* dg\right) = \int_G U_g \delta_x(T) U_g^* dg \in I_x.$$

The two-sided ideal property handles algebraic words, and graph-norm convergence handles the closure. Hence $J_x \subseteq I_x$.

For the reverse inclusion, choose an effective interaction edge $\{r, s\}$ with pair coefficient $-a_e/4$, where $a_e = a \neq 0$ for the single-harmonic profile and $a_e = a_M \neq 0$ on the fixed decaying-profile label λ_M . For every occupation vector $|\nu\rangle$, the untruncated pair-creation term has the component

$$(1 - P_\nu)A_x P_\nu |\nu\rangle \supset -\frac{a_e}{4}\gamma_{\nu,rs}|\nu + e_r + e_s\rangle, \quad \gamma_{\nu,rs} > 0.$$

Therefore

$$\eta_\nu := \|(1 - P_\nu)A_x P_\nu\|^2 > 0, \quad P_\nu \delta_x(P_\nu)^* \delta_x(P_\nu) P_\nu = \eta_\nu P_\nu.$$

Every occupation projection belongs to J_x . If Q_L is an increasing finite sum of these projections, then $Q_L \rightarrow 1$ strongly and $\|T - Q_L T Q_L\| \rightarrow 0$ for every compact T . It follows that

$$J_x = I_x = D_x \cap \mathcal{K}(\mathcal{H}_x).$$

The carrier is infinite dimensional and D_x is unital, so the common ideal is proper. If $XI_x = 0$, then $XP_\nu = 0$ for every occupation projection, and their strong completeness gives $X = 0$; the ideal is essential. A central projection acting as its unit would fix every P_ν , hence would equal the identity and contradict properness. The ideal is therefore not centrally complemented.

J.6.2. Selected-family transport and the interaction-crossing obstruction. For selected subfamilies $S_x \subseteq \Sigma_x^1$ and $S_y \subseteq \Sigma_y^1$, form the graph algebras, bounded envelopes, response coefficient algebras, and response ideals by replacing Σ_x^1 and Σ_y^1 with S_x and S_y . Let $W : \mathcal{H}_y \rightarrow \mathcal{H}_x$ satisfy

$$W \text{Dom } A_y = \text{Dom } A_x \cap W\mathcal{H}_y, \quad A_x W = W A_y.$$

Suppose every $s \in S_x$ has a partner $r_W(s) \in S_y \cup \{0\}$ such that

$$sW = W r_W(s), \quad s^* W = W r_W(s)^*.$$

On the corresponding response coefficient algebras, define

$$\rho_W^D : D_{x,S_x} \rightarrow D_{y,S_y}, \quad \rho_W^D(X) = W^* X W.$$

The partner relations make $W\mathcal{H}_y$ reducing for every algebraic word in the selected generators, so compression is multiplicative on those words. The constraint relation gives

$$W^* \delta_x(T) W = \delta_y(W^* T W), \quad \|W^* T W\|_{1,y} \leq \|T\|_{1,x}.$$

Compression extends to the graph closure and, by operator-norm continuity, to the bounded envelope and response coefficient algebra. It sends every response-square generator to the corresponding response square on the smaller label, and hence

$$\rho_W^D(J_{x,S_x}) \subseteq J_{y,S_y}.$$

This is containment for the selected subfamilies; it gives no ideal equality for a proper selected subfamily.

The natural Fourier-mode inclusion fails these hypotheses across an interaction edge. Let a_j, a_j^* be the annihilation and creation operators and set

$$c_j = a_j + a_j^*.$$

If a refinement adds a mode s adjacent to a retained mode r , the vacuum embedding is $W\psi = \psi \otimes \Omega_s$, while the fine constraint contains $-ac_r c_s/4$. Thus

$$A_{F'} W\psi - W A_F \psi \supset -\frac{a}{4} (c_r \psi) \otimes a_s^* \Omega_s.$$

For $c_r \psi \neq 0$, this component is nonzero and orthogonal to $W\mathcal{H}_F$. The vacuum range is neither invariant nor reducing, and $A_{F'} W \neq W A_F$. The vacuum embedding therefore supplies no graph-norm homomorphism of the required kind.

J.6.3. *Finite commutator families and their kernels.* Write C for the fixed represented constraint in the finite-family calculation. Let

$$\mathcal{F} = \{p_a : a \in A\} \dot{\cup} \{z_b : b \in Z\},$$

where the p_a are mutually orthogonal occupation projections and $[C, z_b] = 0$. Define the commutator Gram operator and the two common kernels by

$$\begin{aligned} G &= \sum_{h \in \mathcal{F}} [C, h]^* [C, h], \\ K^{(1)} &= \bigcap_{h \in \mathcal{F}} \ker[C, h], \\ K^{(\infty)} &= \bigcap_{w \in \text{Alg}^*(\mathcal{F})} \ker[C, w]. \end{aligned}$$

For the single-harmonic connection, the symbol-level off-diagonal constraint couples e_1 -adjacent Fourier modes with amplitude $a/2$. Under that basis normalization, the nonzero spectrum of the symbol-level Gram operator lies in $[a^2/4, \infty)$ at every finite truncation, with equality at the smallest truncation. On the full finite-mode Fock representation, use

$$q_j = \frac{a_j + a_j^*}{\sqrt{2}}.$$

Each classical Fourier coefficient $a/2$ contributes a pair-creation matrix element of size $a/4$, and the number-preserving free term gives the Fock-family lower bound $a^2/8$. The values $a^2/4$ and $a^2/8$ therefore belong to different, explicitly stated normalizations.

For comparison, replace the single-harmonic connection by

$$f(x^1) = \sum_{m \geq 1} \frac{a_m}{m} \sin(mx^1), \quad a_m = a 2^{-(m-1)}, \quad a \neq 0.$$

Its curvature is

$$\sum_{m \geq 1} a_m \cos(mx^1) dx^1 \wedge dx^2 J.$$

On the sparse two-mode sets $\lambda_M = \{0, M e_1\}$, only the M -th harmonic couples the two modes, and G is exactly $a_M^2/2$ times the identity. Its nonzero eigenvalue tends to zero at the rate $4^{-(M-1)}$. Each individual nonzero-amplitude label has positive detection strength, but the sparse family has no uniform positive lower bound. Every cubic cutoff containing all modes up to its cutoff retains the symbol-level lower bound $a^2/4$ because it contains the first harmonic. Finite collections of harmonics with a common nonzero amplitude likewise retain a uniform lower bound, while polynomially decaying amplitudes give the corresponding polynomial decay rate.

The common kernels follow from a direct finite-dimensional calculation. Let $\{e_j\}$ be the distinguished basis, $P_j = |e_j\rangle\langle e_j|$, and let O be the off-diagonal part of C . For $x = \sum_j x_j e_j$,

$$\sum_j \|[C, P_j]x\|^2 = \|Ox\|^2 + \sum_j \|Oe_j\|^2 |x_j|^2, \quad \bigcap_j \ker[C, P_j] = \text{span}\{e_j : Oe_j = 0\}.$$

Every specified single-harmonic family has a nonzero adjacent coupling at each basis vector. On λ_M , both norms $\|Oe_j\|$ equal $|a_M|/2 > 0$, and the same nonvanishing

holds for the specified finite occupation families. Hence

$$K^{(1)} = \{0\}.$$

The entries z_b commute with C and leave this calculation unchanged. Since $\mathcal{F}$ is contained in its algebraic $*$ -algebra, $K^{(\infty)} \subseteq K^{(1)} = \{0\}$; the reverse inclusion is trivial. Consequently

$$K^{(1)} = K^{(\infty)} = \{0\}$$

for every named nonzero-amplitude family. The uniform single-harmonic lower bound and the decaying-profile loss of a uniform lower bound do not change this fixed-label kernel statement.

J.6.4. Declared membership, norm closure, and graph closure. It remains to separate the membership roles used in the two operator-family theorems. On an untruncated finite-mode Fock carrier, let $P_\nu = |\nu\rangle\langle\nu|$ be the occupation projections and let N be the total number operator. For

$$k_n = \frac{(-1)^{\lfloor n/2 \rfloor}}{n+1},$$

define

$$K_{k,L} = \sum_{|\nu| \leq L} k_{|\nu|} P_\nu, \quad L \geq 2.$$

Suppose the declared finite family contains the projections occurring in this sum and that each remaining entry commutes with C . Then $K_{k,L}$ belongs to the algebraic, graph, norm-linear, operator-system, and C^* -generated families determined by that list. It is a finite linear combination of several occupation projections and hence is not one of the distinguished atoms. If $\{r, s\}$ is an effective pair channel with coefficient $-a/4$, then

$$|\langle a_r^* a_s^* \Omega, [C, K_{k,L}] \Omega \rangle| = \frac{|a|}{3} > 0.$$

Both $CK_{k,L}$ and $K_{k,L}C$ extend to finite-rank operators, so this commutator is bounded and finite rank. The operator $K_{k,L}$ is therefore absent from both the distinguished atoms and the declared zero-commutator entries. The declared finite list and each of its generated hulls have different membership.

The infinite diagonal

$$K_k = k(N)$$

separates operator-norm closure from graph closure. Every total-number eigenspace is finite dimensional and $k_n \rightarrow 0$, so K_k is compact and

$$\|K_k - K_{k,L}\| = \sup_{n > L} |k_n| \rightarrow 0.$$

It belongs to the operator-norm-closed linear span, and hence to the C^* -algebra, generated by the occupation projections.

The number-preserving terms of C commute with K_k . Pair creation changes total number from n to $n+2$, for which

$$|k_{n+2} - k_n| = \frac{2(n+2)}{(n+1)(n+3)}, \quad \sqrt{(\nu_r+1)(\nu_s+1)} \leq \frac{n+2}{2}.$$

The product of these factors is at most $4/3$. Pair annihilation obeys the same bound because

$$|k_n - k_{n-2}| = \frac{2n}{(n-1)(n+1)}, \quad \sqrt{\nu_r \nu_s} \leq \frac{n}{2}.$$

Only finitely many mode pairs occur, and $[C, K_k]$ therefore extends boundedly.

To see that this commutator is not compact, let e_m be the occupation vector with occupations m in the two modes r, s and zero elsewhere, and let f_m be obtained by increasing those two occupations by one. The vectors e_m form an orthonormal sequence, while

$$|\langle f_m, [C, K_k] e_m \rangle| = \frac{|a|}{4}(m+1) \left(\frac{1}{2m+1} + \frac{1}{2m+3} \right) \rightarrow \frac{|a|}{4}.$$

Thus $[C, K_k]$ and $[C, K_k]^*[C, K_k]$ are noncompact. With $C = A_x$ on this fixed label, the same sequence gives

$$\liminf_{L \rightarrow \infty} \|\delta_x(K_k - K_{k,L})\| \geq \frac{|a|}{4}.$$

The operator-norm convergence of the finite diagonals is therefore not graph-norm convergence.

These calculations give the three separations used in Theorem 9.4. The norm-generated C^* -algebra contains K_k , while admitting it to a graph family whose response-square ideal is the compact occupation ideal would place the noncompact operator $[C, K_k]^*[C, K_k]$ in that ideal. Every generated family containing the specified atoms contains $K_{k,L}$, while the declared finite family excludes it. The interaction-crossing vacuum inclusion also fails to intertwine the constraints. No single membership family in the stated four-role competitor class has all four roles on these representations.

The response calculations exhibit the field law and its rank dependence. Their quantum, spectral, graph, finite-family, Gaussian, and relational conclusions use the additional represented or relational data stated in each model.

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